

Chapter 4

The Ontological View of Generative Relational Theory

Chapter 4 begins Part II by making explicit the ontological view that underlies Generative Relational Theory (GR) and, through it, Generative Relational Economics (GRE). Part I established the historical conditions under which GRE becomes salient, reconstructed the intellectual lineages from which it draws, and formulated its preliminary disciplinary constitution. Those tasks repeatedly relied on distinctions among objects, relations, processes, representations, subjects, values, institutions, and emergent forms of reality. The present chapter develops the ontological architecture required to use those distinctions with greater precision. Its purpose is neither to establish a final metaphysics nor to reduce heterogeneous domains to a single relational vocabulary. It is to clarify what kinds of reality GR and GRE can speak about, under what commitments, and with what degree of ontological restraint.

The central architecture developed in this chapter distinguishes three registers: *ontological reality*, *relational reality*, and *manifestational reality*. The distinction is analytical rather than a division of the world into three independent substances or three successive stages. Ontological reality concerns the question of what obtains or occurs at the level of reality whose ultimate constitution may remain partly or wholly outside the commitments required by a particular inquiry. Relational reality concerns objects that emerge within generative-relational systems and acquire sufficient persistence and stability over some region or interval of spacetime to exercise causal efficacy within those systems. Manifestational reality concerns the conditions under which something becomes available in a mode of appearance, experience, articulation, measurement, classification, symbolization, or other manifestation. These registers identify different questions concerning reality and thereby allow existence, relational efficacy, manifestation, knowledge, and representation to remain analytically distinguishable.

Relational reality occupies a particularly important position in GR. An object can become relationally real without the framework first establishing that it possesses an independent or fundamental ontological existence. Money, legal status, institutional authority, socially sustained identities, forms of trust, experienced continuity of self, and objects embedded in religious or cultural relations may acquire persistence, stability, and causal efficacy through the systems in which they emerge. Their relational

reality concerns how they are generated and sustained within those systems and how, once stabilized, they participate in the generation of subsequent relations. This claim leaves their ultimate ontological status open where stronger commitment is unnecessary. Relational reality is therefore neither weakened reality nor a synonym for collective belief. It names a distinct analytical status grounded in generative emergence, persistence, stability, and causal participation.

Relational reality also need not await perception. A relational-generative structure may already persist and exercise causal efficacy before a subject becomes aware of it, before it acquires perceptual salience, or before a symbolic vocabulary has been developed through which it can be articulated. This feature becomes important for later GR inquiries into contingency, aesthetic judgment, subjectivity, historical emergence, and the formation of meaning. A contingent configuration, for example, may possess a relational and historical structure before its contingency becomes manifest to a subject and before an aesthetic judgment concerning it occurs. Manifestation can therefore disclose something whose relevant generative conditions precede the event of disclosure. Neither conscious recognition nor symbolization is a general precondition for relational reality.

The distinction between relational and manifestational reality nevertheless does not establish a one-directional sequence. Manifestations can themselves enter generative-relational systems. Once an appearance, judgment, classification, measurement, declaration, image, symbol, institutional record, or other manifestation becomes operative within a system, it may alter expectations, practices, classifications, capacities, constraints, and subsequent relations. Manifestation may thereby contribute to the stabilization, transformation, or dissolution of relational realities. A declaration of trust can alter the relation through which trust continues to develop; a monetary inscription can participate in reproducing monetary relations; a legal classification can reorganize the institutional relations to which it refers; and a manifested self-description can affect later recognition, conduct, and self-formation. Manifestational reality therefore participates in generation rather than occupying only an observational surface outside it.

For this reason, the three registers are better understood through a graph of heterogeneous relations than through a chain of successive layers. Ontological conditions may participate in the emergence of relational structures. Relational structures may condition what can become manifest. Manifestations may reorganize relational structures and, through historically accumulated effects, alter the conditions encountered by later processes. Relations between the registers can be recursive, many-to-many, temporally extended, and partly non-invertible. No universal sequence from ontology to relation to manifestation is presupposed. Particular trajectories may exhibit such an ordering locally, but the architecture itself permits feedback, mediation, historical transformation, and multiple paths among the registers.

A limited heuristic analogy can be drawn with the three-register architecture associated with Lacanian thought. Ontological reality can recall some of the problems posed through the Real, relational reality some functions associated with the Imaginary, and manifestational reality some aspects of the Symbolic. The correspondence is deliberately incomplete. GR's ontological reality is not equivalent to the Lacanian Real; relational reality extends beyond image and identification; and manifestational

reality includes perceptual, affective, bodily, material, institutional, statistical, and technological manifestations that cannot be reduced to symbolic inscription. The useful analogy lies principally in refusing a linear hierarchy among registers and in attending to their mutual mediation. The relation among the registers matters more here than any attempted one-to-one identification among their names.

A further consequence concerns comparison. The fact that two objects satisfy the conditions for relational reality does not establish that their generative structures are commensurable. They may occupy a common metric space, different metric spaces connected through an explicitly constructed mapping, non-metric spaces admitting another form of structural comparison, or relational domains for which no warranted common comparison structure has yet been established. Shared reality-status therefore does not imply shared structure. A perceptual object generated under conditions ordinarily described as hallucinatory and a perceptual object associated with an independently physical object may both become relationally real within relevant systems while differing substantially in their generative organization. The ontological, relational, and manifestational questions concerning them remain distinguishable, and comparison between their structures requires additional justification.

This point becomes consequential when the chapter turns to manifestation mappings and their possible non-invertibility. Similar manifestations may arise from different relational configurations, while similar relational configurations may yield different manifestations under changing bodily, historical, institutional, cultural, technological, or symbolic conditions. Reverse inference from manifestation to relational structure therefore requires explicit epistemic assumptions. Manifestational similarity alone provides insufficient grounds for relational identity, just as relational comparability does not by itself establish ontological identity. These distinctions provide part of the ontological basis for the epistemological problems developed in Chapter 5.

The chapter proceeds by stages. Section 90 states the ontological position of GR and develops the principle of methodological restraint. The subsequent sections introduce the three registers of reality and clarify ontological suspension before returning to change, structure, constraint, and difference as minimal primitives through which stable configurations can be analyzed. The chapter then develops relational reality and manifestational reality in greater detail, examines the generative emergence of manifestation, and studies the mappings through which relational configurations become manifest. Later sections address non-invertibility, symbolization, recursive relations among manifestation and subsequent relational formation, subjecthood, value, meaning, social facts, institutions, process, events, and becoming. Category-theoretic and scale-relative views are introduced where they offer useful formal or structural clarification, while their ontological interpretation remains deliberately limited. The final movement historicizes ontological categories, clarifies the methodological status of relational-generative analysis, considers compatibility with alternative ontological commitments, and states the boundaries of the chapter's claims.

The resulting ontology is intentionally open. GR requires enough ontological articulation to distinguish kinds of reality, analyze their relations, and prevent epistemic or representational categories from silently replacing the realities under study. It

does not require those distinctions to culminate in a closed metaphysical system. The three-register architecture instead provides a disciplined space within which heterogeneous realities can emerge, persist, act, manifest, mediate one another, and transform historically while questions of ultimate ontology remain open where the inquiry does not require their resolution. This ontological view prepares the transition from the reality of generative-relational systems to the epistemic problem of how such realities can be known through necessarily situated manifestations, traces, representations, and interventions.

4.1 Ontological Position

This section establishes the ontological position from which the remainder of Chapter 4 proceeds. Its role is preparatory: before GR distinguishes ontological, relational, and manifestational reality, it must clarify what kind of ontological commitment an open generative-relational research programme actually requires. The argument proceeds in five steps. It first specifies the function of an ontological position in GR, then examines why economic analysis cannot avoid ontological commitments simply by leaving them implicit. It next distinguishes commitments made for a local explanatory task from stronger claims about the ultimate constitution of reality. The final two subsections formulate methodological restraint and explain how GR can make substantive ontological commitments without converting those commitments into a closed metaphysical system. The mode of argument is conceptual and methodological. The aim is to establish conditions for disciplined inquiry rather than to settle the ultimate furniture of reality.

4.1.1 Ontological Position in Generative-Relational Inquiry

Every inquiry presupposes some account of what kinds of objects, processes, relations, differences, events, or structures can enter explanation. A study of prices presupposes that prices can be identified with sufficient stability for the purposes of the analysis. A study of institutions presupposes that institutions can persist through changes in personnel and individual acts. A study of subjects presupposes some basis on which actions, memories, capacities, or recognitions can be attributed to a subject over an analytically relevant interval. A study of causal effects presupposes that differences in the world can be related to subsequent differences under stated conditions. These presuppositions need not constitute a complete metaphysics, yet they carry ontological content because they determine what the inquiry treats as sufficiently real, persistent, and efficacious to bear explanatory weight. (Quine 1948; Fraassen 1980)

GR makes this background explicit because its principal objects are often emergent, historically variable, multiply realized, reflexive, or only partially accessible through observation. The framework studies how objects and capacities arise through generative relations, how stabilized forms participate in later generation, how manifestations mediate subsequent relations, and how the relevant units of analysis can change with scale and history. Such questions become confused when the status of an analytical object is left unspecified. A relation represented in a model, a relation inferred

from evidence, a relation stabilized in a social system, and a relation posited as part of ultimate reality are different claims even when the same word is used for each of them.

Definition 4.1 (Ontological position). An *ontological position* is the explicit set of commitments through which an inquiry specifies what kinds of entities, relations, processes, differences, structures, or events it treats as real enough to enter explanation, together with the scope and strength assigned to those commitments.

Definition 4.1 treats an ontological position as more than a list of objects. Scope and strength matter equally. A claim that money is real for the analysis of exchange, obligation, accounting, and institutional action differs from a claim that money is a fundamental constituent of reality. A claim that a relational configuration has causal efficacy within a particular generative-relational system differs from a claim that relation is the ultimate substance from which all reality is composed. A claim that a self persists with enough stability to organize action differs from a claim that the self possesses an invariant metaphysical essence. GR therefore requires an ontological vocabulary capable of supporting local explanatory commitments while preserving distinctions among different strengths of claim.

This position also separates ontology from immediate observability. Some structures relevant to explanation may be indirectly inferred, historically reconstructed, experimentally perturbed, or manifested only through partial traces. Their ontological status cannot be reduced to whether a subject currently perceives them. Conversely, something may be vividly perceived or symbolically articulated while its underlying ontological status remains unsettled. Chapter 4 will later formalize this distinction through the three registers of reality. At this stage, the important point is methodological: a theory of reality must distinguish the conditions under which something becomes available to inquiry from the conditions under which it becomes causally or structurally relevant to the system being studied.

GR consequently adopts an ontological position because generative explanation needs objects that can persist, interact, constrain, enable, transform, and become historically consequential. At the same time, the framework resists extending each local explanatory success into a general metaphysical conclusion. This combination provides the starting orientation for the chapter: explicit commitment where explanation requires it, explicit suspension where the evidence and analytical task do not warrant further closure.

4.1.2 Ontology and Economic Analysis

Economic analysis makes ontological selections whenever it decides what counts as an economic object, a relevant relation, a unit of action, a form of value, a resource, an institution, a constraint, or a consequence. These selections can remain implicit because many categories are inherited from established practice. Households, firms, states, markets, prices, contracts, preferences, assets, skills, debts, and technologies can appear sufficiently familiar that their status as analytical constructions receives little attention. Yet each category compresses a field of heterogeneous relations into an object that can be compared, measured, modeled, regulated, or acted upon. (Giere 2004)

The ontological significance becomes clearer when categories change. A platform can function simultaneously as a firm, infrastructure, market organizer, data environment, contractual authority, and technological mediator. An artificial system can participate in production, recommendation, bargaining, classification, and coordination while questions of agency, subjecthood, legal personality, and responsibility remain separately unsettled. Knowledge can appear as a private asset, a shared capacity, an institutional memory, a public resource, or a relation embodied in practices and networks. Care can be treated as labor, relation, capability formation, social reproduction, or a mixture whose relevant ontology varies with the research question. Such cases reveal that economic analysis does not merely observe a world already partitioned into self-identifying economic units. It also selects and stabilizes objects for inquiry.

GRE requires particular care because its central questions concern generation across time. An economic relation can alter the capacities of subjects, the availability of future actions, the structure of institutions, the distribution of recognition, the meaning of categories, and the conditions under which later value can be generated. The object relevant at the beginning of a process may therefore differ from the object relevant at its end. A training relation can alter the capabilities through which a worker later participates in production. A credit relation can change both material capacity and future bargaining conditions. An institutional classification can reorganize eligibility and thereby reshape the population to which the classification is subsequently applied. The ontology of the analysis must accommodate objects that are partly constituted through the histories in which they participate.

This historical mutability does not imply that economic objects are analytically arbitrary. An object can be historically generated and still become stable enough to constrain action, coordinate expectations, carry obligations, or produce measurable consequences. The relevant question is therefore not whether an object is timeless, but how its stability is generated, over what interval it persists, through which relations it is reproduced, and what causal difference its presence makes. This is one reason relational reality becomes important later in the chapter: many economically consequential objects derive their practical reality from sustained generative relations without requiring an immediate claim about their ultimate metaphysical status.

Economic representations also participate in the realities they describe. Prices, rankings, credit scores, accounting categories, legal forms, forecasts, and classifications can become inputs into later action. Once institutionalized, a representation may influence resource allocation, access, expectations, strategy, or identity. The economic analyst therefore encounters a reflexive environment in which manifestational and representational forms can become causal components of subsequent relational organization. This feature gives ontology a practical significance. Treating a category as merely descriptive can obscure the process through which its use contributes to reproducing or transforming the object under analysis.

Claim 4.2 (Ontological relevance of economic categories). *An economic category acquires ontological relevance for GRE when its stabilization, transformation, or disappearance makes a causal or structural difference to the generative-relational processes under study.*

Claim 4.2 does not require every economic category to possess the same mode of

reality. A legal corporation, a price, a bodily capacity, an ecosystem, a debt, and an expectation can enter economic explanation through different generative histories and different forms of persistence. Their inclusion in one inquiry does not imply that they share a common substance or a common metric. It establishes only that the analysis has reasons to treat them as causally or structurally relevant within its stated domain.

This pluralism is especially important for GRE because heterogeneous objects can interact without becoming ontologically homogeneous. A legal rule can affect material production; an aesthetic judgment can redirect research attention; a symbolic classification can change access to finance; a bodily limitation can alter bargaining relations; an imagined future can influence present investment. The causal field of economic action therefore crosses domains that may require different descriptive vocabularies. A useful ontology must permit interaction across heterogeneous forms of reality while resisting the inference that interaction proves reducibility to one underlying analytical language.

4.1.3 Locality and Scope of Ontological Commitment

Ontological commitment in GR is local to an explanatory task unless a stronger scope is independently established. Locality here refers to the domain within which a commitment is warranted: a generative-relational system, a historical interval, an analytical scale, a class of mechanisms, or a specified research problem. This principle allows GR to say that an object is real enough to explain a process without requiring the same statement to settle its status across every possible domain of inquiry.

Definition 4.3 (Ontological commitment). An *ontological commitment* is a commitment to treat some object, relation, process, difference, structure, or event as possessing sufficient reality, within a specified domain and for a specified explanatory purpose, that claims about its persistence, efficacy, interaction, or transformation can bear argumentative weight.

Definition 4.3 makes ontological commitment purpose-sensitive without making it arbitrary. The commitment remains answerable to evidence, coherence, explanatory performance, and the consequences of intervention. Its purpose specifies the burden of proof. An analysis of currency circulation may require a robust account of money as an institutionally and relationally stabilized object while remaining neutral about whether monetary value possesses any reality outside such systems. An analysis of subjective continuity may require a stable operational subject over a given interval while leaving open deeper claims about the metaphysics of selfhood. An analysis of religiously organized economic life may need to treat a deity as a relationally real object within the relevant system if the deity organizes obligations, trust, sacrifice, authority, or allocation; that analytical commitment does not itself adjudicate theological claims concerning independent divine existence.

Locality also matters for scale. An object stable at one scale may dissolve into changing components at another. A firm can be treated as a persistent organization in an industry-level analysis while appearing as a changing field of contracts, routines, persons, technologies, and authority relations in an organizational study. A market can function as an analytical object at one scale while a microstructural anal-

ysis resolves it into order books, protocols, matching rules, information flows, and strategic interactions. Neither description automatically invalidates the other. Their adequacy depends on the generative processes each inquiry seeks to explain.

Temporal scope produces a similar effect. Persistence is always persistence over some interval. An institution can be highly stable relative to daily transactions and historically transient relative to centuries. A subject's identity can remain sufficiently continuous for contractual responsibility while changing substantially over a life course. A convention may stabilize coordination for a generation and later disappear. GR therefore avoids treating stability as timeless identity. Stability is a relation between a pattern and an analytical timescale, together with the mechanisms that reproduce the pattern across change.

Proposition 4.4 (Local sufficiency of ontological commitment). *For a generative-relational explanation, the ontological commitment required to support the explanation need extend only as far as the persistence, efficacy, and relations necessary for that explanation, unless additional evidence or argument justifies a stronger claim.*

Proposition 4.4 provides a rule of economy for ontological reasoning. If an account of institutional trust depends on a stabilized relation that changes later action, the inquiry can study that relation without deciding whether trust is reducible to neural states, symbolic structures, expectations, embodied dispositions, or some combination of them at an ultimate level. If a model of knowledge production depends on a research community, shared standards, archives, and infrastructures, those objects can be treated as real components of the explanatory domain without requiring a final ontology of collective entities. Stronger commitments remain possible, but they carry additional argumentative burdens.

Local commitment also protects against a common error in interdisciplinary work. A concept transferred from one domain can be useful without carrying its entire source ontology with it. Field language, network language, dynamical systems, category theory, information, geometry, or computation may disclose structure in an economic problem. Their successful use does not automatically establish that the economic world is fundamentally a field, network, dynamical system, category, informational object, geometric manifold, or computation. The imported representation earns authority over the relations it successfully captures; claims beyond that scope require separate support.

The locality principle prepares the distinction developed in the next sections. Ontological reality, relational reality, and manifestational reality will allow GR to specify different forms of claim without forcing them into a single hierarchy. The present section establishes the prior discipline: the scope of an ontological commitment should remain visible, and movement from a local claim to a stronger claim should be argued rather than assumed.

4.1.4 Methodological Restraint in Ontological Claims

Methodological restraint governs the transition from analytical success to ontological assertion. GR uses strong conceptual resources because generative systems can be difficult to describe with ordinary entity-centered language. Relations can have higher-order structure; constraints can alter the space of possible action; historical

trajectories can produce attractors or path dependence; categories can change the behavior of classified subjects; and manifestations can feed back into the systems from which they emerge. Each of these phenomena may invite formal representations. The quality of a representation, however, does not by itself determine the ultimate ontology of the represented system.

The distinction can be stated through three questions. The first asks whether a representation is useful: does it organize observations, distinguish mechanisms, support inference, or enable intervention? The second asks whether the represented structure is real within the domain: does the structure persist, constrain, enable, or causally alter the system in ways that survive changes in description? The third asks whether the representation identifies the ultimate constitution of what exists. These questions can receive different answers. A network model can be useful even when edges are constructed from imperfect proxies. A relational structure can be causally real even when several mathematical representations capture it equally well. An empirically successful representation can remain silent about ultimate ontology.

Remark 4.5 (Representation and ontological inference). The empirical or explanatory success of a representation can support claims about structures captured by that representation while leaving stronger claims about the ultimate constitution of reality underdetermined.

Remark 4.5 is particularly important for a framework that draws from several formal traditions. A graph can reveal patterns of connectivity, a dynamical system can reveal stability and transition, a field representation can express spatially or relationally distributed influence, and a category-theoretic formulation can clarify mappings among structures. These representations may overlap without being mutually reducible. GR therefore treats formal plurality as evidence that the relation between representation and ontology itself requires analysis.

Methodological restraint also applies to ordinary language. Terms such as “subject,” “institution,” “value,” “power,” “relation,” and “field” can acquire an apparent solidity through repetition. A concept introduced as an analytical convenience may gradually be spoken of as though its boundaries were discovered rather than constructed. Conversely, a familiar category can appear natural even when its persistence depends on legal, technological, cultural, and historical conditions. Ontological discipline requires the manuscript to mark when a term refers to a stipulated analytical unit, an empirically inferred structure, a relationally stabilized object, a manifestation, or a stronger ontological claim.

Caution 4.6 (Ontological inflation). Ontological inflation occurs when an analytical distinction, representation, measurement, or successful local explanation is extended into a stronger claim about reality without an additional argument adequate to that stronger scope.

Caution 4.6 applies in both formal and qualitative inquiry. Quantification can inflate a proxy into a property; interpretive analysis can inflate a recurrent meaning into a universal structure; historical reconstruction can inflate a contingent lineage into necessity; a relational model can inflate explanatory dependence into a claim that only relations exist. Restraint does not weaken inquiry. It preserves the distinction between what the evidence establishes and what remains open.

The same discipline applies to causal language. GR often studies objects that acquire causal efficacy through relational stabilization. Causal efficacy here means that the presence, persistence, transformation, or disappearance of the object makes a difference to subsequent states or possibilities under relevant conditions. It does not require that the object be fundamental in a reductionist hierarchy. Nor does it imply that causality can always be assigned to a single isolated object. In generative-relational systems, causal contribution may be distributed across configurations, histories, interfaces, and feedback processes. The ontological commitment should match the causal claim being made.

Methodological restraint therefore has a positive function. It makes room for strong local explanation, plural representation, and cross-disciplinary borrowing while protecting the framework from premature metaphysical closure. This discipline will become especially important when relational reality is defined later in the chapter. The fact that an object is relationally real will establish a substantive claim about its emergence, persistence, stability, and causal efficacy within a system. It will not by itself settle whether that object is ontologically fundamental, reducible to another level, or comparable with every other relationally real object.

4.1.5 Ontological Commitment without Closure

GR's ontological position can now be stated more precisely. The framework is committed to the reality of change, difference, constraint, structure, persistence, and causal efficacy wherever these are required and warranted by the inquiry. It is committed to treating generative relations as analytically consequential when transformations in relations alter what objects, capacities, manifestations, or possibilities can emerge. It is also committed to the possibility that some emergent objects become sufficiently stable to act back upon the systems through which they arose. These commitments provide the minimum ontological depth necessary for generative-relational explanation.

The framework simultaneously leaves several stronger questions open. GR does not need to determine whether relations are more fundamental than *relata* in every domain. It does not need to identify one privileged formal language in which all reality must be represented. It does not need to decide whether every socially consequential object is reducible to physical processes, whether every physical process is best understood relationally, or whether one universal metric can compare heterogeneous forms of reality. These questions can be addressed where a particular inquiry requires them, but the book does not make their settlement a prerequisite for economic analysis.

This openness is compatible with substantive disagreement. Two researchers can share a GRE analysis of the relational generation of money while holding different metaphysical views about social ontology. Two researchers can agree that a self-model becomes relationally stable and causally consequential while disagreeing about consciousness or personal identity. Two researchers can study how a religious object organizes economic action while differing on its ontological status outside the relational domain under investigation. The research programme remains coherent because its shared commitments concern the generative-relational structures re-

quired for the explanation, while stronger metaphysical claims can remain explicit points of divergence.

The position can therefore be summarized through a principle of graded commitment. Claims should become ontologically stronger only when the research question, evidence, and argument require and support the additional strength. Observation may justify a claim about manifestation. Recurrent and causally consequential organization may justify a claim about relational reality. A further claim about ontological reality requires its own warrant. Movement among these registers cannot be inferred merely from vocabulary or intuitive resemblance. Section 91 develops these distinctions directly.

Proposition 4.7 (Open ontological commitment). *GR permits substantive commitments concerning the reality, persistence, efficacy, and transformation of objects within generative-relational systems while keeping stronger metaphysical questions open whenever those questions are not required for the explanatory task.*

Proposition 4.7 expresses the balance sought throughout this chapter. An ontology too thin to distinguish reality from representation would leave GRE unable to explain how emergent objects become causally consequential. An ontology that converts every useful analytical form into an ultimate metaphysical claim would make the programme unnecessarily brittle and would collapse heterogeneous domains into premature unity. Open ontological commitment occupies the intermediate position: reality claims are made where generative explanation requires them, their scope is stated, and further closure remains an additional task rather than an automatic consequence.

This position also preserves historical revisability. The objects recognized by an inquiry can change as institutions, technologies, categories, and relations change. New forms of economic subjecthood may emerge; old classifications may disappear; new infrastructures may create objects that earlier vocabularies could not represent adequately; improved methods may separate structures previously treated as one or reveal continuities previously missed. An ontological position adequate to generative inquiry therefore includes the possibility that its own categories will require revision. Revisability concerns the framework's descriptions and commitments, while the reality under study may exceed any one historical vocabulary used to describe it.

Section 90 thus establishes the ontological discipline for what follows. GR will make reality claims strong enough to support explanation, causal analysis, and historical inquiry. It will distinguish the scope of those claims, preserve the difference between representation and represented structure, and resist the extension of local success into unsupported metaphysical closure. With this position established, Section 91 turns to the chapter's central architecture: ontological reality, relational reality, and manifestational reality as three distinct but mutually mediating registers.

4.2 The Three Levels of Reality

This section introduces the three-register architecture through which GR distinguishes ontological reality, relational reality, and manifestational reality. The inherited outline uses the word *levels*, but the three are not ordered strata in which

one sits above or below another. They are better understood as registers that answer different questions about reality and that can mediate one another through historically situated generative relations. The distinction is necessary because GR repeatedly encounters objects that are causally effective without requiring a settled claim about their ultimate ontology, as well as manifestations that disclose, transform, or even help stabilize the realities to which they relate. Treating these cases within a single undifferentiated category of “reality” would obscure differences among what obtains, what becomes stably effective in a relational system, and what becomes manifest under particular conditions.

The architecture developed here is deliberately minimal. It does not claim that every philosophical problem can be assigned uniquely to one register, nor that each object possesses only one form of reality. The same object may be considered differently according to the question being asked. Money can be examined through the material and institutional conditions that participate in its existence, through the stabilized relational reality by which it organizes exchange and obligation, and through the perceptual, symbolic, legal, or computational forms in which it becomes manifest. A sense of self can likewise be approached through bodily and ontological questions, through the relational processes that stabilize continuity and identity, and through the forms in which selfhood becomes experienced, narrated, classified, or represented. The registers therefore separate analytical questions without requiring the realities under study to be separable in practice. (Whitehead 1929)

4.2.1 Ontological Reality

The first register concerns reality in its ontological status. GR needs this register because an inquiry can refer to what obtains or occurs without claiming that it has already captured the ultimate constitution of what obtains or occurs. The distinction is particularly important when perception, symbolic representation, institutional classification, or model construction provides only partial access to the processes under investigation. Ontological reality marks the fact that the reality of an event, structure, difference, or process cannot be reduced simply to the manner in which it has become available to a particular observer or system of representation. (Husserl 2012; Merleau-Ponty 2012)

Definition 4.8 (Ontological reality). *Ontological reality* denotes what obtains, occurs, or possesses efficacy insofar as its reality is not constituted merely by its being perceived, manifested, represented, measured, classified, or known within a particular epistemic situation. GR may refer to this register while leaving the ultimate constitution of what is ontologically real under ontological suspension.

Definition 4.8 is intentionally compatible with different metaphysical positions. It does not identify ontological reality with matter, physical substance, mind, relation, process, information, or another privileged category. Nor does it define ontological reality as whatever is permanently inaccessible. Some ontologically relevant structures may be directly observable under suitable conditions; others may be reconstructed from traces, inferred through intervention, or remain only partially accessible. The defining distinction concerns the status of the claim rather than the accessibility of the object: ontological reality is not exhausted by a particular manifestation

or representation of it.

This register therefore gives GR a place to formulate questions that exceed what has presently become manifest. A physical process may continue whether or not it is observed. Historical conditions may constrain later possibilities even when no contemporary subject explicitly recognizes those constraints. Differences in biological, ecological, technological, or material organization may participate in generative processes before they acquire a social classification. In each case, an inquiry may have reasons to refer to a reality whose full ontology remains unsettled while resisting the stronger reduction of that reality to what is presently available in experience or representation.

The ontological register also protects GR from an opposite reduction. The fact that a relational or manifestation phenomenon is historically generated does not by itself imply that all aspects of the phenomenon are exhausted by the history through which it has been interpreted. An ontological question may remain open even after a rich account of relational emergence has been developed. This openness is especially important when different ontological interpretations remain compatible with the same relational evidence. The framework can then preserve the difference rather than forcing premature metaphysical closure.

Ontological reality is thus a register of commitment and restraint at once. GR can acknowledge that inquiry addresses realities rather than only descriptions while limiting its claims to what the research problem and evidence support. Section 92 develops this discipline through ontological suspension. For present purposes, the first register provides the background against which relational reality can be distinguished from both ultimate ontological identity and manifest appearance.

4.2.2 Relational Reality

Relational reality is the central technical category through which GR recognizes objects that become real within generative-relational systems. The category was developed to avoid a recurrent false choice. Many consequential objects are difficult to treat as fundamental or independently existing substances, yet treating them as merely fictional, subjective, or representational fails to explain their persistence and causal effects. Money, institutional authority, debt, trust, socially sustained identities, religious objects, and the experienced continuity of a self can organize action and reshape subsequent relations even when their ultimate ontological status is left unsettled. GR therefore requires a criterion of reality grounded in emergence and efficacy within a relational system.

Definition 4.9 (Relational reality). *Relational reality* denotes an object that emerges within a generative-relational system and, over some region or interval of spacetime, persists with sufficient stability to exert causal efficacy upon that system. Its relational reality consists in its generated and sustained participation in subsequent relations rather than in a prior requirement that it possess an independently established ontological identity.

The word *object* in Definition 4.9 is deliberately broad. It can refer to a stabilized relation, configuration, pattern, institution, capacity, identity, process, or other analytically individuated object when the relevant conditions are satisfied. The defini-

tion is not meant to reify every transient correlation or every label introduced by an observer. Emergence alone is insufficient. A relationally real object must acquire persistence or recurrent stability over an analytically relevant spacetime domain, and its presence must make a causal difference within the generative-relational system in which it has emerged. Sections 93 and 94 will develop these conditions more carefully.

The criterion permits GR to treat money as relationally real without requiring a metaphysical substance called money. Monetary reality is sustained through legal recognition, accounting practices, expectations, exchange relations, institutional infrastructures, material and digital inscriptions, and histories of coordinated action. Once sufficiently stabilized, money affects access to goods, contractual possibilities, investment, labor, debt, expectations, and later relations. Its reality within such systems is therefore not adequately characterized by saying that participants merely believe in it. The belief, inscription, institution, and practice form parts of the generative relations through which the monetary object acquires persistence and causal efficacy.

The same logic can apply to a god within a religiously organized generative-relational system without requiring GR to settle the theological question of divine ontological existence. A divine object may become relationally real insofar as it is stably generated through ritual, memory, doctrine, architecture, collective practice, personal experience, authority, and other relations, and insofar as it causally reorganizes conduct, obligation, temporal practice, resource allocation, affect, identity, or institutional life. The relational claim neither establishes nor denies an independent divine ontology. It concerns the reality generated within the relational system and the causal work that this reality performs there.

A similar distinction applies to the experienced continuity of “I.” A relatively stable sense of self can emerge through bodily continuity, memory, recognition by others, language, institutional identity, habit, affect, and narrative. Once stabilized, this self-object can organize intention, responsibility, expectation, action, and interpretation. Its relational reality can therefore be studied without requiring GR first to decide whether a metaphysically invariant self exists. The example is especially useful because it shows that relational reality is not confined to conventionally external social objects. A reality can be intimate and subjectively consequential while still being relationally generated and historically sustained.

Relational reality also does not require prior perception. A generative-relational structure can persist and act before any subject recognizes it, before it receives a name, and before it is stabilized in symbolic form. This feature is important for the GR treatment of contingency and aesthetic judgment. The structure through which a contingent conjunction becomes possible can already be historically and relationally real before its contingency becomes salient to a subject. An eventual experience of surprise, beauty, significance, or curiosity is then a manifestation within a structure whose relevant generative conditions did not begin with the experience itself. The relational reality of those conditions is therefore conceptually prior to their recognition, even though later recognition can feed back into the system and change what follows.

Claim 4.10 (Manifestation is not required for relational reality). *An object can sat-*

isfy the conditions of relational reality before it is perceived, consciously recognized, linguistically articulated, symbolized, or otherwise manifested to a particular subject.

Claim 4.10 prevents relational reality from collapsing into phenomenology, discourse, or representation. It also leaves room for a stronger recursive thesis developed later in the chapter: although manifestation is not a general precondition for relational reality, manifestations can become relations within the system and can therefore participate in generating, stabilizing, transforming, or dissolving relational realities. The absence of necessary priority in one direction is precisely why the three-register architecture must be understood as a graph rather than a chain.

4.2.3 Manifestational Reality

The third register concerns the reality of manifestation. Something can become present to a subject, instrument, institution, classificatory system, archive, legal process, market interface, or other relational configuration without the manifestation exhausting either the relational structure from which it emerged or the ontological reality to which it may refer. GR therefore treats manifestation as a positive object of analysis rather than as a transparent window through which an already complete reality is simply observed.

Definition 4.11 (Manifestational reality). *Manifestational reality* denotes reality insofar as something becomes available, differentiated, experienceable, perceptible, recordable, articulable, measurable, classifiable, symbolizable, or otherwise present within a particular relational and historical configuration. A manifestation is conditioned by the relations through which this availability occurs, and its reality does not by itself establish the complete ontological or relational structure of what is manifested.

Manifestation is broader than symbolization. A pain can become manifest affectively and bodily before it is named. A pattern can become manifest statistically through an instrument or dataset without being available to unaided perception. An institutional boundary can become manifest through differential access or treatment before its rules are explicitly articulated. A landscape can become manifest aesthetically through a mode of attention that is partly perceptual and affective. A market condition can become manifest through price movements, queues, shortages, or changes in conduct. Symbolic representation is an especially powerful mode of manifestation because language, categories, diagrams, records, and numbers can stabilize and circulate what has become available, but symbolic inscription does not exhaust the manifestational register.

Manifestational reality should also be distinguished from the referential truth of a manifestation. A vivid perceptual flower generated under pathological or altered perceptual conditions can possess manifestational reality: the flower appears, organizes attention, and may provoke action or affect. An ordinary perceptual flower associated with a physically existing plant can likewise possess manifestational reality. The fact that both are manifestations does not establish that their ontological sources or generative structures are identical. Conversely, a difference in ontological interpretation does not make one of the manifestational events analytically nonexistent. GR can therefore investigate the reality and consequences of a manifestation while

keeping separate the further question of what produced it and what ontological status should be assigned to its apparent object.

This distinction becomes important whenever manifestation itself has effects. A classification printed on a document can alter eligibility. A diagnosis can reorganize self-understanding and institutional treatment. A price displayed on a market interface can change subsequent bids. A public declaration of trust can reshape expectations between participants. A representation of a social category can influence how people act toward those classified by it. In such cases, the manifestation is not merely evidence about an antecedent relational reality. Once incorporated into subsequent interactions, it becomes part of the generative environment.

Claim 4.12 (Manifestational mediation). *A manifestation can enter a generative-relational system as an operative relation or object and thereby participate in the generation, stabilization, transformation, or dissolution of subsequent relational realities.*

Claim 4.12 does not imply that every manifestation substantially changes the system in which it occurs. Its significance is structural: the architecture permits such mediation and therefore cannot treat manifestation as only the terminal output of a prior process. The causal importance of any particular manifestation remains an empirical and historical question. Section 100 will later examine recursive relations among reality, manifestation, and representation in greater detail.

Manifestational reality also has a historical dimension. What can become manifest, to whom, through which medium, and with what resolution depends on available sensory capacities, concepts, institutions, instruments, archives, languages, technical infrastructures, norms, and practices of attention. A relation that was difficult to manifest under one historical configuration can become readily measurable or publicly articulable under another. Conversely, changes in classification or institutional practice can render previously salient realities less visible. Manifestation therefore belongs to history rather than constituting a timeless interface between mind and world.

4.2.4 Distinctions among the Three Registers

The three definitions establish different criteria and therefore resist several common reductions. Ontological reality concerns the status of what obtains or occurs without reducing that status to a particular manifestation. Relational reality concerns stable and causally efficacious emergence within a generative-relational system. Manifestational reality concerns becoming available under a particular configuration of relations and mediation. These registers can overlap in concrete cases, but overlap does not erase the difference among the questions they pose.

A relationally real object does not thereby acquire a settled independent ontology. The social efficacy of money, a legal person, a deity, an institutional role, or a stabilized self does not require GRE to decide that the relevant object exists as a fundamental substance. The relational claim can remain both strong and limited: the object has emerged, persists, and causally matters within the system. Conversely, an ontologically real process does not automatically possess relational reality relative to every generative-relational system. Something may exist yet remain causally disconnected

from the system under study, or it may fail to become sufficiently stabilized within that system to constitute the relational object relevant to the inquiry.

Relational reality likewise does not require manifestation, as Claim 4.10 states. Historical constraints, latent institutional structures, unrecognized patterns of dependence, or the generative conditions of a contingent encounter may operate before they are consciously noticed or symbolically articulated. Manifestation is therefore neither the origin nor the universal criterion of relational reality. At the same time, once a manifestation enters subsequent relations, Claim 4.12 allows it to become one of the conditions through which later relational realities are generated. The distinction between the registers thus supports directional asymmetry in particular cases without imposing a permanent hierarchy among them.

Manifestational reality also does not guarantee that the apparent object possesses the ontological status attributed to it by a subject. A manifestation can be vivid, stable, socially consequential, and repeatedly recorded while the ontological interpretation of its source remains uncertain or contested. Nor does manifestational similarity establish relational identity. Two similar experiences, measurements, classifications, or observable patterns may arise from different generative structures. The inverse problem is equally important: similar relational structures can manifest differently under different bodily, institutional, cultural, technical, or historical conditions. Later sections will develop these many-to-one and one-to-many possibilities more formally.

The distinctions also prevent a further assumption: shared relational reality does not entail comparability. Two objects can each satisfy the criteria of relational reality while lacking a justified common metric, ordering, coordinate system, or structure-preserving comparison. They may inhabit different metric spaces, non-metric relational spaces, or domains for which no warranted mapping has yet been constructed. A hallucinated flower and an ordinarily perceived physical flower provide a simple illustration. Both can be relationally consequential within the systems in which they arise, yet their generative structures need not be treated as instances of the same measurable object merely because the word “flower” is used for both manifestations. Establishing a comparison relation is an additional analytical achievement, not an automatic consequence of assigning the same reality-status.

Claim 4.13 (Non-necessary comparability of relational realities). *The fact that two objects possess relational reality does not by itself establish a common metric, ordering, equivalence relation, structural map, or other justified basis of comparison between them.*

Claim 4.13 has methodological consequences throughout GRE. Comparative analysis must specify what is being compared, through which relation, and under what invariances or losses of information. Isomorphism, homomorphism, simulation, statistical resemblance, functional equivalence, and analogical comparison answer different questions and require different assumptions. Where no warranted comparison structure exists, preserving incomparability can be more rigorous than imposing a common measure for analytical convenience. This point becomes especially important when GRE studies heterogeneous values, forms of life, subjectivities, institutions, and modes of manifestation.

The distinctions among registers should therefore be read as safeguards against premature reduction. Ontological uncertainty need not erase relational reality. Relational efficacy need not establish ultimate ontology. Manifestational vividness need not disclose either register completely. Lack of manifestation need not imply lack of relational structure. Shared reality-status need not imply shared geometry. Each of these distinctions preserves a question that would otherwise disappear if “reality” were treated as a single undifferentiated category.

4.2.5 Relations across the Three Registers

The three-register architecture becomes most distinctive when the relations among the registers are considered. A tempting interpretation would treat them as a sequence: an ontological reality first exists, relational structures subsequently form around it, and a final manifestation presents the result to a subject. Such a trajectory may describe some local cases, but GR does not adopt it as the general architecture. The three registers form a graph of heterogeneous and historically variable relations. Different paths can connect them, and what becomes an effect along one path can become a condition along another.

Ontological conditions can participate in the emergence of relational realities. Material resources, bodily capacities, ecological processes, technical infrastructures, spatial arrangements, and other ontologically relevant conditions can constrain or enable the relations through which an object stabilizes. Yet the resulting relational object need not be reducible to those conditions. A monetary system, for example, requires material and technical supports but acquires causal properties through legal, institutional, symbolic, and practical relations that cannot be read directly from the physical medium of a banknote or database entry.

Relational realities can condition manifestation. What becomes salient, nameable, measurable, experientially coherent, institutionally visible, or symbolically available depends partly on the relations already organizing subjects, instruments, categories, institutions, and environments. A relation of trust can change what conduct is interpreted as reassuring or suspicious. An institutional category can determine which differences enter administrative records. A scientific instrument and its associated practices can make a pattern manifest that unaided perception could not distinguish. The manifestation therefore emerges within a relational field rather than arriving from outside it.

The reverse mediation is equally important. Once a manifestation occurs, it can alter the relational field. A statement, image, classification, measurement, judgment, or public performance can be remembered, circulated, institutionalized, contested, or incorporated into later expectations. Through these processes, manifestation can contribute to new persistence and causal efficacy. A relation may thus help generate a manifestation, while the manifestation becomes one of the relations through which a later relational reality is stabilized. Historical time matters because the system encountered after manifestation need not be the system that preceded it.

This recursive possibility is particularly significant for aesthetic judgment and contingency. The relational-generative-historical structure through which a contingent encounter becomes possible can exist before the encounter is apprehended as con-

tingent. When contingency later becomes manifest through surprise, attention, aesthetic judgment, or conceptual recognition, the manifestation can redirect inquiry, memory, valuation, preservation, or future practice. The recognition therefore neither creates the entire preceding structure nor merely reflects it. It enters the history of that structure and can change what the system subsequently generates.

The architecture has a limited heuristic affinity with Lacan's three registers of the Imaginary, the Symbolic, and the Real. Lacanian register theory itself changed substantially over time, and later formulations emphasize mutual dependence among the registers rather than treating them as a simple developmental sequence (Johnston 2026). GR does not adopt Lacan's definitions or identify its registers one-to-one with his. Ontological reality is not the Lacanian Real; relational reality extends well beyond the Imaginary; and manifestational reality includes forms of appearance and mediation that are not reducible to the Symbolic. The useful analogy lies at the architectural level: a plurality of registers can be mutually conditioning, and the relations among them can be more informative than an attempt to place them in a single hierarchy.

Remark 4.14 (Graph rather than chain). Ontological, relational, and manifestational reality should be treated as registers connected by heterogeneous relations rather than as stages in a universal sequence. Their connections may be recursive, many-to-many, historically changing, and only partially invertible. A path observed in one case does not establish the direction or form of mediation in another.

The graph conception also leaves room for missing edges. A research programme need not assume that every ontological condition is relationally relevant to the system under study, that every relational reality becomes manifest, or that every manifestation permits reliable inference toward its generative conditions. Some relations may be weak, historically absent, technically inaccessible, or analytically unjustified. Treating the architecture as a graph therefore does not mean multiplying connections without discipline. It means that connections are objects of inquiry whose existence, direction, stability, and explanatory importance must be established rather than presumed.

The three registers can now support the rest of the chapter. Section 92 clarifies the methodological status of ontological suspension. Section 93 identifies the primitive vocabulary through which emergence and stabilization can be analyzed. Sections 94 and 95 then return separately to relational and manifestational reality, allowing persistence, stability, causal efficacy, forms of manifestation, and participation in subsequent relations to receive fuller treatment. The later analysis of manifestation mappings, non-invertibility, symbolization, and recursive mediation builds on the architecture established here. The purpose of the present section is therefore not to close the ontology of GR but to preserve three analytically distinct questions and to make their interrelation available for disciplined investigation.

4.3 Ontological Suspension

This section develops *ontological suspension* as a methodological discipline within GR. The preceding sections established that generative-relational inquiry requires ex-

plicit ontological commitments while also distinguishing ontological, relational, and manifestational reality. Ontological suspension specifies how those commitments can remain proportionate to the explanatory task. Its role is neither to weaken reality claims nor to replace ontology with epistemic uncertainty. Rather, it allows inquiry to affirm what has been sufficiently established at one register while withholding stronger claims whose resolution is unnecessary, unavailable, or not yet warranted. The section proceeds by distinguishing suspension from denial and agnosticism, then treats suspension as a disciplined practice, examines its use under ontological disagreement, and concludes by clarifying how suspended questions can remain analytically usable without being prematurely settled. (Fraassen 1980; Quine 1948)

4.3.1 Suspension and Denial

Ontological suspension begins with a distinction between declining to affirm a stronger ontological claim and denying that claim. The distinction is elementary but consequential. An inquiry may have sufficient grounds to establish that an object is relationally real while lacking grounds to determine whether that object possesses an independent ontological status. In such a case, GR does not infer nonexistence from the absence of a stronger commitment. It leaves the stronger question open and continues the analysis at the register for which evidence and explanatory need are available.

Consider money. GRE can study money as a relationally real object because monetary forms can emerge through institutional, legal, material, symbolic, and practical relations; persist with sufficient stability; and causally affect allocation, obligation, expectation, production, and future relations. None of these claims requires GRE to settle whether “money” corresponds to an independently existing ontological kind beyond the relations through which monetary reality is sustained. Suspension therefore preserves the reality relevant to the inquiry while withholding a metaphysical conclusion that the analysis does not need.

The same distinction applies to less conventional objects. A deity can possess relational reality within a religious community when the relevant object is historically sustained, structures expectations and conduct, enters institutional practices, organizes temporal and spatial routines, and causally affects subsequent relations. GR can analyze this relational reality without determining whether the deity also possesses ontological reality independent of those relations. Suspension in this case does not entail atheistic denial, theological affirmation, or the reduction of religious experience to error. It identifies the precise ontological question that the inquiry is not presently resolving.

A similar structure can arise in the analysis of selfhood. The experienced continuity of an “I” may be relationally stabilized through embodiment, memory, recognition, language, institutional identity, habit, and historical continuity. This stabilized self can organize action and therefore possess causal efficacy within the system. The corresponding relational claim does not require GR to affirm or deny an invariant metaphysical self. The inquiry can proceed by studying the generative conditions, persistence, transformations, and consequences of selfhood while suspending stronger ontological commitments about what the self ultimately is.

Definition 4.15 (Ontological suspension). *Ontological suspension* is the disciplined withholding of commitment concerning an ontological claim whose resolution is not presently warranted or required for the explanatory task, while preserving and using those weaker or differently scoped reality claims that the inquiry can support.

Definition 4.15 contains two positive components. First, suspension is directed toward a specified claim rather than toward reality in general. Second, suspension preserves other warranted commitments. It is therefore compatible with strong claims at another register. An inquiry may suspend the ontological status of money while affirming its relational reality; suspend the ultimate nature of consciousness while studying manifestational and relational structures of subjectivity; or suspend the metaphysical status of a social category while examining its institutional stability and causal effects.

This claim-specific character prevents ontological suspension from becoming an indiscriminate refusal to decide anything. GR does not treat uncertainty as an intellectual virtue in itself. Where evidence, argument, or explanatory necessity supports a stronger commitment, suspension can be narrowed or lifted. Where a claim has been sufficiently falsified within the relevant domain, denial may be justified. Suspension therefore occupies a distinct methodological position between unwarranted affirmation and unwarranted negation.

The distinction is particularly important for generative inquiry because the objects of interest may cross registers. A manifestational event can have effects even when the ontological status of what appears remains unsettled. An illusory flower perceived under altered physiological conditions can become relationally consequential through attention, movement, fear, memory, communication, or later interpretation. A physically present flower can likewise become relationally consequential through perception and subsequent action. The fact that both can possess relational reality does not erase differences in their ontological conditions. Suspension allows the inquiry to preserve those differences without requiring an immediate final comparison between them.

Claim 4.16 (Suspension does not imply unreality). *Suspending the ontological status of an object does not entail that the object is unreal at other registers. Relational and manifestational reality remain independently assessable under their own conditions.*

Claim 4.16 is foundational for the remainder of the chapter. It prevents a recurrent collapse in which only ontologically fundamental objects are treated as real, while institutionally, psychologically, culturally, or symbolically mediated objects are relegated to a secondary status. GRE instead asks which reality claim is being made, under which conditions, and with what explanatory consequences.

4.3.2 Suspension and Agnosticism

Ontological suspension also differs from general agnosticism. Agnosticism ordinarily describes a position of non-knowledge or unresolved belief concerning a question. Suspension, as used here, is more local, functional, and methodologically organized. A researcher may personally hold a metaphysical view while suspending it within a particular analysis because the view is not required to establish the argument. Conversely, a researcher may remain personally undecided while the inquiry

itself reaches a sufficiently strong local ontological conclusion. GR therefore treats suspension as a property of an analytical procedure rather than as a report of the researcher's private belief state.

This difference matters because a research programme must coordinate inquiry across people who may hold incompatible metaphysical commitments. If every empirical or theoretical analysis required prior agreement concerning the ultimate ontology of persons, institutions, meanings, values, information, or social facts, large portions of inquiry would become hostage to disputes operating at a different level. Ontological suspension creates a disciplined space in which researchers can identify the stronger question, state that it remains open for the purposes of the analysis, and continue working with commitments that are both sufficient and jointly assessable.

Suspension is also compatible with differentiated confidence. The framework need not divide claims into only affirmed, denied, or suspended categories. A claim can be weakly supported, strongly supported, locally adequate, historically contingent, scale-dependent, or open to revision. Suspension concerns the point beyond which the inquiry refrains from extending its commitments. In this sense, it functions as a boundary on inference. The boundary may move as new evidence, conceptual distinctions, or explanatory requirements emerge.

For example, suppose an analysis establishes that a particular institutional category has persistent effects on eligibility, behavior, and resource distribution. The category can be treated as relationally real within the institutional system. Whether that category tracks a natural kind, a socially constructed kind, a historically contingent administrative device, or some combination may remain unresolved. The inquiry can suspend this further ontological classification while retaining the causal and institutional findings. If later research demonstrates that the distinction among these possibilities is itself necessary to explain the observed dynamics, the suspended question becomes active and must be investigated.

This temporality is important. Suspension is not a permanent exile of difficult questions. It is a way of sequencing inquiry so that questions are addressed when their resolution becomes relevant and when the available methods can bear on them. A suspended claim may later become central; a previously central ontological dispute may later become irrelevant to a more local explanatory task. GR therefore understands suspension as revisable and historically situated.

Remark 4.17 (Suspension is inquiry-relative). The same ontological proposition may be suspended in one inquiry and actively investigated in another. Suspension attaches to the relation between a claim and an explanatory task, not permanently to the object itself.

This inquiry-relative character also prevents the framework from treating ontological suspension as a universal solution. Some questions cannot be responsibly deferred. A study of legal responsibility may require a more explicit account of agency than a study of transaction frequency. A study of machine subjecthood may require ontological distinctions that can be suspended in a narrow analysis of machine-mediated market behavior. A study of hallucination may require distinctions between perceptual generation and external physical objects that are unnecessary in an analysis limited to the subsequent behavioral effects of the experi-

ence. The appropriate degree of suspension therefore depends on what explanatory burden the analysis places on its objects.

A further difference from agnosticism concerns positive use. Agnosticism can leave a proposition untouched. Suspension often permits the object associated with that proposition to remain analytically active under another description. One can suspend whether a deity possesses independent ontological reality while analyzing the relational reality of divine authority; suspend whether the self is ontologically fundamental while analyzing self-continuity; suspend whether value exists independently of valuation while studying value as a historically stabilized relational formation. The suspended question remains visible, but inquiry does not need to stop at it.

4.3.3 Suspension as Methodological Discipline

Ontological suspension becomes useful only when it is practiced with discipline. An unspecified statement that “ontology is suspended” can conceal ambiguity just as easily as an unspecified ontological assertion can. GR therefore requires suspension to identify the claim being withheld, the commitments being retained, the reason for suspension, and the conditions under which the question would need to be reopened.

A minimal procedure can be stated in four steps. First, the inquiry identifies the ontological proposition whose stronger resolution is at issue. Second, it states the reality claims that remain necessary for the analysis. Third, it explains why the stronger proposition is not presently required or not sufficiently warranted. Fourth, it tracks whether later stages of the inquiry silently depend on the suspended proposition. These steps make suspension auditable rather than rhetorical.

Definition 4.18 (Disciplined suspension). A suspension is *disciplined* when the inquiry specifies the suspended claim, preserves the commitments required for explanation, states the reason for withholding further commitment, and prevents later inference from covertly presupposing what has been suspended.

The final condition is especially important. An analysis might claim to suspend whether a category is natural or socially constituted, then later interpret its persistence as evidence of naturalness. It might suspend whether an artificial system possesses subjecthood, then assign responsibility to it in a way that presupposes a particular theory of agency. It might suspend whether a market category reflects intrinsic value, then treat observed price as if it directly disclosed such value. These moves would not constitute ontological suspension. They would amount to concealed reintroduction of the stronger claim.

Disciplined suspension therefore requires attention to language. Terms such as “entity,” “agent,” “subject,” “value,” “information,” “institution,” “identity,” and “cause” can carry stronger ontological implications than an analysis intends. GR does not require abandoning ordinary terminology, but it does require sensitivity to when a convenient analytical label begins doing metaphysical work. Local definitions, explicit qualifications, and distinctions among registers can prevent such slippage.

Suspension also has a scalar dimension. An object may be treated as sufficiently

real and stable at one scale while its internal constitution remains suspended. A firm can be analyzed as an organizational actor in a market model even though its internal relations are heterogeneous and contested. A person can be treated as a subject of a legal relation while a theory of personal identity remains open. A currency can be treated as a stable monetary unit for one period while its institutional production, material carriers, and symbolic representations are separately investigated. The relevant question is whether the chosen level of objecthood supports the explanatory task without obscuring transformations that matter to the result.

The same logic applies temporally. A relational object may be stable enough for analysis over one interval and unstable over another. Ontological suspension can prevent the local stability of an object from being mistaken for timeless identity. This is especially important in GRE because many objects of economic analysis are historically generated. Firms change legal form, markets change infrastructure, currencies change institutional backing, identities change classificatory regimes, and technologies alter the relations through which objects acquire causal efficacy. Suspension preserves the distinction between “stable enough here” and “fundamentally invariant.”

Methodological discipline further requires that suspension remain compatible with causal explanation. GR does not use suspension to avoid specifying mechanisms. If an object is treated as relationally real because it exerts causal efficacy, the analysis must still investigate the relations through which that efficacy arises. Saying that the object’s ultimate ontology is suspended does not permit causal vagueness. On the contrary, the weaker metaphysical commitment increases the importance of showing how persistence, stability, and causal participation are generated within the relevant system.

Claim 4.19 (Suspension and explanatory burden). *Ontological suspension reduces unnecessary metaphysical commitment but does not reduce the explanatory burden associated with relational or manifestational claims. The mechanisms, conditions, and scope supporting those claims remain open to investigation.*

Claim 4.19 prevents suspension from functioning as a refuge from difficult analysis. If GRE claims that trust, money, authority, or identity is relationally real, it must still account for how that reality is generated and maintained. Suspension concerns what need not be claimed beyond that analysis, not what need not be explained within it.

4.3.4 Suspension under Ontological Disagreement

Ontological disagreement is common in the domains GRE studies. Social ontology, philosophy of mind, philosophy of science, religious ontology, theories of value, institutional theory, and theories of agency all contain persistent disagreements concerning what kinds of things exist and what makes them real. Generative-relational inquiry requires a way to work across such disagreement without pretending that the disagreements have disappeared.

Ontological suspension provides one such mechanism. When competing ontologies generate the same commitments required for a local analysis, the inquiry can bracket their stronger divergence and examine the shared relational or manifestational structure. For example, researchers may disagree over whether institutions

possess a reality irreducible to individual action. If both accounts nevertheless recognize a stabilized pattern of rules, expectations, roles, and sanctions that causally affects behavior, a GRE analysis can investigate that relational configuration while leaving the deeper disagreement unresolved for the purposes of that task.

This strategy does not imply that ontological disagreement is always irrelevant. Competing ontologies may produce different explanations, identify different objects, or recommend different methods. A strongly individualist ontology and a strongly structural ontology may disagree not only about metaphysical grounding but also about where causal efficacy should be located. In such cases, suspension of the disagreement would remove precisely what the research needs to test. GR therefore distinguishes *non-operative disagreement*, which can be suspended for a particular task, from *operative disagreement*, which changes the explanatory consequences and must remain visible within the analysis.

Definition 4.20 (Operative ontological disagreement). An ontological disagreement is *operative* for an inquiry when competing ontological commitments alter the identification of relevant objects, mechanisms, causal relations, evidence, or explanatory consequences within that inquiry.

Definition 4.20 gives suspension a limit. If an ontological difference changes what the model predicts, what evidence counts, what mechanism is possible, or what intervention follows, the difference cannot be treated as merely philosophical background. Conversely, when competing metaphysical accounts leave the local explanation unchanged, insisting on prior resolution can add controversy without adding explanatory power.

This distinction also supports pluralism within GRE. Researchers can develop different ontological interpretations of the same generative-relational phenomenon while remaining part of a common research programme, provided the differences are explicitly stated and their consequences can be compared. Ontological suspension therefore does not require theoretical homogeneity. It can create a shared analytical surface on which competing deeper accounts become comparable where comparison is possible.

The possibility of comparison, however, must itself be established. Section 91 emphasized that two relational realities need not inhabit a common metric or even a common comparison space. The same caution applies to ontological theories. Two accounts may use incompatible primitive vocabularies or organize reality at different scales. Suspension can permit local cooperation without forcing artificial commensurability. When translation is possible, the mapping should be stated; when it is partial, the loss should be acknowledged; when no justified mapping exists, the disagreement may remain structurally unresolved.

This point is important for cross-cultural and interdisciplinary inquiry. A phenomenon can be embedded in distinct conceptual, religious, scientific, legal, or experiential ontologies whose categories do not align cleanly. GRE should not assume that one vocabulary can always translate another without remainder. Ontological suspension can preserve analytical openness while allowing researchers to study the relational consequences of each framework within its own domain. The aim is not to declare all ontologies equivalent but to avoid manufacturing equivalence before the

conditions of comparison have been established.

Under disagreement, suspension also has an ethical dimension. Prematurely declaring one ontology to be the sole legitimate description can erase forms of experience or institutional reality that remain causally consequential for the people and systems under study. At the same time, treating every ontological claim as equally warranted would abandon standards of evidence and argument. GR therefore seeks a middle position: stronger ontological judgments remain possible, but they are made only when the inquiry requires them and when the grounds for doing so are explicit.

4.3.5 Suspension and Analytical Usability

The practical value of ontological suspension lies in the fact that analytically useful objects need not wait for metaphysical closure. Economic and social inquiry routinely operates with objects whose ultimate status is contested but whose effects are sufficiently stable to investigate. GRE systematizes this practice by distinguishing analytical usability from final ontological settlement.

An object is analytically usable when it can be specified with enough stability, relational location, and causal relevance to support the explanatory task. This criterion is intentionally weaker than ontological fundamentality. A market index can be analytically useful without constituting a fundamental natural kind. A legal person can be analytically useful without resolving metaphysical theories of personhood. A social identity can be analytically useful without assuming that the category is natural, timeless, or internally homogeneous. A manifestational form can be analytically useful even when the underlying generative structure remains only partially known.

Claim 4.21 (Analytical usability under suspension). *An object whose stronger ontological status is suspended can remain analytically usable when its relevant boundaries, persistence, relations, and causal consequences are sufficiently specified for the explanatory task.*

This principle helps GR avoid two symmetrical errors. The first is ontological inflation: treating analytical usefulness as proof of fundamental existence. The second is ontological paralysis: refusing to analyze an object until every metaphysical question surrounding it has been settled. The former converts models into ontology too quickly; the latter makes inquiry dependent on conclusions it may never need. Suspension preserves a working space between them.

Analytical usability is nevertheless conditional. The object must remain adequate to the task. If a category's internal heterogeneity begins to change the result, the category must be refined. If a boundary that was previously harmless becomes causally consequential, the boundary must be reconsidered. If a manifestation once treated as a proxy ceases to track the relational structure of interest, the proxy must be revised. Suspension therefore works together with the revisability of analytical objects.

This revisability becomes particularly important in recursive systems. A category used for analysis can enter the system and alter what it describes. A risk classification can change credit access; a diagnostic label can alter self-understanding and institutional treatment; an economic forecast can change expectations and investment;

an aesthetic judgment can redirect attention and preservation; a legal designation can reorganize rights and obligations. Once the analytical or manifestational form becomes generatively active, its status may need to be reconsidered as part of the relational reality under study. Suspension does not isolate the analyst from this recursion. It requires the inquiry to notice when its representations have become causal components of the system.

The same principle applies to contingency. A contingent relational configuration may become stable enough to exert causal efficacy before any subject recognizes it as contingent. Its relational reality can therefore be analyzed prior to conscious manifestation. If the contingency later becomes aesthetically salient, symbolically articulated, or institutionally recorded, those manifestations may feed back into the relations through which the object persists. Ontological suspension allows the analysis to follow this history without forcing the pre-manifestational structure, the manifested object, and the ontological substrate into a single category.

The method can therefore be summarized as selective commitment. GR commits to what the explanatory task can support, suspends what lies beyond that support, and reopens suspended questions when they become operative. This practice preserves the reality of emergent and causally efficacious objects while resisting the conversion of local explanatory success into universal metaphysical closure.

Remark 4.22 (Suspension and the three-register architecture). Ontological suspension is one of the mechanisms that keeps the three registers analytically distinct. It permits a researcher to affirm relational or manifestational reality while leaving a stronger ontological claim unresolved, and it permits later evidence or argument to modify that suspension without retrospectively invalidating the earlier local analysis.

Ontological suspension thus completes the methodological groundwork established by Sections 90–92. Section 90 showed why GR requires explicit ontological commitments without ontological closure. Section 91 distinguished ontological, relational, and manifestational reality and treated their relations as a graph rather than a chain. The present section has specified how inquiry can move among those registers without making every local reality claim carry the burden of a final metaphysics. Section 93 can now turn from the status of ontological claims to the primitive analytical vocabulary through which emergent stability becomes describable: change, structure, constraint, and difference.

4.4 Primitive Structure and Ontological Emergence

Chapter 4 requires a vocabulary capable of describing emergence before the analysis presupposes fully constituted entities, subjects, relations, institutions, or values. Chapter 3 introduced change, structure, constraint, and difference as a minimal primitive structure for GRE. The present section develops their specifically ontological role. Its objective is to show how these four methodological primitives can support descriptions of emergence and relative stability while remaining compatible with ontological suspension. The discussion considers each primitive separately, then examines their interaction and the formation of stable configurations. The method is constitutive and deliberately weak: the primitives provide conditions for describing how

determinate objects can become analytically available without being declared the metaphysically fundamental constituents of reality.

Definition 4.23 (Methodological primitive). A *methodological primitive* is a distinction provisionally treated as analytically available before the framework constructs richer objects from it. Its primitive status concerns the organization of inquiry and does not, by itself, establish metaphysical fundamentality, temporal priority, indivisibility, or universal explanatory sufficiency.

Definition 4.23 preserves the restraint developed in Sections 4.1–4.3. Change, structure, constraint, and difference may themselves admit further decomposition under another theory or at another scale. A physical account may redescribe a social constraint through material processes; a phenomenological account may differentiate forms of experienced difference; a mathematical representation may replace ordinary structural language with a more specific formal object. GR does not require those approaches to converge on one final primitive basis. The quartet functions as a minimal analytical grammar within the present research programme.

The order in which the four primitives are presented should likewise be interpreted analytically rather than chronologically. GR does not posit a first moment of pure change followed by the later arrival of structure, constraint, and difference. Each becomes intelligible through the others. Change is identified through differences across configurations; structure supplies organization within which such differences can be tracked; constraints condition accessible transformations; and differences become consequential through structures that preserve, amplify, suppress, classify, or redirect them. Ontological emergence is therefore approached through their interaction rather than through a linear derivation from one primitive. (P. W. Anderson 1972; Simon 1962; Nicolis and Prigogine 1977)

4.4.1 Change

Change provides the minimal temporal opening required for emergence. An analysis of emergence must permit a later configuration to differ from an earlier one, or a process to produce distinctions that were previously absent at the selected resolution. GR therefore uses change in a broad sense: a transformation in state, organization, relation, capacity, boundary, rule, classification, or possibility. This formulation allows the identity of what changes to remain partly open. The object need not be fully constituted before the transformation begins.

This point matters for ontological analysis because many economic and social objects are encountered only after a period of formation. A firm may acquire legal identity after organizational relations have already developed. A monetary practice may stabilize before a standardized representation is established. A collective identity may become causally consequential while its boundaries remain contested. If inquiry requires the object to possess a fixed identity before change can be described, the formative process is partly hidden by the analytical vocabulary. Treating change as primitive permits the analysis to track transformations through which the object itself becomes more determinate.

Change can occur at several levels. A configuration can move among states while its constitutive organization remains relatively stable. Its structure can also change,

altering the pattern through which later states become possible. Constraints can be modified so that transitions previously difficult or unavailable become accessible, while other transitions become costly or foreclosed. Differences that were once negligible can become consequential when a new institution, technology, category, or practice makes them operative. A still stronger transformation can alter the mechanisms through which subsequent change is generated. GR therefore distinguishes change in a configuration from change in the generative conditions of that configuration.

The primitive also includes maintenance. Relative stability frequently depends on continuing activity. A legal status persists through records, procedures, recognition, enforcement, and institutional reproduction. A currency remains usable through repeated practices of acceptance, accounting, settlement, and regulation. A self-description may remain stable through memory, bodily continuity, social recognition, and narrative reproduction. In these cases, stability at one analytical level coexists with ongoing transformations at another. Change and persistence consequently form compatible features of a generative-relational system.

Change alone does not identify emergence. Many transformations dissipate without producing a persistent configuration. Others alter an existing object without generating a new one at the relevant scale. The ontological significance of change depends on the organization through which transformations become patterned, retained, and capable of affecting later processes. Structure supplies that second dimension.

4.4.2 Structure

Structure refers to patterned organization sufficiently coherent for distinctions, transitions, and dependencies to be tracked. Its use in GR is intentionally plural. A structure may be spatial, temporal, institutional, legal, symbolic, computational, ecological, organizational, or relational. Several descriptions may apply to the same configuration at different scales or for different research purposes. The concept therefore identifies organization without requiring one privileged formal representation.

Ontological emergence requires structure because a transient difference becomes analytically significant only within some organization that allows it to persist, recur, combine with other differences, or modify future transformations. A repeated exchange practice, for example, becomes more than a collection of isolated transactions when expectations, roles, records, conventions, enforcement, and material infrastructures organize later exchanges. The relevant structure may remain incomplete and historically variable while already giving the process a degree of continuity.

Structure should therefore be separated from rigidity. A configuration can preserve recognizable organization while its constituents change. An institution can maintain procedural continuity while officeholders, rules, technologies, and interpretive practices evolve. A network can retain a characteristic pattern while particular ties form and dissolve. A person can preserve an experienced continuity of self while memories, relations, capacities, and bodily conditions change. Structural stability is always relative to a selected resolution and timescale.

This relativity is especially important for the three-register architecture. A struc-

ture identified at the manifestational register may differ from the structure inferred at the relational register. Repeated observations can display a stable statistical pattern without uniquely determining the generative configuration that produces it. Conversely, a relational structure can remain sufficiently stable while its manifestations vary across contexts. Structure is therefore register-sensitive: the analyst should specify what is organized, at which register, over what interval, and under which observational or representational conditions.

Structure also enables the preservation of history. Earlier transformations can leave traces in current organization through institutional rules, material infrastructures, learned dispositions, network positions, accumulated resources, classifications, or path-dependent expectations. The resulting configuration carries a history without requiring every historical event to remain explicitly represented. In GR, emergence is therefore commonly historical: a stable object can condense prior transformations into a structure that shapes what happens next.

Treating structure as primitive does not establish that structure exists independently of process. The methodological role is narrower. GR requires a way to describe organized regularity before deciding whether a particular case is best understood through entities, relations, processes, fields, or another ontology. Structure supplies that analytical capacity while leaving its deeper constitution open.

4.4.3 Constraint

Constraint introduces asymmetry into the space of possible transformations. A configuration becomes generatively consequential when some transitions are enabled, favored, stabilized, delayed, made costly, prohibited, or rendered practically unavailable relative to others. Constraint therefore refers to conditions that shape the accessible trajectories of a system. Such conditions can be material, institutional, legal, informational, symbolic, bodily, ecological, computational, or relational, and their mechanisms need not be identical.

This primitive is central to emergence because persistence requires more than repeated occurrence. A configuration that can be displaced by arbitrarily small perturbations may appear momentarily without becoming stable at the timescale relevant to the inquiry. Constraints can help maintain a configuration by limiting departures, supporting repair, directing feedback, preserving boundaries, or making particular responses more probable. They can also prevent an incipient configuration from stabilizing. Emergence and dissolution are therefore both partly intelligible through changing constraints.

Constraint includes enabling conditions. A legal form can restrict certain actions while making others possible by defining authority, ownership, liability, or procedural competence. A technical standard can narrow acceptable formats while allowing coordination among previously disconnected participants. A language constrains expression through grammar and convention while enabling complex articulation. An infrastructure conditions where and how interaction occurs while expanding the range of activities that can be sustained. GR accordingly treats constraint through its effect on accessible possibilities rather than equating it with direct prohibition or coercion.

Constraints are often positional. The same institutional arrangement can enable one subject while restricting another; the same market structure can generate opportunity for one firm while creating dependency for another; the same symbolic classification can confer recognition in one setting and produce exclusion in another. The relevant question is therefore which transitions are conditioned for which subjects or configurations, through which mechanism, and over what scale. A general statement that a system is “constrained” is insufficient without this local specification.

Constraint can itself be historically generated. A practice repeated over time may produce expectations that later function as constraints. A manifestation can be recorded, classified, and institutionalized in ways that reshape subsequent possibilities. Material investments can harden earlier decisions into infrastructure. A legal category can stabilize a distinction and thereby affect later subject formation. In these cases, the outcome of one generative process becomes part of the constraint structure governing another. The graph-like architecture of Chapter 4 is therefore visible already at the primitive level: products of prior emergence can enter the conditions of later emergence.

No normative valence follows from constraint alone. A constraint may preserve a valuable capacity, stabilize domination, prevent harm, sustain cooperation, reproduce exclusion, or combine several of these effects. Normative evaluation requires additional concepts. At this stage, constraint identifies how the space of possible transformation acquires shape.

4.4.4 Difference

Difference provides the minimal condition for distinction. GR uses the term to indicate non-equivalence that becomes analytically or causally consequential within a configuration. Difference may concern state, position, timing, capacity, relation, classification, intensity, direction, accessibility, or another dimension selected by the inquiry. The concept remains weaker than inequality, opposition, hierarchy, or conflict. Those more specific relations can emerge from differences but should not be built into the primitive itself.

Difference is required for change to be identifiable, yet the two concepts remain distinct. A difference can coexist at one time between positions or components without representing a temporal transition. Change can then be described when a relevant difference appears, disappears, increases, decreases, or is reorganized across configurations. Structure determines which differences can be represented as belonging to a common organization, while constraint affects which differences can persist or alter later trajectories.

The significance of a difference is relational. A variation that has no effect under one configuration may become decisive under another. A minor difference in documentation can become institutionally consequential when a legal procedure uses that distinction to allocate status. A small variation in network position can matter greatly under a process that amplifies preferential access. An unarticulated bodily sensation can become salient after a new interpretive category is learned. A price difference can remain irrelevant where exchange is impossible and become operative once an infrastructure connects the relevant markets. Difference therefore becomes genera-

tive through the structures and constraints in which it participates.

This point also guards against equating manifestation with the production of difference itself. A difference can be relationally operative before it is perceived, named, or symbolized. A historical configuration may already contain an asymmetry or contingency that shapes subsequent possibilities before any subject recognizes it. Later manifestation can render that difference salient and can subsequently transform the relational configuration, but the manifestation is not necessarily the origin of the operative distinction. This possibility is especially important for GR accounts of contingency, aesthetic judgment, subject formation, and historical recognition.

Difference also raises the problem of comparability. Two differences cannot automatically be assigned magnitudes on a common scale. Some domains admit a justified metric; others support only ordering, classification, structural correspondence, or qualitative distinction; still others presently lack a warranted comparison map. The primitive of difference therefore establishes distinguishability without presuming commensurability. This restraint preserves the possibility that heterogeneous relational realities can each be real while remaining structurally incomparable under the currently available analytical framework.

4.4.5 Interaction among the Four Primitives

The four primitives acquire their explanatory force through interaction. GR does not treat change, structure, constraint, and difference as independent components that can simply be assembled after separate analysis. Each conditions how the others become meaningful. Structure permits differences and changes to be tracked; difference allows structures and transformations to be distinguished; constraints shape which transformations remain accessible; and change can modify the structures and constraints through which later differences become consequential.

A minimal generative description therefore asks how these dimensions co-organize over time. An initially small difference can be amplified when a structure channels repeated interaction through it. The resulting pattern can generate constraints that preserve the difference and make later trajectories dependent upon it. Conversely, a structural change can weaken an existing constraint, permitting previously suppressed differences to become operative and producing a new configuration. Feedback can stabilize the resulting organization, while another process may destabilize it. Emergence is located in this evolving organization rather than in any one primitive taken alone.

Claim 4.24 (Mutual constitution of the primitive structure). *Within GR, change, structure, constraint, and difference function as mutually conditioning methodological primitives. Their analytical separability supports explanation, while their interaction supplies the minimal vocabulary through which the formation, persistence, transformation, and dissolution of more determinate configurations can be described.*

Claim 4.24 also clarifies why the quartet should not be converted into a universal causal sequence. A local analysis may identify a historically specific trajectory—for example, an initial difference followed by structural amplification and later institutional constraint—but another process can begin from an inherited constraint, a structural rupture, or continuing change that gradually generates a distinguishable

difference. The explanatory order is empirical and historical. The primitive architecture provides the vocabulary within which that order can be reconstructed.

The interaction is likewise compatible with multiple registers. A manifestational difference can alter a relational constraint when an articulated classification changes institutional treatment. A relational structure can condition which manifestations become likely or intelligible. An ontological condition, where warranted, can restrict both relational formation and manifestation. Later sections will develop these cross-register relations in greater detail. At the present stage, the important point is that primitive interactions can occur within a register and across registers, and the mapping among them need not be one-to-one.

This framework also accommodates observer dependence without reducing reality to observation. The analyst chooses boundaries, timescales, and distinctions through which change, structure, constraint, and difference are represented. Those representational choices can alter what becomes visible in the analysis. The causal efficacy of the configuration, however, does not depend on the analyst having represented it correctly. GR therefore separates the methodological primitives used to construct an account from the stronger claim that reality itself is exhausted by those primitives.

4.4.6 Emergence of Stable Configurations

Ontological emergence, in the restrained sense required here, concerns the formation of a configuration that becomes sufficiently organized and persistent to be treated as a determinate object of inquiry. Such a configuration can arise when changes are retained through structure, consequential differences are reproduced or transformed, and constraints support a degree of continuity across further change. Stability is relative to timescale, boundary, and analytical resolution. It does not require immobility, permanence, equilibrium, or immunity to transformation.

Definition 4.25 (Relatively stable configuration). A *relatively stable configuration* is an organized pattern that preserves sufficient continuity across a specified region or interval of spacetime for its identity, effects, or transformations to be tracked at the resolution relevant to an inquiry. Its stability may depend on continuing generative activity and can coexist with substantial internal change.

Definition 4.25 deliberately stops short of defining relational reality. Relative stability is necessary for many relationally real objects, yet stability alone is insufficient. The following section will require the further question of causal efficacy within the generative-relational system. A recurring pattern that remains observationally stable but exerts no relevant causal effect may be descriptively important without satisfying the full GR criterion for relational reality. Section 4.4 therefore establishes the primitive conditions through which candidate objects become available for the stronger analysis that follows.

Stable configurations can be generated through heterogeneous mechanisms. Repetition can reinforce expectations; feedback can maintain a range of states; institutional rules can preserve boundaries; material infrastructures can retain earlier decisions; memory can transmit prior distinctions; legal recognition can stabilize roles; symbolic classifications can coordinate action; repair processes can restore a configu-

ration after perturbation; and dependencies can make departure increasingly costly. GR does not assume that one mechanism explains every case. The ontological architecture should remain broad enough to distinguish different stabilization processes while still asking how they participate in generation.

The emergence of a configuration also does not guarantee a unique level of description. The same process may stabilize an object at one scale while remaining highly variable at another. A market can display stable aggregate patterns while individual relations change rapidly. A person's self-experience can preserve continuity while bodily and social conditions evolve. An institution can remain legally continuous while its practical operation is reorganized. Multiple relatively stable configurations can therefore overlap, nest, or intersect without being reducible to one common metric or hierarchy.

The historical dimension is decisive. Once a configuration stabilizes, it can become part of the conditions under which later emergence occurs. A relational object generated at one time can constrain later action; a manifestation can be institutionalized and become a component of subsequent relational structure; a difference can be preserved in records and reactivated under a later classification regime. Emergence therefore changes the generative field in which future emergence takes place. The result is a recursive historical process rather than a sequence in which completed objects simply accumulate alongside one another.

Claim 4.26 (Primitive structure and ontological emergence). *Change, structure, constraint, and difference provide GR with a minimal methodological grammar for describing how relatively stable configurations can emerge, persist, transform, and dissolve. The adequacy of this grammar is assessed through explanatory use, while stronger claims concerning the ultimate ontology of the primitives or the configurations generated through them remain subject to the local requirements of inquiry and to ontological suspension.*

Claim 4.26 locates the quartet within the broader ontological position of this chapter. GR commits to enough reality for emergence, persistence, and causal organization to be investigated. It refrains from converting its starting vocabulary into a final metaphysics. This combination prepares the next step. Once a relatively stable configuration has emerged, the analysis can ask whether it also satisfies the stronger conditions of relational reality: emergence within a generative-relational system, persistence or stability over a relevant region or interval of spacetime, and causal efficacy within that system. The following section develops that criterion directly.

4.5 Relational Reality

Section 4.2 introduced relational reality as a technical category of GR: an object is relationally real when it emerges within a generative-relational system, persists with sufficient stability over a relevant region or interval of spacetime, and exerts causal efficacy upon that system. Section 4.4 then supplied the weaker primitive vocabulary through which candidate objects can become describable before the full criterion is satisfied. The present section develops relational reality in its own right. Its objective is to clarify the distinct work performed by emergence, persistence, stability, causal

efficacy, and participation in subsequent relations, while preserving the separation between relational reality and ontological identity. The method is constitutive rather than reductive: the section specifies conditions under which GR is warranted in treating an object as real within a generative-relational system without claiming that those conditions provide a final metaphysics of the object.

The criterion is deliberately stricter than the claim that something is related to something else. Almost any object can be placed within a descriptive network, and almost any event can be associated with surrounding conditions. Relational reality instead concerns an object whose generation and continuation depend upon a relational domain in which it becomes sufficiently determinate to participate causally in what follows. A transient correlation, an observer's arbitrary grouping, or a label that leaves the system unchanged does not become relationally real merely because it can be represented relationally. The category therefore joins an ontological claim of limited scope to an explanatory burden: the analyst must specify the relevant system, the object that has emerged, the domain over which it persists, the sense in which it remains stable, and the causal differences attributable to its presence. (Emirbayer 1997; Berger and Luckmann 1966; Searle 1995)

Claim 4.27 (Relational reality is system-relative rather than merely observer-relative). *A claim of relational reality is indexed to a generative-relational system and an analytically relevant spacetime domain. This relativity does not reduce the object to an observer's description: once the relevant conditions are satisfied, the object can constrain, enable, or redirect processes independently of whether a particular subject recognizes or represents it.*

Claim 4.27 is important because the word *relational* can otherwise be misread as subjective or perspectival. GR uses the term differently. The system relative to which an object is real can include human subjects, institutions, technologies, material infrastructures, ecological processes, symbolic practices, artificial agents, or other relevant components. A subject's recognition may participate in the system, but recognition is neither a universal origin nor a universal criterion of relational reality. This allows GR to treat pre-perceptual and pre-symbolic structures as relationally real when they already satisfy the stronger conditions developed below.

4.5.1 Relational Emergence

Relational emergence concerns the formation of an object through the organization of relations rather than its simple discovery as an already individuated unit. The phrase does not imply that nothing existed before the object emerged. Components, processes, histories, constraints, and differences may already be present. What emerges is a relatively determinate object at the scale and register relevant to the inquiry: a monetary form, an institutional office, a trust relation, a collective identity, an experienced continuity of self, a market convention, or another configuration that was not previously available in the same organized form.

Definition 4.28 (Relational emergence). *Relational emergence* is the formation of an analytically individuable object through a generative configuration of relations such that the object's organization and later effects cannot be specified adequately by treating the participating relata as independent of the relations through which the object

is generated.

This definition makes emergence stronger than aggregation. A pile of tokens and a currency system can contain some of the same material objects while differing radically in relational organization. A set of persons and an institution can contain the same persons while the institution introduces offices, expectations, procedures, permissions, records, sanctions, and histories that modify what those persons can subsequently do. The relationally emergent object is therefore not identified merely by listing its components. Its identity depends on the organization through which components, positions, practices, and constraints become mutually consequential.

Relational emergence can be gradual, punctuated, recurrent, or threshold-like. A trust relation may thicken through repeated interaction without a uniquely identifiable first instant. A legal status may acquire a conventional date of creation even though the relational conditions that make it operative developed earlier. A collective identity may emerge unevenly across subgroups and institutions. A digital asset may be technically instantiated at one moment while its economic relational reality develops only as practices of recognition, exchange, custody, valuation, and regulation stabilize around it. GR therefore avoids requiring a universal temporal signature for emergence.

The same caution applies to boundaries. An emergent object can possess fuzzy or contested edges while still becoming causally consequential. Institutional membership may be disputed; the practical boundary of a market may shift; a social category may be internally heterogeneous; the experienced continuity of self may vary over time. Relational reality requires enough individuation for the object's persistence and causal role to be investigated, but it does not require perfectly sharp boundaries. Boundary specification is part of the inquiry and may itself be historically generated.

Relational emergence also need not begin with manifestation. A configuration can develop through material, institutional, ecological, computational, or social relations before any subject recognizes it as a distinct object. This point is especially significant for contingency. A historically specific conjunction may already organize possibilities and causal pathways before its contingency becomes salient, aesthetically judged, or symbolically articulated. Later manifestation can disclose and transform such a configuration, but the event of recognition need not be the origin of its relational reality.

Conversely, manifestation can participate in emergence. A declaration, classification, measurement, ritual, image, diagnosis, price, legal inscription, or self-description can become one of the relations through which an object is generated. The graph architecture developed in Section 4.2 therefore applies within relational emergence itself: the existence of pre-manifestational emergence in some cases does not imply that manifestation is causally external to emergence in all cases. Later sections examine this mediation more directly.

4.5.2 Persistence

Emergence alone is insufficient for relational reality because many configurations appear briefly and disappear without acquiring an enduring role in the system. Persistence identifies the temporal and spatial continuation required for an emergent object to become more than an evanescent event at the chosen scale. The relevant in-

terval need not be long in absolute terms. What matters is whether the object remains available across enough of the system's evolution to participate in the processes under study.

Definition 4.29 (Persistence). For a specified generative-relational system and analytical domain, *persistence* is the continued or recurrent availability of an emergent object's identity across a region or interval of spacetime despite changes in some of its constituents, states, manifestations, or supporting relations.

Persistence is therefore compatible with transformation. A corporation can persist while employees, shareholders, strategies, assets, technologies, and offices change. A currency can persist while notes are redesigned, payment infrastructures change, exchange rates fluctuate, and particular users enter or leave the system. A sense of self can persist through changes in memory, body, social position, and narrative. Treating persistence as invariance would make relational reality unable to describe precisely the historically evolving objects for which the concept is intended.

The criterion instead requires an account of continuity. Such continuity can be maintained through legal succession, memory, records, repeated practices, network position, material infrastructure, reciprocal expectation, reproduction of rules, or other mechanisms. Different objects can therefore persist through different processes. GR does not require one universal substance that remains unchanged beneath all variation. The continuity relation itself is an object of analysis.

Persistence can also be intermittent. A ritual order may become active only at recurrent times; an institutional emergency power may remain latent between invocations; a social identity may become salient in some contexts and recede in others; a market convention may disappear locally and reappear under recurring conditions. If the organization that allows recurrence remains sufficiently preserved, relational reality can extend across periods in which the object is not continuously manifest. Persistence should therefore be evaluated relative to the object's mode of existence rather than by imposing uninterrupted perceptual presence as a universal standard.

This point again separates relational reality from manifestation. An institution does not cease to be relationally real whenever no one is looking at its records. A trust relation does not vanish between encounters merely because it is not presently expressed. A contingency-producing structure can remain operative before anyone recognizes the contingency it affords. Persistence concerns the continuation of the relational organization and its capacity to participate in later processes, not continuous appearance to a subject.

The spacetime domain of persistence must nevertheless be specified. An object can be relationally real locally without possessing the same reality everywhere or indefinitely. A currency can cease to function beyond a jurisdiction or historical period. A deity can organize one community's practices without possessing the same relational status in another community. A classification can be institutionally decisive in one administrative system and irrelevant in another. Locality does not weaken the reality claim; it identifies the domain within which the claim is warranted.

4.5.3 Stability

Persistence and stability are related but distinct. Persistence asks whether the object continues or recurs across a relevant domain. Stability concerns whether its organization remains sufficiently coherent under ordinary variation and disturbance for the object to retain explanatory identity. A pattern can persist statistically while being too unstable to function as a recognizable causal object, while a highly stable configuration can exist only over a short but analytically decisive interval. GR therefore retains both criteria.

Definition 4.30 (Relational stability). *Relational stability* is the capacity of an emergent object to retain sufficient organizational coherence under a specified range of endogenous variation, environmental change, perturbation, or reproduction for its identity and causal role to remain analytically trackable.

The phrase *sufficient organizational coherence* is intentionally contextual. There is no universal numerical threshold at which an object becomes relationally real. Relevant tolerance depends on the system, timescale, question, and kind of object. A high-frequency market convention may be stable over minutes while a constitutional institution is assessed over decades. A transient coalition can be stable enough to determine a vote even if it dissolves immediately afterward. A psychological self-relation may fluctuate affectively while retaining a recognizable organization across a life period. The threshold should therefore be justified by the explanatory task rather than imported from an unrelated domain.

Stability also does not imply equilibrium. Some relational realities persist precisely through continual adjustment. A market-making practice can remain stable through rapid repricing; an institution can preserve authority through repeated reinterpretation; a language can persist through continuous lexical and grammatical change; a trust relation can remain viable because participants repair disruptions rather than because disruptions never occur. Dynamic maintenance is therefore fully compatible with relational stability.

Nor does stability imply desirability. Durable domination, exclusion, debt dependency, stigmatization, and distrust can all become highly stable relational realities. The criterion is descriptive and ontological within GR's limited sense: it asks whether a configuration has acquired sufficient persistence and coherence to act within the system. Normative evaluation remains a separate task. This separation is essential for GRE because some of the most consequential economic realities are precisely those whose stability reproduces harmful or unjust relations.

Stability can furthermore be self-enhancing. Once an object begins to affect expectations and action, those effects can generate relations that increase the probability of its continued reproduction. Trust can induce cooperation that produces shared experience, thereby strengthening later trust. Distrust can induce defensive behavior that is interpreted as evidence for further distrust. Institutional authority can coordinate compliance that makes authority easier to reproduce. Capital can alter access to future opportunities in ways that facilitate further accumulation. GR treats such feedback as a possible mechanism of stabilization rather than as a universal property of relational reality.

The opposite process is equally important. Manifestations, interventions, shocks,

reinterpretations, or changes in supporting relations can destabilize an object. A currency crisis can weaken expectations that previously sustained monetary coordination. A legal judgment can reorganize an institutional category. A new historical archive can transform a collective identity. An aesthetic or epistemic manifestation can redirect attention toward previously unnoticed contingencies. Stability is therefore historically produced and historically vulnerable.

4.5.4 Causal Efficacy

Causal efficacy supplies the decisive condition that prevents relational reality from collapsing into descriptive persistence. A pattern can persist because the analyst repeatedly groups observations in the same way while making no difference to the system itself. GR requires more. The emergent object must participate in producing, constraining, enabling, redirecting, amplifying, suppressing, or otherwise modifying subsequent processes within the relevant generative-relational system.

Definition 4.31 (Causal efficacy in relational reality). An emergent object possesses *causal efficacy* within a generative-relational system when its presence, organization, or state participates in producing differences in subsequent states, relations, capacities, constraints, possibilities, or transformations of that system under a warranted causal account.

This definition is deliberately plural about causal methodology. Causal efficacy may be supported through intervention, counterfactual analysis, process tracing, mechanistic reconstruction, temporal comparison, natural experiment, simulation, ethnographic evidence, historical reconstruction, or another method appropriate to the object. GR does not require every causal claim to be expressed through one statistical formalism. It does require the analyst to distinguish causal participation from mere co-occurrence or interpretive convenience.

The relevant effect also need not be direct. Money alters behavior partly through prices, accounting, law, expectations, infrastructures, and obligations. Institutional authority can act through role recognition, procedure, sanction, habit, and anticipatory compliance. A god that is relationally real within a religious system can affect conduct through ritual, obligation, architecture, collective memory, moral interpretation, or institutional organization without requiring the relational claim to settle independent theological ontology. The experienced continuity of “I” can organize plans, responsibility, avoidance, memory, and future self-interpretation through several mediating processes. In each case, the causal object is relationally constituted while its efficacy is distributed through other relations.

Causal efficacy can also be latent. A legal right may not be exercised during a given interval while its recognized availability changes bargaining and expectation. A credit line may remain unused while altering the decisions available to a firm. An institutional veto can shape proposals precisely because actors anticipate it and avoid triggering its use. An object’s causal reality therefore cannot always be inferred from visible episodes of activation. Capacities, constraints, and expectations can be causally effective through the possibilities they open or foreclose.

At the same time, attribution must remain disciplined. Because relational objects are constituted through networks of supporting relations, it may be difficult to iso-

late their causal contribution. GR does not solve this problem by declaring every surrounding condition part of the object. The analyst must justify the individuation used in the causal account and remain open to competing decompositions. Different inquiries may locate causal efficacy at different but compatible scales; other decompositions may produce genuinely operative disagreements. Section 4.3 provides the framework for preserving such disagreements without forcing premature ontological closure.

The requirement of causal efficacy also clarifies why perception is unnecessary for relational reality. A structure can affect trajectories before it is recognized. Institutional exclusions can shape access before a person becomes aware of the classification producing them. A historical arrangement can make some encounters more likely than others before anyone interprets the resulting event as contingent. A biological or material condition can alter social relations before it is symbolized. Recognition may later become causally important, but the earlier efficacy is enough to establish that manifestation was not the origin of every relevant effect.

4.5.5 Participation in Subsequent Relations

Within GR, the most characteristic expression of causal efficacy is participation in subsequent relation formation. Relationally real objects do not merely produce isolated outcomes; they can alter who or what can enter into relation, which relations become possible or costly, how existing relations are interpreted, what capacities become available, which differences become salient, and how future objects can emerge. The reality of the object is therefore historical and generative: what has emerged can become part of the conditions of later emergence.

Claim 4.32 (Historical recursion of relational reality). *Once an emergent object becomes relationally real, it can enter the generative conditions of subsequent processes. It may thereby participate in producing relations and objects that were not part of the configuration through which it originally emerged.*

This recursive possibility gives relational reality particular importance for GRE. Money generated through prior legal, political, technological, and social relations can later organize labor relations, investment, debt, consumption, taxation, and institutional development. A firm generated through contracts, assets, labor, law, and organizational routines can become a subject of later relations and can itself create positions, technologies, products, obligations, and externalities. A category produced through administrative practice can redirect resource allocation and thereby alter the population to which the category is later applied. A reputation generated from past interaction can change access to future interaction. Relationally real objects can thus become relata in later systems while retaining histories of how they were generated.

This does not require the later system to reproduce the original generative conditions unchanged. An object can migrate across contexts, acquire new supports, or undergo partial reconstitution. A monetary instrument can move from one technological substrate to another. A religious symbol can enter commercial or political relations that differ from those through which its earlier significance formed. A self-description can be taken up by institutions and returned to the subject through clas-

sification. Relational reality can therefore accumulate layers of mediation over time.

Manifestational mediation is one important mechanism in this recursion. Once a relational reality becomes named, measured, pictured, priced, narrated, classified, or publicly performed, that manifestation can become part of the relations through which the object subsequently persists and acts. A trust relation can be altered by an explicit declaration of trust; a market can be changed by the publication of a price index; a person's institutional identity can be changed by the issuance of a document that records it. The manifestation is not merely a passive window onto a completed relational reality. It can become a causal component of the system.

The same logic applies to aesthetic judgment and contingency. A contingent configuration can already be relationally real before it is recognized. Once the contingency manifests as surprising, beautiful, significant, or worthy of attention, the judgment can redirect inquiry, preservation, imitation, funding, teaching, or collective memory. The manifestation then modifies the relational history of the object. GR can therefore affirm both pre-manifestational relational reality and later manifestational mediation without imposing a universal temporal chain between them.

Participation in subsequent relations also makes relational reality vulnerable to path dependence. Early stabilizations can influence which later relations become easier to form and which alternatives become progressively harder to realize. This does not imply deterministic lock-in. New relations, manifestations, shocks, interventions, and reinterpretations can reopen possibilities. The point is rather that a relationally real object can alter the field of later generation and therefore cannot be treated as an inert output of its own origin.

4.5.6 Relational Reality without Ontological Identity

The preceding conditions establish a mode of reality without settling ultimate ontological identity. An object can be generated, persistent, stable, and causally efficacious within a relational system even when competing metaphysical accounts of that object remain unresolved. This is not a concession that the object is somehow less real. It is a restriction on what kind of reality has been established.

Claim 4.33 (Relational reality does not entail ontological identity). *Establishing relational reality does not, by itself, establish that the relationally real object possesses an independent, fundamental, or uniquely specified ontological identity. Conversely, ontological existence does not by itself establish relational reality within every generative-relational system.*

Money again offers a simple example. Its relational reality can be demonstrated through the coordinated practices, institutions, expectations, inscriptions, and causal consequences that sustain monetary relations without deciding that money constitutes an ontologically fundamental kind. The experienced continuity of self can be relationally real because it persists and reorganizes action without requiring an invariant metaphysical ego. A divine object can be relationally real within a religious system because it organizes practice and causally affects subsequent relations while GR suspends the independent theological ontology of divinity. In each case, the relational claim is substantive even though a stronger ontological claim remains open.

This separation also accommodates objects whose manifestational structures differ radically. Consider a flower appearing under hallucinatory conditions and a flower encountered under ordinary physical conditions. Either can become relationally real if, within the relevant generative-relational system, it acquires sufficient persistence or recurrent stability and causally affects attention, action, memory, affect, communication, or subsequent relations. Their common eligibility for relational reality does not imply that the two objects possess the same ontological status, arise through the same generative structure, or admit a justified common metric. Shared reality-status is therefore weaker than structural equivalence.

Remark 4.34 (Reality-status and comparability). Two objects can both satisfy the criterion of relational reality while lacking a warranted metric, ordering, equivalence relation, structure-preserving mapping, or other common comparison framework. Comparability is an additional analytical achievement rather than a consequence of shared relational reality.

Remark 4.34 protects the framework from a hidden homogenization. Money, grief, trust, institutional authority, divine presence, selfhood, market price, mathematical beauty, and physical objects may all become relationally real under appropriate conditions, while the structures through which they emerge remain heterogeneous. Some can be compared quantitatively, some only through partial mappings, some through qualitative or structural relations, and some may presently lack a justified common comparison domain. GR therefore separates the question of reality from the question of commensurability.

The same principle limits causal comparison. The fact that two objects affect a system does not imply that their effects can be placed on one scalar measure. Their causal pathways may differ in kind, scale, timescale, or register. GRE should therefore resist universal measures of “relational reality” unless a particular research problem establishes a defensible comparison structure. The criterion identifies whether an object has become stably and causally real within a system; it does not assign a universal magnitude of reality.

Relational reality consequently occupies a precise position within the three-register architecture. It is stronger than mere appearance because manifestation is not required for its existence. It is stronger than descriptive recurrence because causal efficacy is required. It is weaker than a final ontological identification because the ultimate metaphysics of the object can remain suspended. It is historically generative because objects that satisfy the criterion can become conditions of subsequent relation formation. And it remains heterogeneous because shared relational reality does not erase differences in structure, ontology, manifestation, or comparability.

Section 4.5 therefore provides the central ontological criterion that later GRE analyses will repeatedly use. The next section turns to manifestational reality, where the question changes from how an object becomes stably and causally real within a generative-relational system to how realities become available in particular modes of appearance, experience, articulation, measurement, classification, and symbolization. The distinction prepares the chapter to analyze manifestation without treating manifestation as either a transparent copy of relational reality or the universal condition of its existence.

4.6 Manifestational Reality

Section 4.5 established relational reality through a demanding criterion: an object must emerge within a generative-relational system, persist with sufficient stability over a relevant spacetime region or interval, and exercise causal efficacy within that system. The present section turns to a different register. Its objective is to clarify what GR means when something becomes *manifest*, to distinguish manifestation from perceptual appearance alone, and to specify how manifestation can be real without being treated as transparent disclosure of either relational structure or ultimate ontology. The section also clarifies an asymmetry that is central to the three-register architecture. Relational reality can precede manifestation, while manifestations, once they occur, can enter generative-relational systems and mediate subsequent relational realities. The method is therefore register-specific: manifestation is analyzed according to the conditions of appearing, availability, differentiation, and mediation that make something manifest, rather than by importing the persistence criterion of relational reality into the manifestational register.

The term *manifestation* is used broadly but not loosely. GR does not restrict manifestation to what a human subject consciously sees, hears, or names. A phenomenon can become manifest through bodily sensation, affective salience, institutional classification, a legal record, a measuring instrument, a statistical pattern, a computational output, a material trace, a linguistic expression, or another medium through which a difference becomes available in a determinate form. Conscious perception is one important mode of manifestation, and symbolic articulation is another, but neither exhausts the category. The concept is intended to capture the general problem of how something becomes available *in a mode of appearance* within a historically and relationally structured situation. (Husserl 2012; Merleau-Ponty 2012; Gibson 1979)

Definition 4.35 (Manifestation). A *manifestation* is an event, state, or process through which a difference becomes sufficiently differentiated and available within a particular manifestational configuration to be encountered, registered, expressed, recorded, measured, classified, symbolized, or otherwise operative in a mode of appearance.

The phrase *manifestational configuration* is important. Something does not manifest in an abstract vacuum. Manifestation occurs through conditions that can include bodily capacities, sensory organization, instruments, media, languages, classificatory systems, institutional procedures, technical infrastructures, cultural expectations, temporal position, spatial scale, and prior relational histories. These conditions do not necessarily create ex nihilo whatever becomes manifest. They participate in determining the form, resolution, salience, and availability through which it appears.

Definition 4.36 (Manifestational reality). *Manifestational reality* denotes the reality of something insofar as it becomes manifest within a specified manifestational configuration. A claim of manifestational reality establishes that a manifestation occurs or is available in a determinate mode; it does not by itself establish the ultimate ontological identity of what manifests, the complete relational structure from which the manifestation emerges, or the persistence and causal efficacy required for relational reality.

Definition 4.36 gives manifestation its own reality-status without turning manifes-

tation into an inferior copy of a supposedly more authentic domain. A pain can be manifestationally real as pain even while its physiological, psychological, or social constitution remains under investigation. A recorded market indicator can be manifestationally real as an institutional and statistical object even when the relation between the indicator and the economic processes it is intended to represent is contested. A flower appearing under hallucinatory conditions can be manifestationally real as an appearance without this settling whether its ontological or relational structure is equivalent to that of a physically encountered flower. Manifestational reality therefore protects the reality of appearance while keeping open the question of what kind of reality the appearance discloses.

Claim 4.37 (Manifestation does not require conscious recognition). *A manifestation can occur through a bodily, technical, institutional, material, statistical, or symbolic configuration without having already become the object of a subject's reflective awareness. Conscious recognition is one possible uptake of manifestation rather than its universal condition.*

A sensor can register a signal before a human observer inspects the record. An institution can classify an event through an automated procedure before any individual deliberates about the classification. A bodily change can become affectively or physiologically manifest before it is linguistically identified. A material trace can remain available for later interpretation even though no subject was present when it was formed. Claim 4.37 should not be read as making manifestation independent of every relation of availability. A purely postulated state that makes no difference in any manifestational configuration has not yet been shown to be manifest. The point is narrower: the configuration through which something manifests need not center a conscious human observer.

4.6.1 Manifestation and Appearance

Appearance is one of the most intuitive forms of manifestation, yet the two concepts should remain distinct. *Appearance* usually emphasizes how something presents itself within perceptual or phenomenological experience. *Manifestation* is broader. It includes appearance but also extends to forms whose availability depends on inscription, instrumentation, classification, computation, institutional procedure, or other mediating arrangements. GR uses the broader term because many economically consequential realities become available through mechanisms that are neither immediately perceptual nor reducible to an individual subject's phenomenal field.

A price displayed on a screen is perceptually available, but its economic manifestation also depends on pricing procedures, data infrastructures, currency conventions, temporal updating, and the institutional setting in which the number is treated as a price. An unemployment rate can become manifest through a published statistical series even though no individual can perceive the aggregate directly. A legal status can become manifest through a registry entry, certificate, database record, or institutional response. An affective orientation can become manifest through bodily sensation, conduct, speech, or repeated patterns of attention. These cases differ in medium, yet each involves a transition from conditions that are not fully available in the same form to a determinate mode in which some difference can be encountered or acted

upon.

Remark 4.38 (Appearance is a mode of manifestation). Perceptual appearance is treated in GR as an important subset of manifestation rather than as its general model. This avoids making conscious perception the template against which institutional, technical, symbolic, statistical, and material manifestations must be understood.

The distinction also matters for the relation between manifestation and representation. Some manifestations are explicitly representational: a graph, map, account balance, legal document, or verbal statement purports to stand for or organize something beyond the inscription itself. Other manifestations can occur before any such representational intention is established. Pain, surprise, bodily tension, a sensor spike, an anomalous pattern, or a change in behavioral regularity can become manifest without first being encoded as a representation of a known object. Representation is therefore one important organization of manifestation, but manifestation is analytically prior to the claim that a manifested form successfully represents what produced it.

This distinction is especially useful for contingency. A generative-relational and historical configuration can already exist before a subject recognizes a contingent relation within it. When contingency later becomes salient, striking, surprising, beautiful, troubling, or otherwise available to judgment, the event enters the manifestational register in a new way. The manifestation does not create the entire relational history that made the encounter possible. Yet the manifestational event can redirect attention and later action. GR can therefore distinguish the prior relational conditions of an encounter, the manifestational event through which contingency becomes available, and the subsequent relations generated after the encounter.

4.6.2 Stability of Manifestation

The stability requirement developed for relational reality should not be transferred mechanically to manifestational reality. A manifestation can be brief. A flash of pain, an unexpected image, a transient reading, a momentary affect, or a singular institutional announcement may be manifestationally real even if it disappears quickly. The occurrence of manifestation and the persistence of manifestation answer different questions.

Claim 4.39 (Manifestational reality does not require persistent stability). *A manifestation can be real as manifestation without satisfying the persistence and stability conditions required for relational reality. Stability is therefore a variable property of manifestation rather than a universal threshold for entry into the manifestational register.*

This claim prevents the three registers from collapsing into one criterion. If every real manifestation had to persist and exercise causal efficacy, manifestational reality would become only a special case of relational reality. GR instead permits an event to be manifestationally real while remaining fleeting, weakly consequential, or difficult to reproduce. The analytic task is then to ask what kind of stability, if any, the manifestation acquires.

Several forms of manifestational stability can be distinguished. A manifestation

may be *temporally stable* when it remains available over an interval; *recurrently stable* when similar manifestations reappear under comparable conditions; *inscriptionally stable* when a record preserves the manifestation; *cross-contextually stable* when related forms appear across different manifestational configurations; or *procedurally stable* when a standardized practice repeatedly produces a comparable output. These forms should not be treated as equivalent. A persistent database entry, a recurring bodily sensation, and a reproducible measurement can all display stability while relying on different mechanisms.

Stability matters because stabilized manifestations can acquire new analytical and causal capacities. A fleeting observation can be remembered, recorded, replicated, standardized, circulated, or incorporated into a classification. Repeated manifestations can make a pattern easier to recognize. Institutionalized manifestations can become reference points for action. A statistical indicator that is repeatedly produced can influence contracts, policy, expectations, and investment. Under such conditions, the manifestation itself may begin to satisfy the criteria of relational reality: it can emerge as a relational object, persist, stabilize, and exert causal efficacy within the system. This transition is possible rather than automatic.

Remark 4.40 (Stabilized manifestation and relational reality). A stabilized manifestation becomes relationally real only when the full relational criterion is warranted. Repetition, inscription, or durability can contribute to that criterion, but none alone establishes causal efficacy or relational emergence.

This distinction is particularly important in economics, where measurements and classifications can become embedded in the domains they describe. A rating, index, price category, credit score, or accounting classification may begin as a manifestational device. Once actors coordinate around it, institutions incorporate it, and future opportunities are conditioned by it, the manifestation can participate in producing a new relational reality. The framework therefore leaves open a historically important pathway from manifestation to stabilization to causal participation without treating that pathway as inevitable.

4.6.3 Context-Dependent Manifestation

Manifestation is context-dependent because what becomes available, and in what form, depends partly on the configuration through which availability is produced. Context here does not mean merely a subjective frame imposed by an observer. It includes the material, technical, institutional, bodily, linguistic, historical, and relational conditions that organize manifestation.

Definition 4.41 (Manifestational configuration). A *manifestational configuration* is the organized set of conditions through which differences can become available in a particular mode of manifestation. It may include subjects, bodies, instruments, media, institutions, categories, languages, technical procedures, spatial and temporal scales, and historically formed expectations.

Changing the configuration can change the manifestation without requiring that the underlying relational or ontological conditions have changed in the same way. A market process can look different at transaction-level, daily, and annual temporal resolutions. The same institutional relation can appear differently in a legal document,

an ethnographic encounter, an administrative database, and a network representation. A bodily condition can manifest through pain for one subject, an imaging signal for a clinician, and a coded category for an insurer. These differences are not necessarily errors. They reflect distinct manifestational configurations that select, amplify, suppress, aggregate, or transform different aspects of what becomes available.

Context-dependence therefore places a burden on analysis. A manifestation should be reported together with the conditions under which it arose when those conditions materially affect interpretation. A number without a measurement procedure, an image without information about the instrument or scale, a classification without its institutional rule, or a statement without the relational situation of utterance can obscure how the manifestation was generated. This requirement becomes especially important when manifestations are used epistemically, because the analyst may otherwise mistake properties of the manifestational configuration for properties of the relational reality under study.

Claim 4.42 (Context-dependence and constraint). *The context-dependence of manifestation does not imply that manifestations are freely constructed by the manifestational configuration. Manifestations can be jointly constrained by the object, the generative-relational history, and the conditions of manifestation even when their form varies across contexts.*

Claim 4.42 protects the framework from a simple opposition between naive realism and unrestricted constructivism. GR allows that a manifestation can be mediated and context-sensitive while still being constrained by realities not reducible to the mediating context. The exact distribution of constraint is an empirical and theoretical question. In some cases instrumentation dominates the form of output; in others the object strongly constrains what can appear; in still others institutional categories shape which differences become visible at all. The framework does not decide these cases in advance.

Historical context adds another dimension. Manifestational configurations themselves evolve. Categories that once lacked institutional existence can later become standardized. New instruments can make previously inaccessible differences measurable. Changes in language can alter what can be articulated. Political and administrative reforms can transform what gets counted, recorded, or ignored. Digital infrastructures can increase the granularity and frequency with which economic conduct becomes manifest. Manifestational reality is therefore historical not only because the objects manifested can change, but also because the conditions of manifestation acquire histories of their own.

4.6.4 Multiple Forms of Manifestation

The manifestational register is plural. A single relational reality can become available through multiple forms, and these forms need not preserve the same dimensions of the underlying relations. GR therefore resists treating any one manifestational mode as universally privileged.

For analytical purposes, several broad forms can be distinguished. *Perceptual manifestations* include sensory appearances and experienced salience. *Affective manifestations* include felt intensities, moods, attractions, aversions, and embodied orien-

tations. *Behavioral manifestations* concern patterns expressed through action or response. *Material manifestations* include traces, spatial arrangements, artifacts, and physical inscriptions. *Institutional manifestations* include statuses, decisions, records, procedures, and classifications. *Symbolic manifestations* include linguistic expressions, images, signs, narratives, diagrams, and formal notations. *Statistical manifestations* make patterns available through aggregation, estimation, or comparison. *Computational manifestations* include model outputs, classifications, embeddings, simulations, scores, and other forms whose availability depends on computational procedures. These categories can overlap, and no exhaustive taxonomy is intended.

The plurality matters because different manifestations can support different kinds of action. A person may experience economic insecurity affectively before it becomes visible in an income statistic. A statistical indicator can reveal an aggregate pattern that is unavailable to individual perception. A legal category can make a relation institutionally actionable even when the lived experience of participants differs from the category's description. A model output can make a latent structure analytically tractable while remaining opaque to ordinary experience. GR treats these as different manifestational achievements rather than as successive approximations to a single canonical appearance.

Claim 4.43 (Manifestational plurality). *The availability of multiple manifestations of an object does not establish that one manifestation is universally primary, nor does variation among manifestations by itself establish contradiction. Their relations depend on the conditions, scales, and dimensions through which each manifestation is generated.*

Manifestational plurality also means that absence in one mode does not establish absence in all modes. A relation that is not consciously perceived may nevertheless be institutionally recorded. A pattern that is statistically weak at one scale may become visible at another. A social expectation that is rarely articulated explicitly may manifest through repeated conduct. Conversely, strong manifestation in one mode does not guarantee corresponding manifestation elsewhere. A publicly salient narrative can be weakly reflected in measured behavior, while a powerful statistical regularity can remain culturally unnoticed.

This plurality prepares a later problem. If relationally different structures can sometimes give rise to similar manifestations, and if similar structures can sometimes generate different manifestations under different conditions, reverse inference from manifestation to relation becomes non-trivial. The present section does not yet formalize that inverse problem; Sections 96–98 will develop the generative relation and its possible non-invertibility. The point required here is simply that manifestation is many-formed and mediated.

4.6.5 Manifestation without Ontological Disclosure

A manifestation can be genuine without disclosing ultimate ontology. This is a direct consequence of the distinction among the three registers. The fact that something appears, is measured, is recorded, or becomes symbolically articulated establishes a manifestational fact. It does not automatically settle what ultimately exists, what type of entity is responsible for the manifestation, or which ontological account is uniquely correct.

Claim 4.44 (Manifestation does not entail ontological disclosure). *A warranted claim that something has manifested does not by itself warrant a unique claim about the ontological identity, fundamental constitution, or complete generative structure of what manifests.*

The point is stronger than the familiar possibility of perceptual error. Even highly reliable manifestations can underdetermine ontology. A reproducible measurement may be compatible with several theoretical interpretations. A stable institutional record may establish how an organization recognizes a status without resolving the broader ontology of that status. A subject's persistent experience of self can be manifestationally robust while competing accounts of selfhood remain open. A religious manifestation can be experientially and socially consequential while theological ontology remains under suspension. The reality of manifestation and the ontology of its source are therefore distinct questions.

The flower example clarifies the distinction. A flower appearing under hallucinatory conditions and a flower encountered under ordinary physical conditions can both possess manifestational reality as appearances. Their manifestational status does not establish that their relational structures are the same, that their ontological statuses are the same, or that they can be compared through a common metric. The manifested forms may even be phenomenologically similar while being generated through substantially different relational configurations. Manifestational reality preserves the reality of what appears without using similarity of appearance as a shortcut to ontological identity.

Ontological suspension is therefore especially useful in the manifestational register. An inquiry can investigate the conditions, stability, variation, and consequences of manifestations while suspending stronger claims that are unnecessary for the problem at hand. This suspension should remain disciplined. If an ontological difference becomes causally operative for the manifestation or for the question being studied, the inquiry can no longer treat that difference as irrelevant. Section 4.3's distinction between operative and non-operative ontological disagreement thus applies directly here.

The same restraint is needed in formal analysis. A manifestational space may be modeled statistically, geometrically, categorically, or computationally without implying that the modeled coordinates constitute the ontology of what manifests. Formal proximity among manifestations does not establish ontological proximity among their sources. A classifier can group observations efficiently while leaving open whether the grouped cases share a common generative structure. Manifestational tractability is analytically useful, but it should not silently become metaphysical identification.

4.6.6 Manifestational Reality and Relational Reality

Relational reality and manifestational reality are distinct registers whose relations can run in more than one direction. The first distinction is temporal and constitutive: an object can become relationally real before it manifests within a particular configuration. The second is generative: once a manifestation occurs, it can enter the relations through which later realities are stabilized, transformed, or dissolved.

These two claims are compatible because the three-register architecture is a graph rather than a chain.

Claim 4.45 (Pre-manifestational relational reality). *Relational reality does not require prior manifestation to a subject, institution, symbolic system, or measuring apparatus. A relationally real object can already persist, remain stable, and exercise causal efficacy before it becomes manifest within a particular manifestational configuration.*

This claim is especially important for GR's account of contingency and aesthetic judgment. The generative-relational and historical conditions that make a contingent configuration possible can already be operative before the contingency is recognized. When the configuration later becomes manifest as surprising, beautiful, meaningful, or worthy of attention, the judgment belongs to the manifestational history of the object rather than to the origin of all its relational reality. The same structure applies to latent institutional tensions, unrecognized dependencies, or ecological relations whose effects are already causal before they become publicly salient.

Claim 4.46 (Manifestational mediation). *Once a manifestation enters a generative-relational system, it can participate in the generation, stabilization, transformation, or dissolution of relational realities by altering expectations, classifications, practices, capacities, constraints, or subsequent relations.*

A declaration of trust can modify the trust relation it manifests. A public price can alter transactions and expectations. A diagnosis can reorganize a person's institutional and interpersonal relations. A legal judgment can transform the relations whose status it articulates. An aesthetic judgment can redirect attention, preservation, teaching, funding, imitation, and collective memory. In these cases manifestation is not merely an epistemic window onto a completed relational object. It becomes one of the relations through which later reality is generated.

Manifestational mediation does not imply that every manifestation becomes relationally real. A fleeting image can disappear without reorganizing later relations. A measurement can remain unused. A symbol can fail to stabilize a practice. Conversely, a manifestation can acquire relational reality when it becomes sufficiently persistent, stable, and causally efficacious within the relevant system. The distinction is therefore maintained at the level of criteria rather than by assigning each object permanently to one register.

This point also clarifies why the registers should not be pictured as layers through which an object passes once. The same object can occupy different positions relative to different systems and times. A relation may be causally operative before it is manifest; later it may become visible through measurement; the measurement can then alter the relation; a new manifestation can follow from the altered relation; and some manifestations can become stabilized relational objects in their own right. Historical recursion replaces a universal sequence.

Remark 4.47 (Graph architecture of manifestation and relation). The distinction between relational and manifestational reality specifies different criteria of reality, not a fixed ordering of stages. Their connections are historically variable and can include disclosure, mediation, feedback, stabilization, transformation, and loss.

Section 4.6 therefore gives manifestational reality a precise but non-reductive role within GR. Manifestation concerns the modes through which differences become

available in determinate configurations. It can be perceptual without being only perceptual, symbolic without being only symbolic, stable without requiring stability, and causally consequential without being defined by causal efficacy. Its reality can be affirmed while its ontology remains suspended. Most importantly, manifestation neither creates all relational reality nor merely reflects it. The next section develops this relation more closely by asking how manifestation itself can emerge through generative-relational processes and how historically formed conditions govern what becomes available to manifestation.

4.7 Manifestation through Generative Relational Emergence

Section 4.6 defined manifestational reality in terms of the conditions under which differences become available in determinate forms. The present section asks a prior generative question: how do those conditions themselves arise, and through what relations does one rather than another manifestation become possible? The objective is to treat manifestation as an emergent event within historically formed generative-relational systems rather than as a transparent encounter between a finished object and an abstract observer. The section develops the manifestation relation, clarifies how manifestations can be relationally generated, examines the historical formation of manifestation rules, and studies institutional and cultural mediation. It concludes by specifying how manifestation mechanisms can themselves become embedded within relational reality. This formulation preserves the distinction between the registers while explaining why their relations are recursive rather than sequential.

A manifestation can often be described retrospectively as though an object simply possessed an appearance that waited to be observed. GR treats this description as analytically incomplete. What becomes available, in what form, with what salience, at what resolution, and to which subject, instrument, institution, or practice depends on a configuration of relations. Some of these relations may be bodily or material; others may be linguistic, technical, institutional, historical, cultural, economic, or political. Their participation does not imply that they freely invent what manifests. It means that manifestation has generative conditions. These conditions can select, amplify, suppress, translate, discretize, classify, stabilize, or otherwise organize differences into forms that become available within a manifestational configuration. (Giddens 1984; Archer 1995; Bourdieu 1977)

Claim 4.48 (Generative condition of manifestation). *A manifestation is generated through a determinate configuration of relations among what can manifest, the conditions through which it becomes available, and the historical structures that organize that availability. Manifestation therefore cannot in general be reduced either to an intrinsic property of an isolated object or to a free construction by an observing subject.*

Claim 4.48 gives the term *generative* a specific role. It directs inquiry toward the conditions under which a manifestation becomes possible and toward the processes that shape its form. The claim is compatible with ontological realism, constructivism in limited domains, process ontology, phenomenological inquiry, and other positions whose stronger metaphysical differences may remain suspended. Its commitment

is methodological and explanatory: manifested form has a history and a relational organization that can, at least in part, be investigated.

4.7.1 The Manifestation Relation

The most economical way to begin is to distinguish a manifestation from the *manifestation relation*. A manifestation is the available event, state, or form specified in Definition 4.35. The manifestation relation concerns the structured relation through which that availability is produced. It includes whatever configuration is relevant to explaining why a difference becomes manifest in one way rather than another.

Definition 4.49 (Manifestation relation). A *manifestation relation* is a contextually specified relation among a potential source of difference, a manifestational configuration, and an available manifested form, such that the relations among these components participate in generating how the difference becomes manifest within the relevant system.

The term *potential source of difference* is deliberately cautious. GR does not assume that every manifestation can be traced to a single antecedent object whose independent ontology is already established. The source can be a relational configuration, an event, a process, a field of variation, an interaction, or another analytically warranted object. In some inquiries, the source may remain partly unknown while its manifestations are well characterized. In others, the source may itself be relationally real but not ontologically identified. Definition 4.49 therefore avoids building a stronger ontology into the concept than the inquiry has earned.

The manifestational configuration likewise need not be centered on a conscious subject. A telescope, laboratory protocol, legal procedure, market platform, archival system, machine-learning classifier, bodily sensory apparatus, ritual practice, or collective discourse can participate in the relation. Human awareness may enter later, or it may be only one component among several. The relation is consequently broader than perception and broader than representation.

A manifestation relation also need not be deterministic. The same relational configuration can permit several manifestations, and similar manifestations can arise under substantially different configurations. Noise, contingency, path dependence, measurement error, strategic behavior, institutional discretion, and historical variation can all contribute to this multiplicity. Section 98 will later examine the consequences for invertibility. At the present stage, the important point is that the manifestation relation can be probabilistic, underdetermined, multivalued, or only partially reconstructable.

Remark 4.50 (Manifestation relation and causal relation). A manifestation relation can contain causal relations without being reducible to a single causal edge. It may include enabling conditions, constraints, classifications, selection procedures, interpretive conventions, technical transformations, histories of training, and other relations whose combined organization produces an available form.

This distinction matters for GRE because many economic manifestations are composites. A market price is not merely caused by a single underlying value. It becomes manifest through bids and offers, market rules, time windows, transaction infrastruc-

tures, currencies, accounting conventions, data publication, and expectations. An unemployment rate manifests through administrative and statistical categories that distinguish employment, unemployment, labor-force participation, and reference periods. A credit score manifests through variables, models, institutional objectives, data histories, and technical procedures. In each case, what becomes visible to later actors is generated through a manifestation relation whose components can themselves become objects of economic analysis.

The relation can also alter the salience of what becomes manifest. Two systems may contain similar differences while making different differences available for action. An institution that records only monetary transactions can make financial exchange highly manifest while leaving unpaid care weakly represented. A platform interface can make some prices, ratings, or rankings immediately visible while rendering other dimensions costly to discover. A legal procedure can transform a diffuse conflict into a set of admissible categories. Manifestation is therefore partly a problem of organized availability.

4.7.2 Relational Generation of Manifestation

The manifestation relation becomes generative when relations among its components do more than transmit a preformed appearance. They participate in producing the differentiated form through which something becomes available. This does not require the strong claim that manifestation creates the underlying object. It requires only that the manifested form cannot be adequately explained without the relations through which it is generated.

Definition 4.51 (Relational generation of manifestation). *Relational generation of manifestation* occurs when the organization and interaction of elements within a manifestational configuration participate in producing the form, salience, resolution, classification, or availability through which a difference becomes manifest.

Several kinds of generation can occur. A relation may *select* among differences by making some detectable and others practically invisible. It may *transform* a difference by translating it into another medium, as when a physical signal becomes a numerical reading. It may *aggregate* multiple events into a statistical indicator. It may *classify* heterogeneous cases under a legal or administrative category. It may *amplify* weak variation into a salient signal, or *suppress* variation through thresholds and normalization. It may *sequence* events into a narrative or temporal ordering. It may *stabilize* an otherwise fleeting manifestation by inscription, repetition, storage, or institutional recognition. These are different mechanisms and should not be collapsed into a universal operation.

The relational generation of manifestation is especially clear when the same underlying configuration becomes available differently under changed conditions. A bodily process may manifest as pain, as an imaging signal, as a diagnostic code, as a disability category, or as an insurance claim. None of these manifestations is simply identical to the others. Each is generated through different relations, infrastructures, conventions, and purposes. The multiplicity does not make every manifestation arbitrary. It shows that the same object or process can enter several manifestation relations whose outputs differ because their generative conditions differ.

A similar structure appears in economic life. Productive activity can manifest as wage income, firm revenue, national accounts, household time use, ecological burden, or reputational value depending on the relational apparatus through which it is registered. A transaction can manifest simultaneously as a price, a legal transfer, a tax event, a behavioral data point, and a change in inventory. The manifestations overlap but need not coincide. Each selects and organizes a different dimension of the event.

Claim 4.52 (Relational generation without ontological reduction). *The relational generation of a manifestation does not establish that the manifested object is exhausted by the relations that generate its manifestation. A warranted account of manifestation can therefore remain compatible with ontological suspension concerning the ultimate constitution of what manifests.*

This claim prevents two opposite reductions. One reduction treats manifested form as if it transparently revealed an independently fixed object. The other treats the object as if it were nothing more than the manifestation apparatus that renders it available. GR can reject both shortcuts. A scientific measurement can be instrument-dependent without implying that the measured phenomenon is only an instrument effect. A legal identity can be institutionally generated without implying that every relevant aspect of the person is reducible to legal classification. A market category can be constructed through conventions while still producing effects that constrain actors independently of their individual intentions.

The same reasoning is important for contingency. A contingent encounter may become aesthetically or existentially salient only within a historically formed relational field of expectations, memories, concepts, bodily dispositions, and prior experiences. The field participates in generating how contingency manifests. Yet the relevant relational structure need not begin at the moment of conscious judgment. It can already be operative before the subject recognizes the encounter as contingent. Aesthetic judgment can then become a later manifestation within that structure and, once expressed or acted upon, can feed back into subsequent relations. The example illustrates why manifestation is relationally generated without making perception the origin of relational reality.

4.7.3 Historical Formation of Manifestation Rules

Manifestations are not generated only by immediate relations. The relations governing manifestation can themselves have histories. Languages, measurement conventions, administrative categories, scientific instruments, market protocols, aesthetic canons, educational practices, and institutional procedures develop over time. These histories shape what can become manifest, what distinctions are available, and how manifestations are stabilized or interpreted.

Definition 4.53 (Manifestation rule). *A manifestation rule is a historically situated regularity, convention, procedure, threshold, classification, transformation, or selection principle that contributes to determining how differences become available within a manifestational configuration.*

The term *rule* should not be read as requiring explicit codification. Some manifestation rules are written: accounting standards, legal definitions, laboratory protocols,

survey instructions, platform ranking procedures, or diagnostic criteria. Others are tacit: habits of attention, cultural expectations, aesthetic conventions, professional judgment, or learned bodily sensitivities. Still others are embedded materially in instruments and infrastructures. Their common feature is that they shape manifestation with sufficient regularity to be analytically relevant.

Manifestation rules can emerge through deliberate design, cumulative convention, conflict, adaptation, technological change, institutional inheritance, or combinations of these processes. Their histories therefore matter. A category that appears natural at one moment may be the product of earlier administrative choices. A statistical threshold can reflect technical constraints that later become institutionalized. A visual convention can persist long after the original reason for its design has disappeared. Historical formation can thus become sedimented into present manifestation without remaining visible to current users.

Claim 4.54 (Historicity of manifestation rules). *Manifestation rules can be historically generated, reproduced, contested, and transformed. Their present operation therefore does not imply timeless necessity, and changes in these rules can alter what becomes manifest without requiring the underlying generative system to change in the same way or at the same rate.*

This claim is important for historical comparison. An economic object recorded under one classificatory regime may not be directly comparable with an apparently similar object recorded under another. A change in measured unemployment can reflect economic change, classificatory change, or both. The visibility of unpaid labor can change when survey design changes even if household practices remain partly continuous. A new diagnostic category can produce a sharp increase in recorded cases by reorganizing manifestation conditions. Historical analysis must therefore distinguish changes in what is generated from changes in how generation becomes manifest.

The same issue appears in intellectual history and aesthetics. What counts as a salient problem, elegant proof, beautiful form, or meaningful contingency is mediated by historically formed repertoires of attention and judgment. These repertoires do not mechanically determine judgment. They condition what is available for recognition and provide distinctions through which some encounters become expressible. An aesthetic judgment can therefore be both genuinely emergent and historically mediated. GR treats those properties as compatible.

Manifestation rules can also become self-reinforcing. Once an institution measures a phenomenon through a particular category, actors may adapt to the category; resources may be allocated through it; records may accumulate around it; and later inquiry may inherit the same distinction because data already exist in that form. The rule then participates in generating the relational reality it was initially used to manifest. Classification and object formation can become recursively coupled without becoming identical.

4.7.4 Institutional and Cultural Mediation

Institutions and cultures are especially important mediators because they stabilize manifestation relations across persons and time. They supply categories, roles, stan-

dards, symbols, practices, technologies, authorities, and expectations through which manifestations become publicly recognizable and actionable. Their mediation can increase coordination while also making some dimensions more visible than others.

Institutional mediation is evident in legal, administrative, financial, educational, scientific, and organizational settings. A legal institution can transform a history of conduct into a status, claim, liability, or judgment. A central bank can make monetary conditions manifest through rates, statements, balance sheets, and operational procedures. A university can convert heterogeneous intellectual activity into grades, degrees, publications, rankings, and appointments. A firm can make performance manifest through accounting categories and internal metrics. These manifestations often acquire consequences precisely because institutions authorize and reproduce them.

Cultural mediation operates through partially shared repertoires of interpretation, value, symbolism, attention, narrative, and expectation. Cultural mediation can affect which manifestations are immediately legible, which require explanation, which are trusted, and which are ignored. It can also affect the emotional and aesthetic force of manifestation. The same material configuration can be encountered as sacred, ordinary, threatening, beautiful, wasteful, prestigious, or economically valuable under different cultural relations. GR does not infer from this variability that cultural meaning is unconstrained. Material conditions, histories, institutions, and other relations can sharply delimit the range of viable manifestations.

Claim 4.55 (Mediation without passivity). *Institutional and cultural mediation can participate in generating manifestational form while the manifested object, the mediating structures, and the subjects involved remain capable of constraining and transforming one another. Mediation is therefore a generative relation rather than a one-way transmission channel.*

This reciprocal structure becomes visible when manifestations modify the institutions and cultures through which they were generated. A published economic indicator can alter policy debate and market expectations. A court judgment can reshape the categories used in later adjudication. A public aesthetic judgment can redirect preservation, funding, teaching, and imitation. A religious manifestation can reorganize ritual practice or institutional authority. A newly visible form of vulnerability can generate advocacy, legal reform, or changes in professional classification. What manifests through a mediating system can return as a difference within that system.

The possibility of recursive mediation also clarifies why power matters without making power the sole explanation of manifestation. Institutions can privilege some forms of visibility, define admissible evidence, allocate attention, or authorize classifications. Cultural hierarchies can make some voices more legible than others. Yet manifestation can also escape intended control, generate unintended consequences, or reveal differences that established institutions cannot easily assimilate. The manifestational field is structured, but structure does not imply perfect determination.

For GRE, this point is central to the analysis of value. Market price is a powerful manifestation because institutions, currencies, infrastructures, legal rights, expectations, and accounting practices make it actionable across many relations. Its institutional strength does not entail that price exhausts value. Other values may manifest

through care, ecological resilience, symbolic significance, social recognition, lived capability, or forms that remain weakly commensurable with price. Treating one manifestation as exhaustive can therefore become an epistemic and political act with generative consequences of its own.

4.7.5 Manifestation Mechanisms within Relational Reality

The final step is to locate manifestation mechanisms within the relational reality developed in Section 4.5. A manifestation mechanism is not merely an abstract rule for converting one description into another. It is an organized process through which relevant relations participate in producing a manifestation. Instruments, institutions, habits, classifications, algorithms, rituals, sensory systems, and discursive practices can all function as components of such mechanisms.

Definition 4.56 (Manifestation mechanism). *A manifestation mechanism is an identifiable organization of relations and processes through which differences are selected, transformed, amplified, suppressed, aggregated, classified, recorded, expressed, or otherwise made available in a manifestational form.*

Manifestation mechanisms can themselves become relationally real when they satisfy the criteria of relational emergence, persistence, stability, and causal efficacy. A statistical bureau's classificatory practice can persist across administrations and alter resource allocation. A pricing mechanism can become stable enough to organize expectations and transactions. A diagnostic infrastructure can shape treatment pathways and self-understanding. A platform ranking system can influence attention and behavior while being continually reproduced through technical and organizational relations. In such cases, the mechanism is not only a means through which other objects manifest; it becomes an efficacious object within the generative-relational system.

Claim 4.57 (Relational reality of manifestation mechanisms). *When a manifestation mechanism emerges within a generative-relational system, persists with sufficient stability, and exercises causal efficacy within that system, the mechanism can itself possess relational reality independently of whether the objects it manifests share the same ontological or relational status.*

The qualification at the end of Claim 4.57 is important. A measurement institution can be relationally real even when what it attempts to measure remains ontologically contested. A religious interpretive practice can be relationally real without resolving the ontology of the divine object it concerns. A hallucination-generating process can be relationally real even when the manifested flower lacks the same physical ontology as an ordinarily encountered flower. The reality-status of the mechanism and the reality-status of what manifests through it therefore require separate analysis.

This separation also prevents a common inferential shortcut. The persistence and efficacy of a manifestation mechanism do not guarantee the truth of every manifestation it produces. Stable institutions can classify incorrectly. Reliable instruments can be applied outside their calibrated domain. Cultural conventions can stabilize misleading interpretations. Algorithms can reproduce systematic error. Relational reality concerns the mechanism's efficacy within the system; epistemic warrant concerns the adequacy of the manifestation as evidence for another claim. Chapter 5 will

develop this distinction more fully.

At the same time, recurrent manifestation can alter the mechanism itself. Users adapt to metrics, institutions revise categories after disputes, instruments are recalibrated, algorithms are retrained, and cultural expectations change after repeated encounters. The mechanism therefore participates in a recursive history: it generates manifestations, those manifestations enter subsequent relations, and those relations can reconfigure the mechanism. This recursion is one reason the three registers are better represented as a graph of historically evolving relations than as a fixed chain.

Section 4.7 has therefore shifted the analysis from manifestation as an available form to manifestation as a generative-relational achievement. A manifestation relation connects potential sources of difference with historically situated configurations of availability. Relations can generate manifestational form through selection, transformation, classification, amplification, aggregation, and stabilization. The rules governing these operations can themselves be historical products, while institutions and cultures can mediate manifestation without fully determining it. Finally, manifestation mechanisms can become relationally real when they acquire persistence, stability, and causal efficacy. The next section develops the stronger possibility that the mapping through which relational reality becomes manifest can itself be treated as a relationally real object, provided that its own conditions of stability, persistence, and causal efficacy are established.

4.8 Relational Reality of Manifestation Mapping

Section 4.7 treated manifestation as a generative-relational achievement and introduced manifestation mechanisms whose histories can become part of the conditions of later appearance, classification, measurement, and symbolization. The present section takes one further step. It asks whether the relation through which manifestation is generated can itself become an object of relational reality. The answer is affirmative under conditions that are already familiar from Section 4.5: a manifestation mapping can become relationally real when it emerges within a generative-relational system, persists with sufficient stability over a relevant region or interval of space-time, and exercises causal efficacy within that system. This claim does not require the mapping to be ontologically fundamental, mathematically exact, consciously represented, or epistemically correct. It requires the mapping to become sufficiently organized and effective to participate in the subsequent generation of relations and manifestations.

The term *mapping* is used cautiously. It does not presuppose a mathematical function with a fixed domain, codomain, and unique output. Many manifestation mappings are partial, probabilistic, multivalued, path-dependent, historically revised, or only locally stable. A perceptual apparatus, an administrative classification, an accounting convention, a legal procedure, a scientific instrument, an algorithmic classifier, a market protocol, or a linguistic convention can all establish structured correspondences through which differences become manifest. In some cases a formal function is an appropriate representation. In others, a rule, relation, stochastic kernel, institutional procedure, embodied disposition, or family of transformations is

more adequate. The ontological point made here is prior to the choice among those representations. (Giere 2004)

Definition 4.58 (Manifestation mapping). A *manifestation mapping* is a relatively organized relation, rule, process, or mechanism through which differences available within a generative-relational system are transformed, selected, classified, amplified, aggregated, translated, or otherwise rendered into one or more manifestational forms under specified conditions.

Definition 4.58 makes the mapping itself analytically visible. Instead of treating a manifestation as if it simply followed from an antecedent object, inquiry can ask what mediates the transition from a field of differences to the form in which those differences become available. This shift is important because the mediator can acquire a history, a domain of validity, a characteristic stability, and causal consequences of its own. Once this occurs, the mapping is no longer merely a convenient description imposed by an analyst. It can become part of the reality through which the system continues to generate later manifestations.

4.8.1 The Manifestation-Mapping Lemma

The relation between manifestation mapping and relational reality can be stated as a direct consequence of the criterion developed in Section 4.5. The statement is called a lemma because it does not introduce a new criterion of reality; it applies the existing criterion to a higher-order object whose role might otherwise remain implicit.

Lemma 4.59 (Manifestation-mapping lemma). *Let a manifestation mapping be generated within a specified generative-relational system. If, over a relevant region or interval of spacetime, the mapping persists with sufficient stability and exercises causal efficacy in the production, organization, interpretation, or subsequent use of manifestations, then the manifestation mapping qualifies as relationally real relative to that system.*

The lemma is intentionally conditional. A proposed mapping does not become relationally real merely because an analyst can write it down. A transient coincidence, an unused classification scheme, a hypothetical function, or a rule that has not entered the operation of the relevant system may remain an analytical representation rather than a relationally real object. The mapping acquires relational reality when the system itself sustains it sufficiently for it to make a causal difference to what subsequently occurs.

This distinction can be illustrated by comparison between an analyst's model and an institutional classification. An economist may construct a mapping that groups heterogeneous transactions into a set of categories for one exploratory analysis. Unless that mapping enters broader practice, its reality may remain primarily representational. By contrast, when a tax authority, statistical office, payment platform, or accounting regime repeatedly classifies transactions according to a rule that affects reporting obligations, eligibility, prices, audits, or later decisions, the mapping can become relationally real. The classification rule participates in producing subsequent relations. Its effects can persist even when individual actors do not understand the rule in detail.

The same logic applies outside explicit institutions. A learned perceptual discrimination can become stable enough to alter attention and action. A linguistic distinction can guide recognition and social response across repeated situations. A ritualized interpretive practice can channel what members of a community notice and how they respond. An algorithmic ranking rule can shape visibility, participation, and incentives. In each case, the mapping is not merely between two descriptions. It is an operative relation embedded in a system.

Remark 4.60 (Higher-order relational objects). The manifestation-mapping lemma illustrates a broader feature of GR: a relation among objects, or a relation among relations, can itself become an analytically relevant object when it acquires persistence, stability, and causal efficacy. Relational reality therefore does not restrict objecthood to substance-like entities.

This remark is especially important for the graph architecture of the three registers. If mappings among relational and manifestational configurations can themselves become relationally real, then the edges of the graph are not always analytically external to the graph. Some edges can stabilize into objects that later participate in additional edges. A classification relation can become an institution; an interpretive convention can become a norm; a measurement protocol can become an infrastructure; an interface rule can become part of a market's operating environment. The system can therefore generate not only new nodes but also relatively stable relations that subsequently function as nodes at another analytical scale.

4.8.2 Stability of the Mapping

Stability is one of the conditions through which a manifestation mapping can become relationally real, but stability should not be confused with exact invariance. A mapping can remain sufficiently stable while its parameters, categories, thresholds, implementation, or local outputs change. The relevant question is whether enough of its organization persists for the mapping to remain identifiable and causally operative within the inquiry's scale.

Definition 4.61 (Mapping stability). A manifestation mapping exhibits *mapping stability* when variations in its conditions, implementation, or inputs remain compatible with sufficient continuity of its organizing relation for the mapping to retain an identifiable role in generating manifestations over the relevant analytical interval.

This definition allows several forms of stability. A legal mapping from circumstances to status can be stable because courts and agencies reproduce a rule even while interpretations evolve. A market protocol can remain stable while prices and participants change continuously. A perceptual mapping can remain stable across ordinary variation in illumination or bodily state. A machine-learning classifier can preserve a recognizable decision regime across a period of use even if its inputs vary widely. Stability concerns the organization of the mapping, not identity of outputs.

The degree of stability required is domain-relative. A high-frequency trading rule may be relationally significant over milliseconds or hours. A monetary classification can persist for decades. A ritual or linguistic mapping may persist across generations while changing gradually. GR therefore does not impose a universal temporal threshold. The relevant interval is set by the system and the explanatory question. What

matters is that the mapping endure long enough, and coherently enough, to influence subsequent relational development.

Stability also has spatial and institutional dimensions. A classification may be stable within one jurisdiction and absent in another. A cultural mapping may operate within a community without extending beyond it. A measurement protocol may be valid only within a calibrated range. A platform's ranking mechanism can be stable inside one technical environment while changing rapidly after a redesign. Treating stability as local prevents a relationally real mapping from being mistaken for a universal law.

Claim 4.62 (Stability without universality). *A manifestation mapping can be sufficiently stable to possess relational reality within a specified system without being invariant across all times, places, scales, institutions, or domains.*

Claim 4.62 is consequential for economic analysis. Economic categories often appear natural after long institutional stabilization. Wage, price, ownership, debt, unemployment, creditworthiness, productivity, and risk can become highly stable manifestations within particular infrastructures. Their stability can make them causally powerful without establishing that the mappings producing them are timeless or ontologically universal. A category may be deeply real in a relational sense precisely because institutions reproduce it, not because the category exists independently of those institutions.

Mapping stability can also be endogenous to repeated use. When actors anticipate how a system maps differences into manifestations, they can adapt their conduct accordingly. Firms alter reports to fit accounting categories; applicants modify behavior in response to scoring systems; researchers adapt experiments to measurement protocols; users optimize content for ranking systems. Repeated adaptation can stabilize the mapping by making surrounding behavior increasingly compatible with it. It can also destabilize the mapping by exploiting its weaknesses. Stability is therefore an historical outcome rather than a purely formal property.

4.8.3 Persistence of the Mapping

Persistence differs from stability. Stability concerns continuity of organization under variation; persistence concerns continued existence or recurrence of the mapping across the relevant spacetime interval. A mapping can persist while becoming less stable, as when a contested classification remains institutionally present but its interpretation fragments. Conversely, a mapping can be locally stable during each use yet fail to persist because the mechanism is abandoned after a short episode. Relational reality requires attention to both dimensions.

Definition 4.63 (Mapping persistence). A manifestation mapping exhibits *persistence* when the relations, practices, mechanisms, or infrastructures that sustain the mapping continue or recur across a relevant spacetime interval such that the mapping remains available to participate in subsequent manifestation.

Persistence can be maintained through material infrastructure, institutional memory, embodied habit, repeated practice, archived rules, software, education, law, ritual, language, or combinations of these. Different supports can substitute for one

another. A monetary mapping between numerical denomination and purchasing capacity, for example, does not persist through one material carrier alone. It can be reproduced through bank ledgers, legal obligations, payment networks, accounting systems, public expectations, pricing practices, and state institutions. The relational object persists through distributed supports whose composition can change.

This distributed persistence matters because the disappearance of one carrier need not dissolve the mapping. Paper certificates can disappear while account-based records remain. A spoken norm can become codified. An institutional category can migrate from paper forms to databases. A scientific measurement practice can move from one instrument generation to another. Persistence therefore belongs to the generative organization rather than necessarily to a single substrate.

Historical transmission also permits mappings to outlive the conditions under which they first emerged. Categories may continue to organize manifestation after the original rationale has weakened. Administrative systems can preserve distinctions whose historical origin is forgotten. Economic indicators can continue to guide policy after production structures have changed. Such persistence is analytically important because the mapping can remain causally efficacious even when actors no longer possess a clear account of its genealogy.

Claim 4.64 (Persistence without conscious maintenance). *A manifestation mapping can persist relationally without being continuously represented, understood, or intentionally maintained by any single subject, provided that the generative-relational system continues to reproduce the mapping through relevant practices, structures, or infrastructures.*

This claim parallels the pre-manifestational status of relational reality discussed earlier. Just as a relationally real object does not require prior perception, a relationally real mapping does not require a subject who explicitly knows the mapping. Some mappings are tacit, distributed, embodied, or infrastructural. Their reality becomes visible through regularities in what systems make available and through the effects that follow from those regularities.

Persistence also makes the history of mappings empirically investigable. One can trace when a distinction entered an institutional vocabulary, when a metric became standardized, when a legal category acquired force, when an algorithmic rule was deployed, or when a perceptual convention became culturally salient. Such histories can reveal that what later appears to be an immediate manifestation was mediated by a mapping that had itself undergone a process of stabilization and transmission.

4.8.4 Causal Efficacy of the Mapping

Persistence and stability alone do not complete the criterion. A mapping becomes relationally real in the stronger GR sense when it also exercises causal efficacy within the relevant system. Causal efficacy can arise at several points. The mapping can influence which differences become available, how they are organized, how actors respond to them, which resources are allocated, what categories are reproduced, and how later mappings are formed.

Definition 4.65 (Causal efficacy of a manifestation mapping). A manifestation map-

ping has *causal efficacy* when its operation makes a difference to subsequent manifestations, actions, relations, constraints, capacities, allocations, classifications, or trajectories within the relevant generative-relational system.

A mapping's causal efficacy need not be deterministic. A credit score may change the probability of receiving credit without uniquely determining the decision. A diagnostic category may alter the range of available treatments without fixing an individual's course. A market price can reorient behavior while different actors respond differently. A cultural symbol can modify salience and expectation without compelling one interpretation. Probabilistic, enabling, constraining, and path-shaping effects can all be causal in the relevant sense when the mapping makes a counterfactually or dynamically significant difference within the system.

Causal efficacy also distinguishes the reality of a mapping from the accuracy of the manifestation it produces. A mistaken classification can still be causally powerful. A biased indicator can redirect resources. A misleading rumor can alter market behavior. A hallucinated object can affect bodily action and memory. An institution can stabilize an erroneous category whose consequences are nevertheless real. The mapping's relational reality therefore cannot be inferred from epistemic correctness, nor can epistemic correctness be inferred from relational reality.

Claim 4.66 (Reality of mapping and epistemic adequacy). *The relational reality of a manifestation mapping and the epistemic adequacy of the manifestations it produces are distinct questions. A mapping can be persistent, stable, and causally efficacious while remaining inaccurate, biased, incomplete, or inappropriate for a particular inferential purpose.*

This distinction is particularly important in GRE. Economic systems are filled with manifestations that guide action because their mappings have become institutionally real: ratings, prices, scores, indicators, legal statuses, performance measures, rankings, and classifications. Their causal power can induce actors to reorganize behavior around them. Once this occurs, a measurement can partially participate in producing the reality it later appears merely to describe. A performance metric can reorganize work; a risk category can change financing conditions; a ranking can redirect attention; a price index can enter contracts; an eligibility threshold can reshape household decisions. The mapping becomes part of the generative field.

Causal efficacy can also operate through reflexivity. Actors can learn the mapping and strategically act upon it. Once a threshold is known, behavior can cluster around the threshold. Once a ranking formula is inferred, participants can optimize for it. Once a legal classification becomes consequential, actors can reorganize transactions to fall inside or outside its categories. Reflexive response does not make the mapping less real. It often provides evidence of its causal incorporation into the system.

At the same time, causal efficacy can contribute to transformation of the mapping itself. Strategic adaptation can expose failures, institutions can revise rules, and accumulated outcomes can alter normative or technical expectations. A manifestation mapping can therefore be both cause and object of later change. This recursive position reinforces the graph architecture: the mapping mediates manifestation, manifested outcomes alter relations, and those relations can return to modify the mapping.

4.8.5 Manifestation Mapping within Relational Reality

The preceding analysis allows manifestation mapping to be located more precisely within the three-register architecture. A mapping is not a fourth register. It is a relation whose analytical status depends on what it does. Some mappings remain hypothetical or representational devices introduced by inquiry. Others become embedded in generative-relational systems and thereby acquire relational reality. Once embedded, they can mediate what becomes manifest while also being transformed by the manifestations and relations they help generate.

Proposition 4.67 (Relational reality of manifestation mapping). *A manifestation mapping belongs to relational reality when its generative-relational emergence, persistence, stability, and causal efficacy are sufficiently established relative to the system and scale under investigation. Its relational reality does not require that the mapping be consciously represented, mathematically exact, epistemically adequate, ontologically fundamental, or universally valid.*

Proposition 4.67 consolidates the section's argument. It also prevents several conflations. First, the mapping's reality is distinct from the ontological reality of what enters or exits the mapping. A perceptual mechanism can be relationally real whether the manifested object is physical, imagined, hallucinatory, remembered, or ontologically unresolved. Second, the mapping's reality is distinct from the reality-status of a particular manifestation. A durable classification system can produce fleeting manifestations, while a transient mapping can occasionally produce a stable recorded output. Third, the mapping's relational reality is distinct from its normative desirability. Persistent and causally efficacious mappings can reproduce exclusion, distortion, domination, or injustice. Reality does not imply legitimacy.

The distinction is also useful for the earlier example of the hallucinatory flower and the ordinarily encountered physical flower. Both can enter manifestation reality, and both can participate in relationally real consequences. Their manifestation mappings can nevertheless differ substantially. One may involve bodily or neurological conditions that generate a flower-like manifestation without a corresponding ordinary external flower; another may involve interactions among an external physical object, sensory apparatus, environmental conditions, memory, and perceptual organization. GR need not place these mappings in one metric space or claim that their structures are directly comparable. It can instead ask whether each mapping is identifiable, persistent, stable, and causally efficacious within its relevant system. Shared relational reality does not require shared generative architecture.

The same caution applies to comparisons among economic mappings. A price mechanism, a religious valuation practice, a legal recognition procedure, an aesthetic judgment, and an algorithmic score can all become relationally real mappings while lacking a justified common metric. Comparison requires an independently warranted relation of comparison. One can sometimes compare causal effects, stability, information loss, accessibility, or institutional reach, but each comparison constructs a domain and selects dimensions. The common label *relationally real* establishes a criterion of reality rather than a universal coordinate system.

This point becomes important for formalization. If an inquiry represents a mani-

festation mapping as

$$\mathcal{M}_C : X \rightsquigarrow Y, \quad (4.1)$$

where C denotes relevant conditions, Equation 4.1 should be interpreted cautiously: the symbol $\rightsquigarrow$ should be read as a deliberately general mapping relation rather than automatically as a single-valued mathematical function. Depending on the domain, $\mathcal{M}_C$ may be deterministic, stochastic, partial, multivalued, path-dependent, or historically variable. The formal representation becomes useful only after the analyst specifies what X , Y , C , and the mapping relation mean in the problem under study. The notation is therefore a compact placeholder for a family of possible structures rather than an ontological claim that manifestation universally follows one mathematical form.

Remark 4.68 (Mapping and graph architecture). When a manifestation mapping becomes relationally real, the graph connecting relational and manifestational registers contains historically generated mediating structures with their own trajectories. The graph is therefore not merely a diagram of static correspondences; its mediating relations can emerge, stabilize, exercise causal efficacy, and transform.

This dynamic graph helps explain why manifestation cannot be treated as the terminal stage of a chain. A relationally real mapping participates in producing a manifestation; the manifestation can enter subsequent action and representation; those consequences can alter the system that sustains the mapping; and the altered mapping can generate different manifestations later. The history of the mapping becomes part of the history of what can appear.

Section 4.8 has therefore elevated manifestation mapping from an implicit bridge to an explicit object of relational analysis. The manifestation-mapping lemma applies the criteria of relational reality to mappings themselves. Stability concerns continuity of organizing relation under variation; persistence concerns continued or recurrent reproduction across a relevant spacetime interval; causal efficacy concerns the difference the mapping makes to later manifestations and relations. When these conditions are satisfied, the mapping is relationally real even while stronger ontological and epistemic questions remain open. The next section turns to a different problem. Once a mapping has been identified, the availability of a manifestation does not guarantee that the mapping can be reversed. Non-invertibility, multiplicity, and loss of relational information therefore become central to any inference from manifestational reality back toward the structures from which manifestation emerged.

4.9 Non-Invertibility of Manifestation

Section 4.8 established that a manifestation mapping can itself become relationally real when it is sufficiently persistent, stable, and causally efficacious within a generative-relational system. The present section examines a different property of such mappings: whether a manifestation can be used to recover, uniquely and adequately, the relational configuration from which it emerged. The central result is a disciplined account of *non-invertibility*. Manifestation may compress, select, transform, aggregate, or contextually reorganize relational differences. Consequently, a manifestational form can be compatible with several relational configurations,

while a given relational configuration can admit several manifestations. These possibilities limit reverse inference from what appears, is recorded, is measured, or is symbolized to the relational reality that participated in generating it. The argument proceeds through many-to-one manifestation, one-to-many manifestation, manifestational equivalence, relational difference beneath manifestational similarity, and the resulting limits of reverse inference.

The term *non-invertibility* is used here in an analytical rather than universally algebraic sense. Some manifestation mappings can be represented by ordinary functions and studied with familiar notions of inverse functions. Others are stochastic, partial, multivalued, historically variable, institutionally mediated, or dependent on conditions that cannot be held fixed across observations. Section 4.8 therefore represented a general manifestation mapping by Equation 4.1, using $\rightsquigarrow$ to indicate a mapping relation whose exact mathematical form remains to be specified locally. Non-invertibility concerns the information and structure that cannot be uniquely recovered when inquiry moves from a manifestational result back toward the relational configurations compatible with it. (Giere 2004)

A convenient formal device is the *compatibility set*. Let $\mathcal{M}_C$ denote a manifestation mapping under relevant conditions C , let X denote a selected space of relational configurations, and let Y denote a selected space of manifestations. For a manifestation $y \in Y$, define the set of relational configurations compatible with y under C by Equation 4.2.

$$\mathcal{P}_C(y) = \{x \in X \mid x \mathcal{M}_C y\}. \quad (4.2)$$

The notation in Equation 4.2 does not assume that X or Y is metric, that $\mathcal{M}_C$ is deterministic, or that every relevant relational configuration has already been exhaustively specified. It identifies a local inverse problem: given a manifestation and a specified manifestation regime, which relational configurations remain compatible with what has become manifest? When the compatibility set contains several materially different configurations, manifestation underdetermines relational reconstruction.

Definition 4.69 (Manifestational non-invertibility). A manifestation mapping is *non-invertible for a specified inquiry* when the available manifestation, together with the mapping conditions and admissible evidence, does not support unique and warranted recovery of the relevant antecedent relational configuration at the scale required by that inquiry.

Definition 4.69 makes non-invertibility relative to an inquiry, scale, and evidential regime. A mapping can be invertible with respect to one coarse variable while remaining non-invertible with respect to a finer relational history. A measurement may uniquely determine a reported category while failing to determine the heterogeneous trajectories that produced that category. A legal status may uniquely identify the institutionally recognized state of a person while leaving the underlying social, economic, and historical relations radically underdetermined. Conversely, additional manifestations or interventions can sometimes narrow the compatibility set. Non-invertibility therefore concerns the relation between available manifestation and warranted reconstruction rather than an assertion that recovery is impossible under every conceivable extension of evidence.

4.9.1 Many-to-One Manifestation

Many-to-one manifestation occurs when materially different relational configurations generate the same, or sufficiently similar, manifestational form under the relevant conditions. This is the most direct source of non-unique reverse inference. If several relational configurations belong to $\mathcal{P}_C(y)$, observing y establishes compatibility with each of them until further evidence differentiates among the alternatives.

Definition 4.70 (Many-to-one manifestation). A manifestation mapping exhibits *many-to-one manifestation* relative to conditions C when two or more relationally distinguishable configurations are compatible with a common manifestational form, or with forms treated as indistinguishable under the operative manifestational criterion.

The source of many-to-one manifestation can differ across domains. A mapping may aggregate several relational configurations into one category. An instrument may register only one dimension of a multidimensional process. An institutional procedure may intentionally suppress distinctions judged irrelevant to its purpose. A perceptual system may resolve several materially different stimuli into one recognizable form. A statistical summary may retain an average while discarding distributional structure. A market price may summarize heterogeneous beliefs, constraints, liquidity conditions, institutional rules, and strategic positions without uniquely revealing their composition. In each case, the manifestation can remain highly useful while carrying less relational information than the configuration from which it emerged.

Compression is therefore one important mechanism of many-to-one manifestation. Compression should be understood broadly. It includes numerical aggregation, categorical classification, perceptual simplification, institutional standardization, linguistic naming, algorithmic feature selection, and other processes through which a richer relational field is rendered into a lower-dimensional or less differentiated manifestational form. The compressed manifestation can be stable and causally powerful. Its stability does not restore the distinctions removed during manifestation.

Many-to-one manifestation can also arise without deliberate compression. Different histories can converge on similar configurations at the point of manifestation. Distinct causal paths can produce the same observed outcome. Heterogeneous social relations can produce similar behavioral regularities. Different institutional arrangements can yield the same reported indicator over a limited interval. Distinct bodily, perceptual, and environmental configurations can generate similar experiences. The manifestational similarity can therefore result from convergence rather than from a single mechanism that intentionally groups diverse inputs.

The distinction between *same manifestation* and *sufficiently similar manifestation* matters. Exact equality is often unavailable or analytically unnecessary. A measurement system can treat values within a tolerance interval as the same reading. A legal regime can place heterogeneous cases in the same category. A human observer can experience two appearances as indistinguishable despite physical differences. A model can treat two distributions as equivalent for a selected decision rule. The operative equivalence criterion belongs to the manifestation regime and should be stated rather than presumed.

Claim 4.71 (Manifestational underdetermination). *When a manifestation is generated through a many-to-one mapping, the manifestation alone underdetermines the antecedent relational configuration. Reverse inference requires additional evidence, constraints, interventions, or assumptions capable of differentiating among the configurations that remain compatible with the manifestation.*

Claim 4.71 does not imply that reverse inference is futile. It specifies its evidential burden. If the compatibility set can be reduced through additional manifestations, historical records, controlled interventions, cross-context comparison, or independent knowledge of the mapping, reconstruction can become more informative. The point is that the manifestation does not carry a unique antecedent merely by virtue of appearing determinate.

This distinction has particular importance for GRE because economic manifestations often appear in standardized forms. A price, score, balance, wage, category, ranking, or aggregate may be numerically precise while remaining relationally underdetermined. Precision of manifestation and uniqueness of generative history are different properties. A highly precise number can summarize a heterogeneous field of relations, just as a coarse category can sometimes correspond to a relatively specific institutional mechanism. The degree of manifestational precision therefore provides limited information about the invertibility of the mapping.

4.9.2 One-to-Many Manifestation

Non-invertibility also arises in the opposite direction. A single relational configuration can generate several manifestational forms across contexts, observers, instruments, symbolic regimes, times, or scales. Section 4.6 already established that manifestational reality is condition-dependent and can take heterogeneous forms. The present subsection makes the consequence for inversion explicit: variation in manifestation need not imply variation in the relational object at the level under investigation.

For a relational configuration $x \in X$, define its admissible manifestation set under conditions C by Equation 4.3.

$$\mathcal{Y}_C(x) = \{y \in Y \mid x \mathcal{M}_C y\}. \quad (4.3)$$

Equation 4.3 allows the manifestation mapping to be multivalued. Even where a deterministic function is appropriate under fixed conditions, variation in C can generate a family of manifestations for the same selected relational configuration. The analyst must therefore distinguish changes in the relational configuration from changes in the conditions through which that configuration becomes manifest.

Definition 4.72 (One-to-many manifestation). A manifestation mapping exhibits *one-to-many manifestation* relative to the selected analytical scale when one relational configuration can become manifest in more than one materially distinguishable form across admissible manifestation conditions.

The qualification *relative to the selected analytical scale* is essential. What appears to be one relational configuration at a coarse scale may contain fine-grained differences that explain manifestational variation. Conversely, an inquiry may have good

reason to treat those differences as irrelevant to its present object. One-to-many manifestation therefore does not establish that an identical microstate generates arbitrary outputs. It states that the relational identity preserved by the inquiry can coexist with plural manifestations.

A monetary relation illustrates the point. The same institutionalized monetary claim may become manifest as a banknote, an account balance, a payment message, an entry in a ledger, a spoken amount, or a displayed price. These manifestations differ materially and symbolically while participating in a relational structure that can remain continuous across them. Likewise, an institutional status can be manifested through a document, database record, uniform, title, access permission, spatial arrangement, or pattern of deference. A relation of trust can become manifest through speech, action, silence, delegation, disclosure, or patterned expectation. The plurality of manifestations does not by itself establish plurality of the relational object.

One-to-many manifestation also appears in scientific and empirical inquiry. A process can leave different traces under different instruments. A latent relation can generate several observable indicators. A historical event can become available through documents, material remains, memories, later institutional records, and statistical consequences. Each manifestation selects and transforms aspects of the underlying relational history. Their differences can be informative precisely because they arise through different manifestation mappings.

This plurality creates a second inverse problem. If one relational configuration can appear in several ways, then inferential procedures that expect a single canonical manifestation can miss relational continuity beneath manifestational difference. A change in representation can be mistaken for a change in the represented relation. A change in instrument can be mistaken for a change in the measured process. A change in symbolic vocabulary can be mistaken for the disappearance of the phenomenon. Conversely, stable terminology can conceal substantial relational transformation. The ontological and epistemic tasks therefore require attention to both directions of mismatch.

Remark 4.73 (Manifestational plurality and identity). Manifestational plurality does not by itself establish either relational identity or relational difference. Whether several manifestations concern one relationally continuous object depends on the persistence, stability, causal organization, and historical continuity that define the relational object at the relevant scale.

Remark 4.73 prevents an opposite overcorrection. The existence of one-to-many mapping does not license the analyst to declare diverse manifestations identical. The relation among them has to be established. The three-register architecture therefore places a burden on both identification and differentiation: similarity of appearance does not establish relational identity, and difference of appearance does not establish relational difference.

4.9.3 Manifestational Equivalence

Many-to-one manifestation requires a concept for cases in which distinct relational configurations cannot be distinguished within a specified manifestation regime. The appropriate concept is *manifestational equivalence*. The term is deliberately weaker

than ontological identity, relational identity, or structural isomorphism. It describes indistinguishability with respect to a specified manifestation mapping and criterion.

Definition 4.74 (Manifestational equivalence). Two relational configurations are *manifestationally equivalent* relative to a specified manifestation regime when the distinctions between them are not expressed in the manifestational forms that the regime recognizes as relevantly distinguishable for the inquiry at hand.

Manifestational equivalence is therefore indexed by a regime. The regime includes the mapping, conditions, resolution, categories, instruments, symbolic resources, tolerances, and analytical purposes that determine what counts as a relevant manifestational difference. Two configurations can be equivalent under one regime and distinguishable under another. The equivalence can also disappear historically when new instruments, categories, practices, or sensitivities make previously unavailable differences manifest.

This relativity does not make manifestational equivalence arbitrary. A regime can have stable procedures and well-specified criteria. Under those criteria, the equivalence may be reproducible and operationally important. Two chemical states can be indistinguishable to one instrument and distinguishable to another; two institutional practices can receive the same administrative code while producing different lived consequences; two economic trajectories can yield the same quarterly aggregate while diverging in distribution, vulnerability, or future capacity. In each case, equivalence is real relative to a manifestation regime while remaining narrower than identity of the relational configurations.

Proposition 4.75 (Locality of manifestational equivalence). *Manifestational equivalence warrants claims of indistinguishability only relative to the specified manifestation regime and criterion. It does not, by itself, warrant ontological identity, relational identity, equal causal structure, equal historical trajectory, or interchangeability across other manifestation regimes.*

Proposition 4.75 is especially important for comparative analysis. A common indicator can create an equivalence class that is useful for one purpose. Treating the same equivalence class as valid for another purpose requires additional justification. Equal income at one date does not imply equal wealth, security, capabilities, obligations, histories, or future opportunities. Equal test scores do not imply identical learning processes. Equal prices do not imply identical market structures. Equal institutional classifications do not imply identical relations of power. The manifestation criterion selects the dimension of equivalence; it does not silently generalize that equivalence to every other dimension.

Manifestational equivalence can itself become relationally consequential. Once an institution treats distinct cases as equivalent, resources, rights, obligations, attention, or sanctions may be allocated according to the shared manifestation. The equivalence relation then becomes part of the generative-relational system. An administrative category can make heterogeneous persons institutionally interchangeable for one procedure. A credit category can group heterogeneous borrowers and alter their future financing conditions. A ranking band can group organizations and affect later demand. A standardized diagnosis can coordinate treatment while also reorganizing self-understanding and institutional response. In these cases, manifestational equiv-

alence begins as a criterion of indistinguishability and can become part of the relational reality that shapes future trajectories.

This recursive possibility reinforces the graph architecture developed throughout the chapter. Manifestational equivalence can be generated by a relationally real mapping; the manifested equivalence can guide action; those actions can reshape subsequent relations; and the resulting relational changes can alter later manifestation mappings. Equivalence is therefore neither a passive label nor a guarantee of deep sameness. It can be an active component in the historical production of similarity and difference.

4.9.4 Relational Difference beneath Manifestational Similarity

The most important ontological consequence of non-invertibility is the possibility that substantial relational difference persists beneath manifestational similarity. A manifestation can be shared while the relations that generated it differ in organization, history, scale, causal mechanism, or normative significance. GR therefore treats the relational structure beneath a manifestation as an open object of inquiry rather than as something uniquely disclosed by appearance.

The example of the hallucinatory flower and the ordinarily encountered physical flower makes the distinction vivid. Both can enter manifestational reality as a flower-like appearance. Both can become causally consequential for attention, movement, memory, language, or subsequent action. Yet the relational configurations through which the manifestations arise can differ substantially. One may involve bodily or neurological conditions that generate a flower-like percept without the ordinary external object; another may involve an external physical flower, environmental conditions, sensory transduction, perceptual organization, memory, and learned categorization. The fact that both are manifestationally available as a flower does not establish a common generative structure, and GR has no need to presume that the two structures belong to a single metric space.

This example also shows why relational reality and manifestational reality should remain distinct. A hallucinatory flower can participate in relational reality if the generated object persists sufficiently and exerts causal efficacy within the relevant system. The relational reality of that object does not depend on identifying its generative structure with the structure of an ordinarily perceived physical flower. Shared manifestational form and shared relational reality-status therefore remain compatible with structural heterogeneity.

The same pattern appears in less unusual cases. Two organizations can report the same productivity indicator while relying on different labor relations, technologies, environmental externalities, or temporal strategies. Two households can report the same income while occupying different debt structures, care obligations, legal statuses, social networks, or exposure to risk. Two communities can exhibit similar rates of participation while differing in trust, coercion, institutional accessibility, or cultural expectations. Two artificial systems can produce similar outputs while relying on different architectures, data histories, or inference procedures. In each case, manifestational similarity narrows some differences while leaving others unresolved.

Claim 4.76 (Relational remainder beneath manifestation). *For a specified manifestation regime, relational differences that do not alter the recognized manifestation constitute a relational remainder relative to that regime. Such differences can remain causally and historically significant even while they are absent from the manifestational distinction used by the inquiry.*

The term *remainder* does not imply that the omitted relations are secondary in reality. They are a remainder only relative to a particular manifestation regime. A variable excluded from an indicator can be central to later dynamics. A relation invisible to an administrative category can structure access to resources. A historical difference suppressed by aggregation can determine vulnerability under future disturbance. A symbolic system can render one distinction salient while leaving another difficult to articulate. The remainder is therefore an analytical warning against treating the dimensions made manifest as exhaustive of the relational configuration.

Relational remainder also clarifies why greater manifestational detail does not automatically eliminate non-invertibility. Adding variables can reduce the compatibility set, but every representation selects a domain, scale, resolution, and set of distinctions. A richer manifestation can reveal differences that a coarser regime concealed while still leaving other differences unexpressed. Complete inversion would require a specification of what counts as the full relational configuration and a mapping that preserves every distinction relevant to the inquiry. In many social, economic, historical, and perceptual settings, that requirement is stronger than the available evidence can support.

This point should be kept separate from skepticism. GR does not infer that knowledge is impossible because manifestation can be non-invertible. It infers that knowledge claims should be calibrated to the information preserved by the manifestation regime and to the additional evidence used in reconstruction. An inquiry can learn a great deal from a non-invertible mapping. It can identify equivalence classes, exclude incompatible explanations, estimate probabilities, compare mechanisms, reconstruct partial histories, and test interventions. The loss of unique inversion calls for epistemic discipline rather than epistemic paralysis.

4.9.5 Limits of Reverse Inference

Reverse inference begins from a manifestation and attempts to reconstruct relational structures, histories, mechanisms, or objects compatible with it. The analysis above yields a general constraint: the warranted strength of reverse inference depends on the structure of the manifestation mapping, the size and heterogeneity of the compatibility set, the stability of the conditions, and the availability of independent evidence. Manifestation can support inference without transparently revealing its antecedent relational reality.

Proposition 4.77 (Reverse-inference discipline). *An inference from manifestational reality toward relational reality should state the manifestation regime, identify plausible multiplicity in the compatibility set, distinguish assumptions supplied by the analyst from information preserved by the manifestation, and limit the reconstructed claim to the level supported by the available evidence.*

The principle has several consequences. First, a manifestation should not be

treated as a self-interpreting trace. A number, category, statement, image, behavior, or institutional record becomes evidentially informative through knowledge of the relations and mappings that produced it. Second, reconstruction gains strength when independent manifestations arise through partially different mappings. If several traces with different failure modes converge on a compatible relational account, the compatibility set can become narrower. Third, interventions can sometimes distinguish among otherwise manifestationally equivalent configurations. Changing a condition and observing differential response can reveal structure that passive observation leaves hidden. Fourth, historical reconstruction requires attention to mapping change. A category recorded in one period can have a different generative and institutional meaning in another period even when the visible symbol remains constant.

Reverse inference can also be asymmetric across directions. A mapping may reliably predict a manifestation from a relational configuration while providing weak reconstruction from manifestation to configuration. If a broad range of configurations all produce the same output, forward prediction can be stable while inverse identification remains underdetermined. Conversely, a manifestation can sometimes be highly diagnostic of a narrow class of relational structures even when forward manifestation is probabilistic. The epistemic properties of the forward and reverse problems therefore need separate assessment.

This asymmetry is important for causal interpretation. A causal model can identify how a specified relation would affect manifestation under defined conditions without claiming that every observed manifestation uniquely identifies that cause. Similarly, an institution can reliably translate a legal status into a standardized record while the record alone remains insufficient to reconstruct the full path through which the status was obtained. Predictive adequacy in one direction does not guarantee explanatory uniqueness in the other.

Remark 4.78 (Manifestation and disclosure). A manifestation can disclose genuine aspects of relational reality while remaining non-invertible with respect to the relational configuration as a whole. Partial disclosure and non-unique reconstruction are compatible.

Remark 4.78 protects the chapter from two opposed simplifications. Manifestation is more than arbitrary projection because it can be constrained by, and informative about, relational reality. Manifestation is also less than exhaustive disclosure because the mapping can transform or suppress relational differences. The three-register framework therefore allows a manifestation to be real, constrained, causally consequential, and epistemically useful without treating it as a transparent duplicate of what generated it.

The same discipline applies to symbolic and representational forms, which become central in Section 99. Symbolization can preserve some differences, create new distinctions, stabilize categories, and make relations transmissible across time and space. It can also compress, standardize, and reorganize what becomes manifest. A symbol can therefore increase epistemic reach while adding another mapping whose invertibility requires examination. The transition from manifestation to symbolization should consequently be treated as a further generative relation rather than as a neutral copying operation.

Section 4.9 has established non-invertibility as a structural possibility within the three-register architecture. Many-to-one manifestation generates underdetermination because several relational configurations can be compatible with one manifestational form. One-to-many manifestation generates plurality because a relational configuration can become manifest differently across conditions and regimes. Manifestational equivalence identifies local indistinguishability without converting it into ontological or relational identity. Relational remainder names differences that remain outside a specified manifestation criterion while potentially retaining causal and historical significance. Reverse inference is therefore possible under disciplined conditions, but its strength depends on the mapping, the evidence, and the scale of reconstruction. These results prepare the analysis of symbolization, where manifestations can become stabilized, transmitted, classified, and recursively incorporated into later relational reality.

4.10 Symbolization and Manifestation

Section 4.9 established that manifestation can preserve genuine information about relational reality without providing a uniquely invertible image of the relational configuration from which it emerges. The present section introduces a further mediation: symbolization. Its role is to distinguish manifestation from the processes through which manifestations, relations, subjects, values, and other objects are rendered in symbolic form. Symbolization is analytically important because symbols can preserve distinctions across time and space, make heterogeneous objects administratively or economically comparable, coordinate action among actors who do not share immediate experience, and enter institutions as operative records, categories, rules, and representations. These capacities give symbolic forms unusual generative reach. They also make it especially important to avoid treating symbolization as transparent copying. (Cassirer 1955; Bourdieu 1991; Wittgenstein 2009; Espeland and Stevens 1998)

The argument proceeds in six steps. It first separates manifestation from symbolization and defines a symbolic regime. It then examines symbolic representation as a selective relation rather than an identity between symbol and represented object. The following three subsections consider the symbolization of relations, subjects, and value. The final subsection asks when representations themselves become part of relational reality. Throughout, the method remains consistent with the three-register architecture: symbolization can mediate relational reality without being a prerequisite for relational reality, and symbolic efficacy does not establish that the represented object possesses the ontological status attributed to it.

4.10.1 Manifestation and Symbolization

Manifestation and symbolization overlap in many concrete cases but remain distinct analytical operations. A bodily sensation, a sudden interruption of expectation, a visible pattern, an affective response, a material trace, or a behavioral change can become manifest without first being encoded in a symbolic system. Conversely, a symbolic inscription can persist and circulate beyond the immediate manifestation

through which it was first produced. The distinction therefore protects two claims already established in the chapter: relational reality can precede symbolization, and manifestation is broader than symbolic representation.

Definition 4.79 (Symbolization). *Symbolization* is a generative-relational process through which a difference, object, relation, manifestation, possibility, or constructed category is rendered in a symbolic form whose differentiation and use are organized by a symbolic regime.

The definition is deliberately broader than linguistic naming. Numbers, diagrams, legal categories, monetary denominations, account identifiers, rankings, scores, icons, mathematical expressions, database fields, ritual signs, and conventional gestures can all participate in symbolization when their differences are organized within a system of use. A symbol need not resemble what it represents. Its efficacy can arise from the relations through which a community, institution, technical system, or practice distinguishes the symbol, connects it to other symbols, and treats it as relevant to subsequent action.

Definition 4.80 (Symbolic regime). A *symbolic regime* is a relatively organized set of relations, distinctions, conventions, procedures, media, and practices through which symbolic forms are produced, differentiated, interpreted, transmitted, or made operative within a specified generative-relational system.

A symbolic regime can be highly explicit, as in accounting standards, legal codes, programming languages, or administrative taxonomies. It can also be distributed across practice, as in ordinary language, professional conventions, ritual forms, or culturally stabilized categories. The regime may contain formal rules without being reducible to those rules. Institutional enforcement, habitual use, technological infrastructure, historical inheritance, and embodied competence can all contribute to the persistence of symbolic organization.

Symbolization therefore adds a further mapping to the architecture developed in Sections 4.7–4.9. A manifestation can be selected, segmented, named, quantified, classified, or otherwise encoded. Yet symbolization can also target something that has not been fully manifested to a particular subject. An automated system may classify a transaction, assign an identifier, or generate a score without any human observer first experiencing the underlying relation in an immediate perceptual form. Symbolic mediation is consequently not equivalent to conscious interpretation.

The limited Lacanian analogy introduced in Section 4.2 is useful at precisely this point. Lacanian accounts of the Symbolic do not treat symbolic order merely as labels added after an independently completed field of experience; symbolic relations can organize positions, differences, and possibilities within which later experience and subject formation occur (Johnston 2026). GR does not import Lacan's specific account of language, desire, or subjectivity. The relevant architectural insight is narrower: symbolic mediation can be constitutive of subsequent relations rather than only descriptive of antecedent ones. This is one reason the three registers are represented as a graph rather than a chain.

Claim 4.81 (Symbolization is neither necessary nor terminal). *Relational reality does not require prior symbolization, and symbolization does not constitute a terminal stage of manifestation. Once symbolic forms enter subsequent relations, they can participate*

in further manifestation, stabilization, classification, coordination, and transformation.

Claim 4.81 prevents two opposed reductions. A pre-symbolic relation need not be unreal merely because no vocabulary yet captures it. At the same time, a symbolic representation need not remain external to the object it represents. A new legal name, scientific category, market quotation, diagnosis, public label, or institutional identifier can alter the relations through which the represented object subsequently develops. Symbolization belongs inside generative history when it becomes causally operative.

4.10.2 Symbolic Representation

Symbolic representation is one important use of symbolization, but representation should not be treated as identity. A representation selects differences, organizes them through a symbolic regime, and presents them in a form suited to some practice, audience, institution, medium, or inquiry. The represented object can therefore exceed the representation even when the representation is accurate for its stated purpose.

Definition 4.82 (Symbolic representation). A *symbolic representation* is a symbolically organized object, expression, record, or configuration that stands in a specified representational relation to another object, relation, manifestation, possibility, or constructed target within a symbolic regime.

The term *specified* matters. The same inscription can represent different things under different regimes. A number may denote a price, quantity, rank, probability, account balance, legal threshold, or arbitrary identifier. A word can function as a name, category, command, promise, quotation, or fictional designation. An arrow in a diagram can indicate causation, implication, flow, sequence, correspondence, or merely graphical connection. Representation therefore depends on the relations that organize symbolic use rather than on the mark considered in isolation.

Symbolic representation inherits the non-invertibility examined in Section 4.9. Several relational configurations can be represented by the same symbolic category, while one relational configuration can receive different symbolic descriptions under different regimes. The symbolic form can compress history, suppress differences, aggregate relations, or introduce distinctions that were not previously operative. A category can be highly useful while remaining coarse relative to the relational domain. A detailed representation can preserve more distinctions while still leaving a relational remainder.

This selectivity is not necessarily a defect. Representation is often useful precisely because it reduces complexity. A balance sheet excludes many features of an organization so that specified claims and obligations become legible. A map suppresses most properties of a landscape so that spatial relations relevant to navigation become visible. A legal category narrows heterogeneous histories into a form that permits institutional decision. A statistical variable makes comparison possible by holding some distinctions constant. The relevant question is not whether representation reproduces everything, but what it preserves, what it transforms, what it omits, and what later action is organized through it.

Remark 4.83 (Representational adequacy is purpose-relative). A symbolic represen-

tation can be adequate for one inquiry or practice while remaining inadequate for another. Adequacy therefore requires a stated purpose, scale, domain, and set of distinctions rather than an abstract demand for total correspondence.

Purpose-relative adequacy does not make representation arbitrary. A representation can fail because it omits a causally important distinction, misclassifies the object, applies a rule outside its domain, or produces consequences inconsistent with its stated purpose. The point is instead that symbolic adequacy is evaluated through a relation among representation, target, use, and conditions. No single representation is required to exhaust every relation in which an object participates.

Symbolic representation also introduces a distinction between the reality of the representation and the reality of its referent. A record can exist and be causally effective even if the proposition encoded in it is mistaken. A name can organize collective action even when the named object has uncertain ontological status. A false account balance, erroneous risk classification, or inaccurate administrative record can alter access, behavior, and expectations. The symbolic object and the truth of what it represents are therefore analytically separable.

4.10.3 Symbolized Relations

Relations frequently become actionable through symbolization. Contracts, titles, organizational charts, kinship terms, debt records, network links, legal statuses, platform connections, and institutional roles render some relation sufficiently explicit that it can be stored, communicated, enforced, compared, or incorporated into later decisions. Symbolization can thereby increase the persistence and reach of a relation without exhausting the relation itself.

A symbolized relation is not simply a picture of an antecedent connection. The act of recording, naming, or classifying a relation can change the relation's future trajectory. A debt entered into an enforceable ledger differs from an unrecorded expectation. An employment relation formalized by contract acquires rights, obligations, temporal boundaries, and institutional interfaces that can become causally significant. A platform's designation of two accounts as connected can affect information flow, recommendation, visibility, and later interaction. The symbolic form becomes one of the mediations through which the relation is reproduced.

Definition 4.84 (Symbolized relation). A *symbolized relation* is a relation rendered through a symbolic regime in a form that selects, encodes, or organizes some of its differences and that may subsequently participate in the persistence, interpretation, coordination, or transformation of the relation itself.

Definition 4.84 leaves open whether the symbolic form is constitutive, evidential, descriptive, or merely auxiliary in a particular case. A legal marriage status can partly constitute institutional relations that would not exist in the same form without recognition. A diagram of an informal friendship network may instead be an analyst's representation that has little effect on the represented relations. A public label such as "ally," "competitor," or "debtor" can begin descriptively and later alter expectations enough to become generatively important. The role of symbolization is therefore an empirical and historical question.

Symbolization can also stabilize relations by making them portable across contexts. A credential allows a relation of qualification to travel beyond the place where competence was originally demonstrated. A title allows ownership claims to be recognized by actors who did not witness the original acquisition. A contract allows obligations to persist despite changing memories and personnel. A standardized code allows an institutional relation to be processed by distant offices or automated systems. Symbolization can thus extend relational efficacy across spatial, temporal, and organizational discontinuities.

This portability has limits. The symbolic relation can outlive the conditions that initially made it accurate. A category may persist after the underlying relation has transformed. A database can reproduce an outdated status. A contractual label can conceal asymmetries that the formal description does not register. A network edge can imply a homogeneous connection even when the relation changes in intensity, direction, or meaning. Symbolization increases persistence partly by reducing the number of differences that must be reproduced, and that reduction can later become a source of distortion.

4.10.4 Symbolized Subjects

Subjects are especially vulnerable to confusion between relational reality and symbolic representation because modern institutions operate through names, identifiers, roles, statuses, profiles, records, classifications, and credentials. These symbolic forms make subjects legible to systems that could not otherwise coordinate across large populations and long time horizons. They can also become part of the processes through which subject positions are stabilized.

A person can possess relationally effective subjecthood before or beyond any particular representation. Bodily continuity, memory, practical orientation, recognition by others, participation in institutions, and histories of interaction can sustain a subject across changes in name, account, role, or classification. Conversely, a symbolic subject can be generated institutionally even when the deeper ontological status of subjecthood remains unsettled. A corporation, state agency, software account, autonomous system, or legal estate can be treated as a subject for specified operations without GRE inferring that all such subjects share the same metaphysical or moral status.

Definition 4.85 (Symbolized subject). A *symbolized subject* is a subject-position rendered within a symbolic regime through names, identifiers, categories, roles, records, credentials, profiles, or related representational forms that permit the subject to be recognized or acted upon within specified relations.

The definition refers to a subject-position rather than to the complete subject. This distinction is important. An institutional identifier can be necessary for taxation, payment, healthcare, education, migration, or platform participation while representing only a small subset of the relations through which the person exists. A legal person can be highly effective in contracting and ownership while remaining distinct from a human subject. A computational agent can occupy an operational role in a market or organization without the symbolic role settling questions of consciousness, responsibility, rights, or moral standing.

Symbolized subjecthood can nevertheless become causally powerful. A classification can determine access to benefits, credit, employment, insurance, mobility, or legal procedures. A professional title can organize expectations and authority. A public reputation can affect bargaining position. A profile assembled from behavioral traces can shape which opportunities become visible. Once these symbolic forms enter repeated institutional action, they can become part of the relational reality through which the subject's later possibilities are generated.

The possibility of misrepresentation is therefore normatively and economically significant. A mistaken or reductive representation of a subject can still have real effects if institutions act upon it. The subject may be forced to respond to a symbolic object that has acquired causal efficacy independently of its accuracy. This is another instance in which relational reality and epistemic adequacy must remain distinct: a representation can be relationally real in its consequences while being false, incomplete, outdated, or unjust.

4.10.5 Symbolized Value

Economic life depends heavily on symbolic forms of value. Prices, wages, account balances, exchange rates, ratings, scores, rankings, budgets, asset values, accounting categories, and monetary denominations make heterogeneous activities and objects available to calculation and coordination. Yet the presence of a common symbolic scale should not be confused with proof that the underlying values are intrinsically commensurable.

A price is a particularly important example. It can manifest conditions of exchange under a specified market arrangement and can represent a monetary valuation at a particular time. Once displayed, recorded, anticipated, or used as a reference, the price can also affect bargaining, expectations, investment, production, and later prices. The symbolic form therefore does more than report an antecedent value. It can participate in the generation of subsequent valuation relations.

The same is true of non-price representations. A credit score compresses a set of observations and modeling choices into a symbolic scale that can influence lending. A ranking compresses multiple criteria into an ordering that can alter applications, prestige, and resource flows. A performance indicator can reorganize work around what is measured. An accounting category can make one form of contribution legible while leaving another outside the reporting system. Symbolized value can therefore create pathways through which some differences acquire institutional force.

Definition 4.86 (Symbolic commensuration). *Symbolic commensuration* is the construction or stabilization of a common symbolic scale, ordering, category system, or representational coordinate through which heterogeneous objects or relations can be compared for a specified practice, without thereby establishing that their underlying relational structures or values share an intrinsic common metric.

Definition 4.86 connects symbolization to the non-necessary comparability of relational realities established in Section 4.2. Symbolization can generate a practical comparison relation where none was previously available. Monetary units allow unlike goods to enter exchange calculations. Rankings place heterogeneous organizations in a common order. Composite indicators render different observations comparable by

applying weights and normalization rules. These operations can be useful and consequential. Their success as coordination devices does not establish that the compared objects were already elements of one natural metric space.

Claim 4.87 (Symbolic comparability does not establish relational equivalence). *The successful placement of heterogeneous objects within a common symbolic scale or ordering establishes a practical relation of comparison under that regime; it does not by itself establish equivalence, identity, or intrinsic commensurability among the relational realities represented.*

This distinction matters for GRE because symbolic commensuration can redistribute attention and resources. Once a common scale becomes institutionally important, actors may adapt behavior toward improving the represented value. Objects may be redesigned to fit categories. Activities difficult to symbolize may receive less recognition. A symbolic metric can therefore help create the economic field it initially appears merely to measure. The normative evaluation of such a field requires more than checking whether the metric is consistently applied; it also requires asking which relations become visible, which remain outside the scale, and what generative consequences follow from acting through the representation.

4.10.6 Representation within Relational Reality

The preceding subsections converge on a central implication of the three-register architecture: a representation can itself become part of relational reality. The conditions are the same as for other relationally real objects. A symbolic form, symbolic regime, or representation must emerge within a generative-relational system, persist with sufficient stability over a relevant region or interval, and exercise causal efficacy within that system. Symbolicity alone is insufficient; relational efficacy is what makes the representation relationally real.

Proposition 4.88 (Relational reality of representation). *A symbolic representation or symbolic regime qualifies as relationally real relative to a generative-relational system when it is sufficiently persistent and stable to exercise causal efficacy in the generation, coordination, constraint, interpretation, or transformation of subsequent relations within that system.*

Proposition 4.88 allows representation to be treated symmetrically with the manifestation mappings discussed in Section 4.8. An accounting standard can become relationally real because organizations reorganize reporting, investment, and compliance around it. A currency unit can become relationally real because obligations, prices, and expectations are denominated through it. A legal category can become relationally real because institutions repeatedly act upon it. A public narrative can become relationally real when it persists sufficiently to orient collective action. None of these conclusions requires the representation to be ontologically fundamental or epistemically flawless.

The distinction between causal efficacy and truth is particularly important. A representation can be inaccurate and still cause decisions. A rumor can move a market before being corrected. An erroneous database entry can block access. A mistaken classification can alter treatment by institutions. A misleading indicator can redirect resources. The relational reality of the representation concerns its generated persis-

tence and causal participation, not the truth of every proposition associated with it.

Remark 4.89 (Reality of a representation and truth of a representation). The reality of a representation as an operative object and the epistemic adequacy of what it represents are distinct questions. A representation can be relationally real in its effects while remaining incomplete, inaccurate, contested, or false with respect to its target.

This distinction also prevents a converse mistake. Accurate representation does not automatically imply strong relational efficacy. A carefully verified scientific description can remain causally marginal if it does not enter institutions, practices, decisions, or other subsequent relations. Relational reality depends on how the representation participates in the system, not solely on its correspondence to the represented object.

Symbolic forms therefore occupy a distinctive position in the graph among the three registers. They can arise from manifestations, represent relational realities, encode claims about ontological reality, and become relationally real through their own persistence and causal efficacy. They can also condition later manifestations by organizing what actors, institutions, and technical systems are prepared to recognize. The graph is recursive because a representation produced at one moment can alter the mappings, categories, expectations, and relations through which later reality becomes manifest.

The analysis of symbolization also clarifies why representation cannot be added to the three-register framework as a simple fourth layer above manifestation. Representation is a family of mediated relations that can operate across the graph. A symbolic representation may target a manifestation, a relational object, an ontological claim, another representation, or a possible future configuration. It can remain a transient analytical artifact or become a persistent causal object. Its location is therefore determined by the relations in which it participates rather than by a fixed position in a vertical hierarchy.

Section 4.10 has established six points. Manifestation is broader than symbolization. Symbolization occurs within symbolic regimes whose rules and practices can have histories of their own. Symbolic representation is selective and purpose-relative rather than identical with its target. Relations and subjects can be symbolized in forms that subsequently alter their relational development. Symbolized value can generate practical commensurability without proving intrinsic comparability or relational equivalence. Finally, representations themselves can become relationally real when they persist, stabilize, and causally participate in subsequent generation. These results prepare Section 100, where the book turns from individual mediations to the recursive relations among relational reality, manifestation, representation, and the later relations that they help produce.

4.11 Recursive Relations among Reality, Manifestation, and Representation

Sections 4.5–4.10 separated relational reality, manifestation, manifestation mappings, non-invertibility, and symbolic representation so that each could be analyzed without collapsing into the others. The present section recombines those distinctions into a re-

cursive architecture. Its role is to explain how relational realities can condition what becomes manifest, how manifestations can be stabilized or transformed through representation, and how representations can re-enter the generative-relational system to alter subsequent relations. The resulting structure is neither a linear sequence nor a closed cycle. It is a historically evolving graph in which different pathways can become active, inactive, reinforced, interrupted, or redirected under changing conditions. (Merton 1948; Giddens 1984; Arthur 1989)

The section proceeds in five steps. It first examines the movement from relational reality to manifestation while preserving the possibility that much relational reality remains unmanifested. It then considers the movement from manifestation to representation and the selective operations involved in symbolic articulation. The third subsection treats representation as an input into subsequent relations rather than as a terminal description. The final two subsections distinguish recursive stabilization from recursive transformation. This distinction is important because repeated mediation can reproduce an existing relational reality, but it can also change the conditions under which later reality emerges. Recursion therefore concerns the historical consequences of mediation rather than the mere repetition of a fixed circuit.

Definition 4.90 (Recursive generative mediation). *Recursive generative mediation* is a historically situated process in which a manifestation or representation generated within a generative-relational system subsequently becomes an operative condition, input, constraint, capacity, or object within that system, thereby contributing to the generation of later relations, manifestations, or representations.

Definition 4.90 preserves the graph architecture established in Section 4.2. A manifestation need not arise only after a fully completed relational object, and a representation need not remain outside the relation it describes. Once a manifestation or representation acquires causal participation in subsequent processes, it becomes part of the history from which later configurations emerge. The recursion is generative because later states can differ from earlier states; it is historical because the effects of previous manifestations and representations can accumulate, dissipate, conflict, or become institutionalized over time.

For analytical purposes, one possible local description distinguishes a relational configuration $\mathcal{R}_t$, a manifestational configuration $\mathcal{M}_t$, a representational configuration $\mathcal{P}_t$, and a wider set of relevant conditions $\mathcal{C}_t$ at a time or interval indexed by t . A manifestation relation may be written schematically as

$$\mathcal{M}_t = \Phi_t(\mathcal{R}_t, \mathcal{C}_t), \quad (4.4)$$

while a representational process may be written as

$$\mathcal{P}_t = \Sigma_t(\mathcal{M}_t, \mathcal{R}_t, \mathcal{C}_t). \quad (4.5)$$

The subsequent relational configuration can then be represented schematically by

$$\mathcal{R}_{t+1} = \Gamma_t(\mathcal{R}_t, \mathcal{M}_t, \mathcal{P}_t, \mathcal{C}_t). \quad (4.6)$$

These expressions are not proposed as universal equations of social or economic reality. They provide a compact way to mark three analytical facts: manifestation de-

depends on conditions, representation can depend on more than manifestation alone, and later relational configurations can be affected by what has previously manifested or been represented. The mappings Φ_t , Σ_t , and Γ_t may themselves change historically and may, under the conditions established in Sections 4.5 and 4.8, become relationally real objects within the systems in which they operate.

4.11.1 Relational Reality to Manifestation

The first relation concerns how a relationally real object or configuration becomes manifest under specified conditions. Section 4.5 established that relational reality does not require prior manifestation. An institution, trust relation, debt relation, social expectation, symbolic convention, or historically stabilized sense of self can participate causally in a system even when parts of its operative structure remain outside present awareness, observation, or explicit articulation. Manifestation therefore selects from, responds to, or is generated through relational reality without exhausting it.

Equation 4.4 represents this relation only schematically. The manifestation $\mathcal{M}_t$ is generated through the conjunction of a relational configuration and the conditions under which manifestation becomes possible. Those conditions can include bodily capacities, technical instruments, institutional procedures, cultural categories, attention, spatial position, temporal position, data infrastructure, measurement rules, and prior symbolic expectations. A price becomes manifest through a market and quotation mechanism; an organizational hierarchy can become manifest through decisions, titles, reporting lines, or patterns of deference; trust can become manifest through disclosure, reliance, expectation, or vulnerability; a physical process can become manifest through an instrument whose design selects some differences while suppressing others.

The manifestation relation is therefore selective without being arbitrary. A relational reality can constrain the range of manifestations available under a given regime even when multiple manifestations remain possible. Conversely, a manifestational configuration can suppress, aggregate, or fail to disclose differences that remain causally important in the relational domain. The relational remainder introduced in Section 4.9 is especially relevant here: what remains unmanifested under one configuration may become important under another, and later consequences may reveal that an apparently negligible difference had continuing causal significance.

Claim 4.91 (Manifestational partiality). *A manifestation can be genuinely generated by and informative about a relational reality while remaining partial with respect to the relational configuration, history, and unrealized possibilities through which it emerged.*

Claim 4.91 permits a middle position between transparent disclosure and radical disconnection. Manifestation can carry real information. It can support explanation, evidence, coordination, and intervention. Yet the information preserved by a manifestation depends on the mapping conditions through which it arises. Reverse inference therefore remains disciplined by the plurality of relational configurations compatible with the observed manifestation.

This partiality becomes important in cases of contingency and aesthetic judgment.

A contingent event can become salient to a subject at a particular moment, but the conditions through which that event becomes recognizable as surprising, beautiful, significant, or worthy of attention may have been generated historically before the moment of conscious judgment. Prior encounters, categories, practices, expectations, bodily capacities, institutional histories, and other relations can contribute to a manifestational configuration without first appearing to the subject as explicit objects of awareness. The subsequent judgment concerns a manifestation, but the relational-generative structure enabling that manifestation can precede the judgment. The reality of that structure is therefore not created merely by being noticed.

The same point also protects a distinction between manifestation and ontological access. A manifestation can be generated through a relational reality whose own ontological status remains suspended. The manifestation can then become evidence about relational organization without licensing a stronger claim about ultimate ontology. The graph from relational reality to manifestation consequently remains compatible with the local sufficiency of ontological commitment developed in Sections 4.1 and 4.3.

4.11.2 Manifestation to Representation

A manifestation can subsequently become represented, recorded, named, classified, quantified, narrated, diagrammed, encoded, or otherwise symbolized. Section 4.10 showed that this movement introduces another selective relation. Representation can preserve a manifestation across time, transmit it across space, place it in relation to other manifestations, and make it available to institutions or actors who were absent from the original event. These capacities enlarge the reach of manifestation. They also introduce new sources of transformation.

Equation 4.5 therefore allows $\mathcal{P}_t$ to depend on more than $\mathcal{M}_t$ alone. A representation can incorporate background theories, institutional categories, prior records, legal rules, measurement conventions, or relational knowledge that is not contained in the immediate manifestation. A medical image can be represented differently under different diagnostic categories; an observed transaction can be represented differently under accounting, legal, or statistical regimes; a verbal statement can be classified differently depending on institutional context. Representation is thus not a passive second copy of appearance. It is an organized operation within a symbolic regime.

This operation can make previously dispersed manifestations comparable. A price system places heterogeneous goods within a monetary scale. A ranking places organizations within an ordinal relation. A legal record places heterogeneous histories within a procedural category. A scientific variable can transform observations into comparable measurements. Such symbolic commensuration can enable coordination and analysis while also suppressing differences that do not enter the symbolic scheme. The practical success of representation therefore remains distinct from the claim that the represented realities share one intrinsic metric.

Remark 4.92 (Representation can preserve and transform). Representation can simultaneously preserve selected information from a manifestation and transform the conditions under which that information is interpreted, compared, transmitted, and

acted upon. Preservation and transformation are therefore not mutually exclusive descriptions of the same representational process.

This duality is particularly visible when a representation creates a durable record. A transient manifestation can disappear while its symbolic trace persists. Once preserved, the representation can become available to actors, institutions, or technical systems in later contexts that differ substantially from those of the original manifestation. The later use can therefore generate relations that could not have arisen from the original manifestation alone. Historical records, legal precedents, archives, credit histories, scientific publications, accounting entries, and algorithmic profiles all illustrate ways in which representation can extend the temporal reach of a manifestation.

At the same time, what is represented can feed back into what later becomes manifest. Once actors learn the categories under which they are classified, they may adapt behavior, resist the category, seek inclusion, or reorganize self-understanding. Once organizations know the metrics by which they are evaluated, they may redesign processes to improve those metrics. Once a price becomes publicly quoted, expectations can change before the underlying production relation changes. Representation therefore begins to participate in the conditions of later manifestation even before it qualifies as a stable relational reality in its own right.

4.11.3 Representation to Subsequent Relation

The third relation is the point at which recursion becomes explicit. A representation can enter subsequent relations and thereby alter the generative-relational system from which later realities emerge. Section 4.10 established that a representation can itself become relationally real when it persists with sufficient stability and exercises causal efficacy. The present subsection focuses on what follows once such causal participation occurs.

Equation 4.6 represents the later relational configuration $\mathcal{R}_{t+1}$ as depending, where relevant, on prior relational structure, manifestation, representation, and wider conditions. This dependence need not be direct. Representations can act through institutions, expectations, technologies, incentives, memory, attention, authority, or collective coordination. The same representation can also generate different effects in different relational environments. A legal classification has little force without institutions able to recognize and enforce it; a market quotation acts differently under liquid and illiquid conditions; a self-description has different consequences depending on whether others recognize, contest, or ignore it.

The causal role of representation therefore depends on relational uptake. A symbol becomes generatively consequential when actors or systems are organized so that the symbol can affect later processes. This can occur through explicit interpretation, but conscious interpretation is not required. Automated scoring systems, database constraints, software permissions, payment infrastructures, and algorithmic classifications can produce consequences because technical and institutional relations are organized around symbolic states. A representation can therefore enter subsequent relations through human, institutional, material, or computational pathways.

Proposition 4.93 (Representational recursion). *When a representation generated within a generative-relational system becomes causally operative in later relations, the*

representation becomes part of the historical conditions through which subsequent relational and manifestational realities are generated. Its later effects are therefore not external commentary on the system but part of the system's generative history.

Proposition 4.93 has important consequences for economic analysis. A credit score can affect access to credit, and the resulting access or exclusion can alter later financial behavior that contributes to future scores. A market price can affect production and demand, which then affect later prices. An accounting classification can alter organizational investment, which changes the activity later recorded under that classification. A public forecast can affect expectations and decisions that become part of the later economic configuration. In each case, representation and represented reality cannot be analyzed as permanently external to one another.

The same logic applies to subjectivity. A manifested experience can be narrated, named, or institutionally classified. That representation can affect memory, recognition, expectation, and future conduct, thereby participating in the relational conditions through which later subjectivity develops. The self-description is neither simply identical with the subject nor irrelevant to subject formation. It becomes one relation among others through which a relatively continuous subject can be sustained or transformed.

Aesthetic judgment provides another instructive case. A contingent encounter can become manifest as striking or beautiful; the judgment can then be symbolically articulated, recorded, taught, cited, or institutionalized. Such representations can redirect attention, confer prestige, organize preservation, influence research selection, and shape the conditions under which later observers encounter the object. The judgment therefore participates in a history that can alter the relational reality surrounding the object without implying that the object's earlier relational reality depended on having already been judged.

4.11.4 Recursive Stabilization

Recursive mediation can stabilize a relational reality when later manifestations and representations repeatedly reinforce the relations through which that reality persists. Stabilization is not mere repetition. Each iteration can occur under slightly different conditions, involve different actors, or produce different manifestations while nevertheless preserving enough structure for the relational object to remain recognizable and causally effective across time.

Definition 4.94 (Recursive stabilization). *Recursive stabilization* is the process through which manifestations or representations generated by a relational reality subsequently reinforce, reproduce, coordinate, or restore relations that contribute to the persistence and stability of that relational reality over a relevant spacetime interval.

Money provides a familiar illustration. Monetary relations generate prices, account balances, banknotes, receipts, quotations, and other manifestations and representations. These forms then guide exchange, accounting, debt settlement, planning, and expectation. Repeated use contributes to the persistence of the monetary relation itself. The relational reality of money is therefore reproduced partly through manifestations and representations generated within monetary relations and later

re-entering those relations.

Trust can display a similar pattern without requiring explicit symbolic articulation at every stage. Repeated reliable interaction can generate manifestations of trust such as reliance, disclosure, delegation, or reduced monitoring. Some of these manifestations may be represented through promises, reputational records, narratives, or institutional certifications. Those representations can then support further reliance and coordination. The resulting recursive stabilization can make trust more persistent, although the process remains vulnerable to contradiction, changing conditions, and competing relations.

Institutional categories can also stabilize recursively. A category used to allocate rights, resources, or responsibilities can shape behavior around the category. The resulting behavior can then appear to confirm the category's practical importance, leading to further institutional use. The recursion may be administratively useful, but stabilization alone does not establish epistemic adequacy or normative legitimacy. A poorly founded category can become highly stable when institutional relations repeatedly reproduce it.

Claim 4.95 (Stability does not validate the stabilized representation). *Recursive stabilization can increase the persistence and causal efficacy of a representation or relational object without establishing that the representation is epistemically adequate, normatively desirable, or ontologically fundamental.*

Claim 4.95 is especially important for GRE because economic systems frequently reward what they have already made legible. A metric used for allocation can cause actors to optimize for the metric; the resulting concentration of activity can make the metric appear increasingly representative of the field. A market category can attract capital, talent, and institutional attention; the resulting expansion can make the category appear more naturally given than it was at its emergence. Recursive stabilization can therefore generate historical depth and causal reality around initially contingent classifications.

This mechanism also helps explain why relational realities can exhibit self-reinforcing tendencies without becoming immutable attractors. Stabilization can be local, conditional, and reversible. Competing representations, institutional changes, shocks, new technologies, altered expectations, or previously suppressed relational differences can weaken the recursive support. The relevant analytical question is therefore which feedback pathways currently sustain the relational object, over what interval, and under what conditions those pathways may cease to reproduce it.

4.11.5 Recursive Transformation

Recursion can also transform rather than stabilize. A manifestation or representation can introduce a difference that changes the relations through which later manifestations and representations are produced. In such cases, the system does not simply reproduce an existing relational reality more strongly or weakly. The effective configuration of relations, constraints, capacities, categories, or manifestation rules changes.

Definition 4.96 (Recursive transformation). *Recursive transformation is the process*

through which a manifestation or representation generated within a generative-relational system contributes to changing the relations, constraints, capacities, mappings, or symbolic regimes that condition subsequent relational and manifestational realities.

A new representation can transform the field by creating a distinction that later actors begin to organize around. A legal recognition can change who is able to contract, own, inherit, claim protection, or appear before an institution. A new accounting category can redirect investment and alter organizational boundaries. A scientific classification can reorganize research attention and intervention. A public articulation of an experience can provide a vocabulary through which previously dispersed manifestations become connected, compared, and institutionally recognized. In each case, representation contributes to changing the generative conditions rather than merely preserving an antecedent state.

Transformation can also occur through failure or contradiction. A representation may repeatedly fail to account for observed manifestations. Those mismatches can generate pressure to revise the symbolic regime, alter the manifestation mapping, or reconstruct the underlying relational model. The appearance of anomalies is itself conditioned by available representational resources, but once recognized, anomalies can become relationally consequential by reorganizing inquiry and intervention. Recursion therefore includes pathways in which the system revises the means through which it manifests and represents itself.

The distinction between stabilization and transformation should remain analytical rather than absolute. A recursive process can stabilize some relations while transforming others. The introduction of a standardized metric may stabilize a field of comparison while transforming organizational behavior. A legal reform may transform formal rights while stabilizing deeper resource asymmetries. A new identity category may stabilize recognition for some actors while transforming relations of affiliation and exclusion. GRE therefore requires the scale and object of stabilization or transformation to be specified rather than inferred from the existence of recursion alone.

Proposition 4.97 (Historical recursion of the three-register graph). *In a generative-relational system, manifestations and representations can become historical conditions of later relational reality when they enter subsequent relations with sufficient causal efficacy. The three-register architecture is therefore recursively historical: later relations can depend partly on how earlier relational realities became manifest, represented, interpreted, preserved, and acted upon.*

Proposition 4.97 summarizes the principal result of the section. The graph among ontological, relational, and manifestational reality cannot be adequately understood as a static taxonomy. Relational reality can precede manifestation, yet manifestation can alter later relational reality. Representation can preserve or transform manifestation, yet representation can itself become relationally real. Symbolic regimes can mediate what later becomes manifest, and changing relations can alter the symbolic regimes through which earlier realities were understood. Historical time therefore enters not as an external parameter added after the ontology, but as part of the generative conditions through which the relations among registers evolve.

This recursive architecture also clarifies the limited use of the Lacanian analogy introduced earlier in the chapter. The relevant similarity lies less in matching ontological reality, relational reality, and manifestational reality one-to-one with the Real, Imaginary, and Symbolic than in refusing a simple hierarchy in which one register merely reflects another. Symbolic and manifestational forms can reorganize later relations; relations can condition later manifestation; and some dimensions of reality can remain resistant to the available symbolic or manifestational regime. GR preserves this graph-like intuition while assigning its own criteria of relational reality, causal efficacy, persistence, and historical emergence.

Section 4.11 therefore completes the first major movement of Chapter 4. Sections 4.1–4.11 have established an ontologically restrained three-register architecture in which relational realities can emerge before perception or symbolization, manifestations arise through historically formed mappings, representations can preserve and transform what manifests, and recursive mediation can stabilize or transform subsequent relations. The next movement turns from the architecture itself to particular classes of object that occupy it. Section 101 begins with subjects, asking how subjecthood can be distinguished across ontological, relational, manifestational, and institutionally represented forms without presuming that manifestational equivalence establishes subject identity.

4.12 Subjects across the Three Levels

The three-register architecture developed in Sections 4.2–4.11 becomes especially consequential when the analytical object is a subject. Economic, legal, institutional, interpersonal, and technical practices routinely treat subjects as if their identity were straightforward: a person signs a contract, an organization owns an asset, an account receives a payment, an employee performs a role, or an artificial system selects an action. Yet the apparently simple noun *subject* can refer to markedly different realities. It can refer to an ontological claim about what kind of being exists, a relationally stabilized locus of causal participation, a manifestation through which subjecthood becomes perceptible or experienceable, or an institutional representation through which a system identifies and acts upon a participant. The role of this section is to separate these senses while showing how they can recursively mediate one another. Its objective is to provide an ontologically restrained account of subjecthood adequate to GRE without reducing subjects to representations or presuming that every operationally similar subject shares the same underlying structure. The discussion proceeds from ontological subjecthood to relationally effective subjecthood, manifestational subjecthood, institutional representation, human and artificial subjects, and finally manifestational equivalence without subject identity. (Mead 1934; Ricoeur 1992; Goffman 1959; Honneth 1995)

The distinction extends rather than replaces the conception of the economic subject developed in Chapter 3, Section 75. Chapter 3 treated an economic subject as a relatively stabilized locus of economically consequential participation. The present section asks what kind of reality such a locus has and how that reality can diverge from the ways in which the subject appears, understands itself, or is represented by institutions. A worker, corporation, household, state agency, platform account,

or artificial system can therefore be approached through several registers at once. These approaches need not converge into one exhaustive description. Their analytical value lies partly in making divergence visible.

Definition 4.98 (Register-sensitive subjecthood). *Register-sensitive subjecthood* is an analytical treatment of a subject that distinguishes among claims concerning its ontological status, its relationally stabilized and causally effective participation, its modes of manifestation, and its symbolic or institutional representation, while allowing these dimensions to mediate one another historically.

Definition 4.98 does not imply that every empirical inquiry requires four separate models of the same subject. It supplies a discipline of distinction. When the dimensions coincide sufficiently for the research question, a simpler representation can be appropriate. When they diverge, however, treating them as identical can conceal economically and normatively important mechanisms. A person can remain relationally consequential while disappearing from an administrative record; an institutional profile can persist after the relations that originally produced it have changed; a self-description can alter subsequent conduct; an artificial system can occupy a stable operational position without this position settling questions of consciousness or moral personhood. The registers therefore help specify which kind of subjecthood is being claimed in any particular analysis.

4.12.1 Ontological Subjecthood

Ontological subjecthood concerns the strongest question in the section: what, if anything, the subject is independently of a particular manifestation, classification, role, record, or analytical representation. The formulation is intentionally open. Different ontologies can locate subjecthood in an embodied organism, a persisting person, a conscious field, a process of self-organization, a bundle of capacities, a relationally constituted agent, a narrative continuity, or some combination of these and other structures. GR does not need to adjudicate this entire field before investigating economically consequential subjecthood. The principle of ontological suspension established in Section 4.3 therefore applies with particular force.

The analytical importance of the ontological question remains even when it is suspended. A legal person, a remembered self, a behavioral profile, a biological organism, and an institutional role may all be associated with what ordinary language calls the same subject, yet they select different persistence conditions. The organism can persist through loss of a role. A legal identity can persist through substantial biological and psychological change. A role can persist through replacement of its occupant. A digital profile can persist after the person it represents has ceased to participate in the relevant system. These differences indicate that subject persistence cannot be inferred from a single criterion before the relevant register has been specified.

The sense of “I” provides a particularly instructive case. A continuing self-experience can become relationally real when it emerges through bodily, mnemonic, interpersonal, linguistic, historical, and other relations, persists with sufficient stability, and causally organizes later conduct. The existence of such relational reality does not by itself settle whether there is an ontologically independent, indivisible, or permanently self-identical entity corresponding to that experience. Conversely,

ontological suspension concerning the ultimate status of the self does not weaken the causal reality of self-experience when it organizes memory, commitment, expectation, responsibility, attention, or action. The three-register architecture therefore allows inquiry into the reality of selfhood without forcing immediate passage from causal efficacy to metaphysical conclusion.

The same restraint applies to collective subjects. A corporation can own property, employ workers, enter contracts, borrow, sue, be sued, accumulate reputation, and survive replacement of many of its members. These properties make corporate subjecthood highly consequential in relational and institutional registers. Whether the corporation should also be treated as an ontologically irreducible collective subject is a stronger question. GRE can investigate the generative consequences of corporate action while leaving this stronger issue open unless the research problem makes it operative.

Remark 4.99 (Ontological subjecthood and analytical reference). Suspending a final ontology of the subject does not make the term *subject* empty. It requires the analysis to specify the properties, persistence conditions, capacities, and relations relevant to the inquiry rather than treating ordinary-language identity as a complete ontological account.

Remark 4.99 is especially important for interdisciplinary work. Economics, law, psychology, computer science, biology, and political theory can refer to apparently similar subjects while selecting different properties as causally or normatively relevant. GRE can connect these descriptions where warranted, while preserving the possibility that their objects are only partially overlapping.

4.12.2 Relationally Effective Subjecthood

Relationally effective subjecthood concerns a different question: whether a subject has emerged as a sufficiently persistent and stable locus of participation to exercise causal efficacy within a generative-relational system. This formulation applies the relational-reality criterion from Section 4.5 directly to subjecthood. A subject is relationally effective when actions, expectations, obligations, memories, permissions, capacities, risks, or other consequences can accumulate around a recognizable locus and when that locus subsequently affects the development of the system.

Definition 4.100 (Relationally effective subject). A *relationally effective subject* is a subject-position or subject-configuration that emerges within a generative-relational system, persists with sufficient stability over a relevant spacetime region or interval, and exercises causal efficacy within that system through the organization of action, expectation, attribution, capacity, constraint, responsibility, or subsequent relation formation.

The definition permits considerable heterogeneity in how subjecthood is stabilized. Biological continuity can matter for a human subject. Legal registration can matter for a corporation. Memory and recognition can matter for interpersonal identity. Repeated interaction can stabilize a household or community role. An account, mandate, technical identity, and authorization structure can stabilize the operational position of an artificial system. The common analytical feature is not a common substance but the emergence of a causally consequential locus around which relations

become organized.

This criterion also clarifies why relational subjecthood can precede explicit self-awareness or symbolic recognition. An infant can participate in relations of care, expectation, attachment, dependency, and resource allocation before possessing the symbolic capacities required for a mature self-description. A worker can occupy a structurally consequential position before recognizing its wider institutional significance. A population can be differentially affected by an institutional arrangement before a political or scholarly vocabulary exists through which that position is named. Subjecthood in the relational register therefore does not wait for complete manifestation to the subject itself.

Relational subjecthood can nevertheless be transformed through manifestation. A person who becomes aware of being trusted may act differently because that trust has become manifest. A worker who learns how an institution classifies a role can reinterpret obligations or opportunities. A community that acquires a shared description of its position can coordinate around that description. The manifestation does not create the entire relational history *ex nihilo*; it enters that history and can reorganize its subsequent development. Subjecthood therefore exemplifies the recursive mediation described in Section 4.11.

Persistence in this register remains relative to the relevant process. A market transaction may require only short-lived identification. Credit, employment, reputation, liability, insurance, inheritance, or long-term cooperation require longer continuity. Some systems construct subject persistence through identifiers, records, biographies, organizational membership, addresses, credentials, or legal continuity. Others rely more strongly on embodied recognition, memory, or recurring interaction. The duration and mechanism of persistence are therefore empirical properties of the relational system rather than fixed attributes of subjecthood in general.

The relational criterion also permits subjecthood to be distributed. A decision attributed to one organization can depend on committees, software, routines, legal mandates, data infrastructures, professional norms, and external constraints. A subject can remain a useful locus of attribution even when the causal organization supporting that locus is internally heterogeneous. The analytical task is then to distinguish the stabilized subject-position from the wider relational architecture that enables it. Relative individuation and relational dependence can coexist.

Claim 4.101 (Relational subjecthood without exhaustive individuation). *A subject can become sufficiently individuated for causal, institutional, or economic analysis while remaining dependent on a wider relational configuration for the persistence, capacities, and boundaries through which that individuation is sustained.*

Claim 4.101 protects GRE from two opposite simplifications. Subjecthood need not be treated as fully pre-given before relation, and relational dependence need not dissolve every subject into an undifferentiated field. The degree and mechanism of individuation are themselves objects of inquiry.

4.12.3 Manifestational Subjecthood

Manifestational subjecthood concerns the ways in which a subject becomes available within a manifestational configuration. The subject may become manifest to itself, to another subject, to an institution, to an instrument, to a public, or to a computational system. These modes can include bodily appearance, speech, action, affect, memory, narrative, performance, biometric signal, statistical pattern, recorded behavior, visual image, or other forms through which some aspect of subjecthood becomes accessible.

Definition 4.102 (Manifestational subjecthood). *Manifestational subjecthood* denotes the reality of a subject insofar as some aspect of that subject becomes available, distinguishable, experienceable, recordable, measurable, interpretable, or otherwise present within a specified manifestational configuration.

Manifestational subjecthood can be fleeting. A facial expression may reveal a momentary affective state without becoming a persistent identity. A statement may disclose an intention that changes shortly afterward. A measurement may reveal one bodily or behavioral feature while leaving many others inaccessible. The entry condition for manifestational reality therefore differs from the persistence condition required for relational reality. A manifestation can be real as manifestation even when it does not stabilize into a durable subject-position.

The first-person case deserves particular care. Self-awareness is a manifestation of subjecthood, yet the relational conditions from which it emerges can precede the moment of explicit awareness. Memory, bodily continuity, habits, language, recognition by others, prior commitments, and institutional histories can participate in stabilizing a subject before those conditions become thematized by the subject. A reflective judgment such as “this is who I am” can then become a new manifestation that feeds back into the relational system. It can alter conduct, expectation, affiliation, memory selection, and the interpretations supplied by others. The manifestational and relational registers thus become recursively coupled without becoming identical.

Manifestation to others can diverge from manifestation to oneself. A person can experience continuity while being institutionally treated as discontinuous because of lost records or changed identifiers. Conversely, an institution can preserve a stable identity while the subject experiences profound discontinuity in self-understanding. A public performance can stabilize one recognizable persona while private experience remains substantially different. The three-register framework allows these differences to be described without requiring one manifestation to stand automatically as the complete truth of the subject.

Manifestations can also conflict. Spoken self-description, observed behavior, physiological measurement, administrative classification, and third-party testimony may disclose different aspects of a subject and can be produced through different manifestation mappings. Their disagreement does not necessarily imply that one simple hidden subject could be reconstructed by choosing the single correct manifestation. Section 4.9 already established that manifestation is generally non-invertible. The subject case makes the implication concrete: multiple manifestations can remain jointly informative while preserving a relational remainder that no one manifestation exhausts.

Remark 4.103 (Plural manifestation of a subject). Different manifestations of a subject can carry different forms of evidential, practical, or institutional relevance. Their plurality does not by itself establish either contradiction or equivalence because each manifestation can depend on a different mapping, audience, purpose, timescale, and relation to the subject's history.

This plurality becomes especially important in economic life. Creditworthiness, productivity, reliability, vulnerability, preference, risk, need, and identity may each be manifested through different procedures. Economic institutions frequently act on these manifestations as though they captured the subject itself. GRE instead asks which aspect became manifest, through which conditions, and with what consequences for later relations.

4.12.4 Institutionally Represented Subjecthood

Institutionally represented subjecthood is a more specific case in which manifestations are selected, symbolized, recorded, and made operational within an institutional system. A legal name, taxpayer identifier, employee number, account, credit file, eligibility category, citizenship status, corporate registration, professional credential, platform profile, or risk score can become a representation through which institutions recognize and act upon a subject. Such representations are often economically decisive because they determine access, liability, authority, visibility, payment, exclusion, or the accumulation of records across time.

Definition 4.104 (Institutionally represented subject). An *institutionally represented subject* is a symbolically or technically stabilized representation through which an institution identifies, classifies, tracks, authorizes, constrains, rewards, sanctions, or otherwise acts upon a purported subject within a specified domain of operation.

The word *purported* is analytically important. The institution's representation can refer adequately to a relationally effective subject, can refer only partially, can lag behind changes in the subject, can combine several subjects, can split one subject across multiple records, or can persist after the relevant relation has ended. The representation and the represented subject are therefore distinguishable objects even when institutional practice treats them as interchangeable.

Represented subjecthood can itself acquire relational reality. An account profile can influence whether a payment is accepted. A credit record can alter borrowing possibilities. A legal status can determine which contracts become available. A credential can organize access to occupations. A platform category can change visibility and expected conduct. Once these representations become persistent, stable, and causally efficacious within the relevant system, they satisfy the relational-reality criterion in their own right. Their causal reality remains distinct from the accuracy or justice of the representation.

This distinction is especially important where institutional systems act faster than subjects can contest their representations. A classification can shape subsequent opportunities and thereby help generate the behavior that later appears to confirm the classification. A profile can influence which information, prices, tasks, or social contacts become available, changing the relational environment from which later manifestations emerge. The representation can therefore participate in producing the

future subject-position that it was initially intended only to describe.

The recursive structure also permits resistance and revision. Subjects can correct records, challenge classifications, strategically present themselves, acquire new credentials, create alternative institutions, or refuse particular categories. Institutions can redesign identifiers, revise eligibility rules, or abandon representations that produce systematic error. Institutionally represented subjecthood is therefore historically alterable even when its practical effects are substantial.

Claim 4.105 (Representational efficacy and subject accuracy). *The causal efficacy of an institutional representation of a subject can vary independently from the representation's descriptive adequacy to the subject it purports to represent.*

Claim 4.105 extends the distinction developed in Section 4.10. A highly influential profile can be incomplete or mistaken; a highly accurate description can remain institutionally irrelevant. Economic analysis therefore needs separate accounts of representational accuracy, institutional uptake, and causal consequence.

4.12.5 Human and Artificial Subjects

The distinction among registers becomes particularly useful when human and artificial subjects occupy adjacent operational positions. Chapter 3 introduced a working distinction between *human subjecthood* and *artificial operational subjecthood*. The present chapter can now state why that distinction matters ontologically. Two systems can become manifest through similar outputs, occupy similar institutional roles, or even be treated as interchangeable for a bounded task while differing substantially in embodiment, historical formation, dependence relations, vulnerability, ownership, persistence conditions, and the wider structures through which their capacities are generated.

A human subject is relationally situated through bodily existence, care, development, language, memory, social recognition, material dependence, institutional position, and historical participation. Many economically relevant properties of human subjecthood are inseparable from vulnerability and finite embodiment: needs must be met through time, bodily capacities can change, care relations can become necessary, and economic loss can affect lived possibilities. These observations do not provide a complete ontology of the human person. They identify dimensions that are causally and normatively relevant to economic analysis and that should not disappear merely because another system can reproduce a selected manifestation of human performance.

Artificial operational subjecthood can arise when an artificial system becomes a sufficiently stable and causally consequential locus of delegated or institutionally recognized action. A system can select transactions, generate offers, route resources, negotiate within constraints, produce classifications, communicate with counterparties, or maintain a persistent operational identity. Where these functions persist and affect subsequent relations, the artificial system can be relationally real as an operational subject-position. This claim does not require GRE to settle whether the system is conscious, possesses first-person experience, qualifies as a legal person, or has independent moral standing.

The relevant relations surrounding artificial operational subjecthood can include model architecture, training data, deployment infrastructure, ownership, authorization, interface design, objective specification, maintenance, oversight, and liability. These relations can differ radically from those that sustain human subjecthood. Even where a human and an artificial system generate the same answer to a query, execute the same trade, or satisfy the same institutional test, equality of manifestation under that test provides limited information about equality of relational structure.

The same caution applies in the opposite direction. Difference in underlying structure does not imply that artificial systems are economically negligible. A system can occupy a causally central position precisely because institutions, firms, markets, or individuals organize their activity around its outputs. The analytical task is therefore to identify the kind of subject-position that has emerged, the relations that sustain it, the manifestations through which it becomes legible, and the institutions through which its actions are attributed.

Claim 4.106 (Operational subjecthood and ontological restraint). *Attribution of operational subjecthood to an artificial system records a stabilized and causally consequential position within a generative-relational system. It does not, without additional argument, establish experiential, ontological, juridical, or moral equivalence with human subjecthood.*

Claim 4.106 permits GRE to analyze technologically mediated economies without choosing between two unhelpful extremes: treating artificial systems as mere inert tools regardless of their operational role, or treating functional substitution as sufficient evidence of full subject identity. The three-register architecture keeps the empirical consequences of artificial action visible while preserving stronger questions for the domains in which they can be responsibly examined.

4.12.6 Manifestational Equivalence without Subject Identity

The preceding distinctions culminate in the problem of manifestational equivalence. Two subjects can become indistinguishable with respect to a specified manifestation while remaining different in relational organization and with their ontological relation unresolved. The equivalence is therefore local to the manifestation criterion, the conditions under which it is produced, and the purpose for which the comparison is made.

Definition 4.107 (Local manifestational equivalence of subjects). Two subjects are *locally manifestationally equivalent* when, under a specified manifestational configuration and comparison criterion, the manifestations relevant to the inquiry are indistinguishable or sufficiently equivalent for that purpose. Such equivalence does not by itself establish relational identity, ontological identity, equal historical formation, equal capacities outside the comparison domain, or equal normative standing.

The definition applies to human–human, human–artificial, artificial–artificial, organizational, and other comparisons. Two people can produce identical answers while arriving at them through different memories or reasons. Two firms can display the same price while relying on different cost structures and financing relations. Two artificial systems can produce the same output while differing in architecture, training history, ownership, or failure modes. A human and an artificial system can

generate behavior that is indistinguishable through a particular interface while belonging to substantially different relational structures. The manifestation therefore supports an equivalence claim only at the level for which the comparison criterion has been specified.

This point is stronger than the familiar observation that appearances can deceive. A manifestation can be completely accurate with respect to the property it manifests and still remain insufficient for subject identity. If two systems genuinely have the same measured response time, the equality need not be illusory. It simply establishes equality of that measured property. The inference from local equality to global identity requires additional mappings and additional evidence. Non-invertibility therefore arises even when the manifestation itself is reliable.

The relation between an illusory flower and an ordinarily perceived physical flower provides a useful limiting example from the wider architecture of this chapter. Both can become manifest, can enter attention and memory, and can produce subsequent action. Under suitable conditions, the experienced flower can also acquire relational reality through its causal participation in the subject's system. Yet shared manifestational properties such as apparent color, shape, or salience do not establish shared generative structure. The same discipline applies to subjects: similarity in one manifestational domain does not supply a universal metric through which their full relational structures become commensurable.

Manifestational equivalence can nevertheless have powerful institutional consequences. If an institution recognizes subjects only through a narrow interface, subjects that are equivalent under that interface may receive the same permissions, prices, rankings, opportunities, or liabilities. The institution can thereby convert local equivalence into a consequential social fact. Conversely, a representation that distinguishes subjects through an institutionally salient criterion can generate different treatment even when the subjects remain equivalent along many other dimensions. The design of manifestational criteria is therefore itself part of the generative-relational economy.

Claim 4.108 (Subject identity exceeds local manifestational equivalence). *Local manifestational equivalence licenses an equivalence claim only within the specified manifestational domain and comparison criterion. Extension from that equivalence to relational or ontological subject identity requires independent justification.*

Claim 4.108 provides a compact rule for later GRE analysis. It prevents operational tests, institutional classifications, market signals, or behavioral outputs from silently becoming complete ontologies of the subjects they classify. It also preserves the legitimate analytical use of equivalence where a bounded comparison is sufficient. A payment system may need only authorization equivalence; a contractual process may need legal-capacity equivalence; a production model may need task-performance equivalence. The mistake lies in allowing the local criterion to expand beyond the scope it was designed to establish.

Subjects across the three registers can now be understood as historically coupled without becoming reducible to one another. Ontological subjecthood names the strongest and often partly suspended question of what the subject is. Relationally effective subjecthood identifies sufficiently persistent, stable, and causally efficacious

loci of participation. Manifestational subjecthood concerns the modes through which aspects of those subjects become available. Institutional representation stabilizes selected manifestations into symbolic or technical forms through which systems act. Human and artificial subjects can occupy overlapping operational domains while retaining potentially substantial relational and ontological differences. Manifestational equivalence can support bounded coordination and comparison while leaving subject identity open.

This register-sensitive account also prepares the next section. Value and meaning are frequently attributed to, generated by, experienced through, represented for, and circulated among subjects. The same discipline that prevents subject representation from replacing the subject will therefore be required when GRE turns to value and meaning across the three levels.

4.13 Value and Meaning across the Three Levels

Value and meaning are central to economic life because subjects, institutions, and communities act through distinctions concerning what matters, what counts, what is worth preserving or exchanging, and what objects, events, relations, or representations signify. The three-register architecture developed in this chapter permits these phenomena to be examined without collapsing them into either objective properties or subjective reports. The role of this section is to specify how value and meaning can be approached across ontological, relational, and manifestational registers. Its objective is to distinguish the relational reality of value and meaning from stronger claims concerning their ultimate ontological status, normative validity, or semantic truth, while preserving the causal and historical importance of valuation and interpretation. The discussion proceeds from relationally emergent value and meaning to their manifestation, symbolic mediation, partial commensuration, and historical transformation. The method is analytical and reconstructive: the section applies the ontology developed in Sections 4.2–4.11 to evaluative and meaningful phenomena while keeping heterogeneous forms of value and meaning open to later domain-specific analysis. (E. Anderson 1993; Appadurai 1986; Bourdieu 1984; Reber et al. 2004)

Chapter 2, Section 43 reconstructed several existing traditions concerning value, valuation, plural forms of value, and commensurability. The present section has a different task. It asks what kind of reality value and meaning can possess within GR. The answer cannot be supplied by declaring that value is simply subjective, objective, social, or symbolic. A value can be historically generated yet highly stable; it can organize action before being explicitly articulated; it can acquire symbolic representation in prices, rankings, rituals, or categories; and its representation can subsequently alter the relations through which value is generated. Meaning exhibits a parallel complexity. An interpretation can emerge through language, practice, memory, expectation, embodiment, and institutional history; it can become causally consequential before it is formulated propositionally; and its later articulation can transform the field of relations in which it continues to operate.

The resulting framework therefore separates three questions. The ontological question concerns whether value or meaning has a status independent of the par-

ticular relational and manifestational configurations through which it becomes available. The relational question concerns whether evaluative or meaningful structures have emerged with sufficient persistence, stability, and causal efficacy to become relationally real within a generative-relational system. The manifestational question concerns the forms through which value or meaning becomes perceptible, articulable, measurable, symbolizable, or otherwise available under particular conditions. These questions can interact without yielding identical answers.

Definition 4.109 (Register-sensitive value and meaning). *Register-sensitive analysis of value and meaning* distinguishes among their possible ontological status, their relational emergence and causal efficacy, and the particular forms through which they become manifest or represented, while allowing these dimensions to mediate one another historically.

Definition 4.109 preserves two forms of restraint. First, the existence of relationally real value does not by itself establish that whatever is valued is normatively good. Second, the existence of relationally real meaning does not by itself establish that the interpretation is semantically true, referentially successful, or epistemically justified. A society can stabilize harmful forms of valuation. A community can organize action around a false interpretation. These formations can nevertheless be causally real because they structure conduct, institutions, expectations, and subsequent relations. GRE therefore requires a vocabulary capable of studying the reality of value and meaning without treating causal stabilization as moral or epistemic vindication.

4.13.1 Value as Relationally Emergent

Value becomes relationally relevant when something acquires evaluative significance through a generative-relational system and that significance participates in the organization of subsequent action. The relevant relations can include desire, need, use, care, recognition, exchange, scarcity, obligation, attachment, institutional classification, historical memory, ecological dependence, collective identity, or aesthetic orientation. Different forms of value can therefore arise through markedly different structures. The relational account does not require these structures to be reducible to a common mechanism or a single scalar quantity.

Definition 4.110 (Relationally emergent value). *Relationally emergent value* is evaluative significance that emerges within a generative-relational system, persists with sufficient stability over a relevant spacetime region or interval, and exercises causal efficacy by orienting selection, preservation, exchange, allocation, attention, commitment, refusal, or subsequent relation formation.

The definition applies the general relational-reality criterion from Section 4.5. It does not imply that every transient preference is already a relationally real value. A fleeting attraction can become manifest without acquiring sufficient persistence or causal force to organize later relations. By contrast, a durable commitment to a person, a sacred object, a professional norm, a currency, a landscape, an institutional mission, or a form of knowledge can become relationally real when the evaluative orientation survives across situations and affects future conduct.

Relationally emergent value can precede explicit valuation. A child can be embedded in relations of care before possessing a developed vocabulary of value. A com-

munity can sustain practices that preserve a landscape before articulating a formal environmental ethic. A researcher can repeatedly orient attention toward a problem experienced as promising or beautiful before producing an explicit account of why it matters. In these cases, the evaluative structure is already participating in the organization of action even though its later symbolic formulation remains incomplete.

This pre-manifestational possibility is especially important for aesthetic judgment. An object does not need to become explicitly judged as beautiful before the relational-generative-historical conditions that make such a judgment possible begin to exist. The history of training, perceptual formation, disciplinary practice, memory, cultural repertoire, bodily disposition, and previous encounters can already structure susceptibility to a later manifestation of beauty, elegance, significance, or importance. The eventual judgment therefore emerges within a field whose evaluative organization partly precedes conscious recognition.

Relational emergence also permits value to be plural without assuming that every plurality can be converted into one metric. Money, friendship, legal equality, ecological continuity, scientific insight, bodily integrity, cultural memory, and aesthetic experience can all acquire relational reality as forms of value while belonging to different evaluative domains. Some domains support stable conversion procedures; others permit only partial comparison; still others may lack a warranted common ordering for the relevant inquiry. The existence of multiple relationally real values therefore does not itself establish commensurability.

Claim 4.111 (Relational reality and normative validity). *The relational reality of a value establishes that the evaluative formation is sufficiently persistent, stable, and causally efficacious within a relevant generative-relational system. It does not by itself establish the normative validity, justice, desirability, or universal authority of that value.*

Claim 4.111 is indispensable for GRE. Market value, status value, ideological value, discriminatory valuation, and exploitative institutional priorities can become highly stable and causally consequential. Their relational reality is an object of explanation, not a certificate of legitimacy. Normative assessment requires additional arguments concerning justice, capability, freedom, care, ecological consequence, rights, or other relevant standards. The three-register architecture therefore keeps descriptive ontology and normative evaluation connected without allowing one to substitute for the other.

The ontological status of value can remain more open. A moral realist, constructivist, pragmatist, relational theorist, or other philosophical position may offer different accounts of whether some values obtain independently of their historical realization. GR does not need to resolve those disagreements in order to study how value becomes causally effective in economic systems. Ontological suspension is especially useful here because it permits strong claims about the historical reality of valuation while leaving stronger metaethical questions available for separate treatment.

4.13.2 Meaning as Relationally Emergent

Meaning concerns the significance through which an object, event, relation, sign, practice, or possibility becomes situated within a field of interpretation and action.

It can involve semantic reference, practical significance, narrative orientation, institutional interpretation, affective salience, symbolic association, or a subject's understanding of what an encounter means for subsequent conduct. These modes should not be reduced to a single linguistic model. A gesture can carry meaning without being a sentence; a price can signal scarcity or prestige; a ritual can organize belonging; a legal category can determine institutional consequence; and an unexpected event can acquire significance before its interpretation becomes verbally explicit.

Definition 4.112 (Relationally emergent meaning). *Relationally emergent meaning* is significance generated through a relational configuration that persists with sufficient stability and exercises causal efficacy by organizing interpretation, expectation, attention, memory, orientation, communication, or subsequent action within a generative-relational system.

Meaning in this sense can be relationally real before it is fully symbolized. A familiar place can carry significance through repeated embodied and interpersonal relations without that significance being translated into a proposition. A social role can orient expectation before participants articulate its rules. A contingent encounter can become structured as potentially significant through a subject's history before the subject becomes reflectively aware of what the encounter means. Symbolic language may later stabilize and communicate such meanings, but symbolization is one mediating pathway rather than a universal prerequisite.

The distinction between relational meaning and semantic truth is parallel to the distinction between relational value and normative validity. A rumor can acquire meaning and alter behavior even when its propositional content is false. A mistaken diagnosis can reorganize a person's expectations, institutional treatment, and future decisions. A fictional character can possess enduring cultural meaning without being an ontologically existing historical person. A religious symbol can become highly consequential within a community while its associated metaphysical claims remain subject to independent inquiry. The causal reality of meaning therefore concerns what the interpretation does within a system, while truth concerns a different relation whose criteria depend on the relevant epistemic domain.

Claim 4.113 (Meaning and truth). *The relational reality of a meaning records the persistence, stability, and causal efficacy of an interpretive formation. It does not by itself establish the truth, referential adequacy, or epistemic justification of the interpretation through which that meaning is organized.*

Claim 4.113 protects the framework from a form of relational relativism. GR can recognize that meanings are historically and relationally generated while also allowing evidence, logical consistency, causal inquiry, testimony, measurement, and other epistemic practices to constrain claims about truth. Relational generation explains how an interpretation becomes operative; it does not make every operative interpretation equally warranted.

Value and meaning also interact without becoming identical. Something can be meaningful without being positively valued. A traumatic site can carry intense meaning while being associated with aversion. A market price can be valuable as information for one actor while symbolizing exclusion for another. A mathematical conjecture can possess meaning within a disciplinary history before a community attributes

high research value to it. Conversely, something can be valued instrumentally while carrying little broader meaning for the subject who uses it. GRE therefore benefits from treating evaluative orientation and interpretive significance as partially coupled dimensions rather than interchangeable terms.

4.13.3 Manifestation of Value

Relationally emergent value becomes manifest through heterogeneous forms. A price, ranking, vote, award, credential, investment decision, act of preservation, ritual gesture, refusal to sell, allocation of time, emotional response, legal protection, or pattern of care can each manifest evaluative significance under particular conditions. None of these manifestations is automatically exhaustive. A market price can manifest one institutionally stabilized form of valuation while leaving other forms of value unexpressed. A conservation designation can manifest institutional recognition while only partially representing the attachments, ecological dependencies, or historical meanings through which the protected object matters.

Definition 4.114 (Manifestation of value). A *manifestation of value* is a perceptual, behavioral, affective, institutional, symbolic, quantitative, or other form through which an evaluative significance becomes available within a specified manifestational configuration.

The definition preserves the distinction from Section 4.6: manifestation does not require the persistence or stability required for relational reality. A momentary expression of delight can manifest value even if the evaluation does not persist. Conversely, a relationally stable value may remain only partially manifested at any particular moment. The mapping from evaluative structure to manifestation is therefore neither complete nor necessarily invertible.

Economic analysis often relies on manifestations of value because many evaluative processes become observable through choices, bids, prices, quantities, scores, surveys, ratings, or institutional decisions. Such manifestations can be informative, but Section 4.9 implies a discipline of inference. Similar prices can arise from different evaluative structures; similar preferences expressed in a survey can be sustained by different histories; and the same underlying commitment can generate different behavior under changing constraints. An observed manifestation should therefore be treated as evidence about value under specified assumptions rather than as an automatic identity between the measure and the value itself.

Symbolization can further transform the evaluative field. Once a value is assigned a numerical score, price, label, ranking, or category, the symbolic representation can become part of the system through which later valuation occurs. Rankings can redirect attention. Prices can reorganize expectations. Certification can create prestige. Accounting categories can determine which contributions become visible. Repeated symbolic representation can therefore stabilize one evaluative dimension while weakening, obscuring, or transforming others.

This process can create practical commensurability among heterogeneous objects. A price system, index, scoring rubric, or ranking scheme can place previously heterogeneous objects within a shared representational space. Section 4.10 characterized this process as symbolic commensuration. The common metric can become

highly consequential without establishing that the underlying values were intrinsically commensurable before the mapping was imposed or negotiated. GRE should therefore distinguish the existence of a functioning conversion interface from the stronger claim that the converted values share a common evaluative structure.

Remark 4.115 (Manifested value and commensuration). A successful symbolic conversion can make heterogeneous values operationally comparable for a specified purpose. The existence of that comparison does not establish complete equivalence among the relational structures from which those values emerge.

This distinction becomes central for later GRE analysis of allocation, return, exploitation, compensation, and justice. An institution can compensate one loss through money because monetary conversion is legally or practically available while the affected subject continues to experience an irreducible remainder. The existence of exchange does not establish that nothing was lost in translation.

4.13.4 Manifestation of Meaning

Meaning likewise becomes manifest through multiple channels. Speech, writing, gesture, image, bodily orientation, narrative, ritual, classification, architecture, interface design, statistical representation, silence, and patterned action can all make significance available. Manifestational plurality is especially important because the same relational meaning can become differently available across subjects and contexts, while similar symbolic forms can sustain different meanings in different histories.

Definition 4.116 (Manifestation of meaning). A *manifestation of meaning* is a form through which relationally generated significance becomes perceptible, interpretable, communicable, recordable, or otherwise available within a particular manifestational configuration.

The definition makes room for both symbolic and non-symbolic manifestations. A facial expression can make distress manifest before it is named. The spatial organization of a courtroom can manifest authority without requiring a verbal proposition. A recurring economic practice can manifest expectations that participants have never written down. A scientific diagram can render a relation visible in a form unavailable to ordinary perception. Symbolization becomes especially powerful because it allows meanings to be stabilized, transmitted, recombined, compared, archived, and recursively reintroduced into later relations.

The mapping from meaning to manifestation remains context-sensitive and non-invertible. The same sentence can carry different meanings in different histories. Different languages can manifest overlapping meanings without exact structural equivalence. Silence can signify consent in one institutional context and resistance in another. A numerical indicator can be interpreted as evidence of performance, risk, failure, or injustice depending on the relational field in which it appears. Reverse inference from manifestation to meaning therefore requires knowledge of the historical and relational conditions through which the manifestation becomes interpretable.

Meaning can also mediate value. A commodity's price can change when its meaning changes from ordinary good to collectible object. A place can acquire economic value after becoming symbolically associated with heritage. A technology can be-

come desirable because it signifies modernity, status, autonomy, or belonging. Conversely, valuation can reshape meaning by directing preservation, repetition, institutional recognition, and educational transmission toward some interpretations rather than others. The relation between value and meaning is therefore recursive and historically generative.

Aesthetic judgment again offers a useful limiting case. The manifestation of beauty is not reducible to the verbal statement that something is beautiful. Aesthetic salience can appear affectively or perceptually before reflective judgment, while the relational conditions enabling that appearance may have been generated over a much longer history. Once articulated, the judgment can further alter attention, prestige, memory, canon formation, research investment, and transmission. The manifestation therefore both discloses and participates in the future generation of evaluative and meaningful reality.

4.13.5 Historical Transformation of Value and Meaning

Value and meaning are historically transformable because the relations, manifestation mechanisms, representations, subjects, and institutions through which they are generated can change. A value can persist while its manifestation changes. A symbolic representation can persist while the relational value beneath it weakens or shifts. A meaning can survive through translation into new media, or a stable sign can acquire substantially different meanings across periods. Historical analysis must therefore distinguish continuity of representation from continuity of relational structure.

The recursive architecture from Section 4.11 supplies a useful model. At one time, a relationally emergent value or meaning contributes to a manifestation. That manifestation can be represented, recorded, institutionalized, taught, priced, ranked, or contested. The representation then becomes part of the later relational environment in which subjects form new valuations and interpretations. Historical transformation therefore need not occur through replacement of one complete value system by another. It can proceed through gradual changes in the relations among practices, symbols, institutions, subjects, and memories.

Money illustrates the point clearly. Monetary value depends on historically stabilized relations among institutions, legal arrangements, accounting practices, expectations, exchange, and recognition. Monetary symbols then participate in reproducing those relations. A unit of account can persist while its purchasing power, political significance, technological embodiment, and relation to trust change. The representation remains recognizable, yet the relational field through which it is valuable evolves.

The same structure applies to prestige, professional credentials, cultural objects, legal rights, religious symbols, scientific problems, and forms of identity. A credential can remain formally identical while the labor market changes the opportunities it generates. A cultural symbol can retain its visual form while becoming associated with different collective histories. A research problem can acquire greater value after new techniques make it tractable or after a disciplinary community interprets it as theoretically central. Historical change therefore concerns both the objects valued

and the relations through which valuation itself becomes possible.

Claim 4.117 (Historical plasticity of value and meaning). *The persistence of a value or meaning across time can coexist with substantial transformation in the relations, manifestation mechanisms, representations, and subjects through which it is generated and reproduced. Apparent continuity at one register therefore does not establish structural continuity across the others.*

Claim 4.117 also clarifies why GRE should preserve records of category change. If an institution measures “merit,” “risk,” “productivity,” “quality,” or “need” across decades, continuity of vocabulary can conceal changing evaluative structures and meanings. A time series may remain numerically continuous even while the social object represented by the variable changes. Conversely, terminology can change while a deeper relational structure remains comparatively stable. Historical comparability therefore requires reconstruction of the manifestation mappings and valuation practices that connect periods.

The three-register view consequently prevents both reification and dissolution. Value and meaning need not be treated as timeless substances in order to be real. Their historical and relational emergence does not make them causally negligible, arbitrarily subjective, or reducible to discourse. When they acquire sufficient persistence, stability, and causal efficacy, they can become relationally real. When they appear through prices, judgments, classifications, emotions, symbols, practices, or institutions, they acquire manifestational forms whose structure can be studied independently. Questions of ultimate value, normative validity, semantic truth, and ontological fundamentality remain distinguishable and can be addressed where the inquiry requires them.

This register-sensitive account prepares the transition to social facts and institutional reality. Values and meanings frequently become durable precisely because institutions stabilize classifications, expectations, obligations, roles, and authorized interpretations across persons and time. Section 103 therefore turns from evaluative and meaningful reality to the institutional structures through which relational realities can become collectively reproducible and causally persistent.

4.14 Social Facts and Institutional Reality

Social facts and institutions provide some of the clearest cases in which the three-register architecture can be applied to economic reality. Money, property, legal status, organizational office, debt, professional qualification, citizenship, contractual obligation, and corporate personality can organize action with considerable durability even though their efficacy cannot be identified with the material objects through which they are manifested. Chapter 2, Section 40 reconstructed several existing accounts of social facts, institutional stabilization, legal constitution, symbolic authority, and historical revisability. The role of the present section is different. Its objective is to specify how such phenomena can possess relational reality within GR, how institutional stabilization alters the persistence and causal reach of social facts, and how legal, monetary, and organizational realities traverse ontological, relational, and manifestational registers. The method is register-sensitive: the section distinguishes

the material and other ontological conditions on which institutions may depend, the relational configurations through which institutional objects persist and act, and the texts, records, symbols, performances, classifications, and interfaces through which institutional realities become manifest.

The central claim is modest but consequential. Social construction does not imply diminished reality. When a socially generated object persists with sufficient stability and exerts causal efficacy within the generative-relational system that sustains it, it can satisfy the criterion of relational reality developed in Section 4.5. Its relational reality does not settle whether it possesses an independent ontological existence, whether its representation is epistemically adequate, or whether the institution is normatively legitimate. A legal status can be relationally real while unjust. A monetary unit can be relationally real without being an ontologically fundamental object. An organizational classification can be causally effective even when it misdescribes the people to whom it is applied. The ontology of institutional life therefore requires several distinctions that become obscured when “socially constructed” is treated either as a synonym for unreality or as a complete theory of what social objects are. (Berger and Luckmann 1966; Searle 1995; North 1990; Ostrom 2005)

Definition 4.118 (Institutional reality). *Institutional reality* denotes the relational reality acquired by socially generated objects, statuses, rules, roles, classifications, procedures, or organized capacities when they are stabilized through sufficiently persistent configurations of recognition, practice, authorization, record, material support, expectation, enforcement, or resource allocation and thereby become causally efficacious within the relevant generative-relational system.

Definition 4.118 intentionally includes several possible stabilizing mechanisms rather than reducing institutions to one source. Some institutional realities depend strongly on formal authorization. Others are sustained primarily through convention, habit, organizational routine, infrastructural dependence, or mutually adjusted expectations. Many combine several of these mechanisms. The definition also leaves open whether the relevant institutional object possesses an ontological status beyond its relational realization. The analytical burden is first to identify how it is generated, how it persists, what stabilizes it, and how it affects subsequent relations.

4.14.1 Social Facts

A social fact, in the present framework, is not defined merely by the existence of a shared belief. Shared representations can contribute to its formation, but the relational reality of a social fact requires more than mental agreement. The fact must become sufficiently stabilized within a social field to organize conduct, expectations, opportunities, obligations, identities, access, or resource flows. The relevant relations may include mutual recognition, legal authorization, customary practice, material infrastructure, records, sanctions, market coordination, professional routines, or repeated interaction. What is socially factual is therefore tied to a configuration capable of making a difference within the system.

Definition 4.119 (Relationally real social fact). A *relationally real social fact* is a socially generated object, status, rule, distinction, or expectation that persists with sufficient stability across a relevant spacetime interval and exercises causal efficacy

within the social system through which it is sustained.

This definition preserves the general GR criterion while making the social mode of generation explicit. A marriage, debt, academic credential, property title, rank, office, or currency denomination may be relationally real because actors, organizations, databases, legal procedures, infrastructures, and expectations collectively sustain a configuration through which subsequent action is differentiated. The relevant efficacy may occur even when a particular subject is unaware of the fact. A person can acquire a tax liability without having noticed it, be subject to a legal restriction without understanding its doctrinal basis, or hold a claim recorded in an institutional system before becoming aware that the record exists. Subjective apprehension is therefore neither a necessary nor a sufficient condition of relational reality.

The same point helps distinguish social facts from manifestational facts. A passport, contract, certificate, database entry, court order, price display, uniform, title, or public ceremony can manifest an institutional status. Yet the manifested object is not automatically identical with the relational structure through which the status persists. A printed certificate can be destroyed while the qualification remains institutionally recognized through other records and practices. Conversely, a visually convincing certificate can exist without the institutional relations required to make its represented status effective. Manifestation can disclose, transmit, stabilize, or transform a social fact, but it does not exhaust the social fact's relational reality.

Social facts can also precede explicit symbolization in a local context. Repeated patterns of deference, exclusion, obligation, entitlement, or informal authority may become sufficiently stable to shape conduct before participants develop a settled vocabulary for them. Once named, codified, measured, or represented, these patterns can enter the recursive dynamics described in Section 4.11. The new manifestation may render an existing relation more legible, but it can also reorganize that relation by enabling contestation, coordination, enforcement, comparison, or institutional redesign.

Claim 4.120 (Social fact without universal awareness). *The relational reality of a social fact does not require universal recognition or simultaneous awareness among the subjects affected by it. It requires a sufficiently stable generative-relational configuration capable of sustaining the fact and mediating causal consequences within the relevant system.*

Claim 4.120 matters for large institutional systems. Modern societies routinely coordinate relations among participants who possess highly unequal knowledge of the rules, databases, classifications, procedures, and infrastructures that shape their actions. Institutional reality can therefore be distributed across persons, documents, software, offices, routines, and material arrangements rather than residing in a single collective representation. The resulting social fact may be real for the system even though no individual actor possesses a complete representation of the relations that sustain it.

Social facts need not be mutually commensurable merely because they share the same relational-reality criterion. A religious office, a financial claim, a kinship relation, a legal title, and an academic credential may all be relationally real while being generated through different mechanisms and embedded in different comparison

structures. Even objects bearing similar labels across jurisdictions or historical periods may not be structurally equivalent. Cross-institutional comparison therefore requires an explicitly justified mapping rather than an assumption that shared terminology identifies a common metric space.

4.14.2 Institutional Stabilization

Institutionalization concerns the transition from recurrent social relations toward more durable configurations in which expectations, roles, procedures, classifications, sanctions, and resources become mutually reinforcing. The transition is rarely a single event. It can involve repetition, record keeping, formalization, professionalization, infrastructural investment, legal recognition, routinized enforcement, education, socialization, technological embedding, or the accumulation of precedent. Different combinations can produce different degrees and forms of institutional stability.

Definition 4.121 (Institutional stabilization). *Institutional stabilization* is the historical process through which a socially generated configuration acquires sufficient persistence, reproducibility, and causal reliability that its roles, rules, statuses, procedures, or organized capacities can structure subsequent relations beyond the immediate interactions in which they first emerged.

Institutional stabilization can extend a relational reality across time and space. A bilateral promise can depend heavily on the memory and continued willingness of the two participants. A standardized contract recorded within a legal and administrative system can remain efficacious through changes in personnel, location, ownership, or organizational membership. Stabilization therefore changes not only duration but also the architecture through which persistence is achieved. Memory can be externalized into records; authorization can be attached to offices rather than persons; sanctions can be routinized; procedures can be standardized; and material infrastructures can preserve practical continuity across generations of participants.

This extension of persistence should not be confused with immutability. An institution can be highly stable while remaining historically revisable. Indeed, some institutions stabilize procedures through which their own rules can be amended, offices replaced, assets transferred, or statuses terminated. A constitution, corporate charter, professional code, or monetary regime can therefore include relations of authorized transformation within the institutional reality itself. Stability and revisability are compatible when the procedures governing change are themselves sufficiently stabilized.

Institutional reality can also become recursively self-supporting. A rule shapes behavior; repeated behavior creates expectations; expectations increase the cost of deviation; organizations invest in infrastructure built around the rule; records accumulate; professional roles emerge to interpret and administer it; and these developments make the rule more consequential for later participants. Such recursion can deepen relational reality without converting historical contingency into necessity. What has become difficult to change may still have been generated through a path that could have developed differently.

The three-register architecture helps separate several components of stabilization.

A building, server, ledger, biometric device, courtroom, banknote, or communications network may provide material conditions. A role, entitlement, debt, office, legal relation, or monetary claim may be relationally real within the institution. A statute, record, symbol, dashboard, credential, verdict, price, or ceremony may manifest and represent the relevant relation. Institutional persistence often depends on the coupling among these registers rather than on any one of them in isolation.

Remark 4.122 (Institutional object and institutional manifestation). The persistence of an institutional object can survive the loss or replacement of particular manifestations, while changes in manifestation can nevertheless transform the institutional object when the manifestation itself enters the relations that reproduce it.

This remark captures an important asymmetry. A corporation can survive replacement of its logo, offices, employees, directors, databases, or paper records because its relational continuity is distributed across multiple mechanisms. Yet a change in registration, governance rules, legal classification, or ownership records can alter the corporation precisely because these manifestations are institutionally operative. Manifestation is therefore neither merely superficial nor automatically constitutive; its role depends on the generative relations in which it participates.

4.14.3 Legal Reality

Law offers a particularly clear illustration of relational reality mediated through manifestation. Legal systems depend heavily on texts, judgments, filings, signatures, classifications, procedures, and public acts. These are manifestational forms. Their legal efficacy, however, depends on historically stabilized relations among institutions, officials, subjects, records, recognized procedures, interpretive practices, enforcement capacities, and expectations. The written form of a rule is therefore important without being sufficient to explain the reality of law.

Definition 4.123 (Legal relational reality). *Legal relational reality* is the relational reality acquired by legal statuses, rights, duties, powers, disabilities, procedures, and institutional acts when they are sustained through a sufficiently stable legal-institutional configuration and become causally capable of structuring action, authorization, access, sanction, liability, or resource distribution.

A property right illustrates the distinction. The physical object that becomes property does not contain the right as one of its material attributes. The right is sustained through relations among recognized holders, registries, courts, contractual practices, administrative procedures, social expectations, and enforcement capacities. It may be manifested through a title, deed, digital register, possession, boundary marker, or judicial decision. These manifestations can be indispensable to particular legal systems, but the right's relational reality consists in the stabilized configuration through which claims and actions acquire differential consequences.

Legal acts can also generate new relational realities through authorized manifestation. A judgment, registration, marriage declaration, incorporation filing, appointment, or transfer can change what relations exist because the relevant system assigns generative force to the act. The manifestation is then performative in a specific institutional sense: its efficacy arises from its position within an authorized relational configuration rather than from the words or marks alone. The same utterance or

document outside the relevant institutional relations may lack the corresponding effect.

This generative feature helps explain why legal reality cannot be reduced either to symbolic representation or to physical enforcement. A court order can remain legally real during an interval in which enforcement has not yet occurred. Conversely, coercive conduct can occur without possessing the legal status claimed for it. Legal reality therefore involves relations of validity, authorization, interpretation, institutional competence, and expected consequence that can be analytically distinguished from both textual manifestation and material force.

The framework also permits a distinction between legal reality and legal justice. A discriminatory status, exclusionary rule, or procedurally valid decision can become relationally real when institutions sustain and enforce it. Its causal reality does not establish moral legitimacy. Conversely, a normatively compelling claim may lack institutional realization and therefore possess little or no legal relational reality within a particular jurisdiction at a particular time. GRE can study the economic consequences of this gap without treating the existing legal order as normatively self-validating.

Claim 4.124 (Legal efficacy without ontological identity). *A legal object can possess substantial relational and causal reality without requiring the analyst to posit a distinct ontological substance corresponding to the legal category. Its reality can be traced through the relations that generate, stabilize, manifest, interpret, and enforce it.*

This claim illustrates ontological suspension in practice. Competing philosophies of law may disagree about the ultimate nature of legal normativity, institutional facts, or legal obligation. The relational analysis can remain productive across part of that disagreement by identifying how legal objects become effective in historically specific systems. Where deeper ontological commitments alter the explanation, the disagreement becomes operative and should be addressed explicitly rather than prematurely resolved.

4.14.4 Monetary Reality

Money provides an especially useful case because its manifestations are easy to confuse with the relational object itself. Coins, banknotes, account balances, ledger entries, payment messages, and displayed prices are manifestations through which monetary relations become available and actionable. Their material or digital existence is important, but the monetary reality of a unit, balance, or claim depends on a wider relational configuration involving issuers, banks, payment systems, legal rules, accounting practices, expectations, exchange relations, taxation, contracts, and institutional trust.

Definition 4.125 (Monetary relational reality). *Monetary relational reality is the relational reality acquired by monetary units, balances, claims, prices, and payment capacities when they are stabilized within a generative-relational system sufficiently to coordinate valuation, settlement, obligation, exchange, accounting, or resource allocation and to exert causal effects on subsequent economic relations.*

This definition does not require every monetary form to share the same institu-

tional architecture. Commodity money, state-issued currency, bank deposits, community currencies, digital payment balances, and other monetary arrangements can differ in their stabilization mechanisms. The relevant inquiry concerns the relations through which a particular monetary object is recognized, transferred, accounted for, redeemed, taxed, enforced, trusted, or otherwise made consequential. Similar manifestations can conceal different relational structures, just as similar relational functions can be realized through different manifestations.

A banknote therefore offers only a partial view of monetary reality. The note can be materially destroyed without eliminating the currency. A counterfeit note can closely reproduce the manifestation while lacking the institutional relations that normally sustain the represented monetary claim. A digital balance can persist without any particular physical token corresponding to each unit. These examples reinforce the non-invertibility developed in Section 4.9: observation of a manifestation does not by itself reveal the relational structure that gives it monetary efficacy.

Monetary reality also demonstrates manifestational mediation. Prices, balances, ratings, exchange rates, accounting categories, and policy announcements do more than make an antecedent monetary system visible. Once circulated, they can alter expectations, spending, lending, investment, collateral relations, contractual decisions, and institutional responses. A manifested monetary quantity can therefore enter the generative field that helps produce later monetary and economic relations. The symbolic and relational registers remain distinct while becoming recursively coupled.

The familiar question of whether money is “real” is therefore decomposed into more precise questions. The physical substrate of a note or computer system raises one ontological issue. The stabilized monetary relation raises another. The displayed balance or printed denomination concerns manifestation and representation. The causal effects of monetary recognition concern relational reality. Normative questions concerning distribution, exclusion, monetary governance, or justice require further evaluation. The three-register architecture avoids forcing these questions into one binary judgment.

4.14.5 Organizational Reality

Organizations extend institutional reality through coordinated capacities that exceed the continuity of particular members. A university, corporation, ministry, association, or cooperative can retain identity across changes in personnel, offices, technologies, assets, and internal procedures. This continuity is neither purely symbolic nor reducible to a fixed material substrate. It depends on relations that reproduce membership, authority, decision procedures, ownership, records, routines, responsibilities, resources, and external recognition across time.

Definition 4.126 (Organizational relational reality). *Organizational relational reality* is the relational reality of an organized collective configuration that persists through stabilized relations of membership, authority, procedure, resource control, role differentiation, record, and external recognition, thereby acquiring capacities and causal effects that cannot be identified with any single current participant or manifestation.

The definition does not imply that organizations act independently of people, tech-

nologies, infrastructures, or environments. It identifies a level at which organized relations become sufficiently stable that actions can be attributed to an institutional configuration across turnover in the specific relata realizing it. A company can enter a contract before one manager leaves and remain bound after another arrives. A university can award a degree through procedures distributed across departments, registries, committees, information systems, and legal recognition. The organization persists because relevant relations are reproduced, not because an unchanging substance passes through time.

Organizational personality provides a particularly clear example of the distinction among registers. A legal person can be manifested through a registered name, identifier, seal, website, letterhead, bank account, or contractual signature. The legal representation enables coordinated action, but the organization's relational reality includes governance relations, employees, assets, liabilities, routines, authorizations, counterparties, and historical commitments that extend beyond any one symbol. The organization may therefore possess relational continuity even when particular representations change.

At the same time, organizational representations can reshape the organization. A new accounting category can redirect managerial attention. A performance metric can reorganize incentives. A revised organizational chart can redistribute authority if it is institutionally implemented. A mission statement can remain largely ceremonial or become relationally consequential depending on how it enters hiring, budgeting, evaluation, and decision procedures. The causal significance of representation is thus an empirical property of the generative-relational system rather than an automatic consequence of textual existence.

Organizational reality also clarifies why agency attribution requires care. An organization can possess operational capacities that justify treating it as an analytical subject for particular inquiries, while responsibility, experience, intention, and moral standing may remain distributed or contested. Section 4.12 therefore remains relevant: manifestational or operational unity does not establish identity across every dimension of subjecthood. The three-register architecture permits the analyst to study organizational action without assuming that organizational and human subjectivity are equivalent.

4.14.6 Causal Reality of Social Constructions

The examples above converge on a broader ontological point. A social construction can be causally real because construction refers to a mode of generation rather than to an absence of reality. Once a category, status, institution, or representation becomes embedded in stable relations, it can constrain and enable action, redirect resources, allocate rights, organize expectations, change bodies and environments, reproduce inequality, create capabilities, or alter the conditions under which later relations emerge. The historical contingency of its origin does not cancel these effects.

Claim 4.127 (Causal reality of social construction). *A socially constructed object can possess relational reality when the generative-relational system that sustains it gives the object sufficient persistence, stability, and causal efficacy. Social construction therefore describes a possible history and mechanism of emergence rather than a criterion*

of unreality.

This claim does not imply that every social construction becomes relationally real. A proposed category that fails to circulate, an unenforced rule that leaves practice unchanged, an ephemeral rumor, or an institutional reform that remains purely nominal may lack sufficient persistence or causal efficacy. Nor does it imply that the same constructed object has equal reality across systems. A credential can open access in one institutional field and be ignored in another. A currency can be effective within one payment network and practically unusable outside it. Relational reality remains domain-sensitive and historically situated.

The causal effects of social constructions can also outlive the representations through which they were initially generated. A historical classification may disappear from official vocabulary while its accumulated effects remain embedded in property distributions, organizational routines, infrastructures, reputations, or spatial arrangements. Conversely, a representation can survive after the relations that once made it effective have weakened. This temporal divergence between manifestation and relational efficacy is another reason to resist identifying institutional reality with visible symbols alone.

Institutional reality can further alter its own conditions of reproduction. Once a legal category, currency, organization, or administrative classification becomes established, later actors encounter a relational field already structured by it. They invest, learn, comply, resist, innovate, migrate, or organize around the inherited configuration. Their responses can stabilize the institution, transform it, produce new representations, or generate alternative relations. Institutional reality is therefore historical not only because it has an origin, but because each period of reproduction changes the conditions inherited by the next.

This recursive temporality is especially important for GRE. Economic analysis frequently treats institutions as background constraints or parameters. The generative-relational view instead permits institutions themselves to be investigated as historically produced realities that subsequently participate in producing economic subjects, opportunities, values, incentives, resource distributions, and future institutional forms. A property regime, monetary system, labor classification, welfare category, organizational structure, or accounting convention does not merely constrain choices made within an otherwise unchanged field. It can help generate the field within which later choices, relations, and forms of value become possible.

Remark 4.128 (Reality, legitimacy, and truth). The relational reality of an institution establishes neither its normative legitimacy nor the truth of the representations through which it operates. Causal efficacy, epistemic adequacy, legal validity, social recognition, and normative justification remain distinct dimensions that may interact without collapsing into one another.

The remark preserves a central discipline of the three-register architecture. A discriminatory classification may be institutionally real and normatively objectionable. A mistaken risk score may be epistemically poor and causally powerful. An informal norm may be socially effective without legal validity. A legal entitlement may be valid yet weakly realized in practice. A public representation may accurately describe an institution yet have little causal influence. Analytical precision requires these possi-

bilities to remain separately visible.

The ontological significance of social facts and institutions can therefore be stated without metaphysical inflation. Their reality is not diminished by the fact that they emerge historically through relations, and their relational reality does not compel a claim that they exist independently of those relations. They occupy a domain in which stabilized relations acquire causal efficacy, manifestations become operative components of reproduction, and historically generated objects can confront later actors with substantial practical force. This account prepares the next section's turn from relatively stabilized institutional objects toward process, event, and becoming, where persistence itself must be situated within continuing transformation.

4.15 Process, Event, and Becoming

The preceding sections have developed relational reality largely through objects, subjects, values, institutions, manifestations, and representations that achieve sufficient persistence and stability to become causally efficacious. The role of the present section is to clarify the temporal form of such reality. Its objective is to show how GR can analyze processes, events, and becoming without treating persistence as immobility, without reducing events to instantaneous points, and without turning process language into a final metaphysics. The discussion proceeds through five steps. It first specifies processual reality, then develops an account of events, examines stability within becoming, defines relational transformation, and finally considers how historically generated changes can require new ontological categories within an inquiry. The method remains consistent with the chapter's ontological suspension: temporal and processual descriptions are treated as analytically serious claims about organization and causal efficacy while stronger claims about the ultimate temporal constitution of reality remain open unless the inquiry independently warrants them.

The central difficulty is that relational reality requires persistence and stability, while generative-relational systems are continually changing. These conditions are compatible once stability is understood dynamically. A monetary system persists while transactions, balances, technologies, regulations, and participants change. An organization persists while personnel, routines, assets, and strategies are revised. A sense of self can persist while memory, embodiment, social recognition, and self-description develop. The reality of such configurations lies partly in their capacity to reproduce an organized pattern through change. Processual analysis therefore does not displace the criterion of relational reality developed in Section 4.5; it clarifies one way in which that criterion can be satisfied across time. (Whitehead 1929; Mead 1932; Davidson 2001; Sewell 1996)

Definition 4.129 (Processual reality). *Processual reality* denotes the relational reality of an organized pattern of change whose temporally extended configuration persists with sufficient stability and causal efficacy to participate in the subsequent development of the generative-relational system in which it occurs.

Definition 4.129 makes the process itself, rather than any single temporal slice, the relevant analytical object. A process can include changing components, variable relations, discontinuities, and internal differentiation while remaining identifiable at the

scale of inquiry. Its persistence therefore concerns continuity of organization, trajectory, or generative function rather than exact repetition of state. The definition also remains scale-sensitive. What appears as a relatively stable object at one temporal resolution can appear as an active process at another. A firm can be represented as an institutional entity in one model and as a changing configuration of contracts, routines, capacities, expectations, and material flows in another. Both descriptions can be analytically useful when their scale and purpose are explicit.

4.15.1 Processual Reality

Processual reality becomes important whenever the causal significance of an object depends on the organized continuation of change. A market, learning trajectory, supply chain, technological regime, administrative procedure, or relation of trust may be poorly described by a sequence of independent states because later states depend on how earlier configurations were generated. Historical order, memory, accumulated commitments, adaptation, and feedback can therefore be constitutive of the object under study. The process is relationally real when this temporally extended organization persists sufficiently to shape what can occur next.

The distinction between state and process is analytical rather than absolute. A state description compresses a configuration at a selected temporal resolution. A process description retains some of the transformations through which configurations succeed one another. The appropriate representation depends on the research question. A bank balance at a particular time may be adequately represented as a state for one inquiry, while the sequence of transfers, authorizations, interest accruals, obligations, and institutional validations through which that balance was produced may be indispensable for another. GR therefore treats temporal compression as a methodological choice whose omitted relations should remain visible when they matter to explanation.

Processual reality also clarifies why persistence need not mean substrate identity. Some processes persist through turnover in the entities that realize them. A payment network can continue while servers, software versions, account holders, and organizational personnel change. A legal procedure can retain recognizable continuity while statutes, interpretive conventions, and administrative technologies are revised. A language practice can persist through generations of speakers. In such cases, continuity is carried by a sufficiently stable organization of relations and transformations rather than by an unchanging set of relata.

Claim 4.130 (Dynamic persistence). *A relational reality can persist through substantial change in its constituent states, relata, and manifestations when the organization of transformation remains sufficiently stable to preserve causal continuity at the scale relevant to the inquiry.*

Claim 4.130 is intentionally scale-relative. The same history can support continuity at one scale and rupture at another. A corporation may remain legally continuous through a major restructuring even though employees experience the restructuring as the end of an earlier organizational form. A monetary regime may preserve a currency name while changing the institutions, convertibility rules, settlement technologies, or policy mechanisms that sustain it. The analytical task is therefore to specify

which organization persists, for whom or for what system, across which interval, and with what causal consequences.

Processual reality also accommodates rhythms, cycles, and recurrent dynamics without assuming exact periodicity. Many economic processes are reproduced through recurring but variable practices: payroll, taxation, budgeting, maintenance, certification, price revision, inventory replenishment, and contractual renewal. Their relational reality can depend on repeated coordination whose timing and content change within bounded ranges. Stability here is statistical, institutional, or organizational rather than identical repetition. The relevant process persists because enough of its generative structure is reproduced to maintain causal continuity.

Some processes possess several interacting timescales. Regulatory change can occur slowly relative to trading, infrastructural replacement slowly relative to software updates, and demographic change slowly relative to organizational decisions. GR can represent these differences without assuming that one timescale is ontologically fundamental. The important question is how processes operating at different rates constrain, amplify, or reorganize one another. A slowly changing legal framework can stabilize fast transactions; repeated fast transactions can accumulate pressures that eventually transform the legal framework. Temporal structure is therefore relational and potentially recursive.

The processual view has an additional consequence for manifestation. A process may be relationally real before it is manifested as a coherent process to any subject. Individual observations can appear disconnected until a historical pattern becomes recognizable through later comparison, measurement, narration, or modeling. The manifestation of the process can then feed back into it. Once inflation is represented through an index, organizational decline through a performance metric, or ecological change through a monitoring series, subjects may alter behavior in response to the manifestation. The process existed relationally before its explicit representation in such cases, while its later manifestation becomes one of the relations shaping its subsequent development.

4.15.2 Events

Processes unfold through temporally differentiated changes, some of which acquire a distinctive status as events. An event in GR need not be instantaneous, spectacular, or wholly discontinuous. It is better understood as a temporally bounded reconfiguration whose occurrence makes a consequential difference to the development of a relational system. The event can be brief or extended, visible or initially unrecognized, intentionally produced or emergent from distributed interactions. Its analytical importance lies in the reorganization it introduces into subsequent possibilities.

Definition 4.131 (Relational event). A *relational event* is a temporally delimited reconfiguration within a generative-relational system that produces a causally consequential difference in the organization, constraints, capacities, or trajectories of subsequent relations.

Definition 4.131 distinguishes an event from arbitrary temporal segmentation. Any interval can be bounded by an analyst, but eventhood requires a difference that matters to the system under study. Signing a contract can become an event because obli-

gations, permissions, expectations, and future options are reorganized. A payment default can become an event because it changes credit relations, legal possibilities, reputational classifications, and later financing. A policy announcement can become an event when actors reorganize behavior around it. A software release can become an event if it changes operational capacities or participation conditions. The boundary of the event is therefore partly analytical and partly constrained by the causal structure of the system.

Eventhood can be retrospective. A change that initially appears minor can later be recognized as consequential because subsequent relations amplify it. A local technical modification may become a turning point after complementary technologies, regulations, practices, or expectations align around it. Conversely, an occurrence widely represented as momentous can have limited relational consequences if the surrounding system absorbs it without substantial reorganization. Manifestational salience and relational eventhood can therefore diverge.

This divergence matters for historical analysis. Public discourse often privileges events that are visible, narratively convenient, institutionally recorded, or symbolically dramatic. GR allows an analyst to distinguish the manifestational prominence of an occurrence from its relational significance. A highly publicized announcement may remain weakly causal, while a quiet administrative change, altered database schema, revised eligibility rule, or gradual infrastructural threshold may reorganize access and behavior across a large system. Event analysis therefore requires attention to causal consequences as well as to representation.

Events can also emerge without a single author. A financial panic, collective shift in expectations, congestion cascade, labor shortage, reputational collapse, or technological convention may become event-like through distributed interactions. The event is then generated by a relational configuration whose participants may possess only partial knowledge of the overall process. This distributed generation is compatible with the relational-reality criterion because causal efficacy can be system-level even when no actor intends or fully represents the resulting reconfiguration.

Remark 4.132 (Event and manifestation). A relational event can precede its explicit recognition, while a manifested occurrence can acquire additional event-like consequences through the responses its manifestation generates. Eventhood and manifestation therefore interact recursively without being identical.

Remark 4.132 extends the graph architecture developed earlier in the chapter. A threshold crossing may reorganize a system before subjects recognize it. Later measurement, naming, announcement, or narration can then become a second source of causal change. The event's relational reality and its manifestational history should therefore be reconstructed separately when both matter.

4.15.3 Stability within Becoming

The concept of becoming names the continuing production of difference through time. GR uses the term cautiously. It does not require a metaphysical doctrine in which process is declared more fundamental than entity, nor does it imply that every configuration changes at the same rate or in the same manner. Becoming identifies the analytical fact that generative-relational systems can produce new states, relations,

capacities, constraints, meanings, manifestations, and objects through their histories. Stability must therefore be understood within this ongoing production of difference.

A stable relational reality is often maintained through active reproduction. Money remains effective because infrastructures settle payments, institutions enforce claims, subjects continue to price and transact, records are maintained, and expectations are reproduced. An organization remains continuous because roles are reassigned, records updated, resources replenished, decisions implemented, and membership boundaries reproduced. Trust can persist because interaction repeatedly confirms or repairs expectations. Stability is therefore frequently achieved through work, adjustment, feedback, repair, and renewal.

This point motivates a distinction between static and dynamic stability. Static stability would describe limited change around a configuration under specified conditions. Dynamic stability describes persistence of organization through continuing transformation. The distinction is analytical; many systems exhibit mixtures of both. A legal rule may remain textually unchanged while its interpretation evolves. A platform may continually update software while preserving a recognizable participation architecture. A household may maintain a pattern of care through changing tasks, needs, and members. The relevant form of stability depends on what must remain sufficiently organized for causal continuity to persist.

Claim 4.133 (Stability within becoming). *Stability in a generative-relational system can consist in the continued reproduction of an organized relational pattern through change. Persistence therefore remains compatible with becoming when the transformations themselves sustain sufficient continuity of structure, constraint, capacity, or causal orientation.*

Claim 4.133 prevents a false choice between stable reality and historical transformation. The same relational reality can be persistent and continually reproduced. At the same time, reproduction is never guaranteed. Small modifications can accumulate, external conditions can change, internal contradictions can intensify, new relations can enter the system, and earlier stabilizing mechanisms can lose efficacy. Dynamic stability can therefore weaken, recover, or pass into transformation.

The temporal extent of stability also matters. A configuration can be stable over a short interval and unstable over a longer one. A speculative price regime may persist long enough to organize investment decisions even if it later collapses. An emergency institution can be relationally real during a crisis despite its intended temporariness. A temporary identity category can shape access and conduct while it remains institutionally operative. Relational reality therefore does not require indefinite duration. It requires sufficient persistence relative to the causal processes under analysis.

Becoming also complicates identity. If a relational reality changes continuously, the question of whether it remains the same object cannot always be answered by a single invariant criterion. Legal continuity, organizational continuity, functional continuity, experiential continuity, material continuity, and historical continuity may diverge. GR therefore treats identity through change as an inquiry-relative reconstruction. The analyst should specify which continuities matter to the question and which transformations count as preserving or ending the object under study.

This approach is especially relevant to subjectivity. The continuing experience of

“I” can be relationally real even though embodiment, memory, social position, commitments, and self-description change. Its reality need not depend on an immutable ontological core. At the same time, ontological suspension prevents the relational account from proving that no such core exists. GR can therefore analyze continuity of self through changing relations while leaving stronger metaphysical conclusions open.

4.15.4 Relational Transformation

Relational transformation occurs when change modifies the organization through which a relational reality is generated, stabilized, or made causally effective. The concept therefore concerns more than variation in the values of fixed variables. A relation may change in strength, direction, meaning, institutional form, temporal ordering, or scale; relations among relations may be reorganized; new relata may enter; earlier ones may disappear; and the mechanisms through which manifestations feed back into the system may themselves change.

Definition 4.134 (Relational transformation). *Relational transformation* denotes a historically situated change in the organization of a generative-relational system that modifies the conditions through which relational realities emerge, persist, interact, manifest, or exercise causal efficacy.

Definition 4.134 distinguishes transformation from ordinary state change. A price change within an otherwise stable market arrangement may alter a state without transforming the relational architecture. A change in settlement rules, ownership structure, access rights, information infrastructure, or category system can transform the relations through which prices are generated and acted upon. The distinction remains scale-dependent, but it directs attention toward changes in generative conditions rather than toward magnitude alone.

Relational transformation can be cumulative. Repeated small changes can modify expectations, routines, infrastructures, and classifications until a system operates under materially different generative conditions. The absence of a single dramatic event does not preclude transformation. Gradual digitization of payment systems, for example, can change the relations among money, identity, privacy, institutions, and access through accumulated infrastructural revisions. Similarly, gradual adoption of algorithmic evaluation can reorganize employment, credit, insurance, education, or welfare relations even when no single deployment constitutes the entire transformation.

Transformation can also be path-dependent. Earlier relations shape which later changes are feasible, costly, intelligible, or institutionally legitimate. Historical residues can remain operative after the representations associated with them disappear. Infrastructures, legal precedents, accumulated skills, property distributions, organizational routines, and categories can constrain later possibilities. The transformed system therefore carries aspects of its history into new configurations even when its manifestational vocabulary changes substantially.

The recursive architecture of Sections 4.10 and 4.11 becomes especially important here. Representations can become mechanisms of transformation when classifications, measurements, models, narratives, or interfaces alter behavior and institu-

tional response. A newly introduced performance score can reorganize organizational incentives; a legal category can create new rights and duties; a public risk classification can alter investment; an economic indicator can change policy behavior. Once such representations become causally embedded, transformation occurs partly through manifestational mediation.

Relational transformation can also dissolve relational realities. A currency can lose the relations that sustain its practical efficacy. A legal office can be abolished. A social norm can weaken until it ceases to organize conduct. A trust relation can collapse. Dissolution does not imply that the earlier reality was illusory. A relational reality can have been historically real, causally consequential, and later absent. The ontology developed here is therefore temporally indexed: reality within a generative-relational system can emerge, persist, transform, and cease.

Remark 4.135 (Historical reality and dissolution). The later disappearance of a relational reality does not retroactively remove its earlier causal reality. Historical existence and present persistence are distinct questions.

This remark matters for economic history and institutional analysis. Defunct currencies, dissolved corporations, abolished legal statuses, vanished markets, and obsolete technologies can remain causally significant through accumulated effects, inherited infrastructures, debts, memories, distributions, and institutional residues. Their present relational reality may have ended while consequences generated during their persistence continue to structure later systems.

4.15.5 Emergence of New Ontological Categories

Relational transformation can eventually strain the categories through which an inquiry describes what exists. Existing categories may continue to capture some features of a changing system while becoming inadequate for others. New kinds of institutional object, operational subject, property relation, value form, technical agency, or manifestation can arise through combinations of relations that earlier classifications did not anticipate. The emergence of such objects creates an ontological problem for analysis: the investigator must decide whether an existing category can be extended, whether several categories should be distinguished, or whether a new category is warranted.

The phrase *new ontological category* is used here with ontological restraint. It does not mean that the analyst has discovered a new fundamental constituent of reality. It means that the system presents a sufficiently stable and causally consequential form whose existing analytical classification is inadequate for the inquiry. Category formation is therefore a response to emergent reality under specified evidential and methodological conditions. The category can later be revised, divided, merged, or abandoned as the object and the inquiry develop.

Definition 4.136 (Emergent ontological category). An *emergent ontological category* is an analytically warranted category introduced when a historically generated configuration displays sufficiently distinctive persistence, organization, causal capacities, or modes of manifestation that the categories previously available to the inquiry no longer provide an adequate account of the relevant reality.

Definition 4.136 links ontology to historical inquiry without reducing reality to naming. The category is analytically introduced, but the warrant for introducing it is constrained by the properties and effects of the configuration being studied. A new term alone does not create a new ontological category. Conversely, a novel relational reality may begin to operate before a vocabulary for it exists. This preserves the distinction between relational reality and symbolization established earlier in the chapter.

Contemporary artificial systems offer one illustration. The emergence of systems capable of sustained interaction, autonomous-seeming task execution, representation, coordination, and intervention can strain categories built around tools, agents, organizations, and human subjects. GRE need not settle this difficulty by declaring such systems persons or by reducing them to ordinary instruments. Section 4.12 introduced artificial operational subjecthood precisely as a differentiated analytical category whose scope is narrower than human subjecthood. The category becomes useful when it tracks stable capacities and causal roles that existing descriptions obscure, while ontological and normative questions beyond that scope remain suspended.

Economic institutions can generate similar categorical pressure. Digital assets, platform-mediated labor relations, algorithmically administered claims, hybrid public-private infrastructures, and machine-mediated forms of coordination can combine properties previously distributed across several categories. Whether they warrant genuinely new analytical categories depends on the structure of the inquiry. GRE therefore treats category formation as a disciplined response to emergent configurations rather than as conceptual novelty for its own sake.

The same discipline applies to historical categories. A category that once tracked a stable relational reality can lose adequacy as the system changes. Retaining the old name may conceal transformation if its referent has acquired substantially different generative conditions and causal capacities. Conversely, rapid renaming can exaggerate novelty where relational continuity remains strong. Category revision therefore requires comparison of historical organization, persistence, causal efficacy, manifestation, and transformation rather than lexical difference alone.

Claim 4.137 (Category emergence under ontological suspension). *The emergence of a new analytical ontological category can be warranted by historically generated differences in organization and causal efficacy without requiring a claim that the category identifies a new fundamental kind of being.*

Claim 4.137 connects the present section back to ontological suspension. GR can revise what it counts as an object, process, event, subject, institution, or form of value when reality under inquiry changes or when earlier descriptions prove inadequate. This flexibility is compatible with ontological seriousness because revision responds to evidence about persistence, difference, causal efficacy, and generative structure. It is also compatible with metaphysical restraint because local analytical differentiation does not automatically become a universal ontology.

Process, event, becoming, transformation, and category emergence therefore provide a temporal extension of the chapter's three-register architecture. Relational realities can persist through organized change; events can redirect trajectories before or after they become manifest; manifestations and representations can become causal

components of transformation; and new configurations can eventually require revised categories of analysis. Stability and becoming are thus mutually informative rather than mutually exclusive. The result prepares the transition to Section 105, which introduces a category-theoretic view as one possible formal language for describing objects, morphisms, mappings, and equivalences across relational and manifestational domains while preserving the ontological restraint established throughout the chapter.

4.16 A Category-Theoretic View

The preceding sections have developed an ontology in which relational realities emerge, persist, manifest, become represented, and transform through historically structured systems. The role of the present section is to introduce category theory as one possible formal language for describing selected aspects of this architecture. Its objective is not to claim that relational reality is fundamentally categorical. Rather, category-theoretic language can help distinguish objects from mappings, preserve compositional structure, compare different descriptive domains, and make explicit when two paths of mediation are treated as equivalent or when they remain distinct. The section proceeds from objects and morphisms to categories of relational and manifestational reality, then considers functorial manifestation, loss and excess of information, equivalence, symbolic mediation, changing manifestation rules, multiple realization, and diagrammatic path dependence. Throughout, the formal vocabulary is treated as a methodological expedient whose appropriateness depends on whether the relations under study actually support the compositional assumptions imposed by the model.

Category theory is attractive for GR partly because it directs attention away from intrinsic properties alone and toward relations, mappings, transformations, and preservation across descriptions. Yet this attraction requires restraint. A category is a mathematically specified structure with objects, morphisms, identity morphisms, and associative composition. A generative-relational system need not present itself in this form without analytical construction. The investigator must decide what counts as an object, which transformations count as morphisms, when two morphisms are composable, and what form of equality or equivalence is appropriate. These decisions can illuminate structure, but they also select and compress reality. Category-theoretic analysis therefore belongs within the chapter's broader principle of open ontological commitment: it can support rigorous local claims without acquiring metaphysical authority beyond the structures it successfully represents. (Mac Lane 1998; Awodey 2010)

Definition 4.138 (Category-theoretic representation). A *category-theoretic representation* of a generative-relational domain is an analytically constructed category whose objects and morphisms represent selected configurations and admissible relations or transformations in that domain, with identity and composition specified in a manner appropriate to the inquiry.

Definition 4.138 emphasizes representation rather than identity. The category is a model of selected structure. It can preserve relations that matter to the inquiry while

leaving other features unrepresented. This makes category theory especially useful for the three-register architecture, where the same object can be described within different domains and where manifestation, representation, and transformation often involve mappings between domains rather than a single common metric space.

4.16.1 Objects, Morphisms, and Relational Description

A category consists, minimally, of objects and morphisms between them. For objects A , B , and C , morphisms can be written

$$f : A \rightarrow B, \quad g : B \rightarrow C, \quad (4.7)$$

Equation 4.7 introduces the basic source–target notation used in this section. A composite $g \circ f : A \rightarrow C$ is defined when the codomain of f matches the domain of g . Each object has an identity morphism, and composition is associative wherever it is defined. These requirements are simple, but their interpretation in a generative-relational analysis is not automatic.

An object can represent a stabilized relational configuration, an institutional state, a manifestation, a symbolic representation, a process considered at a selected scale, or another analytically bounded unit. A morphism can represent a transformation, translation, admissible relation, institutional operation, manifestation process, or structure-preserving map. The same empirical material can support more than one categorical representation because the choice of object and morphism depends on the analytical question.

For example, firms can be represented as objects and contractual transformations as morphisms in one model. In another model, contractual arrangements themselves can become objects, while legal or organizational transformations act as morphisms among them. A third representation can treat manifestations of those firms—financial reports, registrations, ratings, or platform profiles—as objects within a manifestational category. Category theory therefore does not decide in advance what the economically relevant object is. It disciplines relations among whatever objects the inquiry has justified selecting.

Remark 4.139 (Object selection). Categorical objecthood is an analytical role, not a declaration of fundamental ontological objecthood. The same relational reality can occupy different formal roles under different resolutions and research questions.

This point matters because objectification can conceal internal structure. Treating an organization as a single categorical object may be useful for studying interorganizational transformations while suppressing internal power relations, departments, subjects, and material infrastructures. Treating those internal elements as separate objects can recover some of that structure but may suppress other scales. The categorical model therefore inherits the boundary and scale problems already identified in GRE. Formal clarity does not remove the need to justify what has been made visible and what has been compressed.

4.16.2 Relations as Morphisms and Relations among Relations

The language of morphisms is naturally attractive to a relational framework, but GR should resist a simple identification between every relation and every morphism. In an ordinary category, morphisms possess specified domains and codomains and participate in associative composition. Many empirical relations are symmetric, partial, probabilistic, context-sensitive, historically contingent, or difficult to compose. Some can nevertheless be represented categorically by choosing an appropriate category; binary relations, for example, can themselves be treated as morphisms in a category whose composition is relational composition. In other cases, transformations rather than relations in the broad GR sense provide the more suitable morphisms.

The useful methodological question is therefore whether a selected relational domain supports a compositional representation. When it does, composition can expose how relations among relations generate higher-order consequences. If $f : A \rightarrow B$ and $g : B \rightarrow C$ represent two admissible transformations, then $g \circ f$ represents the result of applying them in sequence. The composite can be analytically important even when neither transformation alone determines the final configuration. Contract formation followed by registration, classification followed by allocation, or measurement followed by administrative decision can all be represented through composition when the domain and codomain assumptions are justified.

Relations among relations can also require higher-order structures. A change in the rule by which one morphism is interpreted, a comparison between two mappings, or an intervention on a manifestation rule is not simply another relation among the original objects. Category theory offers several ways to represent such higher-order structure, including functors and natural transformations. GR need not adopt the most elaborate formalism by default. The important insight is that the relation connecting two objects and the transformation of that relation belong to distinguishable analytical levels.

Claim 4.140 (Compositional warrant). *A categorical representation of a relational domain is warranted only to the extent that the selected objects, morphisms, identities, and compositions preserve structures relevant to the inquiry. Formal composability should not be inferred merely from the presence of empirical relations.*

This claim protects the method against decorative formalization. A category-theoretic diagram becomes informative when its compositions correspond to meaningful sequences or transformations in the domain. Where composition is ambiguous, history-dependent, or only partially defined, the model should represent that limitation rather than assume away the problem.

4.16.3 Categories of Relational Reality

The strict criterion of relational reality developed in Section 4.5 can be combined with category-theoretic representation without identifying the two. Let a category $\mathcal{R}$ represent a selected domain of relational realities. Its objects can stand for relational configurations that satisfy the relevant persistence, stability, and causal-efficacy conditions at the scale of inquiry. Its morphisms can represent transformations, admissible relations, or structure-preserving changes among those configurations.

The resulting category does not contain relational reality in its entirety. It contains only those distinctions that the model has made categorical. A monetary relation, institution, trust relation, self-model, or organization can possess internal histories and causal structures that are absent from the chosen object. The category becomes analytically useful when the omitted features are not required for the question being asked or when they are represented at another level.

A category of relational reality can also be local. Different subsystems can support different object types and transformation rules. A category appropriate to legal relations need not be identical to one used for technological systems, subjective continuity, or financial networks. The possibility of relating these categories through functors can be more informative than forcing all of them into a common metric or a single universal object space.

Definition 4.141 (Relational category). A *relational category* is a category constructed for an inquiry in which objects represent selected relationally real configurations and morphisms represent admissible relations or transformations whose composition preserves structures relevant to that inquiry.

The definition deliberately leaves open what counts as preservation. In one study, preservation can concern legal validity; in another, organizational continuity, causal accessibility, information flow, contractual structure, or some other relation. The category gains explanatory value from the explicitness of these choices rather than from conformity to a single ontology.

4.16.4 Categories of Manifestational Reality

Manifestational reality admits a parallel but distinct categorical treatment. Let $\mathcal{M}$ denote a category whose objects represent manifestations available under a specified manifestational regime and whose morphisms represent admissible transformations, translations, comparisons, or transitions among those manifestations. The objects of $\mathcal{M}$ can include observations, measurements, classifications, reports, symbolic forms, perceptual states, institutional records, or other manifestations selected by the inquiry.

A manifestational category is not merely a category of representations. Section 4.6 distinguished manifestation from symbolization, and Section 4.10 distinguished symbolic representation from manifestational reality more broadly. A bodily response, an observed pattern, a measurement trace, a legal inscription, and a linguistic description can therefore belong to different manifestational subdomains while all remaining manifestations in the broader GR sense.

The categorical distinction between $\mathcal{R}$ and $\mathcal{M}$ provides a disciplined way to express a point developed throughout the chapter: relational and manifestational realities need not share the same objects or the same structure. A relational configuration can support multiple manifestations, and several relational configurations can become manifest in indistinguishable ways under a limited regime. Instead of treating these mismatches as defects in a single space, category-theoretic analysis can represent them as properties of mappings between distinct categories.

Remark 4.142 (Heterogeneous domains). The usefulness of separate relational and

manifestational categories lies partly in permitting heterogeneity. A mapping between domains can be studied without presuming that the domains share a common metric, coordinate system, or ontology.

This is especially important for the non-necessary comparability of relational realities. Two relationally real objects can belong to categories whose internal morphisms, equivalences, and invariants differ substantially. Comparison then requires an explicit functor, correspondence, or other mapping rather than an assumed common scale.

4.16.5 Manifestation through Functorial Mapping

A functor provides one formal model of manifestation across categorical domains. Suppose $\mathcal{R}$ is a relational category and $\mathcal{M}$ a manifestational category. A functor

$$F : \mathcal{R} \longrightarrow \mathcal{M} \quad (4.8)$$

Equation 4.8 names the functor that will serve as the local manifestation model. It assigns each modeled object A in $\mathcal{R}$ an object $F(A)$ in $\mathcal{M}$ and each modeled morphism $f : A \rightarrow B$ a morphism $F(f) : F(A) \rightarrow F(B)$ while preserving identity morphisms and composition.

The preservation of composition is the substantive assumption. If a relational transformation f followed by g is represented by $g \circ f$, functoriality requires the manifestation of that composition to agree with the corresponding composition of manifested morphisms:

$$F(g \circ f) = F(g) \circ F(f). \quad (4.9)$$

Equation 4.9 can be analytically powerful because it identifies a condition under which relational composition is preserved through manifestation. It can also fail. A manifestational regime may erase sequence, reorder operations, introduce threshold effects, or depend on contextual information absent from the source category. Functoriality should therefore be treated as a model property to be investigated rather than a universal fact about manifestation.

Definition 4.143 (Functorial manifestation). *Functorial manifestation* is a model of manifestation in which selected relational objects and morphisms are mapped into a manifestational category while identities and relevant compositional structure are preserved.

This definition does not require manifestation in general to be functorial. It identifies a useful special case. Administrative systems often aspire to such preservation: a legally valid sequence of operations is expected to produce a corresponding sequence of records; an accounting transformation is expected to be reflected in a consistent ledger transformation; a data pipeline is expected to preserve specified relations among inputs and outputs. Failures of preservation can then become analytically visible as breaks between relational process and manifestation.

4.16.6 Non-Faithful Manifestation and Loss of Relational Information

A functor is faithful when it maps distinct morphisms between the same pair of source objects to distinct morphisms in the target. Non-faithfulness therefore provides one precise formal model of relational distinctions that become indistinguishable in manifestation. Suppose $f, g : A \rightarrow B$ are distinct morphisms in $\mathcal{R}$ but

$$F(f) = F(g). \quad (4.10)$$

Equation 4.10 states the relevant collapse: the manifestational category fails, under this model, to preserve the distinction between the two relational transformations.

This formal structure corresponds closely to the relational remainder developed in Section 4.9. Different causal histories, institutional pathways, or social relations can produce the same recorded outcome. Two workers can receive the same wage through substantially different relations of bargaining power and security. Two firms can receive the same credit classification despite different dependency structures. Two subjects can produce the same interface response while differing in histories, capacities, or internal organization. In each case, a manifestational system can preserve an outcome while collapsing distinctions among the morphisms that generated it.

Claim 4.144 (Non-faithfulness and relational loss). *When a manifestation functor is non-faithful, distinct modeled relational transformations can become manifestationally indistinguishable. The resulting loss of distinction concerns the selected manifestation regime and does not imply that the source transformations were relationally identical.*

Faithfulness is only one form of information preservation. A functor can distinguish morphisms while still mapping distinct objects to the same object, depending on the category and model. Conversely, object-level differences can remain visible while important internal relations are lost. GR therefore uses faithfulness as a precise local concept rather than as a complete measure of manifestational adequacy.

The category-theoretic language also clarifies why improving measurement can require more than adding variables. If a regime systematically identifies distinct relational pathways, richer data may leave the collapse intact unless the mapping itself is redesigned. The epistemic problem can therefore reside in the structure of manifestation rather than only in the quantity of manifested information.

4.16.7 Non-Full Manifestation and Unmanifested Relational Possibilities

Fullness requires a different interpretation. A functor $F : \mathcal{R} \rightarrow \mathcal{M}$ is full when every morphism in $\mathcal{M}$ between $F(A)$ and $F(B)$ is the image of at least one morphism in $\mathcal{R}$ between A and B . A non-full functor therefore indicates that the target category contains morphisms between manifested objects that are not induced by morphisms represented in the source category.

This point should be distinguished carefully from an *unmanifested source possibility*. In a strict functor, every source morphism receives an image. Non-fullness therefore does not, by itself, mean that a relational morphism exists but fails to man-

ifest. Rather, it means that the manifestational domain admits transformations or relations that are not accounted for by the modeled source relations. These can arise from measurement procedures, symbolic recombination, institutional processing, artifacts, interpretive operations, or relations introduced by the manifestation system itself.

To represent relational possibilities that are not manifested at all, the model must be modified. One option is to define the manifestation functor only on a subcategory $\mathcal{R}_{\text{vis}} \subseteq \mathcal{R}$ of relational configurations and morphisms that are accessible under the relevant regime. Relational possibilities in $\mathcal{R}$ outside that subcategory remain unmanifested relative to the regime. Another option is to use a partial, lax, stochastic, or correspondence-based structure where strict functorial assignment is inappropriate.

Remark 4.145 (Fullness and invisibility). Non-fullness and relational invisibility are different structural problems. Non-fullness concerns target morphisms lacking source antecedents in the model; unmanifested relational possibilities concern source structure excluded from, undefined under, or inaccessible to the manifestation mapping.

This distinction is methodologically useful because the two problems call for different interventions. If relevant source relations are invisible, the manifestational regime may require expanded access or a different mapping. If the target generates relations not grounded in the source model, the analyst must study the generative contribution of measurement, classification, symbolic processing, or institutional mediation itself. In GR terms, manifestation can add structure rather than merely transmit it.

4.16.8 Isomorphism, Equivalence, and Manifestational Indistinguishability

Category theory offers several notions of sameness, and GR benefits from keeping them distinct. Two objects A and B in a category are isomorphic when there are morphisms between them whose compositions yield the respective identity morphisms. Isomorphism therefore captures sameness of categorical structure relative to the category, not numerical identity. Two categories can be equivalent even when they are not isomorphic as categories, provided they preserve the relevant structure up to natural isomorphism.

Manifestational indistinguishability is weaker. Section 4.9 defined manifestational equivalence locally to a regime. In categorical terms, two relational objects can satisfy $F(A) = F(B)$ or otherwise become isomorphic or observationally equivalent in $\mathcal{M}$ without being isomorphic in $\mathcal{R}$. Likewise, two source categories can generate equivalent manifestations while differing in relational organization.

Claim 4.146 (Manifestational equivalence is not relational isomorphism). *Equivalence or isomorphism in a manifestational category does not, without additional assumptions on the manifestation mapping, establish isomorphism, equivalence, or identity in the relational category.*

The additional assumptions can be strong. A fully faithful functor preserves and reflects important morphism structure between modeled objects, but even full faith-

fulness does not automatically settle every ontological question outside the representation. GR therefore treats categorical equivalence as always relative to a selected formal domain.

This discipline is particularly important for economic equivalence. Equal price, equal legal category, equal score, equal statistical profile, or equal platform status can create manifestational equivalence under a practical regime. Such equivalence can have real effects because actors and institutions respond to it. Yet the equality need not erase differences in histories, dependencies, capacities, vulnerabilities, or generative conditions. Category theory provides precise language for separating equivalence under a mapping from identity of what is mapped.

4.16.9 Functors between Symbolic and Relational Descriptions

Symbolic regimes can also be modeled categorically. Let $\mathcal{S}$ denote a category of selected symbolic objects and transformations: classifications, legal codes, identifiers, prices, scores, textual descriptions, or other symbolic forms. An encoding or representation functor can map selected relational configurations into $\mathcal{S}$, while an interpretation or institutional enactment mapping can connect symbolic objects back to relational consequences.

The two directions should not be assumed to be inverses. A symbolic representation can compress several relational realities into one category, and an interpreted symbol can generate relations that were not present before the symbolization. A credit score, legal status, diagnostic category, occupational code, or monetary denomination can therefore be both a representation of prior relations and a causal input into later relations. The categorical picture is consequently recursive rather than merely representational.

One can schematically distinguish a relational category $\mathcal{R}$, symbolic category $\mathcal{S}$, and later relational category $\mathcal{R}'$ with mappings

$$\mathcal{R} \xrightarrow{E} \mathcal{S} \xrightarrow{I} \mathcal{R}', \quad (4.11)$$

Equation 4.11 separates the two mappings explicitly: E represents a selected encoding or symbolization and I represents the institutional, interpretive, or practical mediation through which symbols enter later relations. The notation does not imply that $\mathcal{R}'$ is determined by symbols alone. Other histories, material conditions, subjects, and constraints can contribute to its generation.

The value of the categorical representation lies in its ability to make preservation questions explicit. Which distinctions survive E ? Which symbolic morphisms become causally operative under I ? Which compositions commute, and which depend on the order or path of mediation? These are empirical and theoretical questions, not consequences of the notation itself.

4.16.10 Natural Transformations and Changing Manifestation Rules

The chapter has repeatedly emphasized that manifestation mappings can change historically. Category theory offers a formal language for comparing two functors that

map the same source category into the same target category. If $F, G : \mathcal{R} \rightarrow \mathcal{M}$ represent two manifestation regimes, a natural transformation $\eta : F \Rightarrow G$ assigns to each source object A a morphism

$$\eta_A : F(A) \rightarrow G(A) \quad (4.12)$$

in a way compatible with the morphisms of $\mathcal{R}$. Equation 4.12 therefore represents the objectwise component of a change between two manifestation regimes rather than an isolated conversion.

In applications, F and G can represent different measurement systems, classification regimes, institutional reporting rules, interfaces, or symbolic conventions. The components η_A then represent transformations between the manifestations produced under the two regimes. Naturality is a strong structural condition: it requires these transformations to cohere with the modeled relations among source objects rather than being arbitrary object-by-object conversions.

Definition 4.147 (Coherent change of manifestation regime). A *coherent change of manifestation regime* is a model in which two manifestation functors are connected by transformations that preserve the relational compatibility conditions represented by a natural transformation.

Not every historical change satisfies this condition. A new accounting standard, legal classification, algorithmic interface, or statistical methodology can reorganize categories so substantially that no natural transformation between old and new regimes provides an adequate model. The failure of naturality can itself be informative: it signals that translation between regimes depends on path, context, exception, or reinterpretation not captured by a uniform structural mapping.

This makes natural transformations useful for analyzing comparability over time. A time series constructed under changing measurement systems may appear continuous while the transformation between regimes fails to preserve the relevant relational structure. Conversely, a carefully calibrated change can preserve enough structure to support meaningful historical comparison. Category theory therefore turns the vague question of whether two regimes are “comparable” into a more explicit inquiry about what transformations preserve which relations.

4.16.11 Multiple Realization and Functorial Equivalence

Multiple realization describes situations in which different underlying relational configurations realize the same or sufficiently equivalent higher-level organization, function, or manifestation. The three-register framework already permits this possibility because manifestational equivalence does not imply relational identity. Category theory offers several ways to formalize the relation without collapsing the realizing systems into one object.

Distinct objects A and B in $\mathcal{R}$ can be mapped to the same manifestational object or to isomorphic objects in $\mathcal{M}$. More strongly, distinct subcategories of relational realization can be mapped into equivalent manifestational structures. This can model cases in which different technologies, organizational arrangements, biological substrates, cultural practices, or computational architectures support similar manifested capacities.

Multiple realization should be distinguished from arbitrary substitution. Two realizations count as equivalent only relative to specified functions, relations, or manifestations. A payment service implemented through different technical infrastructures can appear equivalent to users for a selected transaction set while differing in resilience, surveillance, governance, energy use, or legal dependency. An artificial system and a human subject can produce locally equivalent outputs while differing profoundly in relational histories and forms of subjecthood. The relevant equivalence therefore belongs to the specified category and mapping, not to the entities without qualification.

Claim 4.148 (Locality of functorial equivalence). *Functorial equivalence among realizations is always relative to the categories, morphisms, and invariants selected by the inquiry. It does not entail equivalence with respect to omitted relational, causal, normative, experiential, or ontological structure.*

This claim preserves the book's resistance to premature equivalence. A formal equivalence can be powerful precisely because its scope is explicit. It identifies what is preserved while leaving open what differs.

4.16.12 Commutative and Non-Commutative Diagrams of Representation

One of category theory's most useful habits is to ask whether different compositional paths yield the same result. A commutative diagram represents paths whose composite morphisms are equal within the model. In a representation problem, commutativity can express consistency between direct and mediated descriptions. If a relational configuration can be manifested directly or first symbolized and then translated into a manifestation, commutativity states that the two modeled paths agree.

Such agreement should not be presumed. Economic and institutional systems are often path-dependent. Classification before valuation can yield a different result from valuation before classification. Legal recognition followed by market participation can generate different relations from market participation followed by later recognition. A symbolic description produced before an intervention can affect the intervention itself, while a description produced afterward has a different causal role. In these cases the relevant diagram is non-commutative: the order and path of mediation matter.

Definition 4.149 (Representational path dependence). *Representational path dependence* occurs when two admissible compositional paths connecting corresponding source and target configurations produce distinguishable outcomes within the modeled domain.

Non-commutativity is particularly important for the recursive architecture of Section 4.11. Once a manifestation or representation becomes causally operative, changing the order of manifestation, classification, recognition, or intervention can change the later relational configuration. Diagrammatic non-commutativity therefore provides one formal language for historical mediation.

At the same time, commutativity can be a normative or engineering objective rather than an empirical fact. An administrative system can be designed so that

equivalent procedural paths yield equivalent outcomes; an accounting architecture can seek consistency across transformations; a legal procedure can attempt to preserve rights across different channels. The diagram then encodes a desired invariant whose empirical realization must be tested.

4.16.13 Limits of the Category-Theoretic View

Category theory provides a powerful language of structure, composition, equivalence, and translation, but its usefulness within GR has identifiable limits. First, categorical analysis depends on prior choices of objects and morphisms. These choices can clarify a domain while reifying boundaries that remain contested or historically variable. A category can therefore be internally rigorous and still poorly matched to the relational reality it is intended to represent.

Second, ordinary categories encode composition abstractly but do not automatically represent probability, intensity, duration, geometry, uncertainty, power, cost, or causal strength. Such information can be added through enriched, probabilistic, metric, higher-order, or other categorical structures, but the additional formalism must earn its relevance. GRE should not escalate mathematical complexity merely because a richer category can be defined.

Third, some generative relations admit composition only partially, depend on context, or are themselves transformed through composition. Historical sequence can change the meaning of later operations; institutional exceptions can defeat uniform mapping; stochastic processes can make a single deterministic morphism inappropriate. In these settings, strict categories and strict functors may be approximations. Partial categories, correspondences, profunctors, lax structures, higher categories, or non-categorical models may provide better representations, but no formal family receives automatic priority.

Fourth, categorical equivalence can preserve structure while leaving material, experiential, historical, and normative differences outside the model. A fully faithful representation of selected relations can still omit who bears risk, whose experience is recognized, what resources are consumed, or which histories made the structure possible. Formal invariance therefore does not substitute for substantive economic inquiry.

Remark 4.150 (Methodological status of category theory). Within GR and GRE, category theory is best treated as a language for disciplined structural comparison and composition. Its successful application establishes properties of the represented domain under specified modeling assumptions; it does not by itself establish that reality is fundamentally categorical.

The category-theoretic view nevertheless contributes something distinctive to the chapter. It allows relational and manifestational domains to remain heterogeneous while still being connected through explicit mappings. It distinguishes loss of relational distinctions from excess structure introduced by manifestation. It separates isomorphism, equivalence, and manifestational indistinguishability. It provides a formal language for coherent changes of mapping rules, multiple realization, and path-dependent representation. Most importantly, it makes visible the conditions under which a comparison preserves structure and the points at which such preser-

vation fails.

These contributions prepare the move to Section 106. Category theory can relate heterogeneous domains without requiring a shared metric, but it does not eliminate differences of scale or the possibility that similar higher-level realities are realized through different lower-level configurations. Scale-relative reality and multiple realization therefore require separate treatment. The categorical language developed here will remain available as one analytical resource while the next section returns to the ontological problem itself: how reality can be stable and causally efficacious at one scale while being differently constituted at another.

4.17 Scale-Relative Reality and Multiple Realization

The preceding section introduced category theory as one formal language for relating heterogeneous domains without requiring a common metric or an identity of underlying structure. The role of the present section is to address a more general ontological problem that remains even when no categorical formalism is used: the same generative-relational system can present different relatively stable objects, relations, processes, and functions at different scales, and similar higher-level functions can be realized through heterogeneous lower-level configurations. The objective is to clarify how GR can speak of scale-relative reality without making reality arbitrary, and how it can recognize multiple realization without treating functional similarity as ontological identity. The section proceeds through scale-relative entityhood, scale-relative relationality, multiple realization of similar functions, heterogeneous realization, and cross-scale reality. Throughout, scale is treated as an analytically specified resolution constrained by the organization and causal structure of the system, rather than as a freely chosen perspective detached from what the system can sustain.

Scale enters GR in several forms. It can be spatial, temporal, organizational, institutional, functional, informational, or relational. A household can be treated as an economic subject at one organizational scale while being analyzed as a changing set of interpersonal, legal, affective, material, and monetary relations at another. A firm can appear as a single contracting entity in one model and as a distributed process of governance, routines, assets, labor, software, infrastructures, and expectations in another. A price can be treated as a scalar variable at one analytical resolution while another inquiry reconstructs the market-making, institutional, informational, and symbolic relations through which that price becomes possible. None of these descriptions is automatically more real because it is finer grained. The relevant question is which organization persists and exerts causal efficacy at the scale required by the inquiry. (P. W. Anderson 1972; Bar-Yam 2004; Simon 1962)

This position differs from a simple micro–macro hierarchy. A scale is not defined only by size, and a finer resolution does not necessarily reveal a privileged ontological base. A legal status can operate at an institutional scale that is not reducible to physical size; a norm can persist across many material realizations; a computational function can be implemented through architectures with different internal organizations; and a subject can occupy several overlapping relational scales simultaneously. Scale-relative analysis therefore concerns the resolution at which organization, per-

sistence, constraint, and causal efficacy become analytically distinguishable.

Definition 4.151 (Analytical scale). An *analytical scale* is a specified resolution of a generative-relational system at which selected entities, relations, processes, boundaries, and causal differences are treated as distinguishable for the purposes of an inquiry.

Definition 4.151 makes scale an explicit component of representation. It does not imply that the analyst creates the relevant reality by choosing a scale. The chosen resolution must remain answerable to the system. If a purported object fails to persist, stabilize, or exercise causal efficacy at the selected resolution, treating it as a relationally real entity becomes difficult to justify. Conversely, a configuration can be relationally real even when a finer-grained description reveals continual replacement, turnover, or transformation among its constituents.

4.17.1 Scale-Relative Entityhood

Entityhood in GR is therefore often scale-relative. A configuration can function as an entity when enough of its organization is stabilized to support identification, interaction, and causal participation at a selected resolution. This entityhood does not require that the configuration be simple, internally homogeneous, or materially indivisible. It requires that treating it as a unit preserve causal and relational differences relevant to the inquiry.

Definition 4.152 (Scale-relative entityhood). *Scale-relative entityhood* denotes the status of a configuration that exhibits sufficient persistence, stability, boundary coherence, and causal efficacy to be treated as a distinguishable relational object at a specified analytical scale, even when finer or alternative scales reveal it as internally heterogeneous, processual, or distributed.

A corporation illustrates the point. At one scale it can enter contracts, own assets, incur liabilities, employ workers, issue securities, and survive changes in membership. At another scale it is constituted through legal rules, governance procedures, organizational routines, technologies, buildings, documents, labor, interpersonal relations, and expectations. The higher-level entity does not disappear because its lower-level realization is heterogeneous. At the same time, the lower-level relations cannot be inferred exhaustively from the fact that the corporation persists as a legal and economic entity. Scale-relative entityhood therefore preserves both the causal reality of the higher-level object and the incompleteness of any single-scale description.

The same pattern appears in money. A unit of account can be analyzed as a stabilized relational object within a monetary system, while its realization depends on material tokens, ledgers, settlement infrastructures, legal rules, central-bank practices, commercial banking, expectations, software, and repeated acts of pricing and payment. A monetary unit is neither reducible to any one of these components nor independent of the system that sustains it. Its entityhood is scale-relative because it becomes causally tractable at the level at which monetary claims, prices, debts, and payments operate.

Scale-relative entityhood also applies to subjects. The continuing sense of “I” discussed in Section 4.12 can possess relational reality at a scale of autobiographical con-

tinuity and practical agency while being realized through changing bodily, neural, mnemonic, linguistic, social, and institutional processes. The fact that these lower-level realizations change does not by itself negate the relational reality of the self-model that persists sufficiently to guide action. Conversely, persistence of the practical self does not settle whether an ontologically independent and unchanging self exists. The three-register architecture and ontological suspension remain intact.

Scale-relativity therefore differs from nominalism about entities. GR does not treat entities as mere names imposed on an otherwise undifferentiated flow. Some boundaries are institutionally authorized, materially reinforced, operationally reproduced, or causally consequential. Others are weak, temporary, contested, or analytically convenient. The task is to identify the mechanisms through which entityhood is stabilized and the range over which treating the configuration as an entity preserves explanatory adequacy.

Claim 4.153 (Constraint on scale-relative entityhood). *A scale-relative entity claim is warranted only insofar as the proposed entity exhibits sufficiently stable organization and causal efficacy at the selected scale; changing the scale can alter the adequacy of the entity description without making the underlying inquiry arbitrary.*

Claim 4.153 prevents two opposite errors. One is reductionism, in which a higher-level object is dismissed because it can be decomposed. The other is reification, in which a convenient aggregate is treated as independently real even when its internal heterogeneity materially changes explanation. A national unemployment rate, a firm's productivity, a household preference, or a platform's market share can be useful higher-level objects, but the explanatory adequacy of each depends on whether aggregation preserves the differences relevant to the problem.

Scale-relative entityhood also has a temporal dimension. An entity can be stable at one temporal resolution and transient at another. A coalition can organize a negotiation for several months without becoming a durable institution. A market panic can persist long enough to reorganize credit even if it dissipates rapidly in historical time. A technical standard can remain stable for decades while individual implementations change continuously. Persistence must therefore be evaluated relative to the causal timescales of the relations being studied.

4.17.2 Scale-Relative Relationality

Relations are also scale-relative. A relation observed at one scale can summarize a large number of lower-level interactions, while a lower-level relation can become invisible or irrelevant at a higher resolution. For example, an employer–employee relation can be represented as a contractual relation in one model, while another inquiry decomposes it into wage-setting procedures, supervisory practices, legal rights, informal expectations, technical dependencies, bargaining capacities, and histories of recognition or conflict. Neither representation is automatically superior. Each preserves different relations and suppresses others.

Scale-relative relationality matters because GR places relations at the center of explanation without assuming that all relations must be represented at the finest available resolution. An economic system would become analytically unusable if every molecular, neurological, linguistic, contractual, technological, and institutional inter-

action had to be represented simultaneously. Relational analysis therefore requires principled compression. The question is whether the compressed relation retains the causal and generative structure relevant to the inquiry.

Definition 4.154 (Scale-relative relation). A *scale-relative relation* is an analytically specified relation whose relata, structure, and causal significance are defined at a selected resolution and whose adequacy depends on whether it preserves the relational differences relevant to the inquiry.

A relation at one scale can itself become a relatum at another. A long-term contractual relation between two firms can be treated as one edge in a supply network even though the relation is internally complex. A local credit network can become a component of a regional financial structure. A household can become a relatum in a tax system. An institution can become a node within an international regime. This nesting is not merely representational convenience when the higher-level configuration develops persistence and causal efficacy of its own.

Relations across scales can also constrain one another. A regulatory regime can shape permissible transactions among individual firms; repeated firm-level practices can gradually alter regulatory expectations; a platform's architecture can condition interactions among users; accumulated user practices can produce pressures that change the platform's rules. GR therefore avoids a one-directional picture in which causal explanation always moves from smaller to larger units. Cross-scale causal organization can involve amplification, constraint, feedback, path dependence, and institutional mediation.

The possibility of scale-relative relationality also clarifies why relational realities need not share a common metric. A relation between two persons, a relation between institutions, and a relation between a legal category and an administrative procedure can all be causally important without being distances in one numerical space. Moving across scale can require translation among heterogeneous forms of relation rather than rescaling a single measure. The category-theoretic view of Section 4.16 provides one possible formal language for such translations, but the ontological point is broader: relational connectedness does not imply metric homogeneity.

Remark 4.155 (Relational compression). A higher-scale relation can be analytically legitimate even when it compresses many lower-scale relations, provided that the compression preserves the distinctions necessary for the explanatory or evaluative task and does not conceal lower-scale heterogeneity that materially changes the conclusion.

Remark 4.155 is especially important for GRE. Economic analysis routinely works with households, firms, sectors, markets, institutions, and states. GRE does not reject these units because they are relationally constituted. It asks when their internal constitution matters and when a stabilized higher-scale relation is an adequate object of inquiry. The answer can change across questions. A firm can be a useful unit in a trade model and an inadequate unit in an analysis of workplace power. A household can be useful in a consumption model and inadequate in an analysis of intra-household allocation. Scale selection therefore carries explanatory and normative consequences.

4.17.3 Multiple Realizations of Similar Functions

Scale-relativity becomes especially important when similar higher-level functions are realized through different lower-level configurations. A payment function can be supported by cash, bank deposits, card networks, mobile ledgers, or other infrastructures. A coordination function can be performed through hierarchy, market exchange, professional norms, algorithmic allocation, or combinations of these. A memory function can be distributed across human recollection, written records, databases, institutional routines, and technical archives. Similar manifested capacities can therefore emerge from heterogeneous relational organizations. (Fodor 1974; Kim 1992)

Definition 4.156 (Multiple realization). *Multiple realization* occurs when two or more materially, organizationally, relationally, or otherwise structurally distinct configurations support a sufficiently similar function, capacity, manifestation, or higher-scale relational reality relative to explicitly specified criteria.

Definition 4.156 makes similarity criterion-dependent. Two systems can realize the same payment function for ordinary retail exchange while differing in privacy, settlement finality, resilience, energy demand, governance, legal recourse, accessibility, or exposure to surveillance. Two organizations can produce similar outputs while distributing authority, risk, labor, and recognition differently. Two subjects can produce locally indistinguishable responses while possessing different histories, internal organizations, capacities, or forms of experience. Multiple realization therefore supports local equivalence without implying general equivalence.

This distinction is continuous with the chapter's treatment of non-invertibility. Similar manifestations do not uniquely determine the relational configuration that produced them. Multiple realization provides a structured reason for this underdetermination: different realizations can converge on a similar manifested function. The inverse problem becomes especially difficult when only the higher-level output is observed and the lower-level realization is hidden, changing, or institutionally inaccessible.

Multiple realization also prevents a simple reduction of function to substrate. If the same higher-level function survives replacement of many lower-level components, the function can acquire a degree of relational autonomy at the scale where it is reproduced. This autonomy is relative rather than absolute. The higher-level reality still depends on realization, but it may not depend on one unique realization. A legal office can persist through changes in officeholders, buildings, record systems, and administrative technologies; a currency can persist through changes in physical tokens and settlement infrastructure; an organization can preserve functions through substantial turnover in members and routines.

Claim 4.157 (Locality of realized equivalence). *When distinct configurations realize a similar function or manifestation, the resulting equivalence is limited to the criteria under which the similarity is established. It does not entail equivalence of causal history, internal relation, normative consequence, material dependence, or ontological status.*

Claim 4.157 gives multiple realization the same discipline that Section 4.16 gave formal equivalence. A comparison must state what is preserved. In economics, this prevents functional substitutability from becoming a claim of social or ethical substi-

tutability. An automated system can substitute for a human worker in a specified productive task while changing employment relations, accountability, knowledge distribution, bargaining power, surveillance, or the meaning of participation. Functional similarity is therefore one relation among many.

Multiple realization can also occur historically within one relational reality. A monetary system, educational institution, welfare programme, scientific practice, or communication network can preserve a recognizable higher-level function while its material and organizational realization changes over time. Historical continuity then consists partly in the reproduction of function across changing realizations. Whether the change constitutes continuity, transformation, or replacement depends on which relations and capacities are treated as constitutive at the scale of inquiry.

4.17.4 Heterogeneous Ontological Realizations

The language of multiple realization can tempt a stronger conclusion: that a higher-level reality is independent of the ontological character of its realizers. GR does not require this conclusion. Realizations can be heterogeneous without being irrelevant. Material, biological, technical, institutional, symbolic, and experiential differences can affect which functions are possible, how reliably they are sustained, what costs they impose, and what forms of causal efficacy they acquire.

For this reason, the phrase *heterogeneous ontological realization* is used cautiously. It does not assert that GR has already established a complete ontology of the realizing domains. It marks cases in which the lower-level configurations supporting a higher-level relational reality require substantially different descriptive vocabularies or ontological commitments. A legal norm can depend on texts, institutions, authorized roles, practices, sanctions, interpretations, and embodied subjects. A digital asset can depend on electrical infrastructure, software states, cryptographic operations, legal recognition, organizational governance, and user expectations. A religious object can depend on material artifacts, ritual practice, collective memory, symbolic systems, embodied experience, and institutional continuity. The higher-level relational reality can span these heterogeneous realizations without being identical to any single one.

Definition 4.158 (Heterogeneous realization). A *heterogeneous realization* is a realization in which the sustaining components or processes belong to analytically distinct domains whose relevant structures cannot be adequately represented by a single homogeneous descriptive vocabulary without significant loss.

Definition 4.158 does not imply that the domains are metaphysically incommensurable. It states a methodological condition: the inquiry currently requires more than one descriptive register to preserve the organization relevant to explanation. Ontological suspension remains appropriate when stronger claims about the ultimate relation among those domains are unnecessary.

Heterogeneous realization matters for the three-register architecture because manifestation can occur through one domain while relational stabilization depends on several. A monetary balance may manifest as a number on a screen while its relational reality depends on legal claims, institutional records, technical infrastructures, settlement capacities, and expectations. A person's institutional identity may

manifest through a credential while its effects depend on legal recognition, database entries, organizational practice, social interpretation, and the subject's own history. A manifestationally simple object can therefore possess a relationally heterogeneous realization.

This point also constrains ontological inference. The fact that a higher-level function is multiply realized does not prove that the realizations are ontologically interchangeable. Conversely, the fact that realizations differ ontologically does not prevent them from supporting a locally similar relational reality. GR therefore separates at least three questions: whether a function is preserved, whether the realizing structures are relationally equivalent, and whether their ontological status is the same. These questions can receive different answers.

Heterogeneous realization has normative consequences as well. Two infrastructures that provide similar user-facing functions can distribute environmental cost, labor burden, vulnerability, control, or exclusion differently. Two organizational forms can produce similar outputs while structuring autonomy and authority differently. Ontological and relational heterogeneity can therefore matter even when manifestationally equivalence is strong. GRE must resist the tendency to treat functional output as an exhaustive description of economic reality.

4.17.5 Cross-Scale Reality

The preceding arguments imply that relational reality can be distributed across scales. A higher-scale object can be realized through lower-scale configurations while becoming sufficiently stable to constrain, enable, or reorganize those very configurations. This reciprocal organization is what the present subsection calls cross-scale reality. The term does not designate a separate fourth register of reality. It describes relations among realities whose persistence and causal efficacy are distinguishable at different analytical scales.

Definition 4.159 (Cross-scale reality). *Cross-scale reality* denotes a configuration in which relationally real structures identified at different analytical scales participate causally in one another's generation, stabilization, constraint, or transformation.

A legal institution provides a simple example. Repeated actions, documents, decisions, records, and recognized roles sustain the institution. Once stabilized, the institution constrains which actions count as valid, who possesses authority, what procedures are available, and which future relations can be formed. The higher-scale institution is realized through lower-scale practices, while the lower-scale practices are reorganized by the institution they sustain. A monetary system similarly depends on transactions and records while simultaneously shaping the terms under which transactions are possible. An organization depends on individual actions while roles, procedures, and collective resources constrain those actions.

Cross-scale reality therefore avoids the assumption that causal explanation must terminate at the finest scale. A finer-grained description can reveal mechanisms through which a higher-level structure is realized, but the higher-level structure can still be causally indispensable because it organizes the space of lower-level possibilities. The statement that a rule is implemented through individual actions does not eliminate the causal significance of the rule when actors orient themselves to it, insti-

tutions enforce it, and it structures future choices. The relation between realization and efficacy is recursive rather than eliminative.

This does not require a strong doctrine of “downward causation.” GR can remain neutral about metaphysical debates concerning causal levels. The analytically necessary claim is weaker: descriptions at different scales can capture causally relevant organization that is lost if analysis is restricted to one scale. If changing a higher-level institutional relation systematically changes lower-level possibilities, and if lower-level interactions reproduce or transform that higher-level relation, then a cross-scale account can be explanatorily warranted regardless of how the ultimate metaphysics of causation is later formulated.

Claim 4.160 (Cross-scale causal adequacy). *A cross-scale explanation is warranted when causal differences at one analytical scale cannot be adequately represented without reference to relatively stable organization identified at another scale, and when the relations connecting those scales can be specified sufficiently for the inquiry.*

Claim 4.160 places a burden on cross-scale explanation. Analysts should identify the relations connecting scales rather than invoke “macro” or “micro” levels as unexplained causes. The relevant connectors can include rules, incentives, infrastructures, information, classifications, authority, material dependencies, learning, imitation, resource flows, technical interfaces, or repeated practices. Different problems require different mechanisms.

Scale selection also affects manifestation. A pattern can be manifest at one scale while remaining invisible at another. An individual may experience a series of unrelated administrative decisions while a population-level analysis reveals a systematic distribution. A macroeconomic indicator can display stability while household-level experiences diverge sharply. A platform can appear reliable to users while infrastructural monitoring reveals growing fragility. Manifestational reality is therefore scale-sensitive, and disagreements about “what is happening” can sometimes arise because different observers, institutions, or instruments manifest different scales of the same relational system.

The graph architecture developed earlier in the chapter remains applicable. A manifestation at one scale can enter relations at another scale. A population statistic can reorganize policy; an individual case can become a symbolic event that changes institutional rules; a market index can alter firm behavior; a local failure can trigger system-wide redesign. Scale and register are therefore distinct dimensions. Ontological, relational, and manifestational realities can each be investigated at several scales, and mappings among registers can themselves operate across scale.

The consequence for GRE is methodological rather than merely ontological. Economic inquiry should specify the scale at which an object is claimed to be real, the realization relations that sustain it, the higher- and lower-scale structures omitted by the representation, and the conditions under which equivalence across realizations is asserted. A stable aggregate should not be decomposed merely because decomposition is possible, but neither should aggregation conceal heterogeneity that changes causal or normative conclusions. The appropriate scale is therefore part of the argument.

Scale-relative reality and multiple realization also clarify why ontological cat-

egories can change historically. New technologies, institutions, infrastructures, classifications, and productive arrangements can stabilize objects and functions at scales that were previously unavailable or analytically insignificant. A platform, digital identity, algorithmic market, multinational supply network, or machine-learning agent can become a relatively stable object only under historical conditions that sustain the relations required for its persistence and efficacy. The categories through which an inquiry describes economic reality can therefore acquire or lose salience as the systems themselves change.

This observation prepares Section 107. The next problem is not only that the same reality can be described at several scales, but that the categories available for identifying subjects, objects, relations, capacities, and institutions are themselves historically formed. Scale-relative analysis preserves a plurality of valid resolutions; historical materiality asks how some resolutions and categories become practically consequential in the first place. Together, the two arguments support an ontology that is structured and causally serious without being frozen into a timeless inventory of entities.

4.18 Historical Materiality of Ontological Categories

The preceding sections have progressively loosened the assumption that an ontology for economic inquiry can be given as a timeless inventory of ready-made entities. Relational realities can emerge and dissolve; manifestations can alter the systems they disclose; representations can become causally operative; subjects, institutions, and values can be stabilized through historically specific relations; and the adequacy of an entity description can vary across scale. The role of the present section is to make the historical consequence of these claims explicit. Its objective is to show that the categories through which GR and GRE distinguish subjects, objects, relations, capacities, institutions, and processes can themselves possess a historical materiality. This does not mean that reality is created at will by vocabulary, nor that every ontological distinction is merely conventional. It means that the kinds of relatively stable realities available to economic analysis, the practical boundaries through which they are reproduced, and the categories adequate to describing them can change as productive capacities, institutions, infrastructures, classifications, and forms of social organization change. The discussion proceeds through historical category formation, productive forces and ontological salience, institutional stabilization, transformation of subject and object categories, and the disciplined revisability of ontological categories.

Historical materiality therefore concerns more than the history of ideas about what exists. It also concerns the history of the relational systems within which some distinctions become materially reproducible and causally consequential. The category of a corporation, for example, is not only a word used by analysts. It is connected to legal personality, accounting, ownership, contracting, record keeping, governance, and organizational continuity. The category of a wage worker is similarly connected to historically specific relations of labor, property, contract, production, and reproduction. A category can thus be historically constituted at two levels at once: as a conceptual distinction within inquiry and as a distinction stabilized in the organization of the world being investigated. Keeping these levels

distinct is essential. A historically formed concept can refer badly to reality, while a historically stabilized relational object can remain causally real even when existing concepts represent it inadequately.

Definition 4.161 (Historical materiality of an ontological category). The *historical materiality of an ontological category* denotes the dependence of a category's practical applicability, boundaries, salience, or causal relevance on historically formed generative-relational conditions, including productive capacities, institutions, infrastructures, classifications, practices, and modes of representation.

Definition 4.161 is deliberately weaker than the claim that all ontological reality is historically produced. GR does not need that stronger metaphysical thesis. The definition instead identifies a property of inquiry and of relational reality: some categories become adequate because historical processes stabilize objects or relations that satisfy the category's conditions; some lose adequacy because those objects or relations transform; and some acquire new meanings because the systems through which they operate change. Ontological suspension therefore remains compatible with historical materiality. One can investigate how a category becomes materially and institutionally operative without claiming that every aspect of the reality to which it refers is exhausted by that history.

4.18.1 Historical Formation of Categories

Economic categories have histories because economies have histories. Property, labor, money, household, firm, market, commodity, debt, employment, platform, data, and capital are not analytically interchangeable across all social formations. Some of these terms can be used at high levels of abstraction across long periods, but their concrete relations, legal forms, technical infrastructures, and causal roles can differ substantially. Historical political economy has long emphasized that economic categories can be specific to forms of production and social organization rather than natural kinds that retain an invariant content across history (Marx 1867; Polanyi 1944). GR adopts this historical discipline without requiring that every later ontological claim be reduced to a single theory of historical development.

A category can form historically in several ways. New relational objects can emerge. Existing objects can acquire new causal roles. Institutional boundaries can stabilize distinctions that were previously fluid. Measurement and recording systems can make previously diffuse differences operational. Technical infrastructures can connect dispersed processes into a new functional whole. Political and legal arrangements can create statuses with durable consequences. Repeated representation can also reorganize the object represented, producing recursive stabilization of the category itself. These pathways often overlap, but they should not be collapsed into one mechanism.

Consider the modern firm. Treating the firm as an economic subject is analytically useful because legal, accounting, organizational, technological, and contractual relations make it possible for a changing population of persons and assets to persist as a recognizable unit. The category becomes materially consequential because contracts are signed in its name, liabilities are assigned to it, accounts are kept for it, and future action is coordinated through it. Yet the category does not have one invariant

realization. A family enterprise, public corporation, cooperative, platform company, and multinational group can differ sharply in governance, ownership, scale, internal authority, and technical organization while remaining recognizable under a broader category of firm for some inquiries.

The same applies to labor. “Labor” can designate human activity at a high level of abstraction, but wage labor, coerced labor, unpaid care, self-employment, professional service, platform-mediated work, and machine-assisted production occupy different relational structures. A historical change in the organization of production can therefore alter which distinctions matter to explanation. An ontology of economic activity that treats all productive contribution as one homogeneous object may remain useful for one aggregate calculation while becoming inadequate for questions about authority, reproduction, dependency, or distribution.

Historical category formation should also be distinguished from simple linguistic change. A new word can appear without a corresponding new relational reality, and a relational reality can emerge before a stable vocabulary exists for it. Section 4.5 established that relational reality does not require prior perception or symbolization. The historical formation of an ontological category can therefore lag behind the formation of the reality it later names. Conversely, institutional naming can precede and help stabilize the object named. Legal recognition of a corporate form, administrative classification of a population, or technical standardization of a digital object can contribute to the relations through which the corresponding reality subsequently persists.

Claim 4.162 (Dual historicity of ontological categories). *The historicity of an ontological category can arise both from changes in the reality to which the category is applied and from changes in the conceptual, institutional, and representational practices through which that reality is distinguished; these two histories can interact without being identical.*

Claim 4.162 prevents two reductions. One would treat categories as passive labels that simply follow changes in an independently organized world. The other would treat the world as if it were produced entirely by classification. GR instead permits recursion. A changing economic system can demand new categories, and an institutionalized category can later reorganize the relations it classifies. Hacking’s historical ontology and work on classificatory practices provides a useful precedent for this latter possibility, particularly where classifications interact with the persons and practices organized around them (Hacking 2002). The present argument extends the point cautiously to a wider range of economic objects while preserving the distinction between classificatory mediation and the full generative structure of those objects.

This dual historicity also means that category boundaries can be contested. Disputes about whether a worker is an employee or independent contractor, whether a digital token is money, property, security, or access credential, whether a platform is a marketplace or an employer, or whether an artificial system counts as an economic agent can be partly conceptual and partly institutional. Such disputes often matter because different classifications activate different rights, liabilities, measurements, expectations, and causal pathways. The contest is therefore not merely semantic. Once linked to institutions, a category can become a mechanism through which resources and possibilities are distributed.

4.18.2 Productive Forces and Ontological Salience

Historical materiality becomes especially visible when productive capacities change. The term *productive forces* is used here in a broad and deliberately non-exclusive sense to include techniques, skills, infrastructures, tools, scientific knowledge, organizational capacities, energy systems, computational resources, and other means through which production and coordinated activity become possible. Marxian traditions give productive forces a central place within historically specific relations of production, but GRE need not adopt a deterministic sequence in which technical development alone explains institutional or ontological change. Productive capacities interact with law, ownership, culture, ecology, political authority, knowledge, and social reproduction. Their relevance in this section is narrower: changes in productive capacity can alter which objects, relations, and scales become stable enough to acquire ontological salience for economic inquiry.

Definition 4.163 (Ontological salience). *Ontological salience* denotes the degree to which a kind of object, relation, process, or capacity becomes sufficiently persistent, causally consequential, and differentiable within a historical system that distinguishing it materially improves explanation, intervention, or evaluation.

Ontological salience is not identical to ontological reality. Something can be real while remaining marginal to the inquiry, and something can be highly salient in representation while its underlying reality is weak or mischaracterized. Salience concerns why a distinction becomes difficult to omit without explanatory loss. When railways, electrical grids, telecommunications networks, digital platforms, or large computational systems become infrastructures of economic coordination, categories corresponding to network position, interoperability, platform access, computational capacity, or data dependency can acquire a practical importance that was absent under earlier productive arrangements.

Technical change can also create new scales of stable organization. A geographically dispersed supply chain becomes a tractable economic object only when transport, communication, contracting, standards, finance, and information systems sustain coordination across distance. A real-time algorithmic market presupposes technical and institutional infrastructures through which orders, prices, identities, risk controls, and settlement can be processed at speeds and scales unavailable to earlier markets. A machine-learning agent can become an operationally significant economic object only when computational infrastructures, training processes, interfaces, institutional permissions, and organizational practices support sustained action. These objects are not made real by technology alone; technology participates in the generative-relational configurations that allow them to persist and act.

Productive forces can also alter the salience of older categories. Mechanization changes the relation between labor time, skill, machinery, and output. Automation can separate productive contribution from direct human execution. Digital replication changes the relation between use and scarcity for some informational goods. Remote communication changes the spatial organization of work. Large-scale data processing changes the practical role of classification and prediction. None of these transformations automatically invalidates categories such as labor, property, commodity, or firm, but each can change what must be specified for those categories to

remain analytically adequate.

The relation between productive forces and ontological salience is therefore generative rather than deterministic. New capacities can open a possibility space without fixing the institutional form that will stabilize within it. Similar technologies can be embedded in different ownership structures, legal regimes, organizational forms, cultural practices, and normative orders. Conversely, institutions can delay, redirect, or selectively amplify technical possibilities. GR therefore treats productive forces as one set of historically material conditions among several, not as an autonomous engine that determines the ontology of the economy.

Remark 4.164 (Material possibility and historical stabilization). A productive capacity can make a relational form possible without making that form inevitable. Ontological salience emerges when technical possibility becomes joined to sufficiently stable institutional, organizational, symbolic, ecological, and practical relations to produce durable causal effects.

Remark 4.164 is particularly important for contemporary AI. The existence of a model capable of generating language does not by itself settle whether it constitutes an economic subject, an instrument, an organizational component, an autonomous service, or some combination of these under different conditions. Those categories depend on the relations through which the system is authorized, controlled, represented, contracted with, integrated into workflows, and made responsible or non-responsible for consequences. Technical capability changes the category problem; institutional stabilization helps determine which category becomes operative.

4.18.3 Institutions and Stabilized Reality

Institutions give historical categories durability by linking distinctions to recurrent practices, rules, records, sanctions, rights, expectations, and authorized procedures. Section 4.14 treated institutions as relationally real when they satisfy the full criterion of persistence, stability, and causal efficacy. The present concern is how institutionalization contributes to the historical materiality of the categories through which such realities are recognized and reproduced.

Institutional categories do more than describe. A property title establishes a recognized relation among persons, objects, claims, and authorities. A corporate registration makes a legal subject available for contracting and liability. An employment classification can activate wage rules, insurance obligations, tax treatment, bargaining rights, or exclusions. A national-accounting category can shape how economic activity is measured and compared. A technical standard can determine which objects interoperate. Once these classifications are embedded in durable procedures, their consequences extend beyond the moment of naming.

North's institutional analysis emphasizes that formal rules, informal constraints, and enforcement characteristics structure incentives and economic interaction (North 1990). Searle's account of institutional facts likewise draws attention to how socially recognized statuses can support powers, rights, duties, and functions that are not reducible to the physical properties of the objects involved (Searle 1995). GR need not adopt either framework as a complete ontology of institutions to retain the insight that institutional recognition can stabilize relational realities and make

category boundaries causally consequential.

Institutional stabilization can produce what might be called category infrastructure. Registries, legal codes, accounting systems, databases, credentials, standards, forms, identifiers, professional practices, and administrative routines make categories reproducible across time and across many local interactions. They reduce the need to renegotiate the existence and meaning of a category in each encounter. A corporation need not be recreated conceptually whenever it enters a contract; a recognized currency unit need not be redefined in each payment; a credential can travel across organizations because institutional infrastructures preserve enough of its meaning to make it actionable.

Definition 4.165 (Category infrastructure). A *category infrastructure* is a relatively persistent assemblage of rules, records, standards, practices, technologies, and authorized interpretations through which a classificatory distinction is reproduced and made causally operative across situations.

Category infrastructure helps explain why some classifications become relationally real while others remain ephemeral. A spontaneous label may influence a local interaction but disappear quickly. A classification linked to records, eligibility rules, technical interfaces, and repeated organizational practice can persist and reorganize future relations. Its materiality lies partly in the infrastructures that carry it. This does not make the classification epistemically correct or normatively legitimate. A mistaken, discriminatory, or obsolete category can remain causally powerful when institutions continue to act upon it.

Institutional stabilization also produces inertia. Once organizational routines, contracts, software systems, forms, statistical series, and legal rights are built around a category, revising that category can become costly. Historical categories can therefore outlive the conditions under which they were initially useful. A classification may persist because infrastructures presuppose it, even as the underlying relational reality becomes more heterogeneous. This lag creates a recurring epistemic and political problem: the category remains materially real in its effects while becoming increasingly inadequate as a description.

The recursive architecture of Sections 4.10 and 4.11 is visible here. A category becomes symbolized and institutionalized; institutions then act through the category; those actions reshape subjects, resources, and opportunities; the altered relations generate new manifestations; and subsequent representations may either reinforce or challenge the category. Historical materiality therefore includes the accumulated consequences of prior classifications. Categories carry histories not only because their meanings change, but because earlier meanings leave infrastructures and distributions behind.

4.18.4 Historical Transformation of Subject and Object Categories

The distinction between subject and object is itself historically articulated within economic systems. GR does not imply that every subject is reducible to a role or that every object is created by institutions. It does, however, treat practical subjecthood and objecthood as potentially relationally stabilized. Section 4.12 already distinguished ontological subjecthood, relationally effective subjecthood, manifestational subject-

hood, and institutionally represented subjecthood. Historical materiality adds that the conditions under which particular forms of subjecthood become economically operative can change.

The wage worker, consumer, shareholder, debtor, insured person, welfare recipient, platform user, data subject, content creator, contractor, and machine operator are not simply different names for a timeless economic individual. Each role selects different relations, rights, obligations, capacities, vulnerabilities, and forms of representation. A person can occupy several of these positions simultaneously, and the same person can move among them over time. Historical change can create new subject positions or reorganize the causal importance of old ones.

Classifications of persons can also be recursive. Hacking's analysis of interactive kinds is especially relevant where classifications enter institutions and self-understanding, although GR does not assume that every economic category behaves in this way (Hacking 2002). A credit category can alter borrowing opportunities and future financial behavior. An employment category can alter legal protection and bargaining position. A professional credential can alter access to institutions and the subject's own expected trajectory. A platform reputation score can mediate visibility and opportunity. In each case a representation of the subject becomes part of the relational system through which later subjecthood is produced.

Objects undergo analogous transformations. Land can be a lived environment, territory, ecological system, taxable parcel, commodity, collateral, commons, sacred site, or development asset under different relational configurations. Data can be treated as record, evidence, asset, input, personal information, property-like resource, public infrastructure, or object of governance. A digital account can become an access credential, payment instrument, identity proxy, contractual interface, or bundle of institutional claims. The physical or informational substrate can remain partly similar while the relational object changes because different institutions and practices stabilize different causal roles.

Claim 4.166 (Historical transformation of economic objecthood). *An economic object's relational reality can transform when the relations that stabilize its boundaries, capacities, rights, uses, or causal roles change, even where some underlying material substrate remains continuous.*

Claim 4.166 helps avoid two extremes. One would assume that the stable material substrate fixes the economic object once and for all. The other would imply that a change of description alone creates a new object. GR instead asks whether the change in relations is sufficient to alter persistence, causal capacities, recognized boundaries, or the possibilities generated by the object. Reclassification matters most when it becomes embedded in relations that materially change what can happen next.

Subject and object categories can also exchange analytical roles. A firm can be a subject in a market model and an object of regulation. A worker can be an agent in a labor-supply model and an object of algorithmic classification. An algorithm can be treated as an object owned by a firm, a process embedded in organizational decision making, or an operational subject when it performs sustained actions within authorized interfaces. These shifts do not by themselves settle metaphysical subjecthood. They show that the subject/object distinction used in economic inquiry is partly task-

and relation-dependent, and that historical changes can make previously marginal orientations practically important.

This point becomes particularly significant under automation and AI. A system can manifest outputs associated with human economic agency while remaining embedded in ownership, infrastructural dependence, governance, and technical realization that differ substantially from human subjecthood. GRE therefore needs categories capable of recognizing artificial operational agency without inferring experiential, moral, juridical, or ontological equivalence from functional similarity. The historical emergence of such systems makes the inadequacy of a simple human-agent/tool dichotomy increasingly visible, but the appropriate replacement categories remain open to empirical and normative development.

4.18.5 Ontological Categories as Historically Revisable

If ontological categories possess historical materiality, they must also be revisable. Revisability, however, should not be confused with conceptual voluntarism. Analysts cannot simply redefine an inconvenient reality out of existence. A useful revision must respond to changes in causal organization, evidential access, institutional structure, explanatory purpose, or normative problem. Historical revisability is disciplined by the realities the categories are intended to track.

Definition 4.167 (Historical revisability of ontological categories). The *historical revisability of an ontological category* is the requirement that its boundaries, criteria, and explanatory role remain open to modification when changes in relational organization, manifestation, evidence, institutional practice, or analytical purpose materially alter the adequacy of the category.

Several forms of revision are possible. A category can be subdivided when previously compressed differences become causally significant. Several categories can be joined for a particular higher-scale analysis when their differences cease to matter to the inquiry. A category can change its defining criteria. Its temporal or spatial scope can be narrowed. A formerly central category can become secondary while another becomes more salient. A category can also remain in use while its ontological interpretation becomes more restrained. Revisability therefore does not imply constant replacement; it includes refinement, rescaling, qualification, and explicit limitation.

Historical revisability is especially important where administrative and analytical categories have become infrastructures. Because institutional categories can shape the reality they classify, revision can change the object of inquiry as well as the representation of that object. Altering an employment classification can change rights and organizational incentives. Revising a monetary definition can change reported aggregates and policy interpretation. Changing a diagnostic or eligibility category can reorganize access to resources. The epistemic act of revision can thus become an intervention in relational reality.

This recursive possibility makes provenance important. Analysts should be able to ask when a category was introduced, through which institutions it became operative, what relations it compresses, which populations or objects it excludes, what technical or legal infrastructures depend on it, and how its meaning has changed. Historical ontology in this sense is not an antiquarian supplement to economic analysis. It is

part of understanding what present categories do.

Claim 4.168 (Discipline of historical revision). *Revision of an ontological category is warranted when an alternative distinction preserves or improves causal, relational, manifestational, or evaluative adequacy for the inquiry, and when the reasons for altering the category can be traced to changes in evidence, organization, scale, purpose, or historically material conditions rather than to terminological preference alone.*

Claim 4.168 gives revisability a burden of argument. A new category should show what explanatory difference it makes. An old category should not be discarded merely because it is historically constituted; historical constitution is compatible with durable causal reality. Money, firms, legal persons, markets, and institutional roles can remain indispensable objects of analysis even when their forms are historically specific. Conversely, long use does not guarantee continuing adequacy. A category can be deeply institutionalized and still fail to represent emerging relations that have become causally significant.

The distinction between relational reality and normative validity is again essential. A historically stabilized category can be unjust, exclusionary, or misleading while remaining causally real. Its normative criticism does not erase its effects. Likewise, a normatively attractive category does not become relationally real merely because it would be desirable for institutions to recognize it. GRE must therefore be able to analyze categories that ought to change without pretending that they have already changed, and to analyze emergent categories without treating emergence as justification.

Historical materiality also places a limit on claims of universal ontology within economics. Categories that function well in one historical configuration may require translation elsewhere. “Household,” “property,” “employment,” “market,” or “firm” can refer to relationally different arrangements across institutional and cultural contexts. Comparative analysis therefore needs to establish which structural correspondences justify using a common category and which differences require separate treatment. Shared terminology should not silently produce equivalence.

At the same time, historical materiality does not eliminate the possibility of cross-historical abstraction. Some categories can be useful precisely because they capture recurring relational forms at a sufficiently general level. Exchange, constraint, dependency, production, reproduction, coordination, obligation, and causal efficacy can sometimes travel across contexts more readily than narrower institutional categories. The appropriate level of abstraction depends on the question. GR’s methodological primitives and the three-register architecture are intended to support this movement across levels without claiming that every concrete economic object has a timeless form.

The resulting position is a historically responsive ontology rather than either a fixed catalogue or unrestricted constructivism. Reality constrains categories; categories can mediate reality; institutions can stabilize categories; productive capacities can change what becomes materially possible and salient; and historical transformation can require new distinctions. Ontological suspension remains active throughout: the historical emergence of a relational object does not decide whether it possesses some stronger ontological status, while the absence of a final metaphysics does not

weaken the claim that historically formed objects and categories can be causally real.

This historical perspective prepares the next methodological step. If categories, scales, and even some objects of inquiry are historically conditioned, relational-generative analysis should be treated as a disciplined analytical expedient rather than as a declaration that all reality possesses one final relational form. Section 108 therefore turns from the historical materiality of categories to the methodological standing of relational-generative analysis itself: what it enables, when it is warranted, and where its use should remain provisional.

4.19 Methodological Standing of Relational-Generative Analysis

The preceding sections have developed an ontological architecture in which relational realities can emerge, persist, stabilize, and exercise causal efficacy; manifestations can disclose and transform those realities; representations can become operative within subsequent relations; categories can vary across scale and history; and formal languages can clarify selected structural features without exhausting the realities they represent. The role of the present section is to state the methodological consequence of that architecture for Generative Relational Theory (GR) and Generative Relational Economics (GRE). Its objective is to clarify the standing of relational-generative analysis as a disciplined analytical expedient: a way of selecting relations, histories, transformations, and downstream consequences for inquiry when those selections improve explanation, without converting the analytical selection into a universal metaphysical thesis. The discussion proceeds through relational selection, generative selection, methodological economy, ontological plurality, and the limits of relational-generative abstraction.

The term *expedient* is used here in a technical and non-pejorative sense. An analytical expedient is a constructed mode of access whose adequacy depends on the problem, scale, evidence, and explanatory purpose. Scientific and interpretive inquiry routinely proceed through such constructions: a boundary is drawn around a system, a variable is selected, a comparison class is defined, a temporal interval is chosen, a mechanism is isolated, or a representation is adopted because the resulting description makes some pattern or causal relation intelligible. The construction can be highly disciplined and empirically constrained while remaining different from a claim that reality possesses exactly the same boundaries or form independently of inquiry. (Fraassen 1980; Giere 2004)

Definition 4.169 (Relational-generative methodological expedient). A *relational-generative methodological expedient* is an explicitly scoped analytical construction that represents an object, process, or domain through selected relations, generative histories, constraints, transformations, and downstream effects in order to improve explanation, comparison, interpretation, or intervention, while leaving stronger ontological claims open unless they receive independent warrant.

This definition preserves two commitments simultaneously. Relational-generative analysis can make substantive claims about causal organization, persistence, historical formation, and transformation. At the same time, the analytical grammar through

which these claims are organized remains answerable to the inquiry rather than automatically elevated into the final structure of reality. Relation and generativity therefore function as strong analytical commitments where they carry explanatory work and as revisable selections where another representation better addresses the problem.

This position extends the open ontological commitment established in Section 4.1. Ontological suspension concerns the strength of reality claims; methodological expedience concerns the form through which inquiry approaches those realities. The two disciplines reinforce one another. A researcher can take relations seriously enough to explain causal organization while suspending the stronger proposition that relations are metaphysically prior to every relatum. A researcher can analyze generative histories without assuming that every relevant property is best explained by tracing its production. GR therefore gains explanatory depth from relational and generative reasoning while retaining room for substance-oriented, process-oriented, phenomenological, institutional, formal, or domain-specific descriptions when those descriptions are locally adequate.

4.19.1 Relational Selection

Relational analysis begins by selecting relations as explanatory objects. This selection directs attention toward dependence, coupling, positioning, mediation, recognition, exchange, constraint, coordination, conflict, transmission, inheritance, and other structures through which the capacities or trajectories of an object depend partly on its situation within a wider configuration. The selection is especially useful where isolating an entity would conceal the mechanisms through which its apparent properties are produced or maintained.

A firm provides a familiar example. Treating the firm as a unit can be entirely appropriate for questions about aggregate output, accounting performance, or contractual liability. A relational analysis becomes useful when the inquiry concerns how the firm's capacities depend on employees, suppliers, legal rules, infrastructures, creditors, platforms, customers, data flows, ownership, organizational routines, and political conditions. The relational description does not abolish the firm as an entity. It changes the resolution of the inquiry so that some of the relations constituting or conditioning the entity become analytically explicit.

The same principle applies to subjects. Section 4.12 distinguished ontological, relationally effective, manifestational, and institutionally represented subjecthood. A relational analysis of a worker, consumer, debtor, citizen, researcher, or artificial operational subject asks which capacities are produced, recognized, constrained, or amplified through relations. Such analysis can reveal dependencies that are obscured when the subject is treated as a self-contained bearer of fixed preferences or capabilities. Yet the usefulness of the relational view does not require the stronger claim that the subject possesses no reality apart from relations. The analytical question is whether relational organization contributes materially to the phenomenon under investigation.

Claim 4.170 (Relational selection). *A relational representation is methodologically warranted when differences in relations, positions, interfaces, or configurations make*

a material explanatory difference to the capacities, persistence, manifestation, or transformation of the object under study.

Claim 4.170 places a burden on relational vocabulary. Merely redescribing an entity with the word “relation” adds little. The analysis should identify which relations are relevant, how they are distinguished, what evidence supports their relevance, and what explanatory difference follows from including them. A relation can be direct or mediated, persistent or transient, symmetric or asymmetric, material or symbolic, institutional or interpersonal, observable or inferred. The selection should match the mechanism and scale of the inquiry.

Relational selection also involves boundary choice. Any empirical analysis includes some relations while treating others as background. A study of labor relations may foreground contract, supervision, wage dependence, household reproduction, or platform mediation while leaving distant ecological or geopolitical relations outside the immediate model. The excluded relations do not thereby become unreal. They remain outside the chosen analytical boundary unless evidence shows that they materially alter the phenomenon being explained. The adequacy of a relational analysis therefore depends partly on whether its boundary captures the relations that make a consequential difference to the local question.

This boundary discipline matters because an indefinitely expanded relational description can become analytically empty. Every object can be placed within an enormous history of material, biological, social, institutional, linguistic, and ecological relations. Inquiry requires selection. GR contributes by making the selection explicit and revisable rather than by demanding maximal relational inclusion. The relevant relation is the relation that carries explanatory, interpretive, evaluative, or intervention significance at the scale being studied.

The three-register architecture further constrains relational selection. An analyst should distinguish relations inferred from manifestations from the relational reality being inferred, and should identify where symbolic or institutional representations mediate the evidence. Section 4.9 showed that reverse inference from manifestation can remain underdetermined. Relational analysis therefore gains rigor when it states which relations are directly observed, which are institutionally recorded, which are reconstructed historically, and which are hypothesized as latent generative structures.

4.19.2 Generative Selection

Generative analysis introduces a second selection. It asks how a configuration produces, enables, constrains, stabilizes, transforms, or dissolves subsequent objects, capacities, relations, manifestations, meanings, and possibilities. The focus therefore extends beyond the description of a present relational structure toward the consequences and historical trajectories that the structure helps generate.

Generativity is particularly useful when an apparently static allocation has dynamic consequences. A distribution of income, ownership, educational access, computational capability, legal status, or institutional authority can alter what actors are later able to learn, build, negotiate, represent, preserve, or contest. The initial configuration may therefore matter not only because of what is distributed at one moment

but because it changes the field of later generation. GRE uses this perspective to investigate trajectories of capability, value, power, knowledge, institutional formation, and relational reproduction.

The generative perspective should remain scoped. Some questions are adequately answered by description, classification, or equilibrium comparison. Others require historical explanation of formation. Still others require causal analysis of what follows from a configuration. The value of generative analysis appears when downstream production or transformation is part of the substantive problem. It is therefore a methodological choice oriented toward a class of questions rather than a requirement that every analysis be converted into a genealogy of everything that preceded it.

Definition 4.171 (Generative relevance). A relation, configuration, event, or representation has *generative relevance* for an inquiry when its presence, organization, persistence, or transformation contributes materially to the production, stabilization, modification, or suppression of later states, capacities, relations, manifestations, or possibilities that fall within the inquiry's explanatory scope.

Definition 4.171 connects generativity to causal and historical significance while preserving scale. A minor interaction may have generative relevance for a local relation and little relevance for a national institution. A legal classification may reorganize one market while leaving another largely unaffected. A technological infrastructure may generate new operational capacities without substantially altering some normative relations. The analyst therefore specifies the downstream object and timescale rather than treating generativity as an undifferentiated property.

Generative selection also introduces temporal asymmetry into analysis. A present structure can be understood partly through the trajectories it opens and closes. This does not imply deterministic prediction. Generative systems can contain contingency, feedback, path dependence, critical transitions, heterogeneous responses, and interactions with conditions outside the analytical boundary. The claim that a configuration generates possibilities is therefore compatible with multiple future trajectories. GRE's interest lies in how the configuration changes the distribution, accessibility, or organization of those possibilities.

This distinction is useful for intervention. A policy may improve a manifest outcome at one moment while weakening capacities required for later adaptation. Another intervention may have modest immediate effects while changing institutional relations that support later generation. A generative analysis brings these temporal consequences into view. It remains possible that another method captures the same phenomenon more efficiently; the methodological status of generativity therefore depends on the explanatory gain it provides for the problem at hand.

Generative analysis also requires attention to the history of the mechanisms it studies. Section 4.18 established that categories and causal organizations can themselves possess historical materiality. A generative mechanism observed in one institutional configuration may change when legal rules, productive capacities, classifications, infrastructures, or symbolic regimes change. The method therefore treats mechanisms as potentially historical objects rather than assuming invariant laws merely because a recurrent pattern has been identified.

4.19.3 Methodological Economy and Provisional Use

A useful analytical framework should explain enough while assuming only what the inquiry requires. Relational-generative analysis follows this principle of methodological economy. It introduces relations when relational organization changes the explanation, generativity when downstream formation matters, and stronger formal or ontological commitments only when they provide additional warranted explanatory value. This economy protects the framework from accumulating conceptual machinery that exceeds the question.

Proposition 4.172 (Local sufficiency of relational-generative analysis). *For a specified inquiry, relational-generative analysis is locally sufficient when the selected relations, generative mechanisms, boundaries, scales, and temporal horizons support the explanatory or interpretive task without requiring stronger commitments whose addition would not materially improve the analysis.*

Proposition 4.172 parallels the principle of local ontological sufficiency developed earlier in the chapter. The aim is neither theoretical minimalism for its own sake nor maximal formal completeness. The aim is proportionality between the analytical apparatus and the problem. A study of institutional trust may need historical relations, symbolic mediation, and recursive feedback without requiring category theory. A study of manifestation mappings may benefit from category-theoretic language without requiring a field-theoretic model. A study of pricing dynamics may use quantitative causal models while leaving questions about ultimate social ontology suspended.

Provisional use follows from this proportionality. An analytical representation can be adopted because it works well for a defined purpose and later revised when the purpose, evidence, or system changes. Provisionality concerns the standing of the representation rather than the seriousness of the inquiry. A provisional model may support strong local findings when its assumptions and scope are explicit. Conversely, a model presented with metaphysical confidence can remain weak if its empirical or conceptual fit is poor.

This stance is particularly important for interdisciplinary work. GR draws on economics, political economy, sociology, philosophy, law, complex systems, computation, history, and other domains. Each discipline carries characteristic objects and methods. Methodological economy allows GRE to borrow a representation for a specified task without absorbing the entire ontology of the source discipline. A network can represent interaction structure; a dynamical system can represent temporal evolution; an institutional analysis can represent rule-bound organization; a phenomenological account can clarify manifestation; a historical reconstruction can establish path-dependent formation. Their local contributions can coexist even where their background assumptions differ.

Provisionality also protects cumulative research. A framework that requires every contribution to settle the full ontology of its objects would impose an unrealistic burden and discourage empirical work. Researchers can instead state the level of commitment required by their study. One contribution may establish that a classification has relational reality because it is persistent and causally efficacious. Another may investigate the manifestation mechanisms through which that classification becomes salient. A third may contest its normative legitimacy. A fourth may propose a

stronger ontology. The contributions can accumulate because the shared analytical distinctions remain clear even where metaphysical commitments differ.

Methodological economy therefore favors modularity. Claims about relational reality, manifestation, representation, value, subjecthood, scale, and historical category formation can be connected without requiring that every module be used in every study. GRE functions more effectively as a research programme when investigators can select the components required by the question while preserving compatibility with the larger architecture.

4.19.4 Relational Analysis and Ontological Plurality

Treating relation as an analytical choice also guards against relational reductionism. Relational reductionism would arise if a successful relational explanation were taken to establish that entities, properties, experiences, material structures, or domain-specific mechanisms possess no analytically independent significance. GR has no methodological need for that extension. An entity can be relationally constituted in some respects and still require description at the level of its internal organization. A process can depend on relations while retaining material constraints that are poorly represented through relational vocabulary alone. A manifestation can be relationally generated while possessing phenomenological features that require another mode of access.

The three-register architecture itself supports this plurality. Ontological reality exceeds what any one manifestation or representation captures. Relational reality identifies one criterion of causal reality within generative-relational systems. Manifestational reality identifies conditions of availability. These distinctions already imply that one analytical vocabulary is unlikely to exhaust every relevant question. Relational-generative analysis therefore operates within an ontology that permits plural modes of access.

Caution 4.173 (Relational reductionism). Relational reductionism occurs when explanatory success at the level of relations is extended into the unsupported conclusion that other analytically relevant structures, properties, experiences, or domain-specific mechanisms can be fully replaced by relational description.

Caution 4.173 is distinct from the ontological inflation identified in Section 4.1. Ontological inflation concerns movement from analytical success to stronger reality claims. Relational reductionism concerns compression of heterogeneous explanatory objects into one preferred vocabulary. The two can occur together, but each requires separate discipline.

Consider monetary reality. A relational analysis can explain how legal recognition, expectations, accounting, payment infrastructures, state capacities, and exchange practices stabilize money. This does not make the material and technical properties of payment systems irrelevant. A banknote has physical constraints; a digital payment system depends on hardware, energy, code, security, and communication infrastructure; monetary policy operates through institutionally specific mechanisms. Relational explanation is strongest when it articulates these components rather than translating them into an abstract relation that loses their causal specificity.

The same applies to subjectivity. The relational reality of a continuing sense of self can be analyzed through memory, recognition, bodily continuity, narrative, institutional identity, and interaction. Such analysis leaves room for psychological, neurological, phenomenological, and philosophical accounts that address different dimensions of the same subject. Relational analysis can identify coupling among these dimensions without claiming jurisdiction over every question they raise.

Ontological plurality also preserves the possibility of domain-specific primitives. Physics, biology, law, economics, and phenomenology may require different operational distinctions. GR can investigate relations across those domains while allowing each to retain objects that are locally indispensable. Cross-domain translation therefore becomes an empirical and conceptual task. Similar relational patterns can support analogy without implying that the underlying objects are of the same ontological kind.

This position is consistent with the non-necessary comparability established in Section 4.5. Two relationally real objects can remain structurally incommensurable. A methodological commitment to relation does not create a common metric merely because both objects are represented relationally. The analyst must establish the mapping, equivalence criterion, or comparison space independently. Relational vocabulary supplies a grammar for asking about correspondence; it does not guarantee correspondence.

4.19.5 Limits of Relational-Generative Abstraction

The final methodological task is to identify where relational-generative abstraction loses information or explanatory traction. Every abstraction suppresses detail. The relevant question is whether what has been suppressed matters to the inquiry. GR therefore treats limitations as part of the specification of an analysis rather than as an external objection added after the model is complete.

Several limits recur. First, relation selection can omit materially consequential internal structure. A network representation may capture who interacts with whom while missing the content, temporality, embodiment, or institutional meaning of the interaction. Second, generative analysis can become overly expansive if every remote historical antecedent is treated as part of the mechanism. Third, system boundaries can hide external dependencies that later prove causally important. Fourth, manifestation data can underdetermine the relational structure from which they arose. Fifth, symbolic representations can impose commensuration on realities whose differences matter. Sixth, scale selection can stabilize an entity that dissolves when examined at another resolution. These are not reasons to abandon relational-generative analysis; they are conditions under which its claims require qualification or supplementation.

Definition 4.174 (Abstraction boundary). An *abstraction boundary* specifies the relations, internal structures, scales, timescales, manifestations, and contextual conditions that an analysis treats as outside its active explanatory representation while recognizing that some excluded elements may remain real and potentially causally relevant.

Definition 4.174 gives methodological exclusion a positive form. An analysis can

say what it brackets. A model of market transactions may bracket household care relations; a study of household reproduction may bracket global financial structure; a formal manifestation mapping may bracket embodied phenomenology; an institutional analysis may bracket molecular mechanisms. The boundary becomes problematic when excluded elements materially change the results while remaining unacknowledged. Explicit abstraction boundaries make such revision easier.

A second limit concerns normative inference. Generative success, relational stability, persistence, and causal efficacy remain descriptive or explanatory properties until a normative argument is supplied. A highly stable institution can generate severe exclusion. A powerful symbolic regime can coordinate activity while distorting recognition. A generative process can expand one form of value while destroying another. GRE therefore keeps evaluative judgment distinct from the fact that a system reproduces itself effectively.

A third limit concerns epistemic access. Relational-generative models often infer structures from traces, outcomes, records, classifications, or manifestations. Sections 4.8 and 4.9 established that manifestation mappings can be relationally real and still non-invertible. An analytically elegant relational structure can therefore remain underdetermined by available evidence. Methodological expedience does not license filling this gap with preferred theory. The model should mark where inference depends on assumptions, priors, idealizations, or inaccessible historical conditions.

A fourth limit concerns performativity and recursion. Once a relational-generative representation enters institutions, it can begin to alter the reality it describes. A risk model can change lending; a labor classification can change rights and incentives; a measure of productivity can reorganize work; a theory of markets can influence market design. The analytical expedient can therefore become a manifestational or relational object within the system. Reflexive monitoring is required when representation changes the generative field.

Claim 4.175 (Revisability of abstraction). *A relational-generative abstraction should be revised, supplemented, or replaced when excluded structures, alternative scales, changing historical conditions, new manifestations, or competing representations materially alter the explanatory conclusions for which the abstraction was adopted.*

Claim 4.175 gives methodological pluralism an operational criterion. The question is not whether relational-generative analysis is universally correct. The question is whether it remains adequate for the inquiry and whether another representation captures consequential structure that it omits. Revision can deepen the relational model, combine it with another method, change scale, alter boundaries, or suspend the relational-generative representation altogether for a particular task.

The resulting methodological position is therefore strong without being exclusive. GR and GRE use relations to investigate dependence, organization, mediation, and position; they use generativity to investigate emergence, reproduction, transformation, and future possibility. These selections support substantive claims when the relevant structures are evidenced and causally consequential. Their success does not require a final declaration that reality consists only of relations or that every phenomenon is best understood through its generative history. Relational-generative analysis earns its place through explanatory performance, explicit scope, and capacity for revision.

Section 108 thus clarifies the methodological standing of the framework developed throughout Chapter 4. Relation and generativity function as disciplined analytical selections; ontological suspension preserves room for stronger or alternative ontologies; methodological economy limits conceptual machinery to what the question requires; and explicit abstraction boundaries make the costs of simplification visible. The next section develops the positive implication of this position by examining compatibility with alternative ontological commitments, including substance-oriented, process-oriented, relation-oriented, and domain-specific ontologies.

4.20 Compatibility with Alternative Ontological Commitments

The ontological architecture developed in Chapter 4 is intended to support substantive inquiry without requiring every investigation to adopt one comprehensive metaphysical doctrine. The role of the present section is to specify how Generative Relational Theory (GR) and Generative Relational Economics (GRE) can remain compatible with alternative ontological commitments while preserving the analytical distinctions already established among ontological reality, relational reality, manifestational reality, representation, scale, and historical emergence. Its objective is to replace a vague appeal to pluralism with explicit conditions of compatibility. The discussion proceeds through ontological pluralism, substance-oriented, process-oriented, relation-oriented, and domain-specific ontologies, followed by criteria for translation, coexistence, and operative disagreement.

Compatibility in this context is weaker than identity and stronger than mere tolerance. Two ontological frameworks can be compatible for a specified inquiry when their commitments can coexist without contradiction at the level required by that inquiry, when their vocabularies can be related through an explicit translation or division of explanatory labor, or when disagreement concerning stronger claims remains non-operative for the local analysis. Compatibility can therefore be partial, asymmetric, scale-dependent, or question-dependent. A framework may be compatible with GR for one class of claims while conflicting with it for another. (Whitehead 1929; Quine 1948; Archer 1995)

Definition 4.176 (Local ontological compatibility). Two ontological commitments are *locally compatible* with respect to a specified inquiry when their jointly retained claims do not generate an operative contradiction for the objects, scales, causal relations, manifestations, or explanatory tasks under study, and when any translation between their terms is stated at the level actually warranted.

Definition 4.176 places scope at the center of compatibility. It avoids treating ontological frameworks as packages that must be accepted or rejected in their entirety. A substance-oriented account may provide a useful specification of relatively enduring entities while a process-oriented account explains their transformation. A relation-oriented account may clarify dependence among those entities while a legal ontology specifies institutionally recognized persons, offices, rights, and obligations. The resulting analysis can be coherent when the commitments are assigned clear roles and when a local translation does not silently become a claim of universal equivalence.

This position extends the methodological standing established in Section 4.19. Relational-generative analysis earns its place through explanatory performance, explicit scope, and revisability. Compatibility with alternative ontologies follows from the same discipline. GR can use an ontology strongly where its commitments matter and suspend stronger disputes where they do not. Such suspension preserves disagreement rather than dissolving it. A physicalist, phenomenological, processual, institutional, or relation-oriented account can therefore remain distinct even when each contributes to a shared analysis.

4.20.1 Ontological Pluralism

Ontological pluralism, as used here, concerns the possibility that different domains, scales, questions, or modes of access require different ontological vocabularies. It does not entail that every ontology is equally adequate, that contradictions are harmless, or that reality changes freely with description. Pluralism instead recognizes that an inquiry can encounter heterogeneous kinds of object and that one vocabulary may illuminate some structures while obscuring others.

The three-register architecture already creates a disciplined form of plurality. Ontological reality concerns what obtains independently of a particular manifestation; relational reality concerns what emerges, persists with sufficient stability, and exercises causal efficacy within a generative-relational system; manifestational reality concerns what becomes available under a specified manifestational configuration. These distinctions do not determine whether the ultimate constituents of a domain are substances, processes, relations, fields, events, organisms, informational structures, or something else. They organize questions that can be posed across several such commitments.

Pluralism also appears across scale. Section 4.17 showed that a configuration can be represented as an entity at one scale and as a distributed process or relation at another. The availability of both descriptions does not imply that each is equally useful for every task. A corporation may be treated as a juridical entity for contract and liability, as an organizational process for internal coordination, as a network for supply-chain dependence, and as a historical institution for long-run transformation. Each view can identify real features while selecting a different resolution.

Claim 4.177 (Plurality with constraint). *Ontological plurality is analytically productive when each adopted vocabulary identifies a specified object or scale, preserves the causal and evidential constraints relevant to that object, and exposes rather than conceals the points at which translation across vocabularies becomes uncertain or fails.*

Claim 4.177 makes pluralism answerable to evidence and explanation. A plurality of models does not provide a license to move opportunistically among incompatible claims. Where two ontologies produce different causal expectations, different identity criteria, or different accounts of persistence, the disagreement becomes operative and should be investigated. Where they differ only in stronger metaphysical commitments that do not change the local analysis, ontological suspension can preserve both possibilities.

This approach also guards against premature synthesis. A translation between two ontologies can be useful without establishing that they refer to the same structure

in the same way. The category-theoretic language introduced in Section 4.16 offers one formal illustration: a mapping can preserve selected structures while leaving other structures unmapped. Ontological comparison should be treated similarly. The analyst should identify what is preserved, what changes, and what remains outside the translation.

4.20.2 Substance-Oriented Ontologies

Substance-oriented ontologies give analytical priority to entities that possess some degree of persistence, identity, or bearer-like continuity across change. GR can accommodate such entities when their identification is empirically or conceptually useful. Relational-generative analysis does not require every object to dissolve into relations. It asks whether the capacities, persistence, or transformation of the object depend materially on relations and generative histories.

This compatibility is especially important in economics. Persons, firms, assets, commodities, contracts, infrastructures, and states are frequently treated as units with relatively stable identities. Many economic questions require precisely such units. Accounting, ownership, liability, inventory, demographic measurement, and contractual analysis become difficult when every entity is redescribed at every moment as a fully decomposed relational configuration. A substance-oriented representation can therefore provide an appropriate local ontology for a specified task.

GR adds two questions to that representation. First, what supports the persistence and identity criteria by which the entity remains identifiable? Second, which capacities attributed to the entity are generated or transformed through relations? These questions can deepen a substance-oriented analysis without invalidating the entity. A firm can remain a firm while its productive capacity depends on labor, law, finance, technology, suppliers, and customers. A person can remain an embodied individual while economic agency depends partly on education, recognition, legal status, infrastructure, and social reproduction.

Remark 4.178 (Entity persistence and relational constitution). A relatively enduring entity can be treated as an analytical bearer of properties while some of those properties, capacities, or identity conditions are explained relationally. Compatibility therefore depends on distinguishing the persistence of the entity from the provenance of every feature attributed to it.

This distinction prevents two opposite reductions. Entity-centered analysis can become insufficient when it treats relationally generated capacities as intrinsic and history-free. Relational analysis can become insufficient when it treats the identification of entities as analytically dispensable even where persistent units carry causal, legal, biological, or institutional significance. Local compatibility permits the two descriptions to coexist when their scopes are explicit.

A stronger metaphysical substance doctrine may eventually conflict with a stronger metaphysical relational doctrine. GR need not resolve that conflict in advance. Under ontological suspension, the operative question is whether the dispute changes the causal, historical, or manifestational claims required by the inquiry. If it does, the disagreement should be made explicit. If it does not, the analysis can proceed while preserving the stronger disagreement for another level of inquiry.

4.20.3 Process-Oriented Ontologies

Process-oriented ontologies emphasize becoming, transformation, activity, event, and temporal organization. Section 4.15 already established that GR can treat processes and events as analytically primary where extended transformation carries the explanatory burden. This orientation is highly compatible with generative analysis because generativity concerns how configurations produce, reproduce, transform, and dissolve subsequent configurations over time.

Compatibility nevertheless requires precision. A generative account need not claim that process is metaphysically prior to every object. An entity can be analyzed through the processes that sustain it while retaining an entity-level identity for other purposes. Dynamic persistence provides one bridge. A phenomenon can remain sufficiently stable to count as a relationally real object even when its stability depends on ongoing processes of maintenance, repair, reproduction, recognition, or exchange.

Economic institutions illustrate this point. Money, organizations, markets, property arrangements, and occupational categories persist through repeated practices, legal acts, accounting routines, infrastructural operations, and expectations. Their persistence can therefore be processually sustained. A process-oriented view helps explain how stability is achieved rather than treating stability as a primitive fact. GR can adopt that explanation while continuing to refer to money, organizations, or institutions as relationally real objects at the scale where their causal efficacy is coherent.

Process orientation also supports the historical dimension of GRE. Generative consequences may unfold across multiple timescales, and later relations can incorporate the products of earlier processes. The recursive architecture of Section 4.11 becomes especially useful here: manifestations and representations can enter subsequent processes and alter the conditions of later emergence. Historical becoming is therefore compatible with temporary stabilization.

Claim 4.179 (Process-compatible persistence). *The persistence required for relational reality is compatible with process-oriented ontology when stability is produced through sufficiently continuous or recurrent processes that preserve an object's causal efficacy over the relevant spacetime interval.*

Claim 4.179 avoids equating persistence with immobility. It also preserves the strict relational-reality criterion. A transient process can be manifestationally real without yet constituting a persistent relational object; a recurrent or self-maintaining process may itself satisfy the persistence, stability, and causal-efficacy conditions required for relational reality. The relevant analytical unit can therefore be an enduring object, an organized process, or a coupled object-process description.

4.20.4 Relation-Oriented Ontologies

Relation-oriented ontologies assign strong explanatory or metaphysical significance to relations, relational positions, or structures of dependence. GR has an obvious affinity with this family of views because relational emergence is central to its analytical vocabulary. The affinity remains deliberately limited by ontological suspension. Relational analysis can be indispensable without establishing that relations are uni-

versally prior to relata or that every domain is exhaustively relational.

A relation-oriented ontology becomes especially useful where objects acquire capacities through position, coupling, recognition, dependence, or interaction. Trust, authority, institutional office, market access, indebtedness, legal standing, and organizational role provide examples in which the relevant reality cannot be adequately specified through isolated intrinsic properties alone. The relational configuration contributes directly to what the object can do and to how it affects subsequent processes.

GR's definition of relational reality, however, is more selective than a general declaration that everything is relational. An object becomes relationally real when it emerges within a generative-relational system, persists with sufficient stability over a relevant region or interval of spacetime, and exercises causal efficacy within that system. The criterion therefore identifies a mode of reality within a system rather than asserting a complete metaphysical inventory of the world.

This selectivity allows GR to remain compatible with weaker and stronger relation-oriented ontologies. A weak relational ontology may claim that some properties are essentially relational. A stronger version may treat relations as ontologically fundamental. GR can work with either where the local analysis concerns structures both frameworks recognize. The stronger dispute can remain suspended until it becomes operative.

The distinction between relational reality and manifestational reality is also important here. A relation can be causally operative before a subject perceives or symbolizes it. Conversely, manifestations and symbols can enter the relation and reshape it. Relation-oriented ontology therefore gains a historical and recursive dimension without requiring manifestation to constitute all relational reality. The graph architecture among the registers remains available under several strengths of relational metaphysical commitment.

4.20.5 Domain-Specific Ontologies

Many inquiries rely on ontologies whose terms are specific to a domain. Law distinguishes persons, offices, rights, duties, jurisdictions, instruments, procedures, and recognized acts. Economics distinguishes assets, liabilities, commodities, firms, households, prices, contracts, and forms of capital. Biology, psychology, computation, ecology, and engineering each employ their own objects and identity criteria. These domain ontologies can carry substantial explanatory and practical force even when their relations to ultimate ontology remain unsettled.

GRE therefore requires a disciplined interface between domain-specific ontology and GR's broader analytical architecture. Domain terms should retain the distinctions that make them operationally meaningful. A legal person should not be translated automatically into a network node; a price should not be reduced automatically to a symbol; a corporation should not be reduced automatically to a set of interpersonal relations; an artificial operational subject should not be assumed to share the ontology of a human subject because both can occupy similar economic interfaces.

The three-register architecture can organize these domain terms without replacing

them. A legal status can be analyzed as relationally real when institutional relations stabilize it and it causally affects what actors can do. Its certificates, registry entries, judgments, and identifiers belong to manifestational and representational processes. Questions about the ultimate ontology of personhood, normativity, or institutional facts can remain open where the economic analysis does not require their resolution.

Domain-specific ontologies also matter for intervention. Institutions act through operational categories. Tax systems, courts, firms, welfare agencies, payment networks, and computational platforms require classifications that determine eligibility, ownership, responsibility, access, or transfer. Such categories can become relationally real because they structure subsequent action. Their practical necessity does not guarantee epistemic adequacy or normative legitimacy. GRE can therefore study the generative effects of domain categories while keeping their truth, justice, and ontological fundamentality distinct.

Definition 4.180 (Domain-ontology interface). A *domain-ontology interface* is an explicit mapping between a domain's operational objects and GR's analytical distinctions that preserves the domain-specific identity conditions required for the inquiry while identifying where relational emergence, manifestation, representation, scale, or historical transformation materially affect those objects.

Definition 4.180 treats translation as selective. The objective is to preserve enough structure for cross-domain inquiry without erasing what makes the source domain distinctive. Some translations may be close; others may preserve only a few relations; still others may fail because the relevant objects lack a warranted common comparison space. The non-necessary comparability developed earlier in the chapter therefore applies equally to ontological vocabularies.

4.20.6 Conditions of Compatibility

The compatibility developed in this section can be summarized through a set of conditions. These conditions are methodological criteria for joint use rather than axioms of a final ontology. They specify when alternative commitments can enter a GR or GRE analysis without conceptual collapse.

First, the object and scale of each ontology should be explicit. A vocabulary suited to embodied persons may operate differently from one suited to organizations, legal offices, or computational agents. Second, identity and persistence criteria should be stated where they matter. What counts as the same firm, subject, institution, process, or relation can vary across domains and scales. Third, causal claims should remain attached to the mechanisms recognized by the relevant ontology rather than being translated into a generic relational idiom that loses mechanism-specific content.

Fourth, mappings across ontologies should specify what they preserve. A translation may preserve causal ordering, functional role, legal status, manifestational form, organizational position, or selected relational structure while losing other dimensions. Fifth, residual structure should remain visible. A successful local correspondence does not imply global equivalence. Sixth, stronger disagreements should be classified by whether they are operative for the inquiry. Ontological suspension is appropriate where disagreement does not alter the retained claims; explicit adjudication becomes necessary where competing commitments generate different expect-

tations, identity conditions, or interventions.

Claim 4.181 (Compatibility discipline). *Alternative ontological commitments can be jointly used within GR or GRE when their scopes, identity criteria, causal roles, translation rules, and residual disagreements are explicit, and when stronger metaphysical disagreements are suspended only to the extent that they remain non-operative for the inquiry.*

Claim 4.181 makes compatibility an active analytical task. It requires more than placing several vocabularies side by side. The analyst must specify how they interact and where they diverge. This requirement is especially important in interdisciplinary research, where the same term can carry different identity criteria across law, economics, computation, psychology, or political theory.

Compatibility also has temporal conditions. An ontology that is locally adequate at one historical stage may become less adequate when new entities, infrastructures, practices, or manifestations emerge. Section 4.18 showed that ontological categories can possess historical materiality. The compatibility relation among ontologies can therefore change as well. A distinction that once aligned across institutional and technical domains may later fracture; new interfaces can create forms of subjecthood or organization that require revised mappings.

A final condition concerns normativity. Ontological compatibility does not settle normative compatibility. Two frameworks may describe the same institution coherently while supporting different judgments about legitimacy, justice, welfare, or responsibility. GRE should preserve that distinction. The ontological question concerns what kinds of objects and relations the analysis recognizes and how they persist or act. Normative evaluation requires additional premises and arguments.

The resulting position is a constrained ontological pluralism. GR offers a common analytical architecture for distinguishing ontological, relational, and manifestational reality; tracing generative histories; identifying causal efficacy; and examining recursive mediation. That architecture can interface with substance-oriented, process-oriented, relation-oriented, and domain-specific ontologies without treating them as interchangeable. Compatibility is earned locally through explicit scope, preservation of relevant structure, and disciplined treatment of residual disagreement.

Section 109 therefore extends ontological suspension into a positive method of interdisciplinary coexistence. Alternative ontologies can contribute different resolutions of the same domain, or address different domains altogether, while GR provides a vocabulary for asking how those resolutions connect, where they fail to connect, and when their disagreements become causally or analytically operative. The next section narrows the chapter's claims by specifying the limits of GRE's ontological commitments and the kinds of metaphysical, empirical, and normative conclusions that the present architecture does not by itself establish.

4.21 Limits of GRE Ontological Claims

The ontological architecture developed in this chapter makes substantive claims about emergence, persistence, stability, causal efficacy, manifestation, representation, scale, and historical transformation. Its usefulness depends equally on

specifying the claims that do not follow from those commitments. The role of the present section is therefore to establish the boundary of GRE's ontological claims before the chapter turns from reality to knowledge. Its objective is not to weaken the three-register architecture, but to distinguish what the architecture warrants from stronger conclusions that require additional metaphysical, empirical, epistemic, or normative argument. The discussion proceeds through the boundary of established claims, ontological underdetermination, the limits of formal ontology, limits of cross-domain generalization, and openness to ontological revision.

A theory can become less precise when the scope of its strongest concepts expands faster than the evidence supporting them. This risk is particularly important for a framework that uses terms such as *reality*, *emergence*, *causal efficacy*, and *subjecthood*. GRE therefore treats explicit limitation as part of ontological discipline. A claim that money, an institution, a self-model, a form of trust, or another object is relationally real says something substantive: the object has emerged within a generative-relational system, persists with sufficient stability over a relevant spacetime region or interval, and exercises causal efficacy within that system. The claim does not by itself determine the object's ultimate ontological constitution. Similar restraint applies to manifestational reality, symbolic representation, formal modeling, and cross-scale analysis. (Fraassen 1980; Giere 2004; Longino 1990)

Definition 4.182 (Ontological claim boundary). An *ontological claim boundary* is the explicitly stated limit separating the commitments warranted by a GR or GRE analysis from stronger claims concerning ultimate constitution, exhaustive classification, universal applicability, epistemic truth, or normative validity that require additional argument.

Definition 4.182 makes limitation internal to the method rather than an afterthought. A strong local claim can coexist with restraint concerning stronger questions. GRE can, for example, assert the causal reality of a monetary institution while suspending the question of whether institutional reality is reducible to physical, psychological, informational, legal, or other structures. It can analyze the relational reality of an experienced continuity of self without claiming to have settled the ontology of consciousness. It can treat a religiously mediated object as relationally real within a system when the persistence, stability, and causal-efficacy conditions are satisfied without inferring from that relational reality either the independent ontological existence or non-existence of the object in question.

This distinction is also important for the architecture of the three registers. Ontological reality, relational reality, and manifestational reality organize different questions. They are not asserted to be an exhaustive partition of everything that can exist, nor are they three substances from which reality is composed. Their graph-like relations are likewise analytical commitments rather than a universal causal diagram imposed on every domain. The framework earns explanatory force by preserving these distinctions while allowing their concrete realization to remain open.

4.21.1 Boundary of Established Claims

The first limit concerns what the chapter has actually established. The three-register architecture distinguishes ontological, relational, and manifestational questions; it

specifies a strict criterion for relational reality; it treats manifestation as condition-dependent and potentially recursive; and it allows representations, mappings, institutions, categories, and subjects to become causally operative within generative-relational systems. These commitments provide a disciplined ontology for GRE inquiry. They leave several stronger questions unresolved.

GRE does not establish that relation is the fundamental substance of reality. Relational-generative analysis can be indispensable for a specified economic object while stronger metaphysical questions remain suspended. Section 4.19 therefore treated relational-generative analysis as a scoped methodological commitment, and Section 4.20 showed how this commitment can coexist locally with substance-oriented, process-oriented, relation-oriented, and domain-specific ontologies. The explanatory importance of relations does not by itself establish their metaphysical priority.

The chapter also does not establish that every persistent pattern is relationally real. Persistence and stability are necessary components of the criterion developed in Section 4.5, but causal efficacy within the relevant generative-relational system remains essential. A regularity that merely recurs in an analyst's description may lack the causal participation required for relational reality. Conversely, a relational reality can be dynamically maintained rather than static. Stability concerns maintained organization over an inquiry-relevant region or interval, not immobility.

Causal efficacy does not establish epistemic truth. Section 4.10 showed that representations can become relationally real through their consequences even when they are partial, inaccurate, or false concerning their targets. A classification can alter eligibility, a mistaken belief can reorganize conduct, and an inaccurate metric can redirect resources. Their effects are real without making their representational content true. The same distinction applies to manifestational salience: what becomes strongly visible or institutionally legible need not provide an exhaustive or privileged disclosure of the underlying relational configuration.

Causal efficacy also does not establish normative validity. Money, legal categories, institutional hierarchies, identities, classifications, and value regimes can become highly stable and causally powerful while remaining ethically contested or unjust. Persistence can indicate successful reproduction rather than justification. The ontology developed here therefore supplies objects for normative analysis but does not convert historical survival, institutional recognition, or generative power into moral endorsement.

Nor does relational reality require prior perception, awareness, or symbolization. This point creates a limit in the opposite direction: the absence of manifestation to a particular subject does not justify the conclusion that no relationally real structure exists. Contingency provides an important example. A historical-generative configuration through which a contingent encounter becomes possible can already be relationally organized before that contingency is consciously recognized, aesthetically judged, or symbolically articulated. The later manifestation may reveal, select, amplify, or transform part of that structure without creating the entire relational reality at the moment of perception.

Claim 4.183 (Non-entailment discipline). *Within GRE, persistence does not by itself*

entail relational reality; relational reality does not entail ontological fundamentality; manifestation does not entail exhaustive disclosure; causal efficacy does not entail epistemic truth; and historical stabilization does not entail normative validity.

Claim 4.183 is a compact statement of the chapter's claim boundary. Its purpose is not skeptical. Each term remains capable of supporting strong conclusions within its appropriate domain. The discipline lies in resisting movement from one kind of warranted claim to another without an explicit bridge.

The same restraint applies to comparison. Two objects can both satisfy the criterion for relational reality without occupying a common metric space, a common ordering, or even a justified common comparison domain. The hallucinatory flower discussed earlier and an ordinarily perceived physical flower may each be relationally real in relevant systems while differing radically in generative structure. Shared reality-status therefore establishes neither structural similarity nor commensurability. Any comparison must specify what relation, mapping, invariant, or practical criterion makes the comparison meaningful.

4.21.2 Ontological Underdetermination

The second limit arises because the evidence available to an inquiry may support several ontological interpretations. The non-invertibility of manifestation developed in Section 4.9 provides one route to this result. Different relational configurations can generate similar manifestations, while similar relational configurations can manifest differently under changing conditions. Even when an observed pattern is robust, reconstruction from manifestation toward underlying organization can remain underdetermined.

Ontological underdetermination is broader than manifestational non-invertibility. The same relationally effective organization can sometimes be described through different ontological vocabularies. A firm can be modeled as a juridical entity, an organizational process, a contractually structured nexus, a network of dependencies, or a historically reproduced institution. These descriptions need not be equivalent, and each can reveal structures hidden by the others. Yet available evidence may fail to determine which vocabulary, if any, captures the ultimate constitution of the object.

Definition 4.184 (Ontological underdetermination). *Ontological underdetermination* obtains when the evidence, causal regularities, manifestations, and successful explanatory relations available within a specified inquiry remain compatible with more than one materially distinct ontological interpretation.

Definition 4.184 is inquiry-relative. Additional evidence, a different scale, a new intervention, or a stronger explanatory demand can reduce underdetermination. The concept therefore differs from a permanent declaration that ontological questions are undecidable. It identifies a present relation between evidence and claim strength.

Underdetermination is especially relevant to the examples that motivate ontological suspension. Consider the continuity of the experienced "I." GRE can investigate how bodily continuity, memory, recognition, narrative, institutional identity, and repeated action participate in stabilizing a relationally real self. Such analysis can explain important causal consequences of selfhood while several ontologies of con-

sciousness and personal identity remain compatible with the retained evidence. The relational analysis is not empty because the metaphysical question remains open; neither does explanatory success in the relational domain settle that question automatically.

A similar point applies to money. Monetary reality can be studied through institutional authority, legal designation, accounting systems, expectations, payment infrastructures, conventions, and material tokens. These relations can sustain an object with substantial persistence and causal efficacy. Several ontological interpretations of institutional facts can nevertheless remain available. GRE needs to resolve among them only when their differences become operative for the economic question under study.

Underdetermination also explains why the ontological register cannot be identified with whatever remains after relational and manifestational analysis. Ontological reality is not a residual box containing an unknown substrate. It names the question of what obtains beyond the commitments licensed by a particular manifestation or relational model. In some inquiries, stronger ontological claims can be justified; in others, suspension remains the more disciplined position.

Claim 4.185 (Strength calibration under underdetermination). *When materially distinct ontological interpretations remain compatible with the evidence and explanatory relations available to an inquiry, GRE should calibrate its claims to the structure jointly supported by those interpretations unless the stronger disagreement becomes operative for explanation, prediction, intervention, or evaluation.*

Claim 4.185 preserves both explanatory ambition and metaphysical restraint. It permits the analysis to advance where common structure is well supported while keeping stronger disagreements visible for later investigation.

4.21.3 Limits of Formal Ontology

Formal languages can sharpen ontological reasoning by making identity criteria, mappings, compositional rules, invariants, state spaces, constraints, and transformation structures explicit. This chapter has used such resources selectively, including dynamical language, mapping notation, scale-relative descriptions, and a category-theoretic view. Their precision can expose assumptions that ordinary prose leaves implicit. Their limits are equally important.

A formal representation does not establish that its mathematical objects are the literal constituents of the domain. A graph can represent relational dependence without implying that a society is fundamentally a graph. A state space can represent possible configurations without implying that the represented system possesses that space as an independent metaphysical object. A category can encode objects and morphisms relevant to an inquiry without establishing categorical ontology. A field model can capture distributed interaction without proving that the social domain is fundamentally a field.

This distinction follows from the chapter's treatment of mappings. Formalization selects objects, relations, equivalence criteria, boundaries, and admissible transformations. Those selections can be analytically justified while remaining choices of

representation. Different formalisms can preserve different structures of the same domain. Their comparative success is an empirical and explanatory matter rather than a direct route from mathematical elegance to ontological truth.

Formal ontology also encounters the problem of representation-induced identity. Once an analyst chooses a state description or category of objects, elements that differ outside that formalization may become identical inside it. This can be useful. An economic model may intentionally treat two heterogeneous realizations as equivalent because they perform the same relevant function. Section 4.17 described such local equivalence under multiple realization. The loss of distinction should nevertheless remain visible. Formal equality inside a model can represent a justified abstraction without entailing ontological identity outside it.

Remark 4.186 (Formal exactness and ontological strength). Greater exactness within a formal representation increases clarity concerning the commitments of that representation. It does not, by itself, increase the ontological strength of the claim connecting the representation to the world.

Remark 4.186 is especially important in interdisciplinary work. Mathematical sophistication can make assumptions more precise while leaving the empirical warrant for those assumptions unchanged. A beautifully defined equivalence relation, functor, potential function, attractor, or metric still requires a defensible interpretation linking it to the domain.

Formal systems can also create commensurability that the source domain did not possess in advance. Section 4.10 described symbolic commensuration; mathematical representation can perform an analogous operation. Assigning values to a common coordinate, metric, utility, score, or latent space can make comparison possible for a specified purpose. The resulting comparison may be operationally powerful and even institutionally consequential. Its existence inside the formalism does not prove intrinsic commensurability among the represented realities.

GRE therefore treats formal ontology as an instrument for disciplined representation, inference, and comparison. It can contribute to ontological inquiry by exposing structure, generating testable implications, and revealing invariant or transformational relations. It remains answerable to the domain and to the evidential conditions under which its objects are interpreted.

4.21.4 Limits of Cross-Domain Generalization

A further limit concerns the transfer of ontological categories across domains. GR is intentionally broad enough to support inquiry into economic, legal, political, educational, technological, ecological, and other relational systems. Breadth creates the temptation to assume that concepts established in one domain transfer unchanged to another. GRE resists that inference.

The criterion for relational reality is deliberately general, but its operationalization is domain-sensitive. Persistence for a market convention can have a different temporal scale from persistence for an ecological relation, a legal office, a computational process, or an interpersonal trust relation. Causal efficacy can also take different forms: changing resource flows, altering legal possibilities, constraining behavior,

reorganizing attention, modifying biological processes, or transforming symbolic expectations. The general criterion gains content only through domain-specific mechanisms.

The same limitation applies to subjecthood. Human persons, corporations, states, algorithms, animals, and collective organizations can occupy causally important positions in economic systems. Similarity in economic function or manifestational output does not establish identical forms of subjecthood. Section 4.12 therefore distinguished human subjecthood from artificial operational subjecthood and treated manifestational equivalence as local. Cross-domain generalization should preserve such differences rather than elevate a shared interface into universal identity.

Value and meaning provide another case. A common symbolic scale can create practical commensuration among heterogeneous values, while the underlying value structures remain non-equivalent. Likewise, a concept of meaning developed for linguistic interpretation cannot automatically be extended to every biological, computational, or institutional process that exhibits patterned response. Domain transfer requires an explicit account of what is preserved and what changes.

Definition 4.187 (Cross-domain transfer condition). A *cross-domain transfer condition* specifies the properties, relations, causal roles, identity criteria, scales, and evidential assumptions that must be preserved for an ontological or relational concept to be used across distinct domains without silently changing its meaning.

Definition 4.187 makes generalization a constructive task. A concept can travel, but its travel requires a map. Sometimes the transfer will preserve a substantial common structure. Sometimes only an analogy survives. Sometimes the domains lack a warranted common space of comparison. The possibility of failure is part of disciplined generalization.

This restraint also protects GR from becoming a vocabulary that redescribes everything in the same terms. If every phenomenon is called a relation, field, attractor, emergence, or generative process without domain-specific criteria, explanatory content diminishes. A broad framework remains useful when its general concepts constrain inquiry while allowing local sciences and disciplines to supply mechanisms, measurements, categories, and evidence that cannot be derived from the framework alone.

For GRE, the economic domain remains primary. Connections to law, politics, ecology, psychology, computation, education, and other domains can be necessary because economic relations cross those boundaries. The ontological interface should nevertheless respect the concepts through which those domains identify their own objects. Cross-domain synthesis is warranted where the interfaces are explicit; conceptual annexation is not.

4.21.5 Openness to Ontological Revision

The final limit is temporal. GRE's ontological categories are intended to remain revisable when new evidence, new forms of organization, new scales, new modes of manifestation, or new conceptual distinctions alter what the framework can adequately describe. This openness follows from the historical materiality developed

in Section 4.18 and from ontological suspension developed in Section 4.3. A historically situated research programme should not treat its present ontology as immune to the processes it studies.

Ontological revision can occur at several levels. A previously useful object category can become inadequate when its internal heterogeneity becomes causally important. A relation once treated as background can become a central mechanism. A manifestational distinction can cease to track the relational structure it was designed to represent. A new technological or institutional configuration can generate objects for which existing identity criteria are unstable. A formal model can reveal overlooked invariants, while empirical evidence can show that an assumed invariant is historically contingent.

Revision does not require abandoning earlier concepts whenever novelty appears. Many categories remain useful across substantial historical change. The relevant question is whether the retained category continues to preserve the distinctions needed for explanation, intervention, and evaluation. Revision becomes warranted when anomalies are systematic, when causal mechanisms fall outside the existing object structure, when previously excluded differences become consequential, or when alternative ontological descriptions achieve materially stronger explanatory adequacy.

Claim 4.188 (Ontological revisability). *GRE should revise an ontological commitment when sustained evidence or explanatory failure shows that the commitment no longer preserves distinctions necessary for identifying relevant objects, causal relations, persistence conditions, manifestations, scales, or transformations within the inquiry.*

Claim 4.188 makes revisability conditional rather than fashionable. Ontological novelty is not inferred merely from new terminology, new technology, or a new representation. Revision should answer to changes in explanatory adequacy and to the realities under investigation.

The three-register architecture itself is subject to this principle. Ontological reality, relational reality, and manifestational reality currently provide a productive set of distinctions for the GR research programme. Their usefulness does not grant them immunity from criticism. Future inquiry may require finer subdivisions, different mappings among registers, domain-specific extensions, or alternative concepts that preserve relevant phenomena more adequately. Such revision would be compatible with the framework's present commitments because openness has been built into those commitments from the beginning.

This openness also prevents ontological restraint from becoming intellectual passivity. Suspension is provisional where inquiry can proceed, and stronger ontological questions can be reopened when they become operative. Formalization can be strengthened when measurement and mechanism justify it. Cross-domain concepts can be generalized when preservation conditions are demonstrated. Alternative ontologies can be compared more sharply when they generate different empirical or practical consequences. The boundary established in this section therefore marks a frontier for further inquiry rather than a prohibition against stronger claims.

Section 110 completes the chapter's explicit account of ontological scope. GRE commits to enough reality to analyze generative systems seriously: relations can produce

stable and causally efficacious objects; manifestations and representations can participate recursively in later reality; subjects, institutions, value, meaning, processes, and categories can acquire historically situated forms of reality; and these structures can be compared across scales when warranted. The framework simultaneously leaves ultimate constitution, exhaustive ontology, universal commensurability, complete formal capture, and unrestricted cross-domain transfer open to additional argument. With that boundary in place, the remaining problem is epistemic: given realities that can exceed, precede, and be transformed by their manifestations, how can an inquiry responsibly know them? Section 111 turns to that transition.

4.22 Reality, Access, and Epistemic Reconstruction

The ontological work of this chapter produces an epistemic problem rather than an epistemic solution. GR has distinguished ontological reality, relational reality, and manifestational reality; specified the conditions under which an object may be treated as relationally real; shown that manifestation is generated through historically and relationally formed conditions; and established that manifestations and representations can recursively alter subsequent relational reality. These commitments clarify what kinds of realities an inquiry may address, but they also make clear why knowing those realities cannot be reduced to reading their visible expressions at face value. The role of this final section is therefore to state the epistemic consequences of the three-register architecture and to prepare the research programme developed in Chapter 5. Its objective is not to provide a complete epistemology in advance, but to identify the structure of the problem: reality can exceed manifestation, manifestation can disclose without exhausting, representation can mediate without transparently copying, and inference must reconstruct relations through historically situated traces.

This transition matters because ontological restraint changes what counts as an adequate epistemology. Once GRE suspends stronger claims about ultimate constitution where they are unnecessary, knowledge cannot be defined as possession of a final metaphysical description. At the same time, ontological suspension does not license a retreat to unconstrained interpretation. Relational realities remain capable of persistence, stability, and causal efficacy; manifestations remain constrained by the configurations through which they emerge; and representations can be compared with evidence, revised, and tested against subsequent consequences. The epistemic task is consequently neither transparent access nor arbitrary construction. It is the disciplined reconstruction of realities whose accessibility is mediated, partial, historical, and potentially recursive. (Giere 2004; Longino 1990; Hacking 2002)

The chapter's preceding distinctions also imply that epistemology must preserve the difference between an object and the conditions under which that object becomes knowable. A monetary relation is not identical to an accounting entry that records it. A legal position is not exhausted by the administrative category through which it becomes legible. A configuration of trust may already be relationally real before it is explicitly acknowledged. A contingent structure can shape what later becomes aesthetically salient before a subject recognizes the encounter as contingent or beautiful. In each case, evidence may arise through manifestation, but the evidential form and

the relational reality to which it refers remain analytically distinct. Knowledge therefore requires attention not only to what appears, but also to the relations that make appearance possible and to the transformations introduced by observation, symbolization, measurement, and representation.

Claim 4.189 (Epistemic consequence of the three-register architecture). *If relational reality can precede particular manifestations, if manifestation is generated through condition-dependent relations, and if manifestations and representations can alter subsequent relational reality, then knowledge of a generative-relational system requires inquiry into both the realities under study and the relations through which evidence of those realities becomes available.*

Claim 4.189 gives the transition its basic form. The manifestation relation is not an invisible channel that epistemology can ignore. It becomes part of the object of inquiry. A trace can be informative because of how it was generated; a classification can be evidentially limited because of what its institutional production excludes; a repeated manifestation can acquire diagnostic value because the relevant mapping is sufficiently stable; and the same manifestation can become misleading when the mapping changes historically. Epistemic evaluation therefore extends from the manifested content to the generative conditions of manifestation itself.

4.22.1 Reality and Access

The first epistemic consequence concerns access. The three-register architecture does not assume that what is real is immediately available to inquiry. Ontological reality may remain partly or substantially under suspension. Relational reality can exist without having been perceived, named, measured, or symbolized by a particular subject. Manifestational reality marks a mode in which some aspect of a configuration becomes available under specified conditions, but availability itself is selective. What becomes accessible depends on embodiment, instrumentation, institutions, categories, interfaces, histories of attention, scales of observation, temporal position, and other relations that condition manifestation.

Access is therefore better treated as a relation than as a binary possession. An inquiry can have strong access to one aspect of a relational configuration and weak access to another. It can observe current outcomes while lacking historical trajectories. It can possess detailed administrative records while missing informal relations that organize those records' production. It can measure transactions precisely while having limited access to care, coercion, tacit obligation, fear, trust, or unrecorded exchange. It can encounter a stable manifestation while remaining uncertain about the multiplicity of relational structures compatible with it. The epistemic problem is shaped by the configuration of access rather than merely by the amount of information accumulated.

Definition 4.190 (Relational epistemic access). *Relational epistemic access* denotes the historically and materially situated set of relations through which an inquiry can obtain, preserve, compare, interpret, or intervene upon manifestations and traces relevant to a specified epistemic target.

Definition 4.190 treats access itself as analyzable. It can be widened by new instruments, archives, institutions, languages, computational methods, or relationships

of trust. It can be narrowed by secrecy, destruction of records, strategic concealment, exclusion from institutions, measurement design, linguistic barriers, unequal resources, or the disappearance of the relevant manifestations. Access can also be distributed. Different actors can hold different traces of the same relational configuration, while no single observer possesses a complete view. The conditions under which evidence is collected thus become part of the history of what can later be known.

This conception also separates absence of access from absence of reality. A relation can be hidden, latent, weakly manifested, or manifest through forms unavailable to a particular inquiry. The absence of a recorded manifestation may therefore have several possible interpretations. The relevant relation may not exist; it may exist without producing the expected trace; the trace may have been produced but not preserved; the observer may lack access to it; or the manifestation rule itself may have changed. The epistemic significance of absence depends on the manifestation and recording relations that connect the target to the evidential field.

The converse problem also matters. Abundant access to manifestations does not necessarily yield correspondingly strong access to relational structure. A platform can generate billions of behavioral observations while leaving motivations, external relations, unobserved constraints, institutional conditions, and counterfactual possibilities underdetermined. Administrative systems can accumulate detailed categories whose precision exceeds the adequacy of the ontology they presuppose. Computational abundance can increase the resolution of a manifestational field while reproducing the same blind spots at greater scale. Epistemic strength therefore depends on the relation between evidence and target, not merely on the volume of evidence.

Ontological suspension intensifies rather than weakens this requirement. When GRE declines to settle what an object ultimately is, it must be particularly explicit about what is actually accessible and what inferential bridge connects that access to the claim being made. The framework can know a relational reality through its persistence, causal consequences, historical traces, responses to intervention, or convergent manifestations without pretending that such knowledge exhausts the object's ontology. Knowledge can therefore be substantive while remaining scoped.

4.22.2 Manifestation and Evidential Access

Manifestation supplies a principal interface through which relational realities become evidentially accessible. The term *interface* is useful only if it is not mistaken for a neutral screen. Section 4.7 showed that manifestations emerge through generative-relational conditions, while Section 4.8 showed that sufficiently stable manifestation mappings can themselves become relationally real. An evidential manifestation therefore carries information about a target through a process whose own structure, stability, and history can affect what the evidence means.

Definition 4.191 (Evidential interface). An *evidential interface* is a manifestation relation, or organized set of manifestation relations, through which features of a specified epistemic target become available for observation, recording, comparison, interpretation, or intervention.

An evidential interface can be bodily, linguistic, institutional, legal, statistical, com-

putational, material, or composite. Pain may become accessible through speech, gesture, physiology, clinical examination, or patterns of action. Monetary relations may become accessible through prices, account balances, contractual records, settlement systems, observed exchanges, or institutional interventions. Trust may become accessible through repeated cooperation, expressed commitments, vulnerability, withdrawal, testimony, or responses under changing conditions. None of these manifestations is identical to the relational reality under study, and each preserves some distinctions while transforming or omitting others.

The evidential value of a manifestation consequently depends partly on knowledge of its production. A stable thermometer reading is informative because a sufficiently understood relation connects relevant physical conditions to the displayed value under specified conditions. A legal registry is informative because institutions generate, preserve, and update entries according to rules whose operation can be investigated. A survey answer is informative under a different manifestation regime involving question wording, interpretation, memory, incentives, social desirability, sampling, and response conditions. Treating all three simply as “observations” would conceal important differences in their generative pathways.

This approach also clarifies why directness is often a poor measure of epistemic quality. A manifestation can feel immediate while being generated through complex learned, embodied, institutional, or technological mediation. Conversely, a highly mediated trace can provide strong evidence when the relevant mapping is well characterized and independently tested. GRE therefore gives priority to reconstructing the evidential relation over ranking evidence by an intuitive scale from direct to indirect. The important questions concern what generates the trace, under what conditions, with what stability, and which relational differences the interface can preserve.

Manifestational plurality can strengthen inquiry when partially independent interfaces constrain the same target from different directions. Testimony, behavior, administrative records, historical documents, material traces, and intervention responses may each preserve different features and possess different failure modes. Convergence among them can reduce the compatibility set described in Section 4.9. Yet convergence is not automatic confirmation. Several manifestations can share the same upstream classification system, institutional incentive, database, or cultural convention and therefore reproduce a common error. Epistemic plurality requires attention to dependence among evidential interfaces rather than merely counting the number of observations.

The historical character of interfaces is equally important. The same symbol, indicator, behavior, or institutional record can change meaning when its manifestation relation changes. A category can persist linguistically while its legal criteria are revised. A price series can span periods in which market institutions, currencies, quality standards, or reporting practices differ. An expression of identity can operate differently after new institutions make that expression consequential. Historical epistemology therefore requires reconstruction not only of traces but of the changing rules through which those traces became possible.

Remark 4.192 (Evidence and manifestation). A manifestation becomes evidence for a specified claim only through an inferential account linking the manifestation, its generative conditions, and the epistemic target. Manifestational reality is therefore

broader than evidence, while evidence is manifestation interpreted within a warranted epistemic relation.

Remark 4.192 prevents the ontology of manifestation from silently becoming an epistemology. A manifestation can be real without being evidentially useful for a particular question. It can also be causally powerful while epistemically misleading. The transition to epistemology begins when inquiry asks what a manifestation licenses us to infer.

4.22.3 Representation and Inference

Representation introduces another layer of mediation. Section 4.10 distinguished symbolization from manifestation and showed that symbolic forms can stabilize categories, produce practical commensurability, and become causally effective in their own right. For epistemology, the central problem is that representations do not merely store already settled knowledge. They select variables, categories, units, scales, boundaries, temporal resolutions, equivalence relations, and conventions through which manifestations become inferentially tractable.

An economic model, statistical indicator, legal category, archival description, qualitative code, graph, narrative, database schema, or computational state representation therefore participates in the structure of inference. Each makes some distinctions explicit and leaves others implicit or unavailable. A representation can preserve the relations relevant to one problem while distorting those relevant to another. The adequacy of representation is consequently task-sensitive without becoming arbitrary: the relevant question is whether the representation preserves, transforms, or suppresses differences that matter for the claim under investigation.

The non-entailment discipline of Section 4.21 applies directly here. Formal exactness does not establish ontological identity. Predictive success does not by itself establish unique causal reconstruction. Institutional uptake does not establish truth. A representation can become relationally real because decisions, expectations, rights, or resource allocations are organized around it while remaining epistemically inadequate concerning its target. This possibility is particularly important in economics, where prices, ratings, rankings, risk scores, classifications, forecasts, and models can feed back into the systems they describe.

Inference must therefore distinguish at least three relations: the relation between a relational reality and its manifestations, the relation between manifestations and their representations, and the relation between those representations and the conclusions drawn from them. Weakness or transformation can enter at each stage. A model can faithfully encode a dataset whose measurement process is biased. A dataset can accurately preserve recorded manifestations while those manifestations are weak indicators of the target. A representation can preserve relevant evidence while an inferential procedure overstates what the evidence supports. Epistemic assessment requires locating uncertainty rather than compressing it into a single undifferentiated confidence judgment.

Claim 4.193 (Layered inferential accountability). *An epistemic claim concerning relational reality should remain accountable to the conditions that generated the relevant manifestations, the operations that represented those manifestations, and the inferen-*

tial rules through which the representation supports the claim.

Claim 4.193 does not require every inquiry to reconstruct an unlimited chain of mediation. It requires the chain to be extended far enough that omitted transformations are unlikely to reverse or materially alter the conclusion. This boundary is itself inquiry-relative. A descriptive task may require only a stable reporting relation. A causal claim may require intervention, temporal ordering, mechanism, or comparison among competing generative accounts. A normative decision may additionally require explicit value assumptions, legitimacy conditions, and treatment of affected subjects whose realities are poorly represented by the available data.

Representation can also change the object of inference. Once a category is published, a score assigned, a forecast circulated, or a model institutionalized, agents can respond. The representation then becomes part of the generative-relational system and can alter later manifestations. An epistemology adequate to GRE must therefore track performativity and recursion. Evidence generated after a representation becomes operative may partly reflect the representation itself. The line between observation and intervention can become historically entangled.

This entanglement is not an exceptional defect of social inquiry. It is a structural possibility wherever manifestational and representational forms enter subsequent causal relations. The epistemic task is to identify when that recursion is material. Chapter 5 will therefore treat representation not only as a medium of knowledge but also as a possible generator of later reality.

4.22.4 Inverse Reconstruction

The preceding points converge on an inverse problem. Inquiry commonly encounters manifestations, traces, records, behaviors, symbols, and outcomes and seeks to reconstruct the relational realities, histories, and generative mechanisms compatible with them. Section 4.9 established that manifestation can be non-invertible: multiple relational configurations may produce similar manifestations, while one relational configuration may produce different manifestations under changing conditions. Epistemology must determine what can responsibly be reconstructed despite that non-uniqueness.

The inverse problem is not simply a technical difficulty produced by insufficient data. Additional data can narrow a compatibility set, but non-invertibility can arise from the structure of the manifestation relation itself. Distinct histories may converge on the same current state. Different institutional arrangements may produce the same aggregate statistic. Different motivations can yield the same observed action. Different causal pathways can generate the same outcome. A single relation can manifest differently across cultural, historical, technological, or bodily conditions. Even an extremely detailed manifestational description may therefore leave relational structure underdetermined.

Definition 4.194 (Relational inverse problem). A *relational inverse problem* arises when an inquiry seeks to reconstruct one or more relational configurations, histories, or generative mechanisms from available manifestations and traces for which more than one materially distinct reconstruction remains compatible with the evidence and maintained assumptions.

Definition 4.194 shifts the epistemic aim away from an automatic demand for unique inversion. Sometimes unique reconstruction can be achieved under strong and justified constraints. In other cases, the appropriate epistemic object is a set of plausible relational configurations, an equivalence class, a partially ordered family of explanations, a distribution over possibilities, or a set of invariants shared across competing reconstructions. The form of knowledge should match the identifiability permitted by the evidence.

This point is especially important for historical and social inquiry. A surviving archive is not the past itself. It is a historically generated manifestational field shaped by preservation, destruction, classification, institutional power, material durability, access, and later interpretation. A current institution can preserve traces of earlier relations while also reorganizing them. A subject's present testimony can carry memory, reinterpretation, later categories, and current stakes. Reconstruction therefore requires attention to the history of the evidential interface, not only the content of surviving evidence.

Intervention can sometimes strengthen inverse reconstruction because different candidate relational structures can respond differently when conditions change. Comparative cases, natural variation, experiments, policy changes, historical discontinuities, and perturbations can expose differences hidden under ordinary conditions. Yet intervention itself can transform the system, and the relation between intervention response and antecedent structure requires argument. GRE therefore treats intervention as one epistemic resource among others rather than a universal solution to non-invertibility.

The inverse problem also explains why heterogeneous methods can be epistemically complementary. Historical reconstruction can identify trajectories concealed by cross-sectional data. Ethnographic or interpretive inquiry can recover meanings and relations compressed by administrative categories. Statistical methods can estimate regularities across cases. Network and dynamical models can expose structural dependencies. Computational simulation can test whether proposed mechanisms are capable of generating observed patterns. Legal analysis can reconstruct institutional rules that shape manifestational regimes. None of these methods possesses automatic priority. Their contribution depends on which dimensions of the compatibility set they constrain and which assumptions they introduce.

Claim 4.195 (Epistemic adequacy under non-invertibility). *When unique reconstruction is not warranted, epistemic adequacy consists partly in preserving the plurality of materially compatible relational explanations, identifying the evidence that distinguishes among them, and calibrating claims to the invariants or differences that the available access can support.*

This claim makes uncertainty structurally informative. Multiple compatible explanations indicate where manifestation fails to preserve distinctions, where access is limited, or where additional assumptions are carrying the inference. The objective is not to celebrate indeterminacy but to locate it. An explicit compatibility set can guide new observation, archival search, intervention, comparison, or model revision more effectively than a premature single reconstruction.

The inverse problem also returns us to the example of relational realities that need

not share a common metric space. If two objects are both relationally real, epistemic comparison still requires a warranted comparison relation. Reconstruction cannot assume in advance that heterogeneous domains can be reduced to one metric, one ranking, or one common representation. Sometimes the relevant knowledge concerns structural correspondence; sometimes temporal dependence, functional equivalence, causal interaction, or qualitative difference; sometimes no defensible comparison is yet available. Epistemology must justify the comparison structure rather than inheriting it from the convenience of representation.

4.22.5 Relational Epistemology

The resulting epistemic orientation can now be stated provisionally. GRE requires an epistemology capable of knowing relationally generated, historically changing, multiply manifested, and recursively represented realities without assuming transparent access or final ontological settlement. Such an epistemology must treat evidence, observation, categories, models, archives, and methods as situated within relations while still allowing disciplined claims about realities that exceed those relations of access.

Definition 4.196 (Relational epistemology in GRE). *Relational epistemology in GRE* denotes the study and practice of warranted knowledge concerning relational and generative realities through situated manifestations, traces, representations, comparisons, interventions, and historical reconstructions, together with explicit analysis of the relations that make those forms of access possible and consequential.

Several commitments follow from Definition 4.196. First, knowledge claims are situated without being reduced to perspective. The position of an observer, institution, instrument, archive, or model affects access, but relational reality can constrain and correct the account produced from that position. Second, mediation is epistemically constitutive rather than merely contaminating. Instruments, languages, institutions, categories, and representations can make otherwise inaccessible realities knowable; their mediation must be analyzed rather than imagined away. Third, historical change applies to epistemic relations themselves. Methods, categories, observation systems, and standards of evidence can acquire histories, and their adequacy can change as the realities under study and the interfaces of manifestation change.

Fourth, epistemic plurality can be productive when methods preserve different dimensions of the relational configuration and expose one another's blind spots. This plurality does not mean that all interpretations are equally warranted. Claims remain answerable to evidence, causal structure, internal coherence, historical reconstruction, comparative performance, and the consequences of intervention where relevant. Fifth, uncertainty should be located and structured. The inquiry should distinguish uncertainty about the target, uncertainty about the manifestation relation, uncertainty introduced by representation, uncertainty arising from missing evidence, and uncertainty created by competing explanations.

Sixth, epistemology in GRE is recursively connected to praxis. Research can change what it studies by classifying, measuring, publishing, intervening, allocating attention, or altering institutional expectations. The production of knowledge can therefore generate new relational realities. This fact does not eliminate the possibility of

knowledge; it adds another causal relation that must sometimes be included in the analysis. Epistemic responsibility includes attention to how inquiry enters the generative field.

The three-register architecture developed in this chapter consequently supplies neither a metaphysical closure nor a simple ladder from reality to knowledge. It supplies a structured problem-space. Ontological reality marks questions whose final settlement can remain open. Relational reality identifies emergent, sufficiently persistent and stable, causally efficacious objects within generative-relational systems. Manifestational reality identifies modes through which aspects of those realities become available under particular conditions. Manifestation can precede neither relation nor ontology by conceptual necessity, and manifestation can recursively reshape the relational field once it becomes operative. Symbolic representations can stabilize, compare, compress, transform, and institutionalize manifestations while acquiring causal efficacy of their own. Across these registers, mappings can be partial, plural, historically variable, and non-invertible.

The epistemic problem follows directly. How can an inquiry identify relations that are only partly manifest? How can it distinguish traces from the structures that generate them? How can it learn manifestation rules and detect when those rules change? How can it reason across boundaries and scales, reconstruct the identity and persistence of relations, analyze higher-order or field-like organization, attribute generative effects, and preserve historical trajectories? How should it treat unequal access, strategic concealment, missing records, performative classifications, computational representations, and evidence that changes because the system has learned to respond to observation? These are not secondary methodological details. They are consequences of the ontology developed here.

Chapter 5 takes these consequences as its starting point. It begins with the epistemic problem created by ontological suspension, then develops the inverse problem of manifestation, knowledge of manifestation relations, the distinction between relation and trace, symbolization and representation, performative representation, partial observation, boundary and scale, relational identity, higher-order relations, relational fields, generative relations, attribution, trajectories, uncertainty, comparison, and epistemic revision. The movement of Part II is therefore complete at the level required for transition: Chapter 4 has clarified the kinds of reality GRE permits itself to discuss and the limits of those claims; Chapter 5 asks how warranted knowledge of those realities can be constructed under conditions of mediation, plurality, history, and incomplete access.

The transition can be stated compactly. GRE begins neither with the expectation of transparent reality nor with the conclusion that mediation makes reality inaccessible. It begins with a more demanding proposition: reality and access are related but distinct; the relations between them are themselves historically generated and potentially knowable; and responsible knowledge requires reconstructing those relations with the same care that GRE brings to the economic realities they disclose.

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